Swisher's prime-power supercongruences

Eight conjectures from Table 2

This collection follows the eight targets A, C, D, E, F, G, J and K from the first reductions to the final odd-level E.3 argument. The seven main manuscripts appear in eight entries because the first circulated in two mathematically distinct editions. A separately titled F.3 working proof follows as a supplement.

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First edition. Boundary sums, exact reductions and obstructions.

II **Cubic scaling and two Swisher supercongruences** 16
Revised first manuscript. All levels of C.3 and K.3 in the stated prime ranges.

III **All-level proofs for J.3 and the odd branch of F.3** 39
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Guide to results and prime ranges

The table locates the final theorem statements for the eight targets. Each manuscript retains its own section and equation numbers. A statement described as open in an early paper records its status at that stage.

Target	Range of the statements in the collected proofs	Paper
A.3	Both residue classes for all odd primes, including 3 and 5, at every prescribed level.	IV, V
C.3	All $p > 3$ and $r \geq 1$, modulo p^{4r}.	II
D.3	$p \equiv 1 \pmod 3$ at every level, modulo p^{6r}, and all prescribed levels for $p > 3$, $p \equiv 2 \pmod 3$.	IV, V
E.3	$p > 3$, $p \equiv 2 \pmod 3$, every odd $r \geq 3$, modulo p^{3r-1}. Paper VII treats $r \geq 5$ and VIII completes $r = 3$.	VI–VIII
F.3	$p \equiv 3 \pmod 4$, every odd $r \geq 3$, including $p = 3$. The proved exponent $3(r-1)/2$ attains the target r.	III, S
G.3	All prescribed branches for $p > 3$. The odd supersingular recurrence ($r \geq 3$) also includes $p = 3$.	IV, V
J.3	All $p > 3$ and $r \geq 1$, with the half cutoff and modulus p^{4r}.	III
K.3	All $p > 5$ and $r \geq 1$, modulo p^{4r}, with the sign in the published table.	II

Small primes. At $(p, r) = (3, 1)$ the C.3 and J.3 differences are $-27/16$ and $135/32$, respectively. Both have valuation 3, below the required 4. For the even G.3 recurrence, $G(9) + 27 = 3^5 \cdot 121019/2^{20}$ has valuation 5 where 6 is required. See V, §9.

Relation to published work. The directly proved E.3 and F.3 ranges above are the odd supersingular branches. For other branches see Guo–Zudilin, *Dwork-type supercongruences through a creative q-microscope*, §3, and Guo, *Rocky Mountain J. Math.* 55 (2025), 147–154. The former also establishes B.3 and L.3. For I.3 see Guo, *Int. J. Number Theory* 15 (2019), 29–36. The initial paper cites the separate H.3 manuscript [132] as an input. The reference list for each paper remains with its text.

Editorial record. The first edition was transcribed from its 13-page PDF and checked against the extracted original text. The seven later entries through VIII use the deposited editable sources. Their proof and review claims retain the status stated by the authors. This edition collates the texts and checks the printed cross-references. Five certificate programs for E.3 were rerun. Among the reproduced calculations are 158 higher-level and 56 fifth-level moment values, and nine direct evaluations at $r = 3$.

MANUSCRIPT I

Targeted Reductions for Eight Swisher Congruences

Transfer from Corpus [131] and [132], Exact Obstructions, and Remaining Boundary Sums

Research working note 10 November 2026

Abstract

We investigate (A.3), (C.3), (D.3), (E.3), (F.3), (G.3), (J.3), and (K.3) using the excellent-Frobenius and jet-transport mechanisms in corpus manuscripts [131] and [132]. No new complete resolution of any of these eight conjectures is claimed. The main rigorous outputs are an all-level prefactor congruence and an equivalent residual problem for (K.3), an all-level diagonal congruence and boundary reduction for the odd branch of (F.3), and exact finite block reductions for (C.3) and (J.3). We also prove a parity-sensitive boundary-residual lemma, and give exact counterexamples to several tempting intermediate claims. The counterexamples concern proof transfers, not the original conjectures. Computation is reported separately from proof.

Status. This is an incomplete research note, with explicit proofs of the stated reductions. Historical novelty is not asserted. Classical identities and published input theorems retain their attribution. The corpus manuscripts are examined as proposed mathematical inputs; this note does not certify all of their geometric arguments.

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1 Scope and outcome

The excluded targets are (B.3), (H.3), (I.3), and (L.3). We use Swisher's published Table 2 [1], rather than the initial arXiv version: the signs in (K.3) were corrected in the published table. Throughout, $v_p(p) = 1$ and $(a)_k$ denotes a rising factorial. The natural non-exceptional ranges are stated with each result. We do not infer a current exhaustive literature classification from a failure to locate a later proof.

Target	Outcome of this investigation
A.3	The ordinary target is p^{5r}. A direct first-level Whipple perturbation comparison fails already at $(p, r) = (5, 2)$; the original conjecture is not refuted.
C.3, J.3	Exact terminating evaluations give a common finite three-term obstruction modulo p^{4r}. Its cancellation remains unproved.
D.3	The ordinary target is p^{6r}. No all-level closure is obtained; the boundary lemma also applies to its underlying sixth-power polynomial.
E.3	No completion of the odd supersingular branch is obtained. Its short cutoff does not satisfy the cubic residual precision used in [132].
F.3	The odd-branch diagonal recurrence is proved for every level; an explicit boundary recurrence remains.
G.3	No complete closure is obtained. The fourth-power boundary calculation identifies a transfer hypothesis that still needs repair.
K.3	The entire prefactor recurrence is proved at modulus p^{4r}; the remaining problem is a single p^{3r} residual congruence.

The input corpus snapshot is commit `91c0fa1f0872f070b001249d0ab7c47e3499a907` of https://github.com/QEEDDD/Verifiable-Mathematical-Reasoning-Corpus. We examined the TEX sources of [131], *Excellent Frobenius Lifts and Supercongruences Unified*, and [132], *Swisher H3 Unified Ordinary and Supersingular*. They are not modified here.

2 What the corpus mechanisms do and do not transfer

The modular cusp theorem in [131] (label `B:thm:main`) assumes weight $w \in \{1, 2\}$ and yields precision $p^{(w+1)r}$. The higher-dimensional standard-family theorem yields p^{2r}. An analytic evaluation at a unit argument requires its own integral evaluation map, fixed-point condition, and multiplier identification. These hypotheses cannot be replaced by a formal substitution.

There is an explicit obstruction already proved in [131] (label `E:thm:sh-cube`). Let

$$F_d(u) = \sum_{k\ge 0} \binom{2k}{k}^d u^k, \quad K = p^r, \qquad E_r = F_d(u)F_{K/p}(\sigma(u)) - F_d(\sigma(u))F_K(u),$$

where F_L is truncated at $k < L$ and σ is the excellent lift in that theorem. For $d \ge 4$ and $p > d$,

$$[u^{K+1}]E_r \equiv 2^{2d-1}d(d-3)p^{2r} \pmod{p^{2r+1}}.$$

The right side is nonzero at this precision. Thus the full-series congruence cannot simply be raised from $2r$ to $4r$ or $5r$. For example, $d = 4, p = 7, r = 1$ gives

$$[u^8]E_1 = 27435582641602576, \qquad v_7([u^8]E_1) = 2.$$

Even applying $1 + 4u\,d/du$ leaves an obstruction: its coefficient is multiplied by $4K + 5$, a 7-adic unit. This excludes a naive global derivative upgrade; it does not exclude additional cancellation at a particular argument. This obstruction is from [131], not a new result of the present note.

For clarity, there is an exact weighted transfer identity. Write $D_r = F_r - \rho F_{r-1} \circ \sigma$, $\theta = t\,d/dt$, and $\mu = \theta\sigma/\sigma$. At a valid analytic fixed point t_0, if

$$c\theta\rho(t_0) = \rho(t_0)(\mu(t_0) - 1), \qquad \lambda = \rho(t_0)\mu(t_0),$$

then the product and chain rules give

$$(1 + c\theta)F_r(t_0) - \lambda(1 + c\theta)F_{r-1}(t_0) = (1 + c\theta)D_r(t_0).$$

Differentiation preserves an established integral modulus, but does not itself increase it. The remaining weighted residual needs a separate estimate. Likewise, the [132] Taylor-transport argument requires the precise residual order of its differential equation; Section 7 below calculates what happens at the short odd cutoffs of other families.

3 An exact reduction of Swisher's $(K.3)$

For each odd positive integer N, put $h_N = (N-1)/2$ and define

$$K(N) = \sum_{k=0}^{h_N} (42k+5)\frac{(1/2)_k^3}{k!^3 64^k}, \tag{1}$$

$$B(N) = \frac{12\displaystyle\prod_{j=0}^{N-1}(N+2j)}{2^{3N-2}(h_N!)^2}, \tag{2}$$

$$U(N) = \sum_{n=0}^{N} \frac{(-N/2)_n^2}{n!(-3N/2)_n}. \tag{3}$$

All three quantities are rational numbers. In particular, $K(1) = 5$, $B(1) = 6$, and $U(1) = 5/6$. We use the positive sign in Swisher's published Table 2: with $\chi_p = (-1)^{(p-1)/2}$, the target is

$$K(p^r) \equiv \chi_p p K(p^{r-1}) \pmod{p^{4r}}, \qquad p > 5. \tag{4}$$

The first arXiv version has an erroneous additional minus sign in this entry; the published version corrects it.

Proposition 3.1 (Exact factorization)**.** *For every odd positive integer N,*

$$K(N) = B(N)U(N). \tag{5}$$

Proof. Use the Guillera WZ pair employed by Osburn and Zudilin in their proof of $(K.2)$, namely

$$F(n,k) = (84n^2 - 56nk + 4k^2 + 52n - 12k + 5) \cdot \frac{(-1)^k (1/2)_n (1/2)_{n+k} (1/2)_{n-k}^2}{2^{4n}(n!)^2(2n-k+1)!},$$

$$G(n,k) = \frac{64n^2(-1)^k (1/2)_n (1/2)_{n+k-1} (1/2)_{n-k}^2}{2^{4n}(n!)^2(2n-k)!}.$$

Negative-index Pochhammer symbols are defined by reciprocals, and a reciprocal factorial at a negative integer is zero. The pair satisfies

$$F(n,k-1) - F(n,k) = G(n+1,k) - G(n,k).$$

Moreover $F(n,0) = (42n+5)(1/2)_n^3/(64^n n!^3)$. Summing first over $0 \le n \le h_N$ and then over $1 \le k \le N$ gives, exactly as in Osburn–Zudilin's equation (2.6),

$$K(N) = F(h_N, N) + \sum_{k=1}^{N} G(h_N+1, k).$$

This telescoping identity requires only that N be odd, not prime. Direct simplification gives

$$F(h_N, N) = B(N), \qquad G(h_N+1, k) = B(N)\frac{(-N/2)_{N+1-k}^2}{(N+1-k)!(-3N/2)_{N+1-k}}.$$

Changing the index to $n = N+1-k$ supplies the terms $n = 1, \dots, N$ in (3); the boundary value supplies $n = 0$. □

Theorem 3.2 (All-level congruence for the prefactor)**.** *For every prime $p \ge 5$ and integer $r \ge 1$,*

$$B(p^r) \equiv \chi_p p B(p^{r-1}) \pmod{p^{4r}}, \qquad v_p(B(p^r)) = r. \tag{6}$$

Proof. Set $N = p^r$, $M = p^{r-1}$ and $t = (N-M)/2$, and write

$$V = \prod_{\substack{1 \le j \le h_N \\ p \nmid j}} j, \quad O = \prod_{\substack{1 \le a < N \\ a \ \mathit{modd},\ p \nmid a}} a, \quad W = \prod_{\substack{0 \le j < N \\ p \nmid j}} (N+2j).$$

Removing the factors divisible by p from the numerator and the factorials in (2) yields the exact equality

$$\frac{B(N)}{pB(M)} = \frac{W}{2^{3(N-M)}V^2}. \tag{7}$$

Pairing the factors in W about $2N$ gives another exact equality,

$$W = (-1)^t O^2 \prod_{\substack{1 \le a < N \\ a \ \mathit{modd},\ p \nmid a}} \left(1 - \frac{4N^2}{a^2}\right). \tag{8}$$

We claim that the last product is 1 modulo p^{3r}. Indeed, the inverse-square sum over all units modulo N vanishes modulo N. To see this, choose a unit g with $g^2 \not\equiv 1 \pmod p$; multiplication by g permutes those units and forces the sum to be annihilated by the unit $g^{-2} - 1$. Pairing j with $N-j$ then shows that the inverse-square sum over $1 \le j \le h_N$, $p \nmid j$, also vanishes modulo N. The sum over odd units $a < N$ is the full sum minus one quarter of this half-interval sum, so it vanishes as well. Consequently the linear term in the product in (8) is divisible by p^{3r}; every term involving two or more selected factors is divisible by p^{4r}.

Let $C_N = \binom{N-1}{h_N}$. Its p-free quotient is

$$\frac{C_N}{C_M} = 2^t \frac{O}{V}. \tag{9}$$

Cai's prime-power extension of Morley's congruence [3, Corollary 3] states that

$$\frac{C_N}{C_M} \equiv \chi_p 4^{N-M} \pmod{p^{3r}}.$$

Since $(-1)^t = \chi_p$, substituting (8) and (9) into (7) gives

$$\frac{B(N)}{pB(M)} \equiv \chi_p \left(\frac{C_N}{C_M}\right)^2 2^{-4(N-M)} \equiv \chi_p \pmod{p^{3r}}.$$

The exact quotient in (7) is a p-adic unit. Starting with $B(1) = 6$, induction therefore gives $v_p(B(M)) = r-1$. Multiplication by $pB(M)$ raises the precision to p^{4r} and proves both assertions. □

Proposition 3.3 (Integrality and equivalent remaining congruence)**.** *For $p > 5$ and $r \geq 0$, $U(p^r) \in \mathbb{Z}_p^\times$ and $U(p^r) \equiv 5/6 \pmod{p}$. For every $r \geq 1$, (4) holds if and only if*

$$\boxed{U(p^r) \equiv U(p^{r-1}) \pmod{p^{3r}}.} \tag{10}$$

Proof. For $N = p^r$ and $1 \leq n \leq N$, write the summand in $U(N)$ as

$$f(N,n) = -\frac{N}{6n} \prod_{j=1}^{n-1} \frac{(1 - N/(2j))^2}{1 - 3N/(2j)}. \tag{11}$$

For $j < N$, $v_p(N/j) \geq 1$, so every factor in this product is a p-adic unit congruent to 1 modulo p. It follows that $v_p(f(N,n)) = r - v_p(n) \geq 0$. Only $n = N$ survives modulo p among $n \geq 1$, and that term is $-1/6$ modulo p. Thus $U(N) \equiv 1 - 1/6 = 5/6 \pmod{p}$.

Finally the exact identity (5) gives

$$\begin{aligned} K(N) - \chi_p p K(M) = {} & B(N)(U(N) - U(M)) \\ & + (B(N) - \chi_p p B(M))U(M). \end{aligned}$$

The second term belongs to $p^{4r}\mathbb{Z}_p$ by Theorem 3.2 and the integrality just proved. The valuation of $B(N)$ is exactly r, proving the equivalence. □

Unresolved step. Equation (10) is not proved here. The reduction removes a prefactor whose required precision is now established for all levels; it does not establish the remaining sum congruence or claim novelty for the reformulation. Splitting (11) into p-divisible and p-free indices produces correction products, but termwise estimates do not attain the required exponent $3r$. An additional cancellation in their total sum remains necessary.

3.1 A further harmonic reduction of the K residual

This subsection uses the $U(N)$ of the preceding K reduction. Put

$$P_j(N) = \prod_{k=1}^{j} \frac{(1 - N/(2k))^2}{1 - 3N/(2k)}, \qquad P_0(N) = 1.$$

For every odd positive integer N,

$$U(N) = \frac{4}{3} - \frac{1}{2}P_{N-1}(N) + \frac{1}{3}T(N), \qquad T(N) = N^2 \sum_{n=1}^{N-1} \frac{P_{n-1}(N)}{n^2(1 - 3N/(2n))}. \tag{12}$$

Indeed the nth summand of U is $-NP_{n-1}(N)/(6n)$, and

$$f(x) = \frac{(1 - x/2)^2}{1 - 3x/2}, \qquad f(x) - 1 = \frac{x}{2} + \frac{x^2}{1 - 3x/2}.$$

Summing the resulting difference $P_n - P_{n-1}$ for $1 \leq n < N$ and then adding the $n = N$ term proves (12).

Proposition 3.4. *For $p > 5$, $N = p^r$, and $M = p^{r-1}$,*

$$P_{N-1}(N) \equiv P_{M-1}(M) \pmod{p^{3r}}.$$

Consequently (K.3) is equivalent to $T(p^r) \equiv T(p^{r-1}) \pmod{p^{3r}}$. The last congruence remains unproved.

Proof. Remove the factors indexed by multiples of p from the quotient of the two products. What remains is $\prod_{1\le k<N,\ p\nmid k} f(N/k)$. On the full unit residue set modulo N,

$$H_2 = \sum_{\substack{1\le k<N\\ p\nmid k}} k^{-2} \in p^r\mathbb{Z}_p, \qquad H_1 = \sum_{\substack{1\le k<N\\ p\nmid k}} k^{-1} \in p^{2r}\mathbb{Z}_p.$$

For the first claim multiply the residues by a unit whose square is not 1 modulo p. For the second, pair k with $N-k$ and expand modulo N^2, obtaining $2H_1 \equiv -NH_2 \pmod{N^2}$. Now

$$\log f(x) = \sum_{\ell\ge1} \frac{3^\ell - 2}{\ell 2^\ell} x^\ell.$$

The first two terms summed over the units lie in $p^{3r}\mathbb{Z}_p$ by these harmonic bounds. Every later term has valuation at least $\ell r - v_p(\ell) \ge 3r$. The convergent logarithm and exponential therefore prove the product congruence. Equation (12) and the K equivalence above prove the last assertion, since 3 is a unit. □

4 An exact boundary reduction of the odd branch of (F.3)

Let $p \equiv 3 \pmod 4$ be a prime. For odd $r \ge 1$ put

$$N_r = (p^r - 3)/4, \qquad S_r = \sum_{n=0}^{N_r} (-1)^n (8n+1) \frac{(1/4)_n^3}{(n!)^3}.$$

Define

$$D_r = (8N_r + 1)\frac{(1/4)_{2N_r}}{(N_r!)^2}, \qquad B_r = \sum_{k=1}^{N_r} G(N_r+1, k),$$

where an empty sum is zero and

$$G(n,k) = (-1)^{n+k} 4 \frac{(1/4)_n^2 (1/4)_{n+k-1}}{(n-1)!^2 (n-k)! (1/4)_k^2}.$$

Proposition 4.1. *For every such p, r, $S_r = D_r + B_r$. For odd $r \ge 3$,*

$$v_p(D_r) = r - 1, \qquad D_r \equiv p^2 D_{r-2} \pmod{p^r}.$$

Consequently the odd branch of Swisher's original (F.3) is equivalent to

$$\boxed{B_r \equiv p^2 B_{r-2} \pmod{p^r}} \qquad (r \ge 3 \text{ odd}).$$

This boxed boundary recurrence is not proved here.

Proof. Set $a = 1/4$ and

$$F(n,k) = (-1)^{n+k}(8n+1)\frac{(a)_n^2 (a)_{n+k}}{(n!)^2 (n-k)! (a)_k^2}.$$

Use $1/m! = 0$ for negative integers m, and set $G(0,k) = 0$. The WZ identity

$$F(n, k-1) - F(n,k) = G(n+1,k) - G(n,k)$$

is an exact rational identity. Indeed its four quotients after division by $F(n,k)$ are

$$\frac{F(n,k-1)}{F(n,k)} = -\frac{(a+k-1)^2}{(a+n+k-1)(n-k+1)},$$

$$\frac{G(n+1,k)}{F(n,k)} = -\frac{(a+n)^2}{(2n+a)(n-k+1)}, \qquad \frac{G(n,k)}{F(n,k)} = \frac{n^2}{(a+n+k-1)(2n+a)}.$$

Substitution proves the identity, including the boundary cases by the zero convention. Summing first over $0 \le n \le N$, then over $1 \le k \le N$, gives

$$\sum_{n=0}^{N} F(n,0) = F(N,N) + \sum_{k=1}^{N} G(N+1,k).$$

Since $F(N,N) = (8N+1)(a)_{2N}/(N!)^2$, this proves the decomposition. The WZ pair appears as Lemma 3.1 in Jana–Kalita, *Proof of a supercongruence conjecture of (F.3) of Swisher using the WZ-method*, arXiv:2011.02762v1. That article was withdrawn for errors; only the identity, independently verified above, is used here.

Write $R_r = (1/4)_{2N_r}/(N_r!)^2$. Let Γ_p be Morita's gamma function. Extracting the factors divisible by p twice from the rising factorial and the factorial gives the exact formula

$$\frac{R_r}{p^2 R_{r-2}} = \left(\frac{p^{r-2}}{2} - \frac{5}{4}\right)\left(\frac{p^{r-2}}{2} - \frac{1}{4}\right) \frac{\Gamma_p(p^r/2 - 5/4)\Gamma_p(p^{r-1}/2 + 1/4)}{\Gamma_p(1/4)\Gamma_p(3/4)\Gamma_p((p^r+1)/4)^2\Gamma_p((p^{r-1}+3)/4)^2}. \qquad (*)$$

For completeness, in $(1/4)_{(p^r-3)/2}$ the first extraction leaves $p^{(p^{r-1}-1)/2}(3/4)_{(p^{r-1}-1)/2}$. The second leaves $p^{(p^{r-2}+1)/2}(1/4)_{(p^{r-2}+1)/2}$. Removing the final two factors of this last rising factorial gives the two prefactors in (*). In $N_r!$, the two factorials left are indexed by $(p^{r-1}-1)/4$ and N_{r-2}. Comparing the powers of p gives exactly p^2. The gamma signs in the numerator multiply to $+1$ because $(p^r-3)/2$ and $(p^{r-1}-1)/2$ are even; denominator signs are squared.

All the factors on the right of (*) are p-adic units: $p \neq 5$. Modulo p, its value is

$$\frac{5}{16}\frac{\Gamma_p(-5/4)}{\Gamma_p(1/4)^2\Gamma_p(3/4)^3} = 1.$$

The equality follows from $\Gamma_p(-5/4) = \frac{16}{5}\Gamma_p(3/4)$ and $\Gamma_p(1/4)^2\Gamma_p(3/4)^2 = 1$. These are the gamma functional equation and reflection formula. The base R_1 is a unit: $(p-3)/2 < (3p-1)/4$ is below the first p-divisible factor in $(1/4)_j$, and $N_1 < p$. Induction using (*) proves $v_p(R_r) = r-1$ and $R_r \equiv p^2 R_{r-2} \pmod{p^r}$. Finally, $8N_r + 1 = 2p^r - 5$ is a unit and is congruent to $2p^{r-2} - 5$ modulo p. Multiplying proves both claims for D_r. Subtracting the diagonal recurrence from the exact decomposition proves the stated equivalence. □

A precise obstruction to deleting the boundary. The withdrawn article's Lemma 3.2 asserts $v_p(G((p^r+1)/4,k)) \ge 3(r-1)/2$ throughout the relevant range. For $p = 7, r = 3, k = 13$, this is false:

$$v_7(G(86,13)) = 1, \qquad \frac{G(86,13)}{7} \equiv 34 \pmod{49}.$$

For an entirely integer valuation check, let $A(n) = \sum_{j=0}^{n-1} v_7(4j+1)$ and $C(n) = v_7(n!)$. Then

$$A(86) = 14, \quad A(98) = 16, \quad A(13) = 3, \quad C(85) = 13, \quad C(73) = 11,$$

so $v_7 G = 2(14) + 16 - 2(13) - 11 - 2(3) = 1$. Direct rational arithmetic gives

$$D_3 \equiv 245, \qquad B_3 \equiv 49, \qquad S_3 \equiv 294 \pmod{343}.$$

Thus the boundary is essential, even though the original conjecture is satisfied in this test. These computations diagnose the invalid termwise bound, and do not constitute a proof of the new boundary recurrence.

5 The C.3 and J.3 deformation reductions

For $p > 3$, put $N = p^r$, $n_r = (N-1)/2$, and

$$c_k = (4k+1)\frac{\binom{2k}{k}^4}{256^k}, \qquad j_k = (6k+1)\frac{\binom{2k}{k}^3}{256^k}, \quad C_r = \sum_{k=0}^{n_r} c_k, \quad J_r = \sum_{k=0}^{n_r} j_k.$$

Here $C_0 = J_0 = 1$, and $\varepsilon = (-1)^{(p-1)/2}$. The respective targets are $C_r - pC_{r-1} \in p^{4r}\mathbb{Z}_p$ and $J_r - \varepsilon p J_{r-1} \in p^{4r}\mathbb{Z}_p$. Guo's published all-level result has modulus p^{3r} for both sums (https://arxiv.org/abs/1910.07551).

For $d = 2, 4$, define the finite rational product

$$R_d(k,x) = \prod_{a=0}^{k-1} \frac{1 - x^2/(2a+1)^2}{1 - x^2/[d(a+1)]^2}.$$

Dougall's terminating very-well-poised summation and the terminating identity used in Long's Lemma 4.2, respectively, give for every positive odd integer N the exact identities

$$\sum_{k=0}^{(N-1)/2} c_k R_2(k,N) = N, \tag{13}$$

$$\sum_{k=0}^{(N-1)/2} j_k R_4(k,N) = (-1)^{(N-1)/2} N. \tag{14}$$

The second identity is recorded at https://msp.org/pjm/2011/249-2/pjm-v249-n2-p09-p.pdf, pp. 414–415; replacing the letter p there by an odd integer does not change the terminating evaluation. The first is the specialization $a = b = 1/2$, $c = (1+N)/2$, and terminating parameter $-(N-1)/2$ of Dougall's formula. Its right-hand side is

$$\frac{(3/2)_{(N-1)/2}((1-N)/2)_{(N-1)/2}}{(1)_{(N-1)/2}(1-N/2)_{(N-1)/2}} = N.$$

These exact classical evaluations are not proofs of the desired undeformed recurrences.

5.1 An exact finite separation of the factors divisible by p

The following reduction avoids any unjustified integral Taylor expansion. Write $k = pt + e$, $0 \le e < p$, put $h = (p-1)/2$, $M = p^{r-1}$, and $\delta_e = 1$ if $e > h$, and 0 otherwise. Splitting the products according to divisibility by p proves exactly

$$R_d(k,N) = R_d(t,M)\left(1 - \frac{M^2}{(2t+1)^2}\right)^{\delta_e} U_d(k,N),$$

where

$$U_d(k,N) = \frac{\prod_{0 \le a < k,\ p \nmid 2a+1}(1 - N^2/(2a+1)^2)}{\prod_{1 \le a \le k,\ p \nmid a}(1 - N^2/(da)^2)}.$$

Every denominator in this last product is a p-adic unit. Consequently, with

$$W_d(k) = \sum_{\substack{0 \le a < k \\ p \nmid 2a+1}} \frac{1}{(2a+1)^2} - \frac{1}{d^2} \sum_{\substack{1 \le a \le k \\ p \nmid a}} \frac{1}{a^2},$$

one has the rigorous, uniform expansion

$$U_d(k,N) \equiv 1 - N^2 W_d(k) \pmod{p^{4r}}.$$

The other factors above are integral for $k \le n_r$. In the last block, $t = (M-1)/2$, only $e \le h$ occurs, so the potentially zero extra factor is never included. This proves that this expansion can safely be multiplied by all the remaining factors and summed.

Here is an entirely finite expression for the remaining obstruction. Use $(w,d,\eta) = (c,2,1)$ or $(j,4,\varepsilon)$ and set

$$L_t = R_d(t,M), \quad z_t = \frac{M^2}{(2t+1)^2}, \quad E_t = \{e : 0 \le e < p,\ pt+e \le n_r\},$$

$$V_t = \sum_{e\in E_t} w_{pt+e}, \qquad B_t = \sum_{\substack{e\in E_t \\ e>h}} w_{pt+e}, \qquad H_t = \sum_{e\in E_t} w_{pt+e}(1-z_t)^{\delta_e} W_d(pt+e).$$

Then the exact terminating identities at levels r and $r-1$, followed by the preceding expansion, prove

$$\sum_{k=0}^{n_r} w_k - \eta p \sum_{k=0}^{n_{r-1}} w_k \equiv \sum_{t=0}^{n_{r-1}} (1-L_t)(V_t - \eta p w_t) + \sum_{t=0}^{n_{r-1}} L_t z_t B_t + N^2 \sum_{t=0}^{n_{r-1}} L_t H_t \pmod{p^{4r}}. \tag{15}$$

Indeed the evaluated deformation at level r is $\eta^r N$, the lower-level evaluation multiplied by ηp is the same, and the k-block of the higher-level evaluation is $L_t(V_t - z_t B_t - N^2 H_t)$ modulo p^{4r}. Rearrangement gives the display. All quantities in this congruence are p-integral. Thus either original conjecture is equivalent to the vanishing modulo p^{4r} of its displayed finite obstruction. We have not proved this vanishing. In particular, none of its three summands may be suppressed without a separate bound. For example, at $(p,r) = (5,2)$ their valuations are $(7,7,7)$ for C and $(6,6,6)$ for J, while the valuation of their total is 8 in both cases. At $(5,3)$ the three valuations are $(9,9,10)$ for C and $(10,10,10)$ for J, while both totals have valuation 12. These exact rational checks exhibit the cross-term cancellation that a proof must control; a termwise $4r$ bound is already false in these examples.

5.2 Taylor formulation and its precision limitation

For comparison put

$$a_{r,j}^X = [x^{2j}] \sum_{k=0}^{n_r} w_k R_d(k,x), \qquad X = C, J.$$

For each fixed r, this expansion converges at $x = p^r$: every denominator has p-adic valuation at most $r-1$, so $x^2/[d(a+1)]^2 \in p^2\mathbb{Z}_p$. One obtains the exact convergent identity

$$\Delta_X(r) = -\sum_{j\ge 1} p^{2rj} \left(a_{r,j}^X - \eta p^{1-2j} a_{r-1,j}^X \right).$$

For $r = 0$, take $a_{0,0} = 1$ and $a_{0,j} = 0$ for $j > 0$. The quadratic coefficient is explicitly

$$a_{r,1}^X = \sum_{k=0}^{n_r} w_k \left(\frac{1}{d^2} H_k^{(2)} - \sum_{a=0}^{k-1} \frac{1}{(2a+1)^2} \right).$$

The coefficients need not be integral uniformly in r. Exact rational calculation for $p = 5$ gives $v_5(a_{r,1}^C) = v_5(a_{r,1}^J) = 2,1,0,-1$ for $r = 1,2,3,4$, respectively. Thus simply replacing p by p^r in the first-level integral Taylor argument is invalid. The finite factor separation above is one way to retain correct valuation bookkeeping.

5.3 Finite calculations, not an all-level proof

The program `probe.py` checks the original recurrences for $p \in \{5,7,11,13,17,19,23,29,31\}$ and $1 \le r \le 5$ with $p^r \le 200000$, at precision p^{4r+4}: 36 pairs (p,r) for each conjecture. All passed. Except for $p=7$, the tested defects had exact valuation $4r$; at $p=7$ the C defect had valuation 6 at $r=1$ and $4r+1$ thereafter, while the J defect had valuation $4r+1$ throughout. The additional observed relation

$$\frac{\Delta_C(r)}{p^{4r}} \equiv \frac{7}{6}B_{p-3}, \qquad \frac{\Delta_J(r)}{p^{4r}} \equiv \varepsilon^r \frac{7}{24}B_{p-3} \pmod p$$

is an unproved pattern in this finite sample, not a theorem or an assertion of novelty. The directly computed leading digits are stored in `probe.jsonl`; `validate.py` independently checks all 72 Bernoulli leading-digit comparisons, eight exact rational instances of the finite block identity, and the two negative-valuation examples.

6 An exact obstruction to a direct $(A.3)$ lift

Define, for odd N,

$$A(N) = \sum_{k=0}^{(N-1)/2} (4k+1)(-1)^k \frac{(1/2)_k^5}{k!^5}.$$

The four-root perturbation in Swisher's first-level argument, with p replaced by N, is the following rational number:

$$\widetilde{A}(N) = \sum_{k=0}^{(N-1)/2} (4k+1)(-1)^k \frac{(1/2)_k^5}{k!^5} \frac{\displaystyle\prod_{j=0}^{k-1}\left(1-\frac{N^4}{(2j+1)^4}\right)}{\displaystyle\prod_{j=1}^{k}\left(1-\frac{N^4}{16j^4}\right)}. \tag{16}$$

It is the terminating very-well-poised ${}_6F_5(-1)$ with upper parameters $5/4, 1/2, (1-iN)/2, (1+iN)/2, (1-N)/2, (1+N)/2$ and lower parameters $1/4, 1+iN/2, 1-iN/2, 1+N/2, 1-N/2$. Whipple's transformation and the classical gamma reflection identity give the exact identity

$$\widetilde{A}(N) = (-1)^{(N-1)/2} N \, {}_3F_2\left[\begin{matrix}1/2, (1-N)/2, (1+N)/2 \\ 1+iN/2, 1-iN/2\end{matrix}; 1\right]. \tag{17}$$

The series on the right terminates at $(N-1)/2$. In particular it is a *perturbed* cubic hypergeometric sum; it is not the unperturbed sum in $(H.3)$.

A direct extension of the first-level comparison would require

$$\widetilde{A}(p^r) \equiv A(p^r) \pmod{p^{5r}}.$$

This proposed intermediate statement is false. Exact rational arithmetic at $p=5$, $r=2$ gives

$$\widetilde{A}(25) - A(25) \equiv 7062500 \pmod{5^{10}}, \qquad v_5\big(\widetilde{A}(25) - A(25)\big) = 6. \tag{18}$$

Equivalently, its quotient by 5^6 is 2 modulo 5. This is a counterexample to the proposed intermediate comparison, *not* a counterexample to $(A.3)$.

The calculation can be reproduced without complex arithmetic: if t_k is the kth term of (16) after deleting $4k+1$, then $t_0 = 1$ and

$$\frac{t_k}{t_{k-1}} = -\frac{(2k-1)\big((2k-1)^4 - N^4\big)}{2k\big(16k^4 - N^4\big)}.$$

For the eight tested pairs $p = 5, 13$, $r = 1, 2, 3$, and $p = 17$, $r = 1, 2$, the exact valuation of $\widetilde{A}(p^r) - A(p^r)$ is $r + 4$. These finite observations are not asserted as a general valuation theorem.

There is a corresponding obstacle to replacing the right side of (17) by the unperturbed $(H.3)$ sum. Writing

$$H(N) = \sum_{k=0}^{(N-1)/2} \frac{(1/2)_k^3}{k!^3},$$

the six exact tests $p = 5, 13$, $r = 1, 2, 3$ give

$$v_p\big(A(p^r) - p^r H(p^r)\big) = r + 3, \qquad v_p\big(\widetilde{A}(p^r) - p^r H(p^r)\big) = r + 3.$$

Thus an all-level theorem for H does not by itself control the parameter corrections required for A. Terms with indices divisible by p introduce lower-level denominator valuations, and their corrections must be transported between levels before the exponent $5r$ can be justified.

7 Exact boundary residuals and an odd-level obstruction

The following elementary statement isolates a hypothesis needed to transfer the jet argument of corpus [132]. It is a result about finite hypergeometric polynomials, not a proof of any remaining Swisher conjecture.

Proposition 7.1 (Boundary residual and parity of its valuation)**.** *Let $d \geq 3$, let $p > d$ be a prime with $p \equiv -1 \pmod{d}$, and let $h \geq 1$ be an integer. Set*

$$c_k = \left(\frac{(1/d)_k}{k!}\right)^h, \qquad F_N(z) = \sum_{k=0}^{N} c_k z^k, \qquad \theta = z\frac{d}{dz}, \qquad \mathcal{L}_{d,h} = \theta^h - z(\theta + 1/d)^h.$$

For every $N \geq 0$ there is the exact identity

$$\mathcal{L}_{d,h} F_N(z) = -(N + 1/d)^h c_N z^{N+1}. \tag{19}$$

If $s \geq 1$ is odd and

$$N_- = \frac{p^s - d + 1}{d}, \qquad N_+ = \frac{(d-1)p^s - 1}{d},$$

then

$$v_p\left(\frac{(1/d)_{N_-}}{N_-!}\right) = \frac{s-1}{2}, \qquad v_p\Big((N_- + 1/d)^h c_{N_-}\Big) = \frac{h(s-1)}{2}, \tag{20}$$

$$v_p\left(\frac{(1/d)_{N_+}}{N_+!}\right) = 0, \qquad v_p\Big((N_+ + 1/d)^h c_{N_+}\Big) = hs. \tag{21}$$

If $s \geq 2$ is even and $N_0 = (p^s - 1)/d$, then

$$v_p\left(\frac{(1/d)_{N_0}}{N_0!}\right) = 0, \qquad v_p\Big((N_0 + 1/d)^h c_{N_0}\Big) = hs. \tag{22}$$

Proof. The coefficient recurrence $k^h c_k = (k - 1 + 1/d)^h c_{k-1}$ cancels every interior coefficient in $\mathcal{L}_{d,h} F_N$, giving (19).

For the valuation of $(1/d)_N/N!$, count the factors divisible by $t = p^e$ in $\prod_{j=0}^{N-1}(dj+1)$ and subtract $\lfloor N/t \rfloor$. The least nonnegative solution to $dj + 1 \equiv 0 \pmod{t}$ is

$$j_e = \begin{cases} (t-1)/d, & e \text{ even}, \\ ((d-1)t - 1)/d, & e \text{ odd}. \end{cases}$$

Writing $N = qt + R$ with $0 \le R < t$, the difference of the two counts is 1 if $j_e < R$, and 0 otherwise.

For odd s and $N = N_-$, the remainder for $1 \le e < s$ is

$$R = \begin{cases} (d-1)(t-1)/d, & e \text{ even}, \\ (t-d+1)/d, & e \text{ odd}. \end{cases}$$

Thus the count difference is 1 exactly when e is even. There is no contribution for $e \ge s$, since both $N_- < p^s$ and $d(N_- - 1) + 1 = p^s - 2d + 2 < p^s$. This proves the first identity of (20). Moreover $N_- + 1/d = (p^s - d + 2)/d$ is a p-adic unit because $p > d$. The second identity follows.

For $N = N_+$, its remainder modulo p^e equals j_e for $1 \le e \le s$, so every count difference is zero. No larger power contributes: $N_+ < p^s$, and the numerator factors are less than $(d-1)p^s < p^{s+1}$. Also $N_+ + 1/d = (d-1)p^s/d$, of valuation s. This proves (21).

For even s and $N = N_0$, the remainder again equals j_e for $1 \le e \le s$; there are no larger contributions. Since $N_0 + 1/d = p^s/d$, (22) follows. □

Two exact examples. For $d = 3$, $p = 5$, and $s = 3$, the two cutoffs are $N_- = 41$ and $N_+ = 83$. Their top-coefficient ratios have valuations 1 and 0, respectively. The boundary residual coefficients therefore have valuations h and $3h$; for cubic sums these are 3 and 9. For $d = 4$, $p = 7$, and $s = 3$, the cutoffs are $N_- = 85$ and $N_+ = 257$, with the same respective ratio valuations 1 and 0 and boundary residual valuations h and $3h$.

Consequence for transferring the method. For the unweighted hypergeometric polynomial underlying the short odd cutoffs, the residual has exact valuation $h(s-1)/2$, not hs. Consequently one cannot import the p^{3s} residual hypothesis of corpus [132] unchanged into the cubic cases $d = 3, 4$. The statement also records the boundary valuation for exponents $h = 4$ or $h = 6$ if those polynomials occur in a proposed argument. It does not identify their differential modules or establish a Frobenius congruence for them.

The longer odd cutoff restores the boundary precision hs. A possible repair must therefore establish both the appropriate Frobenius theorem for that polynomial and a sufficiently precise relation between its long and short truncations. Neither follows from the valuation calculation alone. The linear weight and specialization in each Swisher sum also have to be accounted for separately.

8 Verification and the unresolved statements

All finite checks use integer modular arithmetic with explicit p-factor extraction or exact rational arithmetic. No floating-point tolerance is used. The supplied `verify_reductions.py` independently checks 26 K factorizations, eight K prefactor cases, four F diagonal recurrences, five F decompositions, the explicit false F termwise bound, the false A perturbation bridge, and five parameter sets in the parity lemma. Every check passes.

The larger C/J program checks 36 pairs (p, r) for each conjecture, and all 72 original congruences pass. Its separate validator checks the finite block reduction in eight cases and confirms the displayed nonintegral Taylor coefficient. These tests support the calculations; they do not prove the remaining universal statements.

The exact new work target is now especially explicit for K and F:

$$U(p^r) - U(p^{r-1}) \in p^{3r}\mathbb{Z}_p, \quad \text{equivalently} \quad T(p^r) - T(p^{r-1}) \in p^{3r}\mathbb{Z}_p,$$

and

$$B_r - p^2 B_{r-2} \in p^r \mathbb{Z}_p \quad \text{for the F boundary sum and odd } r \ge 3.$$

Neither is established by the preceding product or diagonal congruences. For C and J the displayed finite three-term obstruction must vanish as a sum; its components need not separately have the target valuation. For A, a parameter-deformation correction must be transported between levels. For D, E, and G this investigation has not supplied the required all-level cancellation. These precise limitations are part of the conclusions, not omitted proof steps.

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MANUSCRIPT II

Cubic Scaling and Two Swisher Supercongruences

Complete Proofs of (C.3) and (K.3), with Six Remaining Targets

Research working note 10 November 2026

Abstract

We prove a cubic scaling theorem for finite products of an arbitrary power series $f \in 1 + X\mathbb{Z}_p[[X]]$, for every prime $p > 3$. As a consequence, a truncated generalized hypergeometric sum whose parameters are all proportional to p^r is stable modulo p^{3r} when its scale is divided by p. An exact WZ factorization and Cai's prime-power Morley congruence then prove Swisher's published (K.3), with its full modulus p^{4r}, for every $p > 5$ and $r \geq 1$. A second exact WZ identity converts (C.3) to a weighted convolution; alternating unit-prefix sums and summation by parts prove its full p^{4r} recurrence for every $p > 3$ and $r \geq 1$. We retain the rigorous reductions for (A.3), (D.3), (E.3), (F.3), (G.3), and (J.3), and sharpen the remaining J target to a normalized parity-product stability statement. Those six targets are not claimed solved. The scaling proofs are elementary and do not depend on the geometric claims of corpus [131] or [132].

Status. This revision contains complete proofs of (C.3) and (K.3), including every all-level estimate used in their reductions. The argument has undergone independent internal checks; it has not undergone external peer review. Historical priority is not asserted. Classical identities and published input theorems retain their attribution. The corpus manuscripts were consulted as proposed mathematical inputs; this note does not certify all of their geometric arguments.

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1 Scope and outcome

The excluded targets are (B.3), (H.3), (I.3), and (L.3). We use Swisher's published Table 2 [1], rather than the initial arXiv version: the signs in (K.3) were corrected in the published table. Throughout, $v_p(p) = 1$ and $(a)_k$ denotes a rising factorial. The natural non-exceptional ranges are stated with each result. We do not infer a current exhaustive literature classification from a failure to locate a later proof.

Target	Outcome of this investigation
A.3	The ordinary target is p^{5r}. A direct first-level Whipple perturbation comparison fails already at $(p, r) = (5, 2)$; the original conjecture is not refuted.
C.3	Complete proof at p^{4r}, for $p > 3$ and every $r \geq 1$.
J.3	The endpoint multiplier is eliminated, leaving a single normalized parity-product recurrence. This remains unproved.
D.3	The ordinary target is p^{6r}. No all-level closure is obtained; the boundary lemma also applies to its underlying sixth-power polynomial.
E.3	No completion of the odd supersingular branch is obtained. Its short cutoff does not satisfy the cubic residual precision used in [132].
F.3	The odd-branch diagonal recurrence is proved for every level; an explicit boundary recurrence remains.
G.3	No complete closure is obtained. The fourth-power boundary calculation identifies a transfer hypothesis that still needs repair.
K.3	The entire prefactor recurrence is proved at modulus p^{4r}; the residual recurrence is now also proved at p^{3r}, giving a complete proof of (K.3).

The input corpus snapshot is commit `91c0fa1f0872f070b001249d0ab7c47e3499a907` of https://github.com/QEEDDD/Verifiable-Mathematical-Reasoning-Corpus. We examined the TEX sources of [131], *Excellent Frobenius Lifts and Supercongruences Unified*, and [132], *Swisher H3 Unified Ordinary and Supersingular*. They are not modified here.

2 Cubic scaling for cutoff product sums

Theorem 2.1 (Cubic cutoff-product scaling)**.** *Let $p > 3$ be prime, $\kappa \in \mathbb{Z}_p$, and $f(X) \in 1 + X\mathbb{Z}_p[[X]]$. For L a positive power of p, including $L = 1$, define*

$$\mathcal{S}(L) = 1 + \kappa \sum_{n=1}^{L} \frac{L}{n} \prod_{j=1}^{n-1} f(L/j).$$

Every evaluation of f is at a point of $p\mathbb{Z}_p$, so all products converge and all summands are p-integral. For every $r \geq 1$,

$$\mathcal{S}(p^r) \equiv \mathcal{S}(p^{r-1}) \pmod{p^{3r}}.$$

The theorem requires no value $f(1)$.

Proof. Set $N = p^r$, $M = p^{r-1}$ and write

$$f(X) = 1 + \delta X + \varepsilon X^2 + O(X^3), \qquad \eta = 2\varepsilon - \delta^2, \qquad g(X) = \frac{f(X) - 1 - \delta X}{X^2} \in \mathbb{Z}_p[[X]].$$

If $j < L$, then $v_p(L/j) \geq 1$, so $f(L/j)$ is a well-defined unit and L/n is integral for $1 \leq n \leq L$. This proves the integrality assertion.

Define only the products

$$P_m = \prod_{j=1}^{m} f(M/j) \quad (0 \le m < M), \qquad P_0 = 1,$$

and set $t_m = \kappa(M/m)P_{m-1}$ for $1 \le m \le M$. There is no need to define P_M; the series f need not converge at 1. We have

$$v_p(t_m) \ge r - 1 - v_p(m), \qquad v_p(P_m - P_{m-1}) \ge r - 1 - v_p(m) \quad (1 \le m < M).$$

For $0 \le m \le M$, introduce the complete unit-prefix sums

$$H_m = \sum_{\substack{1 \le j < pm \\ p \nmid j}} j^{-1}, \qquad J_m = \sum_{\substack{1 \le j < pm \\ p \nmid j}} j^{-2}, \qquad H_0 = J_0 = 0.$$

For $m \ge 1$, let $e_m = 1 + v_p(m)$. Complete unit residues modulo p^{e_m} give $J_m \in p^{e_m}\mathbb{Z}_p$: multiplication by a unit whose square is not 1 modulo p proves the vanishing of the sum of inverse squares. Pairing j with $pm - j$ gives $2H_m \equiv -pmJ_m \pmod{p^{2e_m}}$. Hence

$$v_p(H_m) \ge 2e_m, \qquad v_p(J_m) \ge e_m. \tag{1}$$

Let R_m be the product of $f(N/j)$ over $1 \le j < pm$, $p \nmid j$. Let $R_{m,b}$ be the corresponding product with $j < pm + b$, for $0 \le m < M$ and $1 \le b < p$. The expansion

$$f(x) = 1 + \delta x + \varepsilon x^2 + O(x^3), \qquad 2\varepsilon - \delta^2 = \eta,$$

and finite multiplication give

$$R_m \equiv 1 + \delta N H_m + \frac{N^2}{2}(\delta^2 H_m^2 + \eta J_m) \pmod{N^3},$$
$$R_{m,b} \equiv 1 + \delta N \left(H_m + \sum_{a=1}^{b-1} \frac{1}{pm + a} \right) \pmod{N^2}.$$

All omitted coefficients are integral, since the indices in these unit products are prime to p.

Separate the terms of $\mathcal{S}(N)$ according to divisibility of n by p. The defining product decomposition gives exactly

$$\mathcal{S}(N) - \mathcal{S}(M) = \sum_{m=1}^{M} t_m(R_m - 1) + \kappa N \sum_{m=0}^{M-1} P_m \sum_{b=1}^{p-1} \frac{R_{m,b}}{pm + b}.$$

Writing $C_m = H_m^2 - J_m$, the inner sum satisfies

$$\sum_{b=1}^{p-1} \frac{R_{m,b}}{pm + b} \equiv H_{m+1} - H_m + \frac{\delta N}{2}(C_{m+1} - C_m) \pmod{N^2}.$$

It follows that

$$\begin{aligned} \mathcal{S}(N) - \mathcal{S}(M) \equiv{}& \delta N \sum_{m=1}^{M} t_m H_m + \kappa N \sum_{m=0}^{M-1} P_m(H_{m+1} - H_m) \\ &+ \frac{N^2}{2} \sum_{m=1}^{M} t_m(\delta^2 H_m^2 + \eta J_m) + \frac{\kappa\delta N^2}{2} \sum_{m=0}^{M-1} P_m(C_{m+1} - C_m) \pmod{N^3}. \end{aligned} \tag{2}$$

We now estimate these contributions without dividing by δ.

The exact identity $f(x) - 1 = \delta x + x^2 g(x)$ implies, for $1 \le m < M$,

$$\delta t_m = \kappa(P_m - P_{m-1}) - \kappa W_m, \qquad W_m = (M/m)^2 g(M/m) P_{m-1}.$$

The argument M/m belongs to $p\mathbb{Z}_p$, and $v_p(W_m) \ge 2(r - 1 - v_p(m))$. Summation by parts, with the $m = M$ term kept separate, turns the first line of (2) into

$$\kappa N(1+\delta) P_{M-1} H_M - \kappa N \sum_{m=1}^{M-1} W_m H_m.$$

The boundary term belongs to $p^{3r}\mathbb{Z}_p$, since $v_p(H_M) \ge 2r$. Every remaining product satisfies

$$v_p(W_m H_m) \ge 2(r - 1 - v_p(m)) + 2(1 + v_p(m)) = 2r.$$

Thus the entire first line vanishes modulo p^{3r}.

For the third contribution, (1) gives

$$v_p\big(t_m(\delta^2 H_m^2 + \eta J_m)\big) \ge (r - 1 - v_p(m)) + (1 + v_p(m)) = r$$

for all $1 \le m \le M$. It therefore vanishes after multiplication by $N^2/2$.

Finally, $v_p(C_m) \ge 1 + v_p(m)$ for $m \ge 1$, and

$$\sum_{m=0}^{M-1} P_m(C_{m+1} - C_m) = P_{M-1} C_M - \sum_{m=1}^{M-1} (P_m - P_{m-1}) C_m.$$

The boundary term is divisible by p^r, as is every summand on the right. The fourth contribution is therefore also divisible by p^{3r}. This proves the theorem. The argument includes $r = 1$: then $M = 1$, and the sums indexed by $1 \le m < M$ are empty. □

Corollary 2.2 (Hypergeometric parameter scaling)**.** *Let $p > 3$ and $d \ge 1$. Let $\alpha_1, \dots, \alpha_d \in \mathbb{Z}_p$ and $\gamma_1, \dots, \gamma_{d-1} \in \mathbb{Z}_p^\times$. Put*

$$\mathcal{U}(L) = \sum_{n=0}^{L} \frac{\prod_{i=1}^{d} (-\alpha_i L)_n}{n! \prod_{j=1}^{d-1} (-\gamma_j L)_n}.$$

For every $r \ge 1$,

$$\mathcal{U}(p^r) \equiv \mathcal{U}(p^{r-1}) \pmod{p^{3r}}.$$

The products indexed by j are empty when $d = 1$.

Proof. Take

$$\kappa = -\frac{\prod_i \alpha_i}{\prod_j \gamma_j}, \qquad f(X) = \frac{\prod_i (1 - \alpha_i X)}{\prod_j (1 - \gamma_j X)}.$$

For $1 \le n \le L$, cancelling the nonzero integer factors gives exactly

$$\frac{\prod_i (-\alpha_i L)_n}{n! \prod_j (-\gamma_j L)_n} = \kappa \frac{L}{n} \prod_{k=1}^{n-1} f(L/k).$$

Every denominator is nonzero, since its factors are $-\gamma_j L$ and $k - \gamma_j L$ with $0 < k < L$; their valuations are respectively $v_p(L)$ and $v_p(k) < v_p(L)$. The formula also holds if a numerator parameter is zero. Apply Theorem 2.1. □

In particular, the corollary applies to $\sum_{n=0}^{L}(-\alpha L)_n(-\beta L)_n/(n!(-\gamma L)_n)$ for arbitrary $\alpha, \beta \in \mathbb{Z}_p$ and $\gamma \in \mathbb{Z}_p^\times$. No restriction on $\gamma - 1$ or $\gamma - \alpha - \beta$ occurs.

Corollary 2.3 (The K residual)**.** *For $p > 3$ and $r \geq 1$, put*

$$U(L) = \sum_{n=0}^{L} \frac{(-L/2)_n^2}{n!(-3L/2)_n}.$$

Then $U(p^r) \equiv U(p^{r-1}) \pmod{p^{3r}}$.

Proof. Apply Corollary 2.2 with $\alpha_1 = \alpha_2 = 1/2$ and $\gamma_1 = 3/2$. □

A classical specialization. For $f(X) = 1 + X$ and $\kappa = 1$, the sum is $\mathcal{S}(N) = \binom{2N}{N}$, by the hockey-stick binomial identity. Theorem 2.1 therefore contains the prime-power central-binomial instance of Jacobsthal's congruence. The proof uses the classical separation of indices divisible by p and unit-index harmonic cancellation; comparable tools occur in Osburn–Sahu–Straub [8]. No historical novelty of these tools is claimed.

3 A complete proof of Swisher's $(K.3)$

For each odd positive integer N, put $h_N = (N-1)/2$ and define

$$K(N) = \sum_{k=0}^{h_N} (42k+5)\frac{(1/2)_k^3}{k!^3 64^k}, \tag{3}$$

$$B(N) = \frac{12\displaystyle\prod_{j=0}^{N-1}(N+2j)}{2^{3N-2}(h_N!)^2}, \tag{4}$$

$$U(N) = \sum_{n=0}^{N} \frac{(-N/2)_n^2}{n!(-3N/2)_n}. \tag{5}$$

All three quantities are rational numbers. In particular, $K(1) = 5$, $B(1) = 6$, and $U(1) = 5/6$. We use the positive sign in Swisher's published Table 2: with $\chi_p = (-1)^{(p-1)/2}$, the target is

$$K(p^r) \equiv \chi_p p K(p^{r-1}) \pmod{p^{4r}}, \qquad p > 5. \tag{6}$$

The first arXiv version has an erroneous additional minus sign in this entry; the published version corrects it.

Proposition 3.1 (Exact factorization)**.** *For every odd positive integer N,*

$$K(N) = B(N)U(N). \tag{7}$$

Proof. Use the Guillera WZ pair employed by Osburn and Zudilin in their proof of $(K.2)$, namely

$$F(n,k) = (84n^2 - 56nk + 4k^2 + 52n - 12k + 5) \cdot \frac{(-1)^k (1/2)_n (1/2)_{n+k} (1/2)_{n-k}^2}{2^{4n}(n!)^2(2n-k+1)!},$$

$$G(n,k) = \frac{64n^2(-1)^k(1/2)_n(1/2)_{n+k-1}(1/2)_{n-k}^2}{2^{4n}(n!)^2(2n-k)!}.$$

Negative-index Pochhammer symbols are defined by reciprocals, and a reciprocal factorial at a negative integer is zero. The pair satisfies

$$F(n,k-1)-F(n,k)=G(n+1,k)-G(n,k).$$

Moreover $F(n,0)=(42n+5)(1/2)_n^3/(64^n n!^3)$. Summing first over $0\le n\le h_N$ and then over $1\le k\le N$ gives, exactly as in Osburn–Zudilin's equation (2.6),

$$K(N)=F(h_N,N)+\sum_{k=1}^{N}G(h_N+1,k).$$

This telescoping identity requires only that N be odd, not prime. Direct simplification gives

$$F(h_N,N)=B(N),\qquad G(h_N+1,k)=B(N)\frac{(-N/2)_{N+1-k}^2}{(N+1-k)!(-3N/2)_{N+1-k}}.$$

Changing the index to $n=N+1-k$ supplies the terms $n=1,\dots,N$ in (5); the boundary value supplies $n=0$. □

Theorem 3.2 (All-level congruence for the prefactor)**.** *For every prime $p\ge 5$ and integer $r\ge 1$,*

$$B(p^r)\equiv \chi_p p B(p^{r-1}) \pmod{p^{4r}},\qquad v_p(B(p^r))=r. \tag{8}$$

Proof. Set $N=p^r$, $M=p^{r-1}$ and $t=(N-M)/2$, and write

$$V=\prod_{\substack{1\le j\le h_N\\ p\nmid j}} j,\quad O=\prod_{\substack{1\le a<N\\ a\text{ odd},\ p\nmid a}} a,\quad W=\prod_{\substack{0\le j<N\\ p\nmid j}}(N+2j).$$

Removing the factors divisible by p from the numerator and the factorials in (4) yields the exact equality

$$\frac{B(N)}{pB(M)}=\frac{W}{2^{3(N-M)}V^2}. \tag{9}$$

Pairing the factors in W about $2N$ gives another exact equality,

$$W=(-1)^t O^2\prod_{\substack{1\le a<N\\ a\text{ odd},\ p\nmid a}}\left(1-\frac{4N^2}{a^2}\right). \tag{10}$$

We claim that the last product is 1 modulo p^{3r}. Indeed, the inverse-square sum over all units modulo N vanishes modulo N. To see this, choose a unit g with $g^2\not\equiv 1 \pmod p$; multiplication by g permutes those units and forces the sum to be annihilated by the unit $g^{-2}-1$. Pairing j with $N-j$ then shows that the inverse-square sum over $1\le j\le h_N$, $p\nmid j$, also vanishes modulo N. The sum over odd units $a<N$ is the full sum minus one quarter of this half-interval sum, so it vanishes as well. Consequently the linear term in the product in (10) is divisible by p^{3r}; every term involving two or more selected factors is divisible by p^{4r}.

Let $C_N=\binom{N-1}{h_N}$. Its p-free quotient is

$$\frac{C_N}{C_M}=2^t\frac{O}{V}. \tag{11}$$

Cai's prime-power extension of Morley's congruence [3, Corollary 3] states that

$$\frac{C_N}{C_M}\equiv \chi_p 4^{N-M} \pmod{p^{3r}}.$$

Since $(-1)^t = \chi_p$, substituting (10) and (11) into (9) gives

$$\frac{B(N)}{pB(M)} \equiv \chi_p \left(\frac{C_N}{C_M}\right)^2 2^{-4(N-M)} \equiv \chi_p \pmod{p^{3r}}.$$

The exact quotient in (9) is a p-adic unit. Starting with $B(1) = 6$, induction therefore gives $v_p(B(M)) = r-1$. Multiplication by $pB(M)$ raises the precision to p^{4r} and proves both assertions. □

Proposition 3.3 (Integrality and the residual criterion)**.** *For $p > 5$ and $r \geq 0$, $U(p^r) \in \mathbb{Z}_p^\times$ and $U(p^r) \equiv 5/6 \pmod{p}$. For every $r \geq 1$, (6) holds if and only if*

$$\boxed{U(p^r) \equiv U(p^{r-1}) \pmod{p^{3r}}.} \tag{12}$$

Proof. For $N = p^r$ and $1 \leq n \leq N$, write the summand in $U(N)$ as

$$f(N,n) = -\frac{N}{6n} \prod_{j=1}^{n-1} \frac{(1 - N/(2j))^2}{1 - 3N/(2j)}. \tag{13}$$

For $j < N$, $v_p(N/j) \geq 1$, so every factor in this product is a p-adic unit congruent to 1 modulo p. It follows that $v_p(f(N,n)) = r - v_p(n) \geq 0$. Only $n = N$ survives modulo p among $n \geq 1$, and that term is $-1/6$ modulo p. Thus $U(N) \equiv 1 - 1/6 = 5/6 \pmod{p}$.

Finally the exact identity (7) gives

$$\begin{aligned} K(N) - \chi_p pK(M) = {} & B(N)(U(N) - U(M)) \\ & + (B(N) - \chi_p pB(M))U(M). \end{aligned}$$

The second term belongs to $p^{4r}\mathbb{Z}_p$ by Theorem 3.2 and the integrality just proved. The valuation of $B(N)$ is exactly r, proving the equivalence. □

Theorem 3.4 (Swisher's (K.3))**.** *For every prime $p > 5$ and every integer $r \geq 1$,*

$$\boxed{\sum_{k=0}^{(p^r-1)/2} (42k+5)\frac{(1/2)_k^3}{64^k (k!)^3} \equiv (-1)^{(p-1)/2} p \sum_{k=0}^{(p^{r-1}-1)/2} (42k+5)\frac{(1/2)_k^3}{64^k (k!)^3} \pmod{p^{4r}}.}$$

Proof. Corollary 2.3 establishes exactly the residual congruence in Proposition 3.3, for every level. More explicitly, set $N = p^r$ and $M = N/p$. In the exact identity

$$K(N) - \chi_p pK(M) = B(N)(U(N) - U(M)) + (B(N) - \chi_p pB(M))U(M),$$

the first term belongs to $p^{4r}\mathbb{Z}_p$ because $v_p(B(N)) = r$ and $U(N) - U(M) \in p^{3r}\mathbb{Z}_p$. The second belongs to the same ideal by Theorem 3.2 and $U(M) \in \mathbb{Z}_p$. This includes $r = 1$, where $K(1) = 5$ and $U(1) = 5/6$. □

3.1 A further harmonic reduction of the K residual

This subsection uses the $U(N)$ of the preceding K reduction. Put

$$P_j(N) = \prod_{k=1}^{j} \frac{(1 - N/(2k))^2}{1 - 3N/(2k)}, \qquad P_0(N) = 1.$$

For every odd positive integer N,

$$U(N) = \frac{4}{3} - \frac{1}{2}P_{N-1}(N) + \frac{1}{3}T(N), \qquad T(N) = N^2 \sum_{n=1}^{N-1} \frac{P_{n-1}(N)}{n^2(1 - 3N/(2n))}. \tag{14}$$

Indeed the nth summand of U is $-NP_{n-1}(N)/(6n)$, and

$$f(x) = \frac{(1 - x/2)^2}{1 - 3x/2}, \qquad f(x) - 1 = \frac{x}{2} + \frac{x^2}{1 - 3x/2}.$$

Summing the resulting difference $P_n - P_{n-1}$ for $1 \le n < N$ and then adding the $n = N$ term proves (14).

Proposition 3.5. *For $p > 5$, $N = p^r$, and $M = p^{r-1}$,*

$$P_{N-1}(N) \equiv P_{M-1}(M) \pmod{p^{3r}}.$$

Consequently (K.3) is equivalent to $T(p^r) \equiv T(p^{r-1}) \pmod{p^{3r}}$. The last congruence is now a consequence of Theorem 3.4.

Proof. Remove the factors indexed by multiples of p from the quotient of the two products. What remains is $\prod_{1 \le k < N,\ p \nmid k} f(N/k)$. On the full unit residue set modulo N,

$$H_2 = \sum_{\substack{1 \le k < N \\ p \nmid k}} k^{-2} \in p^r \mathbb{Z}_p, \qquad H_1 = \sum_{\substack{1 \le k < N \\ p \nmid k}} k^{-1} \in p^{2r} \mathbb{Z}_p.$$

For the first claim multiply the residues by a unit whose square is not 1 modulo p. For the second, pair k with $N - k$ and expand modulo N^2, obtaining $2H_1 \equiv -NH_2 \pmod{N^2}$. Now

$$\log f(x) = \sum_{\ell \ge 1} \frac{3^\ell - 2}{\ell 2^\ell} x^\ell.$$

The first two terms summed over the units lie in $p^{3r}\mathbb{Z}_p$ by these harmonic bounds. Every later term has valuation at least $\ell r - v_p(\ell) \ge 3r$. The convergent logarithm and exponential therefore prove the product congruence. Equation (14) and the K equivalence above prove the last assertion, since 3 is a unit. □

4 A complete proof of the C.3 recurrence

For an odd positive integer N, write $n = (N - 1)/2$ and

$$b_k = \frac{(1/2)_k}{k!} = \frac{\binom{2k}{k}}{4^k}, \qquad C(N) = \sum_{k=0}^{n} (4k + 1) b_k^4.$$

The following finite identity is exact over the rational numbers:

$$C(N) = 2N^2 b_n^2 \sum_{l=0}^{n} \frac{b_l b_{N-l}}{N - 2l}. \tag{15}$$

To prove it, put $B(s, k) = b_s^2 b_{s+k} b_{s-k}$ for $s \ge k \ge 0$, and set $B(s, k) = 0$ for $k > s$. Define

$$F(s, k) = (4s + 1)B(s, k), \qquad G(s, k) = \frac{16s^2(s + k)}{(2k - 1)(2s + 2k - 1)} B(s, k), \quad k \ge 1.$$

Direct cancellation of successive factorials verifies

$$F(s,k-1)-F(s,k)=G(s+1,k)-G(s,k).$$

The boundary $s=k-1$ also follows by direct substitution, and $G(0,k)=0$. Sum first over $0\le s<h$, then over $1\le k\le h$, where $h=(N+1)/2$. The left side is $C(N)$, because $F(s,h)=0$. On reversing $k=h-l$, the right side becomes

$$\sum_{k=1}^{h}G(h,k)=8h^2b_h^2\sum_{l=0}^{h-1}\frac{b_lb_{N-l}}{N-2l}=2N^2b_n^2\sum_{l=0}^{n}\frac{b_lb_{N-l}}{N-2l},$$

since $b_h=Nb_n/(N+1)$. This proves (15).

Define

$$Q_k(N)=\prod_{j=1}^{k}(1-2N/j)^{(-1)^j},\qquad R_l(N)=\frac{Q_{2l-1}(N)}{1-N/(2l)}\quad(l\ge1),$$

$$U_C(N)=1+\frac{N^2}{2}\sum_{l=1}^{(N-1)/2}\frac{R_l(N)}{l^2}. \tag{16}$$

Using

$$\frac{b_lb_{N-l}}{b_N}=\frac{\binom{N}{l}^2}{\binom{2N}{2l}},$$

and expanding the two binomial products, (15) becomes

$$C(N)=A_C(N)U_C(N),\qquad A_C(N)=2Nb_{(N-1)/2}^2b_N. \tag{17}$$

For $N=p^r$, all factors in $Q_k(N)$ and $R_l(N)$ are p-adic units, because their indices are strictly below N.

4.1 Two harmonic estimates

Let $p>3$. If L is positive and $a=v_p(L)$, the unit square sum

$$\sum_{\substack{1\le j\le L\\ p\nmid j}}j^{-2}$$

is divisible by p^a. This follows by grouping into complete residue systems modulo p^a; the sum of the inverse squares of the units modulo p^a is zero modulo p^a. In particular, for

$$H_l=\sum_{\substack{1\le j\le l\\ p\nmid j}}j^{-2},\qquad e_l=\max\{v_p(l),v_p(l+1),v_p(2l+1)\},$$

one has $v_p(H_l)\ge e_l$. The cases $p^{e_l}\mid l$ and $p^{e_l}\mid l+1$ use the complete-residue estimate (the endpoint $l+1$ is omitted when divisible by p). In the third case, reflection $j\mapsto 2l+1-j$ gives $2H_l\equiv\sum_{1\le j<2l+1,p\nmid j}j^{-2}\equiv0\pmod{p^{e_l}}$. In particular $H_{(p^r-1)/2}\in p^r\mathbb{Z}_p$.

A second estimate concerns an even integer L with $a=v_p(L)\ge1$:

$$S_i(L)=\sum_{\substack{1\le j<L\\ p\nmid j}}\frac{(-1)^j}{j^i}.$$

Then $S_2(L)\in p^a\mathbb{Z}_p$ and $S_1(L)\in p^{2a}\mathbb{Z}_p$. For the first assertion write $j=u+tp^a$, with $1\le u<p^a$ a unit. The number L/p^a is even, so the alternating signs cancel modulo p^a for every residue u. For the second, reflection about $L/2$ preserves the sign and gives

$$2S_1(L)\equiv-LS_2(L)\pmod{p^{2a}}.$$

4.2 The residual factor is stable to three times the level

We prove

$$U_C(p^r) \equiv U_C(p^{r-1}) \pmod{p^{3r}}. \tag{18}$$

Put $N = p^r$, $M = N/p$, and $n = (N-1)/2$. First consider the part of (16) with $p \nmid l$. A direct quotient gives

$$\frac{R_{l+1}(N)}{R_l(N)} = \frac{1 - N/l}{1 - 2N/(2l+1)} \frac{1 - N/(2l)}{1 - N/(2l+2)}.$$

All four factors are units and hence

$$R_{l+1}(N) - R_l(N) \in p^{r-e_l}\mathbb{Z}_p.$$

Summation by parts therefore proves

$$\sum_{\substack{1 \le l \le n \\ p \nmid l}} \frac{R_l(N)}{l^2} = R_n(N)H_n - \sum_{l=1}^{n-1} H_l(R_{l+1}(N) - R_l(N)) \in p^r\mathbb{Z}_p.$$

Multiplication by $N^2/2$ shows that all these indices together contribute zero modulo p^{3r}.

For the remaining indices write $l = pb$, where $1 \le b \le (M-1)/2$. Oddness of p preserves the parity of the factors whose index is divisible by p. Thus, exactly,

$$R_{pb}(N) = R_b(M)\mathcal{A}_b, \qquad \mathcal{A}_b = \prod_{\substack{1 \le j < 2pb \\ p \nmid j}} (1 - 2N/j)^{(-1)^j}.$$

Set $a = v_p(2pb) = 1 + v_p(b) \le r - 1$. Expansion to second order, with an integral remainder because every denominator j is a unit, gives

$$\mathcal{A}_b \equiv 1 - 2NS_1(2pb) + 2N^2(S_1(2pb)^2 - S_2(2pb)) \pmod{p^{3r}}.$$

The preceding harmonic estimates show $\mathcal{A}_b - 1 \in p^{r+2a}\mathbb{Z}_p$. Meanwhile $M^2/(2b^2)$ has valuation $2(r-a)$. Therefore each individual difference

$$\frac{N^2}{2(pb)^2} R_{pb}(N) - \frac{M^2}{2b^2} R_b(M)$$

is divisible by p^{3r}. Together with the unit-index part this proves (18), also for $r = 1$ when the divisible-index part is empty.

4.3 The prefactor and the original C.3 statement

Let $B_N = \binom{N-1}{(N-1)/2}$ and $D_N = \binom{2N}{N}$. Cai's prime-power Morley congruence and Jacobsthal's congruence give, for $N = p^r$, $M = N/p$,

$$\frac{B_N}{B_M} \equiv \varepsilon 4^{N-M} \pmod{p^{3r}}, \qquad \frac{D_N}{D_M} \equiv 1 \pmod{p^{3r}}, \qquad \varepsilon = (-1)^{(p-1)/2}.$$

The first is Cai's Corollary 3 [3]. The second also follows from Theorem 3 of that paper with $u = 2, v = 1$. Since $b_{(N-1)/2} = B_N/2^{N-1}$ and $b_N = D_N/4^N$, these imply

$$\frac{A_C(N)}{pA_C(M)} \equiv 1 \pmod{p^{3r}}.$$

Both $A_C(N)/N$ and $U_C(N)$ are p-integral units. Combining this prefactor congruence with (18) and $v_p(A_C(N)) = r$ gives

$$\boxed{C(p^r) \equiv pC(p^{r-1}) \pmod{p^{4r}} \qquad (p > 3,\ r \geq 1).}$$

This is exactly the original half-cutoff C.3 recurrence. For clarity, the final implication is

$$C(N) - pC(M) = A_C(N)(U_C(N) - U_C(M)) + (A_C(N) - pA_C(M))U_C(M).$$

Both terms belong to $p^{4r}\mathbb{Z}_p$, by the established valuations. The argument includes $r = 1$, with $A_C(1) = U_C(1) = C(1) = 1$. The restriction $p > 3$ is essential: $C(3) - 3 = -27/16$ is not divisible by 3^4.

5 What the corpus mechanisms do and do not transfer

The modular cusp theorem in [131] (label `B:thm:main`) assumes weight $w \in \{1, 2\}$ and yields precision $p^{(w+1)r}$. The higher-dimensional standard-family theorem yields p^{2r}. An analytic evaluation at a unit argument requires its own integral evaluation map, fixed-point condition, and multiplier identification. These hypotheses cannot be replaced by a formal substitution.

There is an explicit obstruction already proved in [131] (label `E:thm:sh-cube`). Let

$$F_d(u) = \sum_{k\geq 0} \binom{2k}{k}^d u^k, \quad K = p^r, \qquad E_r = F_d(u)F_{K/p}(\sigma(u)) - F_d(\sigma(u))F_K(u),$$

where F_L is truncated at $k < L$ and σ is the excellent lift in that theorem. For $d \geq 4$ and $p > d$,

$$[u^{K+1}]E_r \equiv 2^{2d-1}d(d-3)p^{2r} \pmod{p^{2r+1}}.$$

The right side is nonzero at this precision. Thus the full-series congruence cannot simply be raised from $2r$ to $4r$ or $5r$. For example, $d = 4, p = 7, r = 1$ gives

$$[u^8]E_1 = 27435582641602576, \qquad v_7([u^8]E_1) = 2.$$

Even applying $1 + 4u\,d/du$ leaves an obstruction: its coefficient is multiplied by $4K + 5$, a 7-adic unit. This excludes a naive global derivative upgrade; it does not exclude additional cancellation at a particular argument. This obstruction is from [131], not a new result of the present note.

For clarity, there is an exact weighted transfer identity. Write $D_r = F_r - \rho F_{r-1} \circ \sigma$, $\theta = t\,d/dt$, and $\mu = \theta\sigma/\sigma$. At a valid analytic fixed point t_0, if

$$c\theta\rho(t_0) = \rho(t_0)(\mu(t_0) - 1), \qquad \lambda = \rho(t_0)\mu(t_0),$$

then the product and chain rules give

$$(1 + c\theta)F_r(t_0) - \lambda(1 + c\theta)F_{r-1}(t_0) = (1 + c\theta)D_r(t_0).$$

Differentiation preserves an established integral modulus, but does not itself increase it. The remaining weighted residual needs a separate estimate. Likewise, the [132] Taylor-transport argument requires the precise residual order of its differential equation; Section 9 below calculates what happens at the short odd cutoffs of other families.

6 An exact boundary reduction of the odd branch of (F.3)

Let $p \equiv 3 \pmod 4$ be a prime. For odd $r \geq 1$ put

$$N_r = (p^r - 3)/4, \qquad S_r = \sum_{n=0}^{N_r} (-1)^n (8n+1) \frac{(1/4)_n^3}{(n!)^3}.$$

Define

$$D_r = (8N_r + 1) \frac{(1/4)_{2N_r}}{(N_r!)^2}, \qquad B_r = \sum_{k=1}^{N_r} G(N_r + 1, k),$$

where an empty sum is zero and

$$G(n,k) = (-1)^{n+k} 4 \frac{(1/4)_n^2 (1/4)_{n+k-1}}{(n-1)!^2 (n-k)! (1/4)_k^2}.$$

Proposition 6.1. *For every such p, r, $S_r = D_r + B_r$. For odd $r \geq 3$,*

$$v_p(D_r) = r - 1, \qquad D_r \equiv p^2 D_{r-2} \pmod{p^r}.$$

Consequently the odd branch of Swisher's original (F.3) is equivalent to

$$\boxed{B_r \equiv p^2 B_{r-2} \pmod{p^r}} \qquad (r \geq 3 \text{ odd}).$$

This boxed boundary recurrence is not proved here.

Proof. Set $a = 1/4$ and

$$F(n,k) = (-1)^{n+k} (8n+1) \frac{(a)_n^2 (a)_{n+k}}{(n!)^2 (n-k)! (a)_k^2}.$$

Use $1/m! = 0$ for negative integers m, and set $G(0,k) = 0$. The WZ identity

$$F(n, k-1) - F(n,k) = G(n+1,k) - G(n,k)$$

is an exact rational identity. Indeed its four quotients after division by $F(n,k)$ are

$$\frac{F(n,k-1)}{F(n,k)} = -\frac{(a+k-1)^2}{(a+n+k-1)(n-k+1)},$$

$$\frac{G(n+1,k)}{F(n,k)} = -\frac{(a+n)^2}{(2n+a)(n-k+1)}, \qquad \frac{G(n,k)}{F(n,k)} = \frac{n^2}{(a+n+k-1)(2n+a)}.$$

Substitution proves the identity, including the boundary cases by the zero convention. Summing first over $0 \leq n \leq N$, then over $1 \leq k \leq N$, gives

$$\sum_{n=0}^{N} F(n,0) = F(N,N) + \sum_{k=1}^{N} G(N+1,k).$$

Since $F(N,N) = (8N+1)(a)_{2N}/(N!)^2$, this proves the decomposition. The WZ pair appears as Lemma 3.1 in Jana–Kalita, *Proof of a supercongruence conjecture of (F.3) of Swisher using the WZ-method*, arXiv:2011.02762v1. That article was withdrawn for errors; only the identity, independently verified above, is used here.

Write $R_r = (1/4)_{2N_r}/(N_r!)^2$. Let Γ_p be Morita's gamma function. Extracting the factors divisible by p twice from the rising factorial and the factorial gives the exact formula

$$\frac{R_r}{p^2 R_{r-2}} = \left(\frac{p^{r-2}}{2} - \frac{5}{4}\right)\left(\frac{p^{r-2}}{2} - \frac{1}{4}\right) \frac{\Gamma_p(p^r/2 - 5/4)\Gamma_p(p^{r-1}/2 + 1/4)}{\Gamma_p(1/4)\Gamma_p(3/4)\Gamma_p((p^r+1)/4)^2\Gamma_p((p^{r-1}+3)/4)^2}. \qquad (*)$$

For completeness, in $(1/4)_{(p^r-3)/2}$ the first extraction leaves $p^{(p^{r-1}-1)/2}(3/4)_{(p^{r-1}-1)/2}$. The second leaves $p^{(p^{r-2}+1)/2}(1/4)_{(p^{r-2}+1)/2}$. Removing the final two factors of this last rising factorial gives the two prefactors in (*). In $N_r!$, the two factorials left are indexed by $(p^{r-1}-1)/4$ and N_{r-2}. Comparing the powers of p gives exactly p^2. The gamma signs in the numerator multiply to $+1$ because $(p^r-3)/2$ and $(p^{r-1}-1)/2$ are even; denominator signs are squared.

All the factors on the right of (*) are p-adic units: $p \neq 5$. Modulo p, its value is

$$\frac{5}{16}\frac{\Gamma_p(-5/4)}{\Gamma_p(1/4)^2\Gamma_p(3/4)^3} = 1.$$

The equality follows from $\Gamma_p(-5/4) = \frac{16}{5}\Gamma_p(3/4)$ and $\Gamma_p(1/4)^2\Gamma_p(3/4)^2 = 1$. These are the gamma functional equation and reflection formula. The base R_1 is a unit: $(p-3)/2 < (3p-1)/4$ is below the first p-divisible factor in $(1/4)_j$, and $N_1 < p$. Induction using (*) proves $v_p(R_r) = r-1$ and $R_r \equiv p^2 R_{r-2} \pmod{p^r}$. Finally, $8N_r + 1 = 2p^r - 5$ is a unit and is congruent to $2p^{r-2} - 5$ modulo p. Multiplying proves both claims for D_r. Subtracting the diagonal recurrence from the exact decomposition proves the stated equivalence. □

A precise obstruction to deleting the boundary. The withdrawn article's Lemma 3.2 asserts $v_p(G((p^r+1)/4, k)) \geq 3(r-1)/2$ throughout the relevant range. For $p = 7, r = 3, k = 13$, this is false:

$$v_7(G(86,13)) = 1, \qquad \frac{G(86,13)}{7} \equiv 34 \pmod{49}.$$

For an entirely integer valuation check, let $A(n) = \sum_{j=0}^{n-1} v_7(4j+1)$ and $C(n) = v_7(n!)$. Then

$$A(86) = 14, \quad A(98) = 16, \quad A(13) = 3, \quad C(85) = 13, \quad C(73) = 11,$$

so $v_7 G = 2(14) + 16 - 2(13) - 11 - 2(3) = 1$. Direct rational arithmetic gives

$$D_3 \equiv 245, \qquad B_3 \equiv 49, \qquad S_3 \equiv 294 \pmod{343}.$$

Thus the boundary is essential, even though the original conjecture is satisfied in this test. These computations diagnose the invalid termwise bound, and do not constitute a proof of the new boundary recurrence.

6.1 A normalized finite identity for the remaining odd (F.3) branch

For $N \geq 0$, define

$$S_N = \sum_{k=0}^{N} (-1)^k (8k+1)\frac{(1/4)_k^3}{(k!)^3}, \qquad D_N = (8N+1)\frac{(1/4)_{2N}}{(N!)^2}.$$

Then the following identity is exact:

$$S_N = D_N V_N, \qquad V_N = \sum_{n=0}^{N} \frac{(-N-1/4)_n^2}{n!(-2N-1/4)_n}. \qquad (19)$$

To prove it, use the verified WZ identity of the preceding reduction, including its last boundary term, to write $S_N = \sum_{k=1}^{N+1} G(N+1,k)$. Its final term is $G(N+1,N+1) = D_N$. Reversing the index gives, for $0 \le n \le N$,

$$\frac{G(N+1,N+1-n)}{D_N} = \frac{(-N-1/4)_n^2}{n!(-2N-1/4)_n}.$$

This follows directly from $(a)_{m-n}/(a)_m = (-1)^n/(1-a-m)_n$, so summing proves (19) without any limiting argument.

Let $p \equiv 3 \pmod 4$, $r \ge 3$ odd, $L = p^r$, $M = p^{r-2}$, and $N_X = (X-3)/4$. The already proved diagonal statement says

$$v_p(D_{N_L}) = r-1, \qquad \frac{D_{N_L}}{p^2 D_{N_M}} \in 1 + p\mathbb{Z}_p.$$

Consequently a sufficient remaining theorem, stated with all its integrality obligations, is

$$V_{N_{p^s}} \in \mathbb{Z}_p \quad (s \ge 1 \text{ odd}), \qquad V_{N_L} - V_{N_M} \in p\mathbb{Z}_p \quad (r \ge 3 \text{ odd}). \tag{20}$$

It is also necessary for the full odd branch of (F.3): its first-level S_{N_p} and D_{N_p} are integral and D_{N_p} is a unit, so the original recurrence inductively gives the displayed integrality and normalized congruence. Thus (20), jointly over all odd levels, is equivalent to that branch. It remains unproved here.

The precise parameter orbit. In the normalized sum the upper parameter is $a_L = 1/2 - L/4$ and the lower parameter is $c_L = 5/4 - L/2$. For Dwork's dash $x' = (x+t)/p$, $0 \le t < p$, $x + t \in p\mathbb{Z}_p$, two dash steps give

$$a_L \longmapsto \frac{1}{2} - \frac{L}{4p} \longmapsto \frac{1}{2} - \frac{M}{4} = a_M,$$
$$c_L \longmapsto \frac{3}{4} - \frac{L}{2p} \longmapsto \frac{1}{4} - \frac{M}{2} = c_M - 1.$$

The representatives used are $(p-1)/2$ at both upper steps, $(3p-5)/4$ at the first lower step, and $(p-3)/4$ at the second. The integer shift in the lower parameter cannot be discarded.

There is an exact contiguous relation that accounts for this shift. For $a = -N-1/4$, $c = -2N-1/4$ and $t_n = (a)_n^2/[n!(c)_n]$, the coefficient recurrence implies

$$\frac{3}{4}\sum_{n=0}^{N} n t_n + a^2 V_N = \frac{1}{16} t_N.$$

Indeed $n(c+n-1)t_n = (a+n-1)^2 t_{n-1}$; summing and subtracting the shifted sums leaves the terminal contribution. Therefore

$$\sum_{n=0}^{N_M} \frac{(a_M)_n^2}{n!(c_M-1)_n} = -\frac{(M+1)^2}{3(1-2M)} V_{N_M} + \frac{t_{N_M}}{3(1-2M)}. \tag{21}$$

This finite relation is valid as a rational identity. It does not by itself prove a Frobenius congruence for the sums.

Why direct Lucas reduction does not close the problem. Although the desired normalized sums are experimentally integral, their summands need not be integral. For instance, at $p = 7, r = 3$, $N_L = 85$, the minimum valuation of a summand in V_{85} is -1. A termwise reduction modulo p is therefore invalid. A proof must establish cancellation of those negative-valuation terms before using any finite-field reduction or Dwork comparison. The pure scaling lemma for parameters all proportional to p^r does not apply directly to a_L, c_L, whose nonzero constant parts and lower-parameter shift are displayed above.

7 The C.3 and J.3 deformation reductions

For $p > 3$, put $N = p^r$, $n_r = (N-1)/2$, and

$$c_k = (4k+1)\frac{\binom{2k}{k}^4}{256^k}, \qquad j_k = (6k+1)\frac{\binom{2k}{k}^3}{256^k}, \quad C_r = \sum_{k=0}^{n_r} c_k, \quad J_r = \sum_{k=0}^{n_r} j_k.$$

Here $C_0 = J_0 = 1$, and $\varepsilon = (-1)^{(p-1)/2}$. The respective targets are $C_r - pC_{r-1} \in p^{4r}\mathbb{Z}_p$ and $J_r - \varepsilon p J_{r-1} \in p^{4r}\mathbb{Z}_p$. Guo's published all-level result has modulus p^{3r} for both sums (https://arxiv.org/abs/1910.07551).

For $d = 2, 4$, define the finite rational product

$$R_d(k,x) = \prod_{a=0}^{k-1} \frac{1 - x^2/(2a+1)^2}{1 - x^2/[d(a+1)]^2}.$$

Dougall's terminating very-well-poised summation and the terminating identity used in Long's Lemma 4.2, respectively, give for every positive odd integer N the exact identities

$$\sum_{k=0}^{(N-1)/2} c_k R_2(k,N) = N, \tag{22}$$

$$\sum_{k=0}^{(N-1)/2} j_k R_4(k,N) = (-1)^{(N-1)/2} N. \tag{23}$$

The second identity is recorded at https://msp.org/pjm/2011/249-2/pjm-v249-n2-p09-p.pdf, pp. 414–415; replacing the letter p there by an odd integer does not change the terminating evaluation. The first is the specialization $a = b = 1/2$, $c = (1+N)/2$, and terminating parameter $-(N-1)/2$ of Dougall's formula. Its right-hand side is

$$\frac{(3/2)_{(N-1)/2}((1-N)/2)_{(N-1)/2}}{(1)_{(N-1)/2}(1-N/2)_{(N-1)/2}} = N.$$

These exact classical evaluations are not proofs of the desired undeformed recurrences.

7.1 An exact finite separation of the factors divisible by p

The following reduction avoids any unjustified integral Taylor expansion. Write $k = pt + e$, $0 \le e < p$, put $h = (p-1)/2$, $M = p^{r-1}$, and $\delta_e = 1$ if $e > h$, and 0 otherwise. Splitting the products according to divisibility by p proves exactly

$$R_d(k,N) = R_d(t,M)\left(1 - \frac{M^2}{(2t+1)^2}\right)^{\delta_e} U_d(k,N),$$

where

$$U_d(k,N) = \frac{\prod_{0\le a<k,\ p\nmid 2a+1}(1 - N^2/(2a+1)^2)}{\prod_{1\le a\le k,\ p\nmid a}(1 - N^2/(da)^2)}.$$

Every denominator in this last product is a p-adic unit. Consequently, with

$$W_d(k) = \sum_{\substack{0\le a<k \\ p\nmid 2a+1}} \frac{1}{(2a+1)^2} - \frac{1}{d^2}\sum_{\substack{1\le a\le k \\ p\nmid a}} \frac{1}{a^2},$$

one has the rigorous, uniform expansion

$$U_d(k,N) \equiv 1 - N^2 W_d(k) \pmod{p^{4r}}.$$

The other factors above are integral for $k \le n_r$. In the last block, $t = (M-1)/2$, only $e \le h$ occurs, so the potentially zero extra factor is never included. This proves that this expansion can safely be multiplied by all the remaining factors and summed.

Here is an entirely finite expression for the defect. For C its vanishing at the target precision follows from Section 4; for J it remains open in this note. Use $(w,d,\eta) = (c,2,1)$ or $(j,4,\varepsilon)$ and set

$$L_t = R_d(t,M), \quad z_t = \frac{M^2}{(2t+1)^2}, \quad E_t = \{e : 0 \le e < p,\ pt+e \le n_r\},$$

$$V_t = \sum_{e\in E_t} w_{pt+e}, \qquad B_t = \sum_{\substack{e\in E_t \\ e>h}} w_{pt+e}, \qquad H_t = \sum_{e\in E_t} w_{pt+e}(1-z_t)^{\delta_e} W_d(pt+e).$$

Then the exact terminating identities at levels r and $r-1$, followed by the preceding expansion, prove

$$\sum_{k=0}^{n_r} w_k - \eta p \sum_{k=0}^{n_{r-1}} w_k \equiv \sum_{t=0}^{n_{r-1}} (1-L_t)(V_t - \eta p w_t) + \sum_{t=0}^{n_{r-1}} L_t z_t B_t + N^2 \sum_{t=0}^{n_{r-1}} L_t H_t \pmod{p^{4r}}. \tag{24}$$

Indeed the evaluated deformation at level r is $\eta^r N$, the lower-level evaluation multiplied by ηp is the same, and the k-block of the higher-level evaluation is $L_t(V_t - z_t B_t - N^2 H_t)$ modulo p^{4r}. Rearrangement gives the display. All quantities in this congruence are p-integral. Thus either original conjecture is equivalent to the vanishing modulo p^{4r} of its displayed finite obstruction. Its vanishing for C now follows from the convolution proof above; its vanishing for J remains unproved. In particular, none of its three summands may be suppressed without a separate bound. For example, at $(p,r) = (5,2)$ their valuations are $(7,7,7)$ for C and $(6,6,6)$ for J, while the valuation of their total is 8 in both cases. At $(5,3)$ the three valuations are $(9,9,10)$ for C and $(10,10,10)$ for J, while both totals have valuation 12. These exact rational checks exhibit the cross-term cancellation that a proof must control; a termwise $4r$ bound is already false in these examples.

7.2 Taylor formulation and its precision limitation

For comparison put

$$a_{r,j}^X = [x^{2j}] \sum_{k=0}^{n_r} w_k R_d(k,x), \qquad X = C, J.$$

For each fixed r, this expansion converges at $x = p^r$: every denominator has p-adic valuation at most $r-1$, so $x^2/[d(a+1)]^2 \in p^2\mathbb{Z}_p$. One obtains the exact convergent identity

$$\Delta_X(r) = -\sum_{j\ge 1} p^{2rj} \left(a_{r,j}^X - \eta p^{1-2j} a_{r-1,j}^X \right).$$

For $r = 0$, take $a_{0,0} = 1$ and $a_{0,j} = 0$ for $j > 0$. The quadratic coefficient is explicitly

$$a_{r,1}^X = \sum_{k=0}^{n_r} w_k \left(\frac{1}{d^2} H_k^{(2)} - \sum_{a=0}^{k-1} \frac{1}{(2a+1)^2} \right).$$

The coefficients need not be integral uniformly in r. Exact rational calculation for $p = 5$ gives $v_5(a_{r,1}^C) = v_5(a_{r,1}^J) = 2, 1, 0, -1$ for $r = 1, 2, 3, 4$, respectively. Thus simply replacing p by p^r in the first-level integral Taylor argument is invalid. The finite factor separation above is one way to retain correct valuation bookkeeping.

7.3 Finite calculations, not an all-level proof

The program `probe.py` checks the original recurrences for $p \in \{5,7,11,13,17,19,23,29,31\}$ and $1 \le r \le 5$ with $p^r \le 200000$, at precision p^{4r+4}: 36 pairs (p,r) for each conjecture. All passed. Except for $p=7$, the tested defects had exact valuation $4r$; at $p=7$ the C defect had valuation 6 at $r=1$ and $4r+1$ thereafter, while the J defect had valuation $4r+1$ throughout. The additional observed relation

$$\frac{\Delta_C(r)}{p^{4r}} \equiv \frac{7}{6}B_{p-3}, \qquad \frac{\Delta_J(r)}{p^{4r}} \equiv \varepsilon^r \frac{7}{24}B_{p-3} \pmod p$$

is not proved here. The J relation is a specialization of Sun's 2019 conjecture, quoted by Mao–Sun [7]; see the attribution below. The directly computed leading digits are stored in `probe.jsonl`; `validate.py` independently checks all 72 Bernoulli leading-digit comparisons, eight exact rational instances of the finite block identity, and the two negative-valuation examples.

7.4 A WZ boundary form for the J sum

For a positive odd integer N, write $n=(N-1)/2$ and

$$J(N) = \sum_{a=0}^{n} (6a+1)\frac{\binom{2a}{a}^3}{256^a}.$$

The classical WZ pair used by Mao–Sun [7, Section 3] is $F(a,k) = (6a-2k+1)B(a,k)$ and $G(a,k) = 16a^2B(a,k)/(2a+2k-1)$, where

$$B(a,k) = \frac{(1/2)_a(1/2)_{a+k}(1/2)_{a-k}}{4^a(a!)^2(a-k)!(1/2)_k}.$$

The convention is $F=G=0$ for $a<k$. Direct cancellation of adjacent factors verifies $F(a,k-1) - F(a,k) = G(a+1,k) - G(a,k)$. Summing first over $0 \le a \le n$ and then over $1 \le k \le n+1$ gives

$$J(N) = \sum_{k=1}^{n+1} G(n+1,k).$$

Put

$$E_N = \frac{N}{2^{3N-3}}\binom{2N-1}{N-1}\binom{N-1}{(N-1)/2}, \qquad T_N = \sum_{l=0}^{(N-1)/2} \frac{(1/2)_l(-N/2)_l}{l!(1/2-N)_l}.$$

Then the exact identity

$$\boxed{J(N) = E_N T_N}$$

follows by reversing $k = n+1-l$. Indeed $G(n+1,n+1) = E_N$ and the ratio of successive reversed terms is

$$\frac{(2l+1)(N-2l)}{(2N-2l-1)(2l+2)}.$$

This is the ratio of the displayed ${}_2F_1$ summands.

For $N=p^r$, $M=N/p$, and $p>3$, both binomial factors in E_N/N are p-adic units by Lucas's theorem. Let

$$U_{p,r} = \frac{E_N}{pE_M} \in \mathbb{Z}_p^\times.$$

Consequently the original J.3 statement is exactly equivalent to

$$T_N - \varepsilon U_{p,r}^{-1} T_M \in p^{3r}\mathbb{Z}_p, \qquad \varepsilon = (-1)^{(p-1)/2}.$$

The known p^{3r} result for J gives only p^{2r} precision in this last display. No p^{3r} proof for this transformed sum is established here. The boundary identity combines the endpoint and WZ correction before estimating them, and lowers the hypergeometric order, but does not by itself supply the missing precision.

7.5 Prior source of the refined J conjecture

The leading Bernoulli digit observed computationally is a special case of an existing stronger conjecture. Mao–Sun, arXiv:2109.09877, p. 2, quote Z.-W. Sun's 2019 Conjecture 22: for every positive odd m,

$$\frac{16^{m-1}}{(pm)^4\binom{m-1}{(m-1)/2}^3}\big(J(pm)-\varepsilon pJ(m)\big)\equiv\varepsilon\frac{7}{24}B_{p-3}\pmod p.$$

Their paper proves the case $m=1$. With $m=p^{r-1}$, Lucas's theorem and Fermat's theorem turn this conjecture into

$$\frac{J(p^r)-\varepsilon pJ(p^{r-1})}{p^{4r}}\equiv\varepsilon^r\frac{7}{24}B_{p-3}\pmod p.$$

Thus the observed all-level Bernoulli pattern should be attributed to that conjecture, and should not be described as a new conjecture.

7.6 An exact parity-product embedding

The half-length sum can be embedded in a full-length product sum, but the resulting factors depend on parity. Define

$$f_{\mathrm{odd}}(x)=\frac{1}{1-2x},\qquad f_{\mathrm{even}}(x)=1-x,\qquad P_k(N)=\prod_{j=1}^{k}f_{j\bmod 2}(N/j).$$

For $0\le k<N=p^r$, every argument N/j belongs to $p\mathbb{Z}_p$, so these products are integral units. Duplication of the factors gives

$$\frac{(1/2)_l(-N/2)_l}{l!(1/2-N)_l}=-\frac{N}{2l}P_{2l-1}(N)\qquad(l\ge1).$$

Since $P_{2l}-P_{2l-1}=-(N/(2l))P_{2l-1}$, one obtains

$$\boxed{T_N=\sum_{k=0}^{N-1}(-1)^kP_k(N).}$$

This has exactly N terms. Under a split $j=pm+b$ with odd p, the factors indexed by multiples of p reproduce the same parity pattern at $M=N/p$. The unit factors still have alternating linear coefficients -1 and 2, however, so the constant-function cutoff-product scaling theorem cannot be applied without proving a separate parity-dependent version. No such extension is asserted here.

7.7 The J endpoint multiplier can be eliminated

Put $B_N=\binom{N-1}{(N-1)/2}$ and $D_N=\binom{2N-1}{N-1}$. For $N=p^r$, $M=N/p$, and $p>3$, the prime-power form of Morley's congruence gives

$$B_N/B_M\equiv\varepsilon\,4^{N-M}\pmod{p^{3r}},\qquad\varepsilon=(-1)^{(p-1)/2}.$$

This is Cai's Corollary 3 [3]. Cai's Theorem 3 with $n = N$, $u = 2$, $v = 1$ gives $D_N/D_M \equiv 1 \pmod{p^{3r}}$. Consequently the exact WZ prefactor

$$E_N = ND_NB_N/2^{3N-3}$$

satisfies

$$\frac{E_N}{pE_M} \equiv \varepsilon 2^{M-N} \pmod{p^{3r}}.$$

Both E_N/N and T_N are p-adic units. For the former this follows from Lucas's theorem. For the latter, each term after the initial 1 in the exact product formula below is divisible by p. Thus Swisher's full J.3 recurrence is equivalent to

$$T_{p^r} \equiv 2^{p^r-p^{r-1}}T_{p^{r-1}} \pmod{p^{3r}}.$$

Equivalently, $2^{-p^r}T_{p^r}$ must be stable modulo p^{3r}. This eliminates all gamma or factorial prefactors; the remaining period-two product stability is still to be proved.

7.8 A terminating Euler coefficient transform for the J residual

For odd N, set $h = (N-1)/2$ and $c = 1/2 - N$. Define

$$T_N = \sum_{l=0}^{h} \frac{(1/2)_l(-N/2)_l}{l!(c)_l}.$$

Euler's hypergeometric transformation gives the formal identity

$${}_2F_1(1/2, -N/2; c; z) = (1-z)^{-N/2}{}_2F_1(-N, -h; c; z).$$

The second hypergeometric series is a polynomial of degree h. Since a partial sum of Taylor coefficients through degree h is the coefficient of z^h after division by $1 - z$, it follows that

$$\begin{aligned} T_N &= [z^h](1-z)^{-N/2-1}{}_2F_1(-N, -h; c; z) \\ &= \frac{(1+N/2)_h}{h!}{}_3F_2\left[\begin{matrix}-h, -h, -N \\ c, c\end{matrix}; 1\right]. \end{aligned}$$

Indeed, the coefficient ratio needed in the convolution is

$$\frac{(1+N/2)_{h-j}/(h-j)!}{(1+N/2)_h/h!} = \frac{(-h)_j}{(c)_j}.$$

The resulting ${}_3F_2$ is zero-balanced: the sum of its lower parameters equals the sum of its upper parameters. It does not meet the usual Saalschütz balance condition, which requires their difference to be 1. Its parameters $-h = (1-N)/2$ and $c = 1/2 - N$ also retain constant half shifts. Consequently this exact transform does not by itself bring the J residual within the pure parameter scaling theorem. No all-level congruence for this transformed sum is claimed here.

8 An exact obstruction to a direct $(A.3)$ lift

Define, for odd N,

$$A(N) = \sum_{k=0}^{(N-1)/2} (4k+1)(-1)^k \frac{(1/2)_k^5}{k!^5}.$$

The four-root perturbation in Swisher's first-level argument, with p replaced by N, is the following rational number:

$$\widetilde{A}(N)=\sum_{k=0}^{(N-1)/2}(4k+1)(-1)^k\frac{(1/2)_k^5}{k!^5}\frac{\displaystyle\prod_{j=0}^{k-1}\left(1-\frac{N^4}{(2j+1)^4}\right)}{\displaystyle\prod_{j=1}^{k}\left(1-\frac{N^4}{16j^4}\right)}. \tag{25}$$

It is the terminating very-well-poised ${}_6F_5(-1)$ with upper parameters $5/4, 1/2, (1-iN)/2, (1+iN)/2, (1-N)/2, (1+N)/2$ and lower parameters $1/4, 1+iN/2, 1-iN/2, 1+N/2, 1-N/2$. Whipple's transformation and the classical gamma reflection identity give the exact identity

$$\widetilde{A}(N)=(-1)^{(N-1)/2}N\,{}_3F_2\left[\begin{matrix}1/2,(1-N)/2,(1+N)/2\\ 1+iN/2,1-iN/2\end{matrix};1\right]. \tag{26}$$

The series on the right terminates at $(N-1)/2$. In particular it is a *perturbed* cubic hypergeometric sum; it is not the unperturbed sum in $(H.3)$.

A direct extension of the first-level comparison would require

$$\widetilde{A}(p^r)\equiv A(p^r)\pmod{p^{5r}}.$$

This proposed intermediate statement is false. Exact rational arithmetic at $p=5$, $r=2$ gives

$$\widetilde{A}(25)-A(25)\equiv 7062500\pmod{5^{10}},\qquad v_5\big(\widetilde{A}(25)-A(25)\big)=6. \tag{27}$$

Equivalently, its quotient by 5^6 is 2 modulo 5. This is a counterexample to the proposed intermediate comparison, *not* a counterexample to $(A.3)$.

The calculation can be reproduced without complex arithmetic: if t_k is the kth term of (25) after deleting $4k+1$, then $t_0=1$ and

$$\frac{t_k}{t_{k-1}}=-\frac{(2k-1)\big((2k-1)^4-N^4\big)}{2k\big(16k^4-N^4\big)}.$$

For the eight tested pairs $p=5,13$, $r=1,2,3$, and $p=17$, $r=1,2$, the exact valuation of $\widetilde{A}(p^r)-A(p^r)$ is $r+4$. These finite observations are not asserted as a general valuation theorem.

There is a corresponding obstacle to replacing the right side of (26) by the unperturbed $(H.3)$ sum. Writing

$$H(N)=\sum_{k=0}^{(N-1)/2}\frac{(1/2)_k^3}{k!^3},$$

the six exact tests $p=5,13$, $r=1,2,3$ give

$$v_p\big(A(p^r)-p^rH(p^r)\big)=r+3,\qquad v_p\big(\widetilde{A}(p^r)-p^rH(p^r)\big)=r+3.$$

Thus an all-level theorem for H does not by itself control the parameter corrections required for A. Terms with indices divisible by p introduce lower-level denominator valuations, and their corrections must be transported between levels before the exponent $5r$ can be justified.

9 Exact boundary residuals and an odd-level obstruction

The following elementary statement isolates a hypothesis needed to transfer the jet argument of corpus [132]. It is a result about finite hypergeometric polynomials, not a proof of any remaining Swisher conjecture.

Proposition 9.1 (Boundary residual and parity of its valuation)**.** *Let $d \geq 3$, let $p > d$ be a prime with $p \equiv -1 \pmod{d}$, and let $h \geq 1$ be an integer. Set*

$$c_k = \left(\frac{(1/d)_k}{k!}\right)^h, \qquad F_N(z) = \sum_{k=0}^{N} c_k z^k, \qquad \theta = z\frac{d}{dz}, \qquad \mathcal{L}_{d,h} = \theta^h - z(\theta + 1/d)^h.$$

For every $N \geq 0$ there is the exact identity

$$\mathcal{L}_{d,h} F_N(z) = -(N + 1/d)^h c_N z^{N+1}. \tag{28}$$

If $s \geq 1$ is odd and

$$N_- = \frac{p^s - d + 1}{d}, \qquad N_+ = \frac{(d-1)p^s - 1}{d},$$

then

$$v_p\left(\frac{(1/d)_{N_-}}{N_-!}\right) = \frac{s-1}{2}, \qquad v_p\Big((N_- + 1/d)^h c_{N_-}\Big) = \frac{h(s-1)}{2}, \tag{29}$$

$$v_p\left(\frac{(1/d)_{N_+}}{N_+!}\right) = 0, \qquad v_p\Big((N_+ + 1/d)^h c_{N_+}\Big) = hs. \tag{30}$$

If $s \geq 2$ is even and $N_0 = (p^s - 1)/d$, then

$$v_p\left(\frac{(1/d)_{N_0}}{N_0!}\right) = 0, \qquad v_p\Big((N_0 + 1/d)^h c_{N_0}\Big) = hs. \tag{31}$$

Proof. The coefficient recurrence $k^h c_k = (k - 1 + 1/d)^h c_{k-1}$ cancels every interior coefficient in $\mathcal{L}_{d,h} F_N$, giving (28).

For the valuation of $(1/d)_N/N!$, count the factors divisible by $t = p^e$ in $\prod_{j=0}^{N-1}(dj+1)$ and subtract $\lfloor N/t \rfloor$. The least nonnegative solution to $dj + 1 \equiv 0 \pmod{t}$ is

$$j_e = \begin{cases} (t-1)/d, & e \text{ even}, \\ ((d-1)t - 1)/d, & e \text{ odd}. \end{cases}$$

Writing $N = qt + R$ with $0 \leq R < t$, the difference of the two counts is 1 if $j_e < R$, and 0 otherwise.

For odd s and $N = N_-$, the remainder for $1 \leq e < s$ is

$$R = \begin{cases} (d-1)(t-1)/d, & e \text{ even}, \\ (t - d + 1)/d, & e \text{ odd}. \end{cases}$$

Thus the count difference is 1 exactly when e is even. There is no contribution for $e \geq s$, since both $N_- < p^s$ and $d(N_- - 1) + 1 = p^s - 2d + 2 < p^s$. This proves the first identity of (29). Moreover $N_- + 1/d = (p^s - d + 2)/d$ is a p-adic unit because $p > d$. The second identity follows.

For $N = N_+$, its remainder modulo p^e equals j_e for $1 \leq e \leq s$, so every count difference is zero. No larger power contributes: $N_+ < p^s$, and the numerator factors are less than $(d-1)p^s < p^{s+1}$. Also $N_+ + 1/d = (d-1)p^s/d$, of valuation s. This proves (30).

For even s and $N = N_0$, the remainder again equals j_e for $1 \leq e \leq s$; there are no larger contributions. Since $N_0 + 1/d = p^s/d$, (31) follows. □

Two exact examples. For $d = 3$, $p = 5$, and $s = 3$, the two cutoffs are $N_- = 41$ and $N_+ = 83$. Their top-coefficient ratios have valuations 1 and 0, respectively. The boundary residual coefficients therefore have valuations h and $3h$; for cubic sums these are 3 and 9. For $d = 4$, $p = 7$, and $s = 3$, the cutoffs are $N_- = 85$ and $N_+ = 257$, with the same respective ratio valuations 1 and 0 and boundary residual valuations h and $3h$.

Consequence for transferring the method. For the unweighted hypergeometric polynomial underlying the short odd cutoffs, the residual has exact valuation $h(s-1)/2$, not hs. Consequently one cannot import the p^{3s} residual hypothesis of corpus [132] unchanged into the cubic cases $d = 3, 4$. The statement also records the boundary valuation for exponents $h = 4$ or $h = 6$ if those polynomials occur in a proposed argument. It does not identify their differential modules or establish a Frobenius congruence for them.

The longer odd cutoff restores the boundary precision hs. A possible repair must therefore establish both the appropriate Frobenius theorem for that polynomial and a sufficiently precise relation between its long and short truncations. Neither follows from the valuation calculation alone. The linear weight and specialization in each Swisher sum also have to be accounted for separately.

10 Verification and the six remaining targets

The C and K proofs are symbolic and hold for all permitted primes and levels. Finite computation is an additional check, not a substitute for any step of the proof. The supplied `audit_blocks.py` checks all complete-prefix harmonic bounds, both exact summation-by-parts identities, and the valuations of the linear, quadratic, and expansion-remainder terms. All 13 cases pass, including $(p, r) = (5, 4)$, $(7, 4)$, and $(11, 3)$; the largest cutoff is $7^4 = 2401$. The separate `verify_scaling.py` performs 126 exact rational tests of the general theorem, including a zero linear coefficient, a function singular at 1, and $\kappa = 0$. It also tests the original K.3 and its residual in 12 cases. All pass. An independent `audit_c.py` checks the C factorization, every full/half-prefix harmonic bound, every alternating-prefix bound, each divisible-index product identity, the exact summation by parts, and the prefactor congruence in the same 13 prime/level cases. All pass, with maximum cutoff $7^4 = 2401$. The C WZ identity was also verified symbolically, and its finite convolution was checked for every odd N from 1 to 101.

The original `verify_reductions.py` checks 26 exact K factorizations, eight K prefactor cases, four F diagonal recurrences, five F decompositions, the false F termwise bound, the false A perturbation bridge, and five parameter sets in the parity lemma. Every check passes. The C/J validator retains its 72 original congruence and Bernoulli-digit comparisons and eight finite block identity checks. These computations remain supporting evidence only for the refined Bernoulli-digit statements and the unproved J recurrence.

For J.3 the remaining statement is now precisely

$$2^{-p^r} T_{p^r} - 2^{-p^{r-1}} T_{p^{r-1}} \in p^{3r} \mathbb{Z}_p.$$

Its factors depend on parity. Theorem 2.1 assumes one common function f and does not establish this variant. For the odd F branch, the remaining boundary target is

$$B_r - p^2 B_{r-2} \in p^r \mathbb{Z}_p \qquad (r \geq 3 \text{ odd}).$$

For C, the earlier finite three-term obstruction now vanishes as a consequence of Section 4; its components need not individually have the target valuation. For A, parameter-deformation corrections must be transported between levels. For D, E, and G, the required all-level cancellation has not been supplied. Thus C.3 and K.3 are the two targets completed in this revision; the other six remain outside the proved conclusions of this note.

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MANUSCRIPT III

All-Level Proofs for Swisher J.3 and the Remaining Odd Branch of F.3

Research working manuscript 10 November 2026

Abstract

We prove two all-level supercongruences from Swisher's prime-power table. For J.3, an elementary telescoping identity, a general half-cutoff product-scaling lemma, and Cai's prime-power Morley congruence give the original modulus p^{4r} for every prime $p > 3$. For the remaining odd branch of F.3, creative microscoping with an adapted short cutoff gives the stronger modulus $p^{3(r-1)/2}$ for every prime $p \equiv 3 \pmod 4$ and odd $r \geq 3$, hence the required p^r. We explicitly account for the parameter denominator, cyclotomic multiplicities, and the integral specialization at $q = 1$. The E.3 odd branch and the A.3, D.3, and G.3 families remain outside the conclusions. Exact computations accompany the proofs. This is a research working manuscript; no external review or historical priority is claimed.

1 Statement and scope

For a positive odd integer N, define

$$J(N) = \sum_{k=0}^{(N-1)/2} (6k+1)\frac{\binom{2k}{k}^3}{256^k} = \sum_{k=0}^{(N-1)/2} (6k+1)\frac{(1/2)_k^3}{(k!)^3\,4^k}.$$

Here $(a)_0 = 1$, $(a)_k = a(a+1)\cdots(a+k-1)$, and $J(1) = 1$. Congruences between rational numbers are interpreted in $\mathbb{Z}_p$.

Theorem 1. *For every prime $p > 3$ and every integer $r \geq 1$,*

$$\boxed{J(p^r) \equiv (-1)^{(p-1)/2} p J(p^{r-1}) \pmod{p^{4r}}.}$$

This is the half-cutoff J.3 recurrence in the published table of Swisher [1]. The proof covers both residue classes modulo 4 and includes the first level $r = 1$. It does not replace the half-cutoff by a full cutoff. For this J.3 proof, the substantive imported arithmetic result is Cai's prime-power Morley congruence [2, Corollary 3]. The finite identity and the residual scaling estimate are proved below.

The second main conclusion concerns the exact remaining odd branch of F.3. Write

$$F_r(p) = \sum_{k=0}^{(p^r-3)/4} (-1)^k (8k+1)\frac{(1/4)_k^3}{(k!)^3} \qquad (p \equiv 3 \pmod 4,\ r \text{ odd}).$$

We prove in Section 5 that

$$\boxed{F_r(p) \equiv p^2 F_{r-2}(p) \pmod{p^{3(r-1)/2}} \qquad (r \geq 3 \text{ odd}).}$$

Since $3(r-1)/2 \geq r$, this attains the original odd-branch target. This includes $p = 3$. Ordinary primes and even levels of F.3 are not reproved here; the theorem addresses the remaining range specified for this research round.

2 The exact finite factorization

For $n \geq 0$ put $b_n = (1/2)_n/n!$ and

$$q(n,k) = \frac{(-n-1/2)_k^2}{(k!)^2}, \qquad V_n = \sum_{k=0}^{n} q(n,k), \qquad B_n = \frac{(2n+1)b_n}{4^n}.$$

We prove $B_n V_n = \sum_{k=0}^{n} (6k+1) b_k^3/4^k$. For $n \geq 1$ define

$$R(n,k) = \frac{4k^2(4k-6n-3)}{(2n+1)^3}.$$

The two term ratios

$$\frac{q(n,k+1)}{q(n,k)} = \frac{(2k-2n-1)^2}{4(k+1)^2}, \qquad \frac{q(n-1,k)}{q(n,k)} = \frac{(2n+1-2k)^2}{(2n+1)^2}$$

show, by direct polynomial cancellation, that

$$q(n,k+1)R(n,k+1) - q(n,k)R(n,k) = q(n,k) - \frac{8n}{2n+1} q(n-1,k).$$

Summing over $0 \leq k \leq n$, and using

$$R(n,0) = 0, \qquad q(n-1,n) = b_n^2, \qquad q(n,n+1)R(n,n+1) = \frac{1-2n}{2n+1} b_n^2,$$

gives

$$V_n - \frac{8n}{2n+1} V_{n-1} = \frac{6n+1}{2n+1} b_n^2.$$

Since $B_n/B_{n-1} = (2n+1)/(8n)$, multiplication by B_n yields

$$B_n V_n - B_{n-1} V_{n-1} = (6n+1) \frac{b_n^3}{4^n}.$$

The initial value is $B_0 V_0 = 1$, proving the asserted identity by induction. Taking $N = 2n+1$ gives

$$J(N) = \frac{N}{4^{N-1}} \binom{N-1}{(N-1)/2} \sum_{k=0}^{(N-1)/2} \frac{(-N/2)_k^2}{(k!)^2}.$$

This certificate proves the identity over the rational numbers, without an appeal to convergence or cancellation of terminating parameters.

3 A half-cutoff square-weighted scaling lemma

Let $p > 3$, $f(X) \in 1 + X\mathbb{Z}_p[[X]]$, and $\kappa \in \mathbb{Z}_p$. For an odd prime power $L = p^s$ put

$$R_k(L) = \prod_{j=1}^{k-1} f(L/j), \qquad V_f(L) = 1 + \kappa L^2 \sum_{k=1}^{(L-1)/2} \frac{R_k(L)}{k^2}.$$

The empty sum at $L = 1$ is zero. All product arguments lie in $p\mathbb{Z}_p$, and all terms of $V_f(L)$ are integral. The following congruence is the general scaling theorem:

$$V_f(p^r) \equiv V_f(p^{r-1}) \pmod{p^{3r}} \qquad (r \geq 1). \tag{1}$$

Write $N = p^r$, $M = N/p$, and $n = (N-1)/2$. Define

$$H_k = \sum_{\substack{1 \le j \le k \\ p \nmid j}} j^{-2}.$$

Complete unit residue sums show $v_p(H_k) \ge v_p(k)$. Also $H_n \in p^r\mathbb{Z}_p$: reflection $j \mapsto N-j$ identifies the two half sums modulo p^r, while their complete sum is zero modulo p^r. Here the inverse-square sum on the units modulo p^a is zero modulo p^a for every $a \ge 1$, since $p > 3$. Since $f(X) - 1 \in X\mathbb{Z}_p[[X]]$,

$$v_p(R_{k+1}(N) - R_k(N)) \ge r - v_p(k).$$

Summation by parts therefore gives

$$\sum_{\substack{1 \le k \le n \\ p \nmid k}} \frac{R_k(N)}{k^2} = R_n(N)H_n - \sum_{k=1}^{n-1} H_k(R_{k+1}(N) - R_k(N)) \in p^r\mathbb{Z}_p.$$

After multiplication by κN^2, the unit-index contribution is zero modulo p^{3r}.

For a remaining index $k = pb$, $1 \le b \le (M-1)/2$, splitting the product into unit and divisible indices gives the exact identity

$$R_{pb}(N) = R_b(M)A_b, \qquad A_b = \prod_{\substack{1 \le j < pb \\ p \nmid j}} f(N/j).$$

Set $a = v_p(pb) = 1 + v_p(b) \le r - 1$. For $L = pb$ define

$$h_i(L) = \sum_{\substack{1 \le j < L \\ p \nmid j}} j^{-i}.$$

Then $h_2(L) \in p^a\mathbb{Z}_p$ by complete residue sums, and reflection $j \mapsto L - j$ gives

$$2h_1(L) \equiv -Lh_2(L) \pmod{p^{2a}}, \qquad h_1(L) \in p^{2a}\mathbb{Z}_p.$$

Write $f(X) = 1 + dX + eX^2 + O(X^3)$. A finite multiplication, with an integral remainder, gives

$$A_b \equiv 1 + dNh_1(L) + \frac{N^2}{2}\bigl(d^2h_1(L)^2 + (2e - d^2)h_2(L)\bigr) \pmod{p^{3r}}.$$

Consequently $A_b - 1 \in p^{r+2a}\mathbb{Z}_p$: the displayed linear, quadratic, and remainder terms have valuations at least $r + 2a$, $2r + a$, and $3r$, respectively. Since $v_p(M^2/b^2) = 2(r-a)$, each term

$$\kappa\frac{N^2}{(pb)^2}R_{pb}(N) - \kappa\frac{M^2}{b^2}R_b(M)$$

belongs to $p^{3r}\mathbb{Z}_p$. This proves (1), including $r = 1$, for which the divisible-index sum is empty.

3.1 Application to the transformed J sum

Take $f(X) = (1 - X/2)^2$ and $\kappa = 1/4$. Then

$$V_f(N) = \sum_{k=0}^{(N-1)/2} \frac{(-N/2)_k^2}{(k!)^2},$$

because for $k \geq 1$

$$\frac{(-N/2)_k^2}{(k!)^2} = \frac{N^2}{4k^2} \prod_{j=1}^{k-1} (1 - N/(2j))^2.$$

The exact finite transformation proved separately gives

$$J(N) = B(N)V_f(N), \qquad B(N) = \frac{N}{4^{N-1}} \binom{N-1}{(N-1)/2}. \tag{2}$$

Cai's prime-power Morley congruence [2, Corollary 3] implies

$$B(p^r) \equiv \varepsilon p B(p^{r-1}) \pmod{p^{4r}}, \qquad \varepsilon = (-1)^{(p-1)/2}.$$

More explicitly, for $N = p^r$ and $M = p^{r-1}$, Cai's result is

$$\frac{\binom{N-1}{(N-1)/2}}{\binom{M-1}{(M-1)/2}} \equiv \varepsilon 4^{N-M} \pmod{p^{3r}}.$$

The central binomial coefficients here are units: for example, $\binom{N-1}{h} = \prod_{j=1}^{h} (N-j)/j$ is a product of units, since $v_p(N-j) = v_p(j)$ for $1 \leq j < N$. Thus $v_p(B(N)) = r$, and the displayed congruence is equivalent to $B(N)/(pB(M)) \equiv \varepsilon \pmod{p^{3r}}$. Combining this with (1) yields

$$\boxed{J(p^r) \equiv \varepsilon p J(p^{r-1}) \pmod{p^{4r}} \qquad (p > 3,\ r \geq 1).}$$

The finite transformation and the scaling proof are independent of any conjectural deformation estimate.

For clarity, the final error splits as

$$\begin{aligned} J(N) - \varepsilon p J(M) = {} & B(N)(V_f(N) - V_f(M)) \\ & + (B(N) - \varepsilon p B(M))V_f(M). \end{aligned}$$

The two terms lie in $p^{4r}\mathbb{Z}_p$ separately. This completes the proof of Theorem 1; no inference from a finite numerical range is used.

4 What the general lemma adds

The product-scaling estimate is not restricted to the particular quadratic polynomial needed for J.3. For any fixed $\lambda \in \mathbb{Z}_p$,

$$Q_\lambda(p^r) := \sum_{k=0}^{(p^r-1)/2} \frac{(-\lambda p^r)_k^2}{(k!)^2} \equiv Q_\lambda(p^{r-1}) \pmod{p^{3r}}.$$

Indeed take $f(X) = (1 - \lambda X)^2$ and $\kappa = \lambda^2$; the case $\lambda = 0$ is immediate. More generally, f may be any power series in $1 + X\mathbb{Z}_p[[X]]$, with the same p^{3r} modulus. Convergence of each factor is automatic because $p^r/j \in p\mathbb{Z}_p$ throughout its finite product. The proof uses only inverse-square unit sums, reflection, and summation by parts.

5 The remaining odd branch of F.3

Let $(x;q)_k = \prod_{i=0}^{k-1}(1-xq^i)$ and $[m]_q = (1-q^m)/(1-q)$. Products with several first arguments mean the product of the respective shifted factorials. Define

$$T_L(a,q) = \sum_{k=0}^{L} c_k(a,q), \qquad c_k(a,q) = (-1)^k[8k+1]_q q^k \frac{(aq, q/a, q, -q^2; q^4)_k}{(aq^4, q^4/a, q^4, -q^3; q^4)_k}.$$

The parameter a is an indeterminate throughout every root-of-unity calculation. In particular, parameter denominators do not vanish there.

Theorem 2. *Let $p \equiv 3 \pmod 4$ be prime and $r = 2s+1 \geq 3$. Put $N = (p^r-3)/4$ and $M = (p^{r-2}-3)/4$. Then*

$$\sum_{k=0}^{N}(-1)^k(8k+1)\frac{(1/4)_k^3}{k!^3} \equiv p^2 \sum_{k=0}^{M}(-1)^k(8k+1)\frac{(1/4)_k^3}{k!^3} \pmod{p^{3s}}.$$

Since $3s \geq 2s+1 = r$, this implies the odd branch of Swisher F.3.

5.1 The terminating evaluation

We use the following terminating very-well-poised basic hypergeometric evaluation, with $v = 4t+1$:

$$T_t(q^v, q) = T_t(q^{-v}, q) = (-1)^t q^{-2t}[v]_q. \tag{3}$$

Here is an exact general identity from which it follows:

$$\sum_{k=0}^{t}[2mk+1]_q \frac{(q^{1-v}, q^{1+v}, q/c, q; q^m)_k}{(q^{m-v}, q^{m+v}, cq^m, q^m; q^m)_k} c^k q^{(m-2)k} = (c/q)^t \frac{(q^2/c; q^m)_t}{(cq^m; q^m)_t}[v]_q, \quad v = mt+1. \tag{4}$$

To specify the input identity completely, the terminating very-well-poised ${}_6\phi_5$ summation is

$$\sum_{k=0}^{t} \frac{1-AQ^{2k}}{1-A} \frac{(A,B,C,Q^{-t};Q)_k}{(Q, AQ/B, AQ/C, AQ^{t+1};Q)_k}\left(\frac{AQ^{t+1}}{BC}\right)^k = \frac{(AQ, AQ/(BC);Q)_t}{(AQ/B, AQ/C;Q)_t}.$$

Take $Q = q^m$, $A = q$, $B = q^{1+v}$, $C = q/c$, and $v = mt+1$. The argument becomes cq^{m-2}. On the right, reversal of the two finite products with negative powers gives

$$\frac{(q^{m+1};q^m)_t}{(q^{m-v};q^m)_t} = (-1)^t q^{mt(t-1)/2+t}[v]_q,$$

$$(cq^{m-v-1};q^m)_t = (-c)^t q^{-mt(t-1)/2-2t}(q^2/c;q^m)_t,$$

which proves (4) from that standard summation. This is also the specialization used in Guo–Zudilin, *Dwork-type supercongruences through a creative q-microscope*, arXiv:2001.02311 [3], formula (3.23) and the proof of Theorem 3.6. Its substitution $m = 4, c = -q^{-1}$ gives (3). Equation (4) is an exact identity at the two parameter specializations, rather than a congruence.

Lemma 3 (Root blocks)**.** *Suppose $h > 1$, $h \equiv 1 \pmod 4$, and ζ is a primitive hth root of unity. Write $t = (h-1)/4$. For $b \geq 0$ and $0 \leq j < h$,*

$$\lim_{q\to\zeta} c_{bh+j}(a,q) = \begin{cases} (-1)^b \dfrac{(1/4)_b}{b!} c_j(a,\zeta), & 0 \leq j \leq t, \\ 0, & t < j < h. \end{cases} \tag{5}$$

Furthermore $c_{t-j}(a,\zeta) = -c_j(a,\zeta)$ for $0 \leq j \leq t$. Consequently, if $L \bmod h \geq t$, then $T_L(a,q)$ vanishes modulo $\Phi_h(q)$, in the local ring with coefficient field $\mathbb{Q}(a)$.

Proof. Set

$$U_k = \frac{(aq, q/a; q^4)_k}{(aq^4, q^4/a; q^4)_k}, \qquad Q_k = \frac{(q; q^4)_k}{(q^4; q^4)_k}, \qquad V_k = \frac{(-q^2; q^4)_k}{(-q^3; q^4)_k}.$$

In a complete block of length h, the numerator and denominator of U each run over the same root residues, so their ratio is one. The analogous full-block ratio for V is one because both products equal 2. In Q, the numerator zeros occur at indices $t+uh$ and the denominator zeros at indices $(u+1)h-1$. Their limiting ratio in the uth block is $(4u+1)/(4u+4)$; the products of nonzero factors cancel by permutation of the nontrivial roots. Thus b full blocks contribute $(1/4)_b/b!$. The external $(-1)^k q^k$ contributes $(-1)^b$. A final partial block with $j > t$ has one extra numerator zero. This proves (5).

For completeness, reversing the products in the first short block gives, with all the following expressions evaluated at $q = \zeta$,

$$\frac{U_{t-j}}{U_j} = q^{6j-4t(t+1)}, \qquad \frac{Q_{t-j}}{Q_j} = (-1)^t q^{3j-2t(t+1)}, \qquad \frac{V_{t-j}}{V_j} = q^{j-2t^2-t}.$$

These follow by replacing the reversed factors q^{h-u} by q^{-u}; all denominators here are nonzero for generic a. Also $[8(t-j)+1]_q = -q^{-8j-1}[8j+1]_q$. Combining these identities and the factor $(-1)^{t-2j}q^{t-2j}$ from the external sign and power yields

$$c_{t-j}(a, q) = -q^{-(2t+1)h} c_j(a, q) = -c_j(a, q).$$

The short sum is zero, and (5) proves the assertion for every cutoff whose last block includes this short block. □

5.2 Parameter interpolation and the common denominator

Set $n = p^2$, so $n \equiv 1 \pmod 8$, and write

$$N = nM + \frac{3(n-1)}{4}, \qquad \Delta(a, q) = T_N(a, q) - q^{(1-n)/2}[n]_q T_M(a, q^n).$$

For $0 \le j \le M$, let $A_j = (4j+1)n$. At $a = q^{\pm A_j}$, the left series terminates at $nj + (n-1)/4 \le N$, while the right series terminates at j. By (3), the left value is $(-1)^{nj+(n-1)/4} q^{(1-A_j)/2}[A_j]_q$. The right value is $q^{(1-n)/2}[n]_q(-1)^j q^{-2nj}[4j+1]_{q^n}$. They agree since $(n-1)/4$ is even and $[n]_q[4j+1]_{q^n} = [A_j]_q$. Therefore

$$P(a, q) = \prod_{j=0}^{M} (1 - aq^{A_j})(a - q^{A_j})$$

divides the parameter numerator of Δ. Explicitly a common parameter denominator is

$$D(a, q) = (aq^4, q^4/a; q^4)_N.$$

It also clears the right denominator: for $1 \le i \le M$, the exponent $4ni$ is among the exponents $4j$ with $1 \le j \le N$. Hence $a^N D\Delta$ is a polynomial in a with coefficients in $\mathbb{Q}(q)$ and is divisible by P in $\mathbb{Q}(q)[a]$. The leading coefficient of P is a signed power of q.

5.3 Preserving an extra cyclotomic factor

Fix an even $\ell \in \{2, 4, \dots, 2s\}$, and put $h = p^\ell$. Let $R_h = \mathbb{Q}[q, q^{-1}]_{(\Phi_h)}$. For any odd integer d, the ratio $Q_k = (q; q^4)_k/(q^4; q^4)_k$ has nonnegative Φ_d valuation: in each period d its numerator zero occurs at some index $0 \le j_0 < d$, whereas its denominator zero occurs at index $d-1$. The factors $1 + q^{4i+3}$

have no zero at an odd-order root. The same facts apply to q^n at a primitive p^ℓ root (at $\ell = 2$ the limit is $q^n = 1$ and Q_k has the finite value $(1/4)_k/k!$). Thus $a^N D\Delta \in R_h[a]$.

At a primitive hth root, the left sum vanishes by Lemma 3, since

$$N \bmod h = \frac{3(h-1)}{4} \geq \frac{h-1}{4}.$$

For $\ell = 2$, the right side vanishes due to $[n]_q$ and the remaining sum is finite for generic a. For $\ell > 2$, its base q^n has order h/n, and $M \bmod (h/n) = 3(h/n-1)/4$, so Lemma 3 again applies. Consequently $a^N D\Delta \in \Phi_h R_h[a]$.

Since P has unit leading coefficient in R_h, polynomial division shows that $H = a^N D\Delta/P$ belongs to $R_h[a]$. Reducing the equality $PH = a^N D\Delta$ modulo Φ_h, the polynomial P is nonzero (indeed its leading coefficient remains a unit). The residue ring is a polynomial ring over a field and has no zero divisors; hence $H \in \Phi_h R_h[a]$. Specializing $a = 1$ gives

$$\operatorname{ord}_{\Phi_h} \Delta(1,q) \geq 1 + \operatorname{ord}_{\Phi_h} P(1,q) - \operatorname{ord}_{\Phi_h} D(1,q). \tag{6}$$

5.4 Exact root counts

Put $u = h/n = p^{\ell-2}$. Then $u \equiv 1 \pmod 4$ (including $u = 1$), and

$$\#\{0 \leq j \leq M : u \mid 4j+1\} = \lfloor M/u \rfloor + 1 = \lfloor N/h \rfloor + 1.$$

Indeed the first solution in a period is $(u-1)/4$, while $M \bmod u = 3(u-1)/4$; for $u = 1$ both residues are zero. The last floor equality follows from $N = nM + 3(n-1)/4$ and $n \cdot 3(u-1)/4 + 3(n-1)/4 = 3(h-1)/4 < h$. Therefore

$$\operatorname{ord}_{\Phi_h} P(1,q) = 2(\lfloor N/h \rfloor + 1), \qquad \operatorname{ord}_{\Phi_h} D(1,q) = 2\lfloor N/h \rfloor.$$

Equation (6) gives at least three factors for every even ℓ, i.e.

$$\operatorname{ord}_{\Phi_{p^{2j}}} \Delta(1,q) \geq 3 \quad (1 \leq j \leq s). \tag{7}$$

5.5 The specialization at $q = 1$

This point is needed in addition to the local multiplicities. At $a = 1$ the Pochhammer portion of each summand is $Q_k^3 V_k$. The preceding zero-count argument shows that after cancellation Q_k has no odd-order cyclotomic denominator. Its remaining denominator factors come from $1 - q^{4i}$, and so are even-order cyclotomic polynomials; their values at 1 are 1 or 2. Likewise every factor of $1 + q^{4i+3}$ has even order and value at 1 equal to 1 or 2. Thus there is a Laurent polynomial denominator $B(q) \in \mathbb{Z}[q, q^{-1}]$ such that $B(1)$ is a power of 2, $A(q) = B(q)\Delta(1,q) \in \mathbb{Z}[q, q^{-1}]$, and B has no odd-order cyclotomic factor. This remains true for the right side after substituting $q \mapsto q^n$ because n is odd: an odd power cannot turn an even-order root into an odd-order root, and the value at $q = 1$ of the substituted denominator is unchanged. By (7), A is divisible over $\mathbb{Q}[q, q^{-1}]$ by $\prod_{j=1}^{s} \Phi_{p^{2j}}(q)^3$. Since these factors are monic, Gauss's lemma gives the same divisibility over $\mathbb{Z}[q, q^{-1}]$. Their values at $q = 1$ are all p. Finally,

$$T_L(1,1) = \sum_{k=0}^{L} (-1)^k (8k+1) \frac{(1/4)_k^3}{k!^3}, \qquad q^{(1-n)/2}[n]_q\big|_{q=1} = p^2.$$

Since $B(1)$ is a p-adic unit, (7) proves Theorem 2.

A An exact representation for E.3 and F.3

The following identity provides an auxiliary integral representation. It is not used to replace the cyclotomic proof of F.3 above, and it does not close the remaining E.3 branch.

Proposition 4. *For $m \geq 0$ and a parameter α at which the expressions are defined, put*

$$S_\alpha(m) = \sum_{k=0}^{m} (-1)^k \frac{2k+\alpha}{\alpha} \frac{(\alpha)_k^3}{(k!)^3}.$$

Then the following identity of rational functions holds:

$$\boxed{S_\alpha(m) = \frac{(1+\alpha)_m}{m!} \sum_{k=0}^{m} \frac{(-m-\alpha)_k(\alpha)_k}{(k!)^2}.}$$

In particular it holds for $\alpha = 1/3$ and $\alpha = 1/4$.

Proof. Set $q(m,k) = (-m-\alpha)_k(\alpha)_k/(k!)^2$ and $R(m,k) = -k^2/(m+\alpha)^2$. Direct cancellation gives

$$q(m,k+1)R(m,k+1) - q(m,k)R(m,k) = q(m,k) - \frac{m}{m+\alpha} q(m-1,k).$$

This is an identity of rational functions, so it suffices to check it at generic parameter values. Write $b_m = (\alpha)_m/m!$ and $W_m = \sum_{k=0}^{m} q(m,k)$. The two endpoint identities are

$$q(m-1,m) = (-1)^m b_m^2, \qquad q(m,m+1)R(m,m+1) = (-1)^m b_m^2.$$

Summing the certificate for $0 \leq k \leq m$ therefore yields

$$W_m - \frac{m}{m+\alpha} W_{m-1} = (-1)^m \frac{2m+\alpha}{m+\alpha} b_m^2.$$

Now $P_m = (1+\alpha)_m/m!$ satisfies $P_m/P_{m-1} = (m+\alpha)/m$ and $P_m = (m+\alpha)b_m/\alpha$. Multiplication by P_m gives precisely the mth summand of $S_\alpha(m)$ as an increment. The initial value is $P_0 W_0 = 1$. □

If $\alpha \in \mathbb{Z}_p$, every term of W_m is p-integral: each factor $(\beta)_k/k! = (-1)^k\binom{-\beta}{k}$ lies in $\mathbb{Z}_p$ for $\beta \in \mathbb{Z}_p$. This follows, for example, by approximating $-\beta$ by nonnegative integers in $\mathbb{Z}_p$ and using continuity of the fixed binomial polynomial.

For the short odd-level branches the parameters become

	α	m	$-m-\alpha$
$E.3$	$1/3$	$(p^r-2)/3$	$(1-p^r)/3$
$F.3$	$1/4$	$(p^r-3)/4$	$1/2-p^r/4$

Here E.3 has $p \equiv 2 \pmod 3$, F.3 has $p \equiv 3 \pmod 4$, and $r \geq 3$ is odd. In contrast to the J residual, these upper parameters are not both divisible by p^r. Consequently the square-product scaling theorem proved above cannot simply be applied. The original targets for these two branches are respectively

$$S_{1/3}((p^r-2)/3) \equiv p^2 S_{1/3}((p^{r-2}-2)/3) \pmod{p^{3r-1}},$$
$$S_{1/4}((p^r-3)/4) \equiv p^2 S_{1/4}((p^{r-2}-3)/4) \pmod{p^r}.$$

The F.3 target has been proved in Section 5. The E.3 target remains unproved here: its precision $3r-1$ is not supplied by the integral transformation or by the F.3 argument. No closure of A.3, D.3, E.3, or G.3 is claimed in this manuscript.

B Verification and reproducibility

The accompanying archive contains the single source file, this PDF, and exact-arithmetic verification scripts with their recorded output. The checks include the finite J identity, the telescoping certificates, all harmonic-prefix and product-error bounds used in the proof, the Morley prefactor precision, and the final J recurrence. The principal scaling audit tests $p = 5, 7$ through $r = 4$, $p = 11$ through $r = 3$, and $p = 13, 17, 19$ through $r = 2$, for 17 prime-power cases. The general E/F identity is checked at $\alpha = 1/2, 1/3, 1/4$ and $0 \le m \le 30$. For F.3, the archive additionally records exact rational checks of the stronger integer congruence, an exact calculation modulo Φ_9^3 (with nonzero remainder modulo Φ_9^4), and a separate calculation modulo Φ_{49}^3 over $\mathbb{F}_{1009}$ that exercises cyclotomic factor cancellation in a denominator. The finite-field check is only a diagnostic, not a characteristic-zero proof. These tests can detect implementation and transcription errors; the all-level conclusions rest on the preceding proofs.

The manuscript is independent of any unverified geometric lifting assertion in earlier research notes or corpus entries. The present proof of J.3 uses only the explicitly stated elementary lemmas and the published Cai congruence. The F.3 proof uses the standard terminating very-well-poised summation and the explicit local-ring argument above. A separate assessment of historical priority and external mathematical review remains appropriate.

References

[1] H. Swisher, *On the supercongruence conjectures of van Hamme*, Research in the Mathematical Sciences **2**, article 18 (2015). doi:10.1186/s40687-015-0037-6. See the published prime-power conjecture table, entry J.3.

[2] T. Cai, *A congruence involving the quotients of Euler and its applications (I)*, Acta Arithmetica **103**(4) (2002), 313–320. Publisher's full text. See Corollary 3 for the prime-power Morley congruence.

[3] V. J. W. Guo and W. Zudilin, *Dwork-type supercongruences through a creative q-microscope*, arXiv:2001.02311v4 (2020). arXiv record. See formula (3.23) and the proof of Theorem 3.6 for the terminating evaluation and the creative-microscoping framework. The short odd cutoff and its denominator bookkeeping are proved here.

MANUSCRIPT IV

Supersingular Advances for Swisher's A.3, D.3 and G.3

A second-level congruence, a complete D.3 residue branch, and every odd level of a G.3 residue branch

Research manuscript 10 November 2026

Abstract

We establish three precisely delimited results for Swisher's prime-power supercongruences. For A.3 we prove the second-level congruence modulo p^8 for every prime $p \equiv 3 \pmod 4$. For D.3 we prove the complete $p \equiv 2 \pmod 3$, $p > 3$, branch, including both parities of the level. For G.3 we prove the complete odd-level branch for primes $p \equiv 3 \pmod 4$. The all-level estimates behind the latter results are $v_p(D_r) \geq 5\lfloor r/2 \rfloor$ and $v_p(G_r) \geq 3\lfloor r/2 \rfloor$. The proofs combine exact terminating hypergeometric evaluations, cyclotomic interpolation at colliding nodes, and a finite interval partition respecting prime-power digits. The first odd level requires separate local block identities. No ordinary-prime all-level result is claimed here, and the odd supersingular branch of E.3 remains unresolved.

Status. This is a research manuscript supported by internal cross-checks and reproducible exact arithmetic. It has not undergone external peer review. No claim of historical priority is made.

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1 Statements and exact scope

We use $(a)_k = \prod_{j=0}^{k-1}(a+j)$, $(a)_0 = 1$, and the valuation v_p normalized by $v_p(p) = 1$, with $v_p(0) = +\infty$. All sums in this paper are rational numbers; congruences refer to their images in $\mathbb{Z}_p$. The labels A.3, D.3 and G.3 refer to Section 7 of Swisher [1]. We call the indicated nontrivial residue classes *supersingular* only as convenient terminology for the branches of those conjectures.

For odd N, put

$$A(N) = \sum_{k=0}^{(N-1)/2} (-1)^k (4k+1) \frac{(1/2)_k^5}{k!^5}.$$

For primes $p > 3$ with $p \equiv 2 \pmod 3$, put

$$D_r = \sum_{k=0}^{d_r} (6k+1) \frac{(1/3)_k^6}{k!^6}, \qquad d_r = \begin{cases} (p^r-1)/3 & r \text{ even}, \\ (p^r-2)/3 & r \text{ odd}. \end{cases}$$

For primes $p \equiv 3 \pmod 4$, put

$$G_r = \sum_{k=0}^{g_r} (8k+1) \frac{(1/4)_k^4}{k!^4}, \qquad g_r = \begin{cases} (p^r-1)/4 & r \text{ even}, \\ (p^r-3)/4 & r \text{ odd}. \end{cases}$$

The displayed definitions include $D_0 = G_0 = 1$.

Theorem 1.1 (Second supersingular level of A.3)**.** *For every prime $p \equiv 3 \pmod 4$,*

$$A(p^2) \equiv p^4 \pmod{p^8}.$$

Theorem 1.2 (Complete supersingular branch of D.3)**.** *Let $p > 3$ be prime with $p \equiv 2 \pmod 3$. For every $r \geq 1$,*

$$v_p(D_r) \geq 5\lfloor r/2 \rfloor.$$

Moreover, all the supersingular assertions of D.3 hold:

$$\begin{aligned} &D_2 \equiv 0 \pmod{p^4}, \\ &D_r \equiv p^4 D_{r-2} \pmod{p^{2r+1}} && (r \geq 4 \text{ even}), \\ &D_r \equiv -p^5 D_{r-2} \pmod{p^{2r}} && (r \geq 3 \text{ odd}). \end{aligned}$$

Theorem 1.3 (Every odd supersingular level of G.3)**.** *Let $p \equiv 3 \pmod 4$ be prime. For every $r \geq 1$,*

$$v_p(G_r) \geq 3\lfloor r/2 \rfloor.$$

For every odd $r \geq 3$,

$$G_r \equiv -p^3 G_{r-2} \pmod{p^{r+1}}.$$

The prime $p = 3$ is included.

The valuation estimates do not assert exact valuations. They suffice for the original high-level targets because both terms in the proposed recurrence then vanish at the required precision. At $r = 3$ the leading term and sign require extra work. This distinction is essential: the same valuation argument does not give the stronger even-level target in G.3.

Label	Established in this manuscript	Not established here
A.3	$p \equiv 3 \pmod 4$, $r = 2$, modulo p^8	Higher supersingular levels and ordinary higher-level targets
D.3	Every required level for $p > 3$, $p \equiv 2 \pmod 3$	Ordinary-prime higher-level target modulo p^{6r}
G.3	$p \equiv 3 \pmod 4$, every odd $r \geq 3$	Supersingular even-level recurrence in general; ordinary higher levels
E.3	An exact short-cutoff deformation is recorded in the companion bundle	The remaining odd-level target modulo p^{3r-1}

The table describes the scope of this manuscript, not a claim that all omitted base cases are unknown in the literature. Earlier work in the surrounding project and the excluded labels are not re-proved here. The proofs below are independent of the all-level Frobenius and period-map results in the motivating corpus [5].

Proof organization. Sections 2–4 prove Theorem 1.2. The even-level argument uses the specific published parametric input of Ni–Wang [2], reproduced by Wei [3]; the subsequent interval and cubic-block arguments are given explicitly. Sections 5–6 prove Theorem 1.3 from terminating summation formulas and gamma reflection. Section 7 proves Theorem 1.1, using a shifted WZ identity and removal of a four-root deformation. The standard gamma estimates used there are specified from Long–Ramakrishna [4].

2 An even-level divisibility theorem for (D.3)

Write $(z;Q)_k = \prod_{j=0}^{k-1}(1 - zQ^j)$, and use a list of arguments for a product of such symbols. Put $[m]_q = (1-q^m)/(1-q)$ and

$$c_k(a,b;q) = [6k+1]_q q^{3k} \frac{(aq, q/a, bq, q/b; q^3)_k (q;q^3)_k^2}{(aq^3, q^3/a, bq^3, q^3/b; q^3)_k (q^3;q^3)_k^2}, \qquad S_L(a,b;q) = \sum_{k=0}^{L} c_k(a,b;q).$$

For a positive integer $h > 1$ coprime to 3, let v_h denote the valuation at the irreducible polynomial $\Phi_h(q)$ on $\mathbb{Q}(q)$. The same symbol denotes its Gauss extension when independent parameters are present. All congruences in this section are in this local sense.

The short-block input. We use the following published parametric congruence, with its exact cutoff. If $h \equiv 2 \pmod 3$ and $t = (2h-1)/3$, then

$$S_t(a,b;q) \equiv 0 \pmod{[h]_q \Phi_h(q)}. \tag{2.1}$$

This is Ni–Wang, *Two q-supercongruences from Watson's transformation*, Rev. R. Acad. Cienc. Exactas Fís. Nat., Ser. A Mat. **116** (2022), Article 30, Theorem 2.3, with $d = 3, r = 1$. The exact specialization is reproduced as equation (2.12) in Chuanan Wei, *q-Supercongruences from Jackson's ${}_8\phi_7$ summation and Watson's ${}_8\phi_7$ transformation*, J. Combin. Theory Ser. A **204** (2024), 105853, https://arxiv.org/abs/2104.07025. Since $t < h$, all the displayed denominator factors remain Φ_h-units when $b = 1$ (with a generic). Consequently (2.1) implies $v_h(S_t(a,1;q)) \geq 2$ without a loss of precision upon that specialization. For $h \equiv 1 \pmod 3$ and $t = (h-1)/3$, we also use the simpler short-block congruence $S_t(1,1;q) \equiv 0 \pmod{\Phi_h(q)}$. It follows either by term reversal, or from Ni–Wang Lemma 2.2 as reproduced in Wei equations (2.7)–(2.9); here again $t < h$ makes all the specialized denominators units.

A uniform parametric block lemma. If $h \equiv 2 \pmod 3$, $t = (2h-1)/3$ and $B \geq 0$, then

$$v_h(S_{Bh+t}(a,1;q)) \geq 2. \tag{2.2}$$

Here a is an independent parameter. We give the shift argument in detail, since a short-block result alone would not imply this claim. Set $Q = q^3$ and define

$$A_j(x,a,b;q) = \frac{1-x^2q^{6j+1}}{1-q}\, q^{3j} \frac{(x^2q, axq, xq/a, bxq, xq/b, q; Q)_j}{(x^2q^3, axq^3, xq^3/a, bxq^3, xq^3/b, q^3; Q)_j},$$

$$\mathcal{A}_t(x,a,b;q) = \sum_{j=0}^{t} A_j(x,a,b;q).$$

These expressions are regular near $x = b = 1$ in the Φ_h-local ring with a generic. In particular, their denominators at $x = b = 1$ are units, since $j \leq t < h$. We claim that

$$\mathcal{A}_t(x,a,b;q) \equiv 0 \pmod{\Phi_h(q)} \tag{2.3}$$

as a rational function of the independent parameters x, a, b, and with denominators regular at $x = b = 1$. To verify this claim, apply the terminating balanced Jackson ${}_8\phi_7$ summation with base Q, central parameter $A = x^2q$, and its five additional upper parameters

$$b_1 = axq, \quad c = xq/a, \quad d = bxq, \quad e = xq^{1+2h}/b, \quad f = q^{1-2h} = Q^{-t}.$$

The balancing condition is exact: $b_1cdef = x^4q^5 = A^2Q$. Modulo $\Phi_h(q)$ the resulting finite sum, multiplied by $(1-A)/(1-q)$, is precisely $\mathcal{A}_t(x,a,b;q)$. Jackson's product side is

$$\frac{1-A}{1-q} \frac{(AQ, AQ/(b_1c), AQ/(b_1d), AQ/(cd); Q)_t}{(AQ/b_1, AQ/c, AQ/d, AQ/(b_1cd); Q)_t}.$$

It contains the numerator factor $(q^2;Q)_t$, which vanishes modulo Φ_h: its factor with index $(h-2)/3$ is $1-q^h$. The four denominator arguments are $xq^3/a, axq^3, xq^3/b, q/(bx)$. They stay units near $x = b = 1$, because the first occurrence of $1-q^{mh}$ in $(q;Q)_j$ is at index t, outside the product for $j \leq t$. This proves (2.3), including the regularity assertion. In particular,

$$v_h(\partial_x \mathcal{A}_t(1,a,1;q)) \geq 1. \tag{2.4}$$

For a block index $u \geq 0$, put $x_u = q^{3uh}$ and

$$E_u = q^{3uh} \frac{(aq, q/a, q, q; Q)_{uh} (q;Q)_{uh}^2}{(aq^3, q^3/a, q^3, q^3; Q)_{uh} (q^3;Q)_{uh}^2}.$$

Each complete block of h factors in $(q;Q)/(q^3;Q)$ has one numerator zero and one denominator zero; hence E_u is a Φ_h-unit for generic a. For $0 \leq j \leq t$, direct cancellation of shifted factorials gives

$$c_{uh+j}(a,1;q) = E_u A_j(x_u,a,1;q) R_j(x_u;q),$$

where

$$R_j(x;q) = \frac{(xq;Q)_j^2}{(q;Q)_j (x^2q;Q)_j} \frac{(q^3;Q)_j (x^2q^3;Q)_j}{(xq^3;Q)_j^2}.$$

All these factors are regular near $x = 1$; direct differentiation gives $R_j(1;q) = 1$ and $\partial_x R_j(1;q) = 0$. Since $v_h(x_u - 1) \geq 1$ when $u > 0$, it follows that $R_j(x_u;q) \equiv 1 \pmod{\Phi_h^2}$; the assertion is exact for $u = 0$. Equations (2.1), (2.4) and Taylor expansion now give

$$\mathcal{A}_t(x_u,a,1;q) \equiv \mathcal{A}_t(1,a,1;q) + (x_u - 1)\partial_x \mathcal{A}_t(1,a,1;q) \equiv 0 \pmod{\Phi_h^2}.$$

Thus every block prefix from uh through $uh + t$ has valuation at least two. In the rest of a block, $t < j < h$, the quotient $(q;Q)_{uh+j}/(q^3;Q)_{uh+j}$ has valuation one; its fourth power occurs in $c_{uh+j}(a,1;q)$. Consequently those remaining terms even have valuation at least four. Adding the complete blocks and the final prefix proves (2.2).

Interpolation at colliding nodes. Under the same assumptions on h, t, B, we prove

$$v_h\big(S_{Bh+t}(1,1;q)\big) \geq 4. \tag{2.5}$$

Put $L = Bh + t$ and

$$D(a) = (aq^3, q^3/a; Q)_L, \qquad F(a) = a^L D(a) S_L(a, 1; q).$$

Then $F(a)$ is a polynomial in a with coefficients in the Φ_h-local ring. Indeed, D clears all the parameter denominators, the factor a^L clears all negative powers of a, and the remaining $(q;Q)/(q^3;Q)$ quotients are regular at Φ_h. By (2.2) every coefficient of F has valuation at least two (equivalently, its Gauss valuation is at least two).

For $0 \leq i \leq B$ define

$$A_i = (3i+2)h, \qquad t_i = (A_i - 1)/3 = ih + t, \qquad \alpha_{i,+} = q^{A_i}, \quad \alpha_{i,-} = q^{-A_i}.$$

The nodes are distinct in $\mathbb{Q}(q)$, and $A_i \equiv 1 \pmod 3$. At either node the sum terminates at $t_i \leq L$, and the exact Jackson evaluation is

$$S_L(q^{\pm A_i}, 1; q) = [A_i]_q \left\{ \frac{(q^2;Q)_{t_i}}{(q^3;Q)_{t_i}} \right\}^3. \tag{2.6}$$

For completeness, this is Jackson's identity with central parameter q, other upper parameters $q, q, q, q^{1+A_i}, q^{1-A_i}$ and base Q; the factor $(q^4;Q)_{t_i}/(q;Q)_{t_i}$ telescopes to $[A_i]_q$. No parameter denominator vanishes identically because $3 \nmid A_i$. The same evaluation is the specialization $b = 1$ of Wei equation (2.3), with the termination exponent allowed to be any positive integer $A_i \equiv 1 \pmod 3$. The quotient in braces has valuation one: it has $i+1$ numerator zeros and i denominator zeros. Therefore the right side of (2.6) has valuation four. At each node, $v_h(D(\alpha_{i,\pm})) = 2B$, since among $1 \leq j \leq L$ precisely B integers satisfy $h \mid 3j$. It follows that

$$v_h(F(\alpha_{i,\pm})) \geq 2B + 4. \tag{2.7}$$

Let $P(a) = \prod_{i=0}^{B} (a - \alpha_{i,+})(a - \alpha_{i,-})$. Division by this monic polynomial gives $F = PH + R$, with $\deg R < 2B + 2$. Since the coefficients of P are integral in the local ring, those of H and R still have valuation at least two. Thus

$$v_h(P(1)H(1)) \geq (2B+2) + 2 = 2B + 4.$$

The nodes all approach 1 modulo Φ_h, so ordinary unit-separated interpolation is unavailable. Nevertheless, the precise Lagrange weights at the point 1 are units:

$$R(1) = \sum_{\alpha} F(\alpha) \prod_{\beta \neq \alpha} \frac{1-\beta}{\alpha - \beta}.$$

For every node β, $v_h(1-\beta) = 1$, and for any two distinct nodes α, β we have $v_h(\alpha - \beta) = 1$, since their nonzero exponents and exponent differences are multiples of h. Every displayed Lagrange weight consequently has valuation zero. By (2.7), $v_h(R(1)) \geq 2B + 4$, and hence $v_h(F(1)) \geq 2B + 4$. Finally $v_h(D(1)) = 2B$; dividing proves (2.5).

The complementary roots. If $h \equiv 1 \pmod 3$, $t = (h-1)/3$, and $B \geq 0$, then

$$v_h\big(S_{Bh+t}(1,1;q)\big) \geq 1. \tag{2.8}$$

Indeed, the short block has valuation at least one by the short-block input above. In a block $uh + j$, $0 \leq j \leq t$, the shifted quotient factors reduce modulo Φ_h to the unshifted ones, multiplied by a common regular block factor. The weight and q^{3k} reduce to their unshifted values too. For $t < j < h$, each summand has six numerator zeros in excess of its denominator zeros, and so is zero modulo Φ_h. This proves (2.8) block by block.

Specialization and the original conjecture. Let $p > 3$ be prime with $p \equiv 2 \pmod 3$, let $s \geq 1$, and put $L = (p^{2s} - 1)/3$. For $h = p^\ell$, $1 \leq \ell \leq 2s$, the residue of L modulo h is $(2h - 1)/3$ if ℓ is odd, and $(h - 1)/3$ if ℓ is even. Equations (2.5) and (2.8) therefore give the rational-function congruence

$$S_L(1, 1; q) \equiv 0 \pmod{\prod_{\substack{1 \leq \ell \leq 2s \\ \ell \text{ odd}}} \Phi_{p^\ell}(q)^4 \prod_{\substack{1 \leq \ell \leq 2s \\ \ell \text{ even}}} \Phi_{p^\ell}(q)}.$$

There is no loss of p-adic precision at $q = 1$. To see this explicitly, the reduced denominator of each $(q; q^3)_k/(q^3; q^3)_k$ has only cyclotomic factors whose orders are divisible by 3. For any order d coprime to 3, the numerator's single zero in a period occurs before the denominator's zero, so its valuation is nonnegative. For $3 \mid d$, a remaining factor $\Phi_d(1)$ is either 1 or 3; thus a common reduced denominator for the finitely many summands has value a power of 3 at 1. After clearing this denominator, the numerator lies in $\mathbb{Z}[q]$. Division by the monic integral cyclotomic product above preserves integral coefficients. Since $\Phi_{p^\ell}(1) = p$, taking $q \to 1$ proves

$$\boxed{v_p\left(\sum_{k=0}^{(p^{2s}-1)/3} (6k + 1)\frac{(1/3)_k^6}{k!^6}\right) \geq 5s.} \tag{2.9}$$

Denote this sum by D_{2s}. In particular $D_2 \equiv 0 \pmod{p^5}$, which includes the original base target modulo p^4. For $s \geq 2$, both D_{2s} and $p^4 D_{2s-2}$ are divisible by p^{4s+1}, because $5s \geq 4s + 1$ and $4 + 5(s - 1) = 5s - 1 \geq 4s + 1$. Hence

$$\boxed{D_{2s} \equiv p^4 D_{2s-2} \pmod{p^{4s+1}} \qquad (s \geq 2),}$$

which proves the full even supersingular branch of (D.3). The next two sections supply the odd levels. The ordinary-prime branch is outside the present theorem.

3 All supersingular levels: a digit-interval argument

Retain $c_k(1, 1; q)$ and the notation of the preceding proof. Fix a prime $p > 3$ with $p \equiv 2 \pmod 3$, and put

$$n = p^2, \qquad d = (n - 1)/3, \qquad t_j = (n^j - 1)/3 = dn^{j-1} + t_{j-1} \quad (j \geq 1), \qquad t_0 = 0.$$

Let

$$P_j(q) = \prod_{\substack{1 \leq \ell \leq 2j \\ \ell \text{ odd}}} \Phi_{p^\ell}(q)^4 \prod_{\substack{1 \leq \ell \leq 2j \\ \ell \text{ even}}} \Phi_{p^\ell}(q), \qquad P_0 = 1.$$

For every integer $u \geq 0$, define the interval

$$I(u, j) = \{un^j, un^j + 1, \ldots, un^j + t_j\}.$$

The preceding proof implies the following shifted-prefix statement:

$$P_j(q) \;\Big|\; \sum_{k \in I(u,j)} c_k(1, 1; q). \tag{3.1}$$

Divisibility here and below is after clearing a denominator in $\mathbb{Z}[q]$ whose value at 1 is a power of 3. Indeed, for any $h = p^\ell$, $\ell \leq 2j$, the right endpoint $un^j + t_j$ has the same residue modulo h as t_j, namely the appropriate short-block cutoff in the preceding proof. The endpoint $un^j - 1$ has residue

$h-1$. The congruences in that proof also hold for this latter residue: the extra terms beyond the short cutoff have at least six numerator zeros in excess of denominator zeros. Subtracting the two initial sums gives (3.1); for $u=0$ the subtracted sum is empty.

For an interval of length n^j, also introduce the sets

$$\begin{aligned} B(u,j) &= \{un^j, \dots, (u+1)n^j - 1\}, \\ T(u,j) &= \{un^j + t_j + 1, \dots, (u+1)n^j - 1\}, \\ J(u,j) &= \{un^j, \dots, un^j + 2t_j\}. \end{aligned}$$

At level zero, $B(u,0) = J(u,0) = I(u,0) = \{u\}$ and $T(u,0)$ is empty. The following are exact disjoint interval partitions for $j \geq 1$:

$$B(u,j) = I(u,j) \,\dot\cup\, T(u,j), \tag{3.2}$$

$$T(u,j) = T(nu+d, j-1) \,\dot\cup \dot\bigcup_{c=d+1}^{n-1} B(nu+c, j-1), \tag{3.3}$$

$$J(u,j) = I(u,j) \,\dot\cup\, T(nu+d, j-1) \,\dot\cup \dot\bigcup_{c=d+1}^{2d-1} B(nu+c, j-1) \,\dot\cup\, J(nu+2d, j-1). \tag{3.4}$$

These identities follow immediately from $t_j = dn^{j-1} + t_{j-1}$ and $2t_j = 2dn^{j-1} + 2t_{j-1}$. In (3.4), every piece other than $I(u,j)$ lies in the tail $T(u,j)$.

Apply these rules recursively, stopping at every set $I(v,k)$. They give finite disjoint partitions of both $B(u,j)$ and $J(u,j)$ into such sets. A leaf $I(v,k)$ has undergone exactly $j-k$ descents, through tails at levels $k+1, k+2, \dots, j$. For a level ℓ on that path, every index m in the leaf has remainder greater than t_ℓ modulo n^ℓ. Since $n^\ell \equiv 1 \pmod 3$, it follows that

$$v_{\Phi_{n^\ell}}\left(\frac{(q;q^3)_m}{(q^3;q^3)_m}\right) = 1.$$

Consequently each summand belonging to that leaf, and hence its sum, has a factor $\Phi_{n^\ell}(q)^6$ for every such ancestor level. Combining these distinct high-order factors with (3.1) gives

$$P_k(q) \prod_{\ell=k+1}^{j} \Phi_{p^{2\ell}}(q)^6 \;\Big|\; \sum_{m \in I(v,k)} c_m(1,1;q). \tag{3.5}$$

There is no assumption of integral comaximality between different cyclotomic polynomials here. Clear a common denominator whose value at 1 is a power of 3. The integral numerator is divisible over $\mathbb{Q}[q]$ by every indicated irreducible factor; hence it is divisible by their product. Since that product is monic and integral, its quotient is again integral.

Taking $q \to 1$ in (3.5) shows that every leaf contributes a rational number of p-adic valuation at least

$$5k + 6(j-k) = 5j + (j-k) \geq 5j.$$

Therefore the original hypergeometric sums over each of $B(u,j)$ and $J(u,j)$ have valuation at least $5j$, uniformly in $u \geq 0$.

Finally write $p = 3m+2$. For an odd level $r = 2s+1$,

$$\frac{p^{2s+1} - 2}{3} = mn^s + 2t_s.$$

The full summation interval is the disjoint union of $B(0,s),\dots,B(m-1,s)$ and $J(m,s)$. Thus, together with the even-level theorem, we have proved

$$\boxed{v_p(D_r)\ge 5\lfloor r/2\rfloor \qquad (p>3,\ p\equiv 2 \pmod 3,\ r\ge 1).} \tag{3.6}$$

For odd $r=2s+1\ge 5$, both D_r and p^5D_{r-2} have valuation at least $5s\ge 4s+2=2r$. Hence

$$\boxed{D_r\equiv -p^5D_{r-2} \pmod{p^{2r}} \qquad (r\ge 5 \text{ odd},\ p\equiv 2 \pmod 3).}$$

At this stage the only supersingular recurrence still to prove is the odd level $r=3$, treated in the next section. At that level (3.6) proves only $D_3\in p^5\mathbb{Z}_p$; the remaining required leading congruence is $D_3\equiv -p^5D_1 \pmod{p^6}$.

4 The odd third level of D.3 for all supersingular primes

Let $p>3$ be prime, $p\equiv 2 \pmod 3$. Write

$$m=(p-2)/3,\quad \tau=(2p-1)/3,\quad q=p^2,\quad n=(q-1)/3=pm+\tau,\qquad b_k=(1/3)_k/k!.$$

We prove

$$\sum_{k=0}^{(p^3-2)/3}(6k+1)b_k^6\equiv -p^5\sum_{k=0}^{(p-2)/3}(6k+1)b_k^6 \pmod{p^6}. \tag{4.1}$$

This is precisely the odd $r=3$ instance of Swisher D.3.

4.1 An exact cubic Dougall model

For $N\equiv 1 \pmod 3$, put $t=(N-1)/3$ and define

$$T_3(N)=\sum_{k=0}^{t}(6k+1)b_k^6\prod_{i=0}^{k-1}\frac{1-N^3/(3i+1)^3}{1+N^3/(3i+3)^3}.$$

Let $\omega^2+\omega+1=0$. Dougall's terminating very-well-poised ${}_7F_6$ summation, with central parameter $1/3$ and the five other upper parameters

$$(1-N)/3,\quad (1-N\omega)/3,\quad (1-N\omega^2)/3,\quad 1/3,\quad 1/3,$$

gives

$$T_3(N)=\frac{(4/3)_t((2-N)/3)_t((2+N\omega)/3)_t((2+N\omega^2)/3)_t}{(1+N\omega/3)_t(1+N\omega^2/3)_t(1)_t((1-N)/3)_t}. \tag{4.2}$$

Indeed the five parameters have sum $1+2/3$, as required by the terminating balanced Dougall identity, and $(1-N)/3=-t$. The factors with the three cubic roots combine to the displayed cubic deformation. Equivalently, reversal of the two real shifted factorials yields the rational identity

$$T_3(N)=N\frac{(1/3)_t(2/3)_t}{(t!)^2}\prod_{j=0}^{t-1}\frac{(3j+2)^2-N(3j+2)+N^2}{(3j+3)^2-N(3j+3)+N^2}. \tag{4.3}$$

These are exact finite identities.

4.2 A uniform first-digit cancellation

Define, for $z \in p\mathbb{Z}_p$,

$$L(z) = \sum_{b=0}^{\tau} (6(z+b)+1) \left(\frac{(1/3+z)_b}{(1+z)_b} \right)^6 .$$

Then

$$L(z) \in p^3\mathbb{Z}_p \qquad (z \in p\mathbb{Z}_p). \tag{4.4}$$

First $L(0) \in p^3\mathbb{Z}_p$. To see this apply the cubic model with $N = 2p$ and $t = \tau < p$. Every factor $3i+1$ and $3i+3$ with $0 \le i < \tau$ is a p-adic unit: a possible multiple p has the wrong residue modulo 3, and the numerator endpoint $2p$ is excluded. Hence $L(0) - T_3(2p) \in p^3\mathbb{Z}_p$. In (4.3), $(1/3)_\tau/\tau!$ is a unit, $(2/3)_\tau/\tau!$ has valuation one, and the quadratic-product ratio has valuation two. The latter assertion follows because its only numerator index divisible by p is $j = m$, where the quadratic is $3p^2$, whereas every denominator is a unit. Therefore $v_p(T_3(2p)) = 4$, proving the required bound for $L(0)$.

Put $u(z) = (1/3+z)_\tau/(1+z)_\tau$. Reversing the finite products gives the exact rational identity

$$L(z) = -u(z)^6 L(-z-2p/3).$$

Thus $L(-2p/3) \in p^3\mathbb{Z}_p$. At the midpoint $z = -p/3$, product reversal gives $u(-p/3) = (-1)^\tau$, so $L(-p/3) = 0$ exactly. All denominators in $L(z)$ are units for $z \in p\mathbb{Z}_p$. Consequently the Taylor expansion of $L(px)$ modulo p^3 is a polynomial of degree at most two in x, with coefficients in $\mathbb{Z}_p$. Its values at the three points $0, -1/3, -2/3$ vanish modulo p^3. The Vandermonde determinant of these points is a p-adic unit, proving (4.4).

4.3 The unit two-digit blocks

For $0 \le c \le m$, define

$$K_c = \sum_{\substack{0 \le j \le n \\ b_{qc+j} \in \mathbb{Z}_p^\times}} (6(qc+j)+1) \left(\frac{b_{qc+j}}{b_c} \right)^6 .$$

For $0 \le C \le m$ put $h = 3C+1$ and

$$E_c(h) = \prod_{i=0}^{c-1} \frac{1 - h^3/(3i+1)^3}{1 + h^3/(3i+3)^3}.$$

Cubing permutes $\mathbb{F}_p^\times$. Thus each $E_c(h)$ for $c \le C$ is a unit: a numerator zero would require $h = 3i+1 \pmod p$, which is impossible for $i < C$; a denominator zero would require $h + 3i + 3 = p$, impossible modulo 3. The positive sum in the latter case is less than $2p$.

We claim

$$T_3(qh) \equiv \sum_{c=0}^{C} b_c^6 E_c(h) K_c \pmod{p^6}. \tag{4.5}$$

The unit indices in the large cutoff are exactly

$$k = qc + pa + b, \qquad 0 \le c \le C, \quad 0 \le a \le m, \quad 0 \le b \le \tau.$$

This follows by two applications of the generalized-binomial Lucas criterion; the dash parameters are $1/3 \mapsto 2/3 \mapsto 1/3$. All other summands have $b_k^6 \in p^6\mathbb{Z}_p$ and can be omitted: their deformation factors are units by the same cubic-injectivity argument for the factors divisible by p^2 and are 1 modulo p^3 for all remaining factors.

For a unit index as displayed, the deformation factors divisible by p^2 before cancellation give exactly $E_c(h)$. The factors with valuation one give a product $P_{c,a}(h)$ which is 1 modulo p^3 and is independent of b. Indeed numerator multiples of p have indices $\tau + pt$ and denominator multiples have indices $p-1+pt$; both lists have exactly $pc+a$ entries below k, before removing those divisible by p^2. Finally the factors with valuation zero are 1 modulo p^6. For fixed a, c, the sum of the undeformed summands over the low digit b equals

$$b^6_{qc+pa} L(qc+pa) \in p^3\mathbb{Z}_p$$

by (4.4). Multiplication by $P_{c,a}(h) - 1 \in p^3\mathbb{Z}_p$ therefore changes it by an element of $p^6\mathbb{Z}_p$. This proves (4.5).

4.4 Two-digit scaling of the exact model

For every $0 \le C \le m$, $h = 3C+1$, one has

$$T_3(p^2h) \equiv -p^5 T_3(h) \pmod{p^6}. \tag{4.6}$$

Here are the product counts and sign in full. Work in the unramified quadratic extension containing ω and use (4.2), with $(4/3)_t = N(1/3)_t$. The large cutoff is $t = p^2C + n$. For a fixed integral β, extraction of the p-factors from $(\alpha + p^2h\beta)_t$ twice gives

$$(\alpha + p^2h\beta)_t = p^{L_\alpha + C}(\alpha + h\beta)_C U_{\alpha,\beta}, \quad L_{1/3} = L_1 = pC + m, \quad L_{2/3} = pC + m + 1,$$

where $U_{\alpha,\beta}$ is a unit and its residue modulo p is independent of β. For $\alpha = 1/3, 2/3, 1$, the first zero indices are respectively $\tau, m, p-1$; the new parameters after one extraction are $2/3 + ph\beta, 1/3 + ph\beta, 1 + ph\beta$. The second extraction has exactly C terms in each case. Substitution in (4.2) leaves $T_3(h)$, p^2 from the leading N, and p^3 from the three numerator parameters $2/3$ versus the three denominator parameters 1. Hence $T_3(p^2h) = p^5 T_3(h) U$.

The two $1/3$ unit products cancel modulo p. The other three unit-product ratios are identical; complete residue blocks cancel, and their remaining portions are

$$U \equiv (AB)^3 \pmod{p}, \qquad A = \frac{\prod_{0 \le j \le 2m,\ j \ne m}(j + 2/3)}{(2m+1)!}, \qquad B = \frac{(1/3)_{m+1}}{m!}.$$

The numerator residues in AB, after clearing 3^{p-1}, are exactly $1, \dots, p-1$. Wilson's and Fermat's theorems give

$$AB \equiv -\frac{1}{m!(2m+1)!} = (-1)^m = -1 \pmod{p},$$

since m is odd. Thus $U \equiv -1 \pmod{p}$. The coarse sum $T_3(h)$ is integral because all its cubic deformation denominators are units, as already proved. This establishes (4.6).

4.5 Triangular recovery and the original sum

The exact small-scale identity

$$T_3(h) = \sum_{c=0}^{C} (6c+1) b_c^6 E_c(h)$$

together with (4.5) and (4.6) yields

$$\sum_{c=0}^{C} b_c^6 E_c(h) \{K_c + p^5(6c+1)\} \equiv 0 \pmod{p^6}.$$

The coefficient $b_C^6 E_C(h)$ is a unit. Induction on C therefore proves $K_c \equiv -p^5(6c+1) \pmod{p^6}$ for every $c \le m$. Finally $(p^3-2)/3 = qm + 2n$. In each p^2 block the unit low-digit indices are exactly $j = pa + b$ with $a \le m$ and $b \le \tau$, hence satisfy $j \le n$. The additional final tail $n < j \le 2n$ therefore contains no unit b_{qm+j}, and all its summands lie in $p^6\mathbb{Z}_p$. Thus the original sum partitions into the stated unit blocks plus summands divisible by p^6. It is congruent to $\sum_{c=0}^{m} b_c^6 K_c$, which proves (4.1).

5 The first odd supersingular instance of G.3

Let $p \equiv 3 \pmod 4$ be prime and put

$$S(n) = \sum_{k=0}^{n} (8k+1) \left(\frac{(1/4)_k}{k!} \right)^4 .$$

Then

$$S((p^3-3)/4) \equiv -p^3 S((p-3)/4) \pmod{p^4}. \tag{5.1}$$

For $p \ge 7$, write $m = (p-3)/4$, $t = (3p-1)/4$, $b_j = (1/4)_j/j!$, $c_j = (3/4)_j/j!$. We use Morita's gamma function, the continuous extension to $\mathbb{Z}_p$ of

$$\Gamma_p(n) = (-1)^n \prod_{\substack{1 \le j < n \\ p \nmid j}} j \qquad (n \ge 1).$$

It is unit-valued and satisfies

$$\frac{\Gamma_p(x+1)}{\Gamma_p(x)} = \begin{cases} -x & x \in \mathbb{Z}_p^\times, \\ -1 & x \in p\mathbb{Z}_p, \end{cases} \qquad \Gamma_p(x)\Gamma_p(1-x) = (-1)^{a_0(x)},$$

where $a_0(x)$ is the representative of x modulo p in $\{1, \dots, p\}$.

We require the following elementary local polynomial property. For every $a \in \mathbb{Z}_p$, the function $\Gamma_p(a+pX)$ has, modulo p^4, a polynomial representative of degree at most three whose coefficient of X^j belongs to $p^j\mathbb{Z}_p$. The same is true of finite products and unit-valued quotients of these functions. Here is a direct proof, avoiding any assertion about the coefficients of an infinite Taylor series. The functional equation gives

$$\frac{\Gamma_p(a+p(X+1))}{\Gamma_p(a+pX)} = -\prod_{\substack{0 \le i < p \\ a+i \in \mathbb{Z}_p^\times}} (a+i+pX).$$

The right side lies in $1+p\mathbb{Z}_p$. Put $C = -\prod_{a+i\in\mathbb{Z}_p^\times}(a+i)$ and $A_j = \sum_{a+i\in\mathbb{Z}_p^\times}(a+i)^{-j}$. For $j = 1, 2, 3$ these complete unit power sums are zero modulo p. The logarithm of the displayed ratio consequently satisfies

$$\log C + pA_1X - \frac{p^2A_2}{2}X^2 \pmod{p^4}.$$

The cubic term is in $p^4\mathbb{Z}_p$, and all higher terms are also in $p^4\mathbb{Z}_p$ for $p \ge 7$. Summing this difference equation for nonnegative integer X gives

$$\log \frac{\Gamma_p(a+pX)}{\Gamma_p(a)} \equiv X \log C + pA_1 \frac{X(X-1)}{2} - \frac{p^2A_2}{2} \frac{X(X-1)(2X-1)}{6} \pmod{p^4}.$$

The degree-j coefficients on the right belong to $p^j\mathbb{Z}_p$. Exponentiation preserves this coefficient grading modulo p^4; continuity extends the result to all $X \in \mathbb{Z}_p$. Products and unit-valued inverses

preserve it as well. Throughout the proof, congruences between the resulting polynomials may be read coefficientwise: since their degrees are at most three and $p \geq 7$, evaluation at $X = 0, 1, 2, 3$ has a unit Vandermonde determinant. Thus congruence at all $X \in \mathbb{Z}_p$ implies the corresponding coefficient congruence.

Define analytic functions on $\mathbb{Z}_p$ by

$$F(X) = -\frac{\Gamma_p(1/4 + pX)}{\Gamma_p(1/4)\Gamma_p(1 + pX)}, \qquad J_0(X) = -\frac{\Gamma_p(3/4 + pX)}{\Gamma_p(3/4)\Gamma_p(1 + pX)},$$

$$L(X) = \sum_{a=0}^{t} (8a + 1 + 8pX) \left(\frac{(1/4 + pX)_a}{(1 + pX)_a} \right)^4, \quad H(X) = F(X)^4 L(X),$$

$$V(X) = \sum_{b=0}^{m} (8b + 3 + 8pX) \left(\frac{(3/4 + pX)_b}{(1 + pX)_b} \right)^4, \quad J(X) = J_0(X)^4 V(X).$$

All finite-product denominators are units. Also $F(0) = J_0(0) = 1$. Reversing the finite sums, then applying $\Gamma_p(x)\Gamma_p(1 - x) = (-1)^{a_0(x)}$, gives the exact reflections

$$H(X) = -H(-X - 3/4), \qquad J(X) = -J(-X - 1/4). \tag{5.2}$$

Here the gamma reflection signs disappear after taking fourth powers. For example the reversed finite-product ratio in L is

$$\frac{(-t - pX)_a}{(3/4 - t - pX)_a} = \frac{(1/4 - p(X + 3/4))_a}{(1 - p(X + 3/4))_a}.$$

Moreover $F(X)((1/4 + pX)_t/(1 + pX)_t)/F(-X - 3/4)$ has fourth power one. The proof for J is identical, with t replaced by m.

Set $L_0 = L(0)$ and $V_0 = V(0)$. We first establish

$$L_0 \equiv 3p\frac{(1/2)_t}{t!} \pmod{p^3}, \qquad V_0 \equiv p\frac{(3/2)_m}{m!} \pmod{p^2}. \tag{5.3}$$

Use the terminating very-well-poised Dougall identity

$$\sum_{k=0}^{M} \frac{2k + A}{A} \frac{(A)_k (B)_k (C)_k (-M)_k}{k!(1 + A - B)_k (1 + A - C)_k (1 + A + M)_k} = \frac{(1 + A)_M (1 + A - B - C)_M}{(1 + A - B)_M (1 + A - C)_M}.$$

With $(A, B, C, M) = (1/4, 1/4, (1 + 3p)/4, t)$, its left side is

$$\sum_{a=0}^{t} (8a + 1) \frac{(1/4)_a^2 ((1 - 3p)/4)_a ((1 + 3p)/4)_a}{(a!)^2 (1 - 3p/4)_a (1 + 3p/4)_a},$$

and its right side simplifies exactly to $3p(1/2)_t/t!$. The displayed sum minus L_0, modulo p^3, is

$$\frac{9p^2}{16} \sum_{a=0}^{t} (8a + 1) b_a^4 R_a, \qquad R_a = \sum_{j=0}^{a-1} \left((j + 1)^{-2} - (j + 1/4)^{-2} \right).$$

This latter sum vanishes modulo p: reversal $a \mapsto t - a$ gives $b_{t-a}^4 \equiv b_a^4$, $8(t - a) + 1 \equiv -(8a + 1)$, and $R_{t-a} \equiv R_a$. The last identity follows by reversing the summands in R_t, which also gives $R_t \equiv 0$. The analogous Dougall specialization with $(A, B, C, M) = (3/4, 3/4, (3 + p)/4, m)$, multiplied by 3, differs

from V_0 by a multiple of p^2 and evaluates exactly as $p(3/2)_m/m!$. This proves (5.3). In particular, $L_0 \in p^2\mathbb{Z}_p$ and $V_0 \in p\mathbb{Z}_p$.

The Taylor property and the first reflection in (5.2) now imply that there is $E \in \mathbb{Z}_p$ such that

$$H(X) \equiv \frac{L_0}{3}(8X+3) + p^3E\big((8X+3)^3 - 9(8X+3)\big) \pmod{p^4}. \tag{5.4}$$

Indeed H modulo p^2 is affine with constant term zero, so its reflection forces its linear coefficient to vanish too. Thus H is coefficientwise divisible by p^2. Modulo p^4 it has degree at most three, its cubic coefficient is divisible by p^3, and reflection makes it an odd polynomial in $8X+3$. Evaluating at $X=0$ fixes the remaining linear coefficient. Similarly,

$$J(X) \equiv (8X+1)V_0 \pmod{p^2}. \tag{5.5}$$

We next prove the uniform block identity, for each integer $c \geq 0$,

$$\sum_{j=0}^{p^2-1} (8p^2c + 8j + 1)\left(\frac{b_{p^2c+j}}{b_c}\right)^4 \equiv -p^3(8c+1) \pmod{p^4}. \tag{5.6}$$

Write $j = pb + a$, $0 \leq a, b < p$. Removing numerator and denominator factors divisible by p twice shows that only $a \leq t$, $b \leq m$ can contribute modulo p^4; all other ratios b_{p^2c+j}/b_c are in $p\mathbb{Z}_p$. On the contributing rectangle one has exactly

$$b_{pB+a} = F(B)c_B\frac{(1/4+pB)_a}{(1+pB)_a}, \quad B = pc+b,$$

$$c_{pc+b} = J_0(c)b_c\frac{(3/4+pc)_b}{(1+pc)_b}.$$

Consequently the block in (5.6) is congruent to

$$J_0(c)^4\sum_{b=0}^{m}\left(\frac{(3/4+pc)_b}{(1+pc)_b}\right)^4 H(pc+b).$$

The p^3E term from (5.4) vanishes in this expression modulo p^4: reducing its coefficient modulo p, the reversal $b \mapsto m-b$ fixes c_b^4 and negates $8b+3$, hence annihilates every odd polynomial in $8b+3$. The remaining expression is

$$\frac{L_0}{3}J(c) \equiv \frac{L_0V_0}{3}(8c+1) \pmod{p^4}.$$

Finally (5.3) and Wilson's theorem yield

$$\frac{L_0V_0}{3} \equiv p^2\frac{(1/2)_t}{t!}\frac{(3/2)_m}{m!} \equiv -p^3 \pmod{p^4}.$$

For completeness, after removing the unique factor p in $\binom{2t}{t}$, the last unit quotient is

$$\frac{1}{p}\frac{(1/2)_t}{t!}\frac{(3/2)_m}{m!} \equiv -\frac{((2m+1)!)^2}{(t!)^2(m!)^2\,4^{t+m}} \equiv -1 \pmod{p}.$$

Here $t+m = p-1$, $(t!m!)^2 \equiv 1$, and $((2m+1)!)^2 \equiv 1$ since $2m+1 = (p-1)/2$ is odd. This proves (5.6).

The cutoff $(p^3-3)/4 = p^2m + (3p^2-3)/4$ contains all contributing low-digit pairs in each block $c = 0, \ldots, m$, because their largest index is $pm + t = (p^2-1)/4$. All omitted or added terms have fourth powers divisible by p^4, and b_c is a unit for $0 \leq c \leq m$. Multiplying (5.6) by b_c^4 and summing proves (5.1). For $p = 3$, direct evaluation gives

$$S(6) = \frac{19474365987782658225}{18446744073709551616} \equiv 54 = -27 \pmod{81}.$$

Since $S((3-3)/4) = S(0) = 1$, this proves (5.1) for $p = 3$ as well.

6 Every odd supersingular level of G.3

For an odd prime $p \equiv 3 \pmod 4$, set

$$G_{2s} = \sum_{k=0}^{(p^{2s}-1)/4} (8k+1)\frac{(1/4)_k^4}{k!^4}, \qquad G_{2s+1} = \sum_{k=0}^{(p^{2s+1}-3)/4} (8k+1)\frac{(1/4)_k^4}{k!^4}.$$

We prove

$$v_p(G_r) \geq 3\lfloor r/2 \rfloor. \tag{6.1}$$

Together with (5.1), namely $G_3 \equiv -p^3 G_1 \pmod{p^4}$, this proves the full odd supersingular branch of G.3:

$$G_r \equiv -p^3 G_{r-2} \pmod{p^{r+1}} \qquad (r \geq 3 \text{ odd},\ p \equiv 3 \pmod 4). \tag{6.2}$$

Write $(z;Q)_k = \prod_{i=0}^{k-1}(1-zQ^i)$, with a list of arguments denoting the product of the corresponding symbols. Set $Q = q^4$, $[j] = (1-q^j)/(1-q)$ and

$$g_k(a;q) = [8k+1]q^{2k}\frac{(aq, q/a, q, q; Q)_k}{(aq^4, q^4/a, q^4, q^4; Q)_k}, \qquad T_L(a;q) = \sum_{k=0}^{L} g_k(a;q).$$

The terminating very-well-poised ${}_6\phi_5$ identity is

$$\sum_{k=0}^{t} \frac{1-AQ^{2k}}{1-A}\frac{(A, B, C, Q^{-t}; Q)_k}{(Q, AQ/B, AQ/C, AQ^{t+1}; Q)_k}\left(\frac{AQ^{t+1}}{BC}\right)^k = \frac{(AQ, AQ/(BC); Q)_t}{(AQ/B, AQ/C; Q)_t}.$$

For $v = 4t+1$, substitute $A = q$, $B = q^{1+v}$, $C = q$ and $Q = q^4$. The argument of the sum is q^2, and reversing the negative powers on the product side gives

$$T_L(q^{\pm v}; q) = q^{-t}[v]\frac{(q^2; Q)_t}{(q^4; Q)_t} \qquad (L \geq t). \tag{6.3}$$

In detail, the product side before reversal is $(q^5, q^{3-v}; Q)_t/(q^{4-v}, q^4; Q)_t$; $(q^{3-v}; Q)_t/(q^{4-v}; Q)_t = q^{-t}(q^2; Q)_t/(q; Q)_t$ and $(q^5; Q)_t/(q; Q)_t = [v]$. The numerator parameter $q^{1-v} = Q^{-t}$ terminates the series; no denominator vanishes identically since $v \equiv 1 \pmod 4$.

Cyclotomic prefix congruences. Let $h > 1$ be odd, let $\epsilon \in \{1, 3\}$ satisfy $h \equiv \epsilon \pmod 4$, and set

$$t = (\epsilon h - 1)/4, \qquad L = Bh + t \quad (B \geq 0), \qquad w_h = \begin{cases} 1 & h \equiv 1 \pmod 4, \\ 2 & h \equiv 3 \pmod 4. \end{cases}$$

We claim

$$v_{\Phi_h}(T_L(1;q)) \geq w_h. \tag{6.4}$$

Work in the local ring of $\mathbb{Q}(q)$ at $\Phi_h(q)$. Set

$$D(a) = (aq^4, q^4/a; Q)_L, \qquad F(a) = a^L D(a) T_L(a;q).$$

Then F is a polynomial in a over this local ring: D clears the parameter denominators, a^L clears negative powers, and $(q;Q)_k/(q^4;Q)_k$ has nonnegative Φ_h valuation for every k. Indeed in each period the numerator zero precedes the denominator zero.

For $0 \leq i \leq B$ put

$$v_i = (4i+\epsilon)h, \quad t_i = (v_i - 1)/4 = ih + t, \quad \alpha_{i,\pm} = q^{\pm v_i}.$$

These $2B+2$ nodes are distinct. At each node (6.3) has valuation w_h. To see this, the quotient $(q^2;Q)_{t_i}/(q^4;Q)_{t_i}$ has valuation zero when $h \equiv 1 \pmod 4$, and one when $h \equiv 3 \pmod 4$: the numerator zero occurs at index $(h-1)/2$, while the denominator zero occurs at index $h-1$. The factor $[v_i]$ has valuation one in both cases. Also

$$v_{\Phi_h}(D(\alpha_{i,\pm})) = 2B,$$

since among $1 \le j \le L$ exactly B integers satisfy $h \mid 4j$. Therefore $v_{\Phi_h}(F(\alpha_{i,\pm})) \ge 2B + w_h$.

Let $P(a) = \prod_{i=0}^{B}(a-\alpha_{i,+})(a-\alpha_{i,-})$. Monic division gives $F = PH + R$ with integral coefficients and $\deg R < 2B+2$. Hence $v_{\Phi_h}(P(1)H(1)) \ge 2B+2$. The exact Lagrange formula is

$$R(1) = \sum_{\alpha} F(\alpha) \prod_{\beta \ne \alpha} \frac{1-\beta}{\alpha-\beta}.$$

Every factor in every displayed weight has valuation zero: $1-\beta$ and $\alpha-\beta$ both have valuation exactly one. Thus $v_{\Phi_h}(R(1)) \ge 2B + w_h$. Since $w_h \le 2$ and $v_{\Phi_h}(D(1)) = 2B$, division proves (6.4). The same congruence holds at cutoff $Bh+h-1$, because every extra term past $Bh+t$ has valuation at least four after $a=1$.

Integral specialization. After reduction, the denominator of each $(q;q^4)_k/(q^4;q^4)_k$ has only cyclotomic factors of even order. All such factors evaluate at $q=1$ to either 1 or 2. Consequently a common denominator for all the specialized terms $g_k(1;q)$ has value a power of two at 1. After clearing it, all cyclotomic divisibilities below are divisibilities of integral polynomials. Products of distinct monic cyclotomic factors can therefore be combined before setting $q=1$, without any assertion of integral comaximality.

The digit-interval partition. Put $n=p^2$, $d=(n-1)/4$, $t_j=(n^j-1)/4$ and $t_0=0$. Define

$$P_j(q) = \prod_{\substack{1\le \ell \le 2j \\ \ell \text{ odd}}} \Phi_{p^\ell}(q)^2 \prod_{\substack{1\le \ell \le 2j \\ \ell \text{ even}}} \Phi_{p^\ell}(q), \qquad P_0 = 1.$$

For every $u \ge 0$ let

$$I(u,j) = [un^j, un^j + t_j] \cap \mathbb{Z}.$$

Subtracting two initial prefix congruences from (6.4) gives

$$P_j(q) \;\Big|\; \sum_{k \in I(u,j)} g_k(1;q). \tag{6.5}$$

Indeed for $h=p^\ell$, $\ell \le 2j$, the endpoint un^j+t_j has residue $(3h-1)/4$ if ℓ is odd and $(h-1)/4$ if ℓ is even. The left preceding endpoint un^j-1 has residue $h-1$, covered by the last sentence after (6.4); for $u=0$ the subtracted sum is empty.

Also define the full block, tail, and three-quarter prefix

$$\begin{aligned} B(u,j) &= [un^j, (u+1)n^j - 1] \cap \mathbb{Z}, \\ T(u,j) &= [un^j + t_j + 1, (u+1)n^j - 1] \cap \mathbb{Z}, \\ J(u,j) &= [un^j, un^j + 3t_j] \cap \mathbb{Z}. \end{aligned}$$

At level zero $I = B = J = \{u\}$ and T is empty. For $j \ge 1$, $t_j = dn^{j-1} + t_{j-1}$ gives the disjoint partitions

$$B(u,j) = I(u,j) \,\dot\cup\, T(u,j),$$

$$T(u,j) = T(nu+d, j-1) \,\dot\cup \bigcup_{c=d+1}^{n-1} B(nu+c, j-1),$$

$$J(u,j) = I(u,j) \,\dot\cup T(nu+d, j-1) \,\dot\cup \bigcup_{c=d+1}^{3d-1} B(nu+c, j-1) \,\dot\cup J(nu+3d, j-1).$$

Every non-I piece lies in its parent tail. Recursing and stopping at each $I(v,k)$ partitions both $B(u,j)$ and $J(u,j)$ into finitely many such leaves. A leaf at depth $j-k$ lies in a tail at every ancestor scale $\ell = k+1, \dots, j$. Thus each index in it has remainder greater than t_ℓ modulo $p^{2\ell}$, and each summand carries the additional factor $\Phi_{p^{2\ell}}(q)^4$. Combining these ancestor factors with (6.5) yields

$$P_k(q) \prod_{\ell=k+1}^{j} \Phi_{p^{2\ell}}(q)^4 \;\Big|\; \sum_{a \in I(v,k)} g_a(1;q).$$

All factors here have distinct orders. Clearing the power-of-two specialization denominator and dividing by the monic integral product justifies setting $q = 1$. Each leaf then has p-adic valuation at least

$$3k + 4(j-k) = 3j + (j-k) \geq 3j.$$

The same is true for the sums over $B(u,j)$ and $J(u,j)$.

Finally, writing $p = 4m+3$, the odd cutoff satisfies

$$(p^{2s+1} - 3)/4 = mn^s + 3t_s.$$

Its interval is the union of $B(0,s), \dots, B(m-1,s)$ and $J(m,s)$, giving $v_p(G_{2s+1}) \geq 3s$. The even level follows directly from (6.4), since $P_s(1) = p^{3s}$. This proves (6.1). For odd $r = 2s+1 \geq 5$, both G_r and $p^3 G_{r-2}$ have valuation at least $3s \geq 2s+2 = r+1$, so their signed recurrence follows. Equation (5.1) completes (6.2), including $p = 3$. For that prime the interval decomposition above remains valid with $m = 0$, so the final interval consists only of $J(0,s)$.

7 The second supersingular level of A.3

Write $b_k = (1/2)_k / k!$ and

$$A(N) = \sum_{k=0}^{(N-1)/2} (-1)^k (4k+1) b_k^5 \qquad (N \text{ odd}).$$

Theorem 7.1. *For every prime $p \equiv 3 \pmod 4$,*

$$\boxed{A(p^2) \equiv p^4 \pmod{p^8}.}$$

Thus the entire $r = 2$ supersingular branch of Swisher's A.3 holds.

7.1 A shifted short-block divisibility lemma

For $p \geq 7$, $p \equiv 3 \pmod 4$, put $h = (p-1)/2$ and define

$$F_p(x) = \sum_{j=0}^{h} (-1)^j (4j+1+4x) \left(\frac{(x+1/2)_j}{(x+1)_j} \right)^5.$$

This rational function has a power series in $\mathbb{Z}_p[[x]]$.

Lemma 7.2. *For $x \in p\mathbb{Z}_p$, $F_p(x) \in p^3\mathbb{Z}_p$ and $F_p'(x) \in p^2\mathbb{Z}_p$. Moreover, for $0 \le a \le h$,*

$$F_p((h-a)p) \equiv -F_p(ap) \pmod{p^4}.$$

Proof. For any integer t, set

$$R_t(x) = \frac{(x+1/2)_t}{(x+1)_t}, \qquad c_k = \frac{(3/4)_k}{(1/4)_k}.$$

Negative indices mean reciprocals; more explicitly, for $d \ge 1$,

$$R_{-d}(x) = \frac{\prod_{i=0}^{d-1}(x-i)}{\prod_{i=1}^{d}(x+1/2-i)}.$$

The shifted WZ pair is

$$\mathcal{F}(j,k;x) = (-1)^{j+k}(4x+4j+1)c_k R_j(x)R_{j+k}(x)^2R_{j-k}(x)^2,$$
$$\mathcal{G}(j,k;x) = 8(x+j)(-1)^{j+k}\frac{(2k-1)c_k}{4k-1}R_j(x)R_{j+k-1}(x)^2R_{j-k}(x)^2.$$

Adjacent-factor cancellation verifies the rational-function identity

$$\mathcal{F}(j,k-1;x) - \mathcal{F}(j,k;x) = \mathcal{G}(j+1,k;x) - \mathcal{G}(j,k;x).$$

The displayed formulas give a direct verification: divide the identity by $\mathcal{F}(j,k;x)$ and substitute the adjacent Pochhammer ratios, then clear denominators. Since this is an identity of rational functions, it remains valid when a numerator factor vanishes. Summing over $0 \le j \le h$ and $1 \le k \le h+1$ gives

$$F_p(x) = \sum_{j=0}^{h}\mathcal{F}(j,h+1;x) + \sum_{k=1}^{h+1}\big(\mathcal{G}(h+1,k;x) - \mathcal{G}(0,k;x)\big). \tag{7.1}$$

First take $x = ap$, $0 \le a \le h$. Every expression in (7.1) is well-defined. The elementary factor counts give

factor	lower bound on its p-valuation
$R_t(x),\ 0 \le t \le h$	0 (in fact a unit)
$R_t(x),\ h+1 \le t \le p-1$	1
$R_p(x)$	0
$R_{-d}(x),\ 1 \le d \le h$	1
$R_{-(h+1)}(x)$	0

Indeed, the positive factors acquire the numerator $x+p/2$ at $t = h+1$ and the denominator $x+p$ at $t = p$. The negative factors contain x in their numerator and first acquire a nonunit denominator $x - p/2$ at $d = h+1$. The factors $a+1$ and $a-1/2$ are units for $0 \le a \le h$; the possible extra divisibility of $a+1/2$ only improves the bounds. If $a = 0$, some factors vanish exactly, which also improves the estimates.

There is exactly one factor of p in c_{h+1}, so $v_p(c_{h+1}) = 1$. For

$$d_k = \frac{(2k-1)c_k}{4k-1} \quad (1 \le k \le h+1),$$

we have $v_p(d_k) \ge 0$ and $v_p(d_{h+1}) = 2$. The only possible nonunit $4k-1 = p$ cancels the final numerator factor in c_k; no denominator factor $4i+1$ in c_k is divisible by p, since $p \equiv 3 \pmod 4$ and $4i+1 < 2p$.

These counts show that every term on the right of (7.1) belongs to $p^3\mathbb{Z}_p$. For $\mathcal{F}(j, h+1; x)$, the interior indices $0 < j < h$ even have valuation at least 5, while each endpoint has valuation at least 3. For the upper $\mathcal{G}$ boundary, $k \le h$ supplies one factor from R_{h+1} and two from R_{h+k}; at $k = h+1$ the two missing factors are supplied by d_{h+1}. For the lower boundary, $k \le h$ supplies x and R_{-k}^2; at $k = h+1$ use x and d_{h+1}. Hence $F_p(ap) \in p^3\mathbb{Z}_p$ for $0 \le a \le h$.

Write $F_p(x) = \sum_{j\ge 0} f_j x^j$ with $f_j \in \mathbb{Z}_p$. Modulo p^3, the three evaluations $a = 0, 1, 2$ say that $f_0 + pf_1 a + p^2 f_2 a^2$ vanishes at three distinct residues. The Vandermonde determinant is 2, a unit, so

$$f_0 \in p^3\mathbb{Z}_p, \qquad f_1 \in p^2\mathbb{Z}_p, \qquad f_2 \in p\mathbb{Z}_p.$$

All higher terms then imply the asserted divisibility of $F_p(x)$ and $F_p'(x)$ throughout $p\mathbb{Z}_p$.

Reverse the defining sum with $j \mapsto h - j$. The exact identity is

$$F_p(x) = R_h(x)^5 F_p(-p/2 - x),$$

because h is odd. Also $R_h(x) \equiv b_h \equiv -1 \pmod p$. Since both short sums are divisible by p^3, this gives $F_p(-p/2 - x) \equiv -F_p(x) \pmod{p^4}$. For $x = ap$ the difference between $(h-a)p$ and $-p/2 - ap$ is $p^2/2$. Taylor expansion, using $F_p' \in p^2\mathbb{Z}_p$ and integrality of all the other Taylor coefficients, changes F_p by a multiple of p^4. This proves the last assertion. □

7.2 Removing the four-root deformation at the second level

Let $N = p^2$ and put

$$D_k(N) = \frac{\prod_{j=0}^{k-1}(1 - N^4/(2j+1)^4)}{\prod_{j=1}^{k}(1 - N^4/(16j^4))}, \qquad T(N) = \sum_{k=0}^{(N-1)/2} (-1)^k (4k+1) b_k^5 D_k(N).$$

All the factors in $D_k(N)$ are integral units, and $D_k(N) - 1 \in p^4\mathbb{Z}_p$. Write $k = ap + b$ with $0 \le b < p$. If $b > h$, the permitted cutoff forces $a < h$, and the central-binomial factor b_k has valuation 1. Hence these terms contribute to $A(N) - T(N)$ only modulo p^9. For the remaining indices, $0 \le a, b \le h$, splitting the p-divisible factors from the others gives

$$D_{ap+b}(p^2) \equiv 1 + p^4 L_a \pmod{p^8}, \qquad L_a = \sum_{j=1}^{a} \frac{1}{(2j)^4} - \sum_{j=0}^{a-1} \frac{1}{(2j+1)^4}.$$

The singular numerator indices are exactly $j = up + h$, $0 \le u < a$; the singular denominator indices are $j = up$, $1 \le u \le a$. All other factors are 1 modulo p^8, and products of two singular corrections are already multiples of p^8.

Define the weighted block

$$S_a = \sum_{b=0}^{h} (-1)^{ap+b} (4ap + 4b + 1) b_{ap+b}^5 = (-1)^a b_{ap}^5 F_p(ap).$$

Lemma 7.2 gives $S_a \in p^3\mathbb{Z}_p$. Lucas's congruence gives $b_{ap} \equiv b_a \pmod p$, and reversal of the half-binomial products gives $b_{h-a} \equiv -b_a \pmod p$. Together with the last assertion of the lemma and the oddness of h, these imply

$$S_{h-a} \equiv -S_a \pmod{p^4}.$$

On the other hand,

$$L_{h-a} \equiv L_a \pmod p.$$

To see this, write $L_a = \sum_{j=1}^{2a}(-1)^j/j^4$. Reflection $j \mapsto p-j$ reverses parity and preserves inverse fourth powers modulo p, so $L_h \equiv 0$ and $L_h - L_{h-a} \equiv -L_a$. Since h is odd, pairing a with $h-a$ has no fixed point. Consequently

$$\sum_{a=0}^{h} L_a S_a \equiv 0 \pmod{p^4}.$$

It follows that

$$A(p^2) - T(p^2) \equiv -p^4 \sum_{a=0}^{h} L_a S_a \equiv 0 \pmod{p^8}. \tag{7.2}$$

7.3 The explicit second-level model

For every odd positive integer L, define $T(L)$ by the same finite deformation sum, replacing N by L. Its upper parameters are $5/4, 1/2, (1-L)/2, (1+L)/2, (1-iL)/2, (1+iL)/2$ and its lower parameters are $1/4, 1+L/2, 1-L/2, 1+iL/2, 1-iL/2$. Whipple's terminating ${}_6F_5(-1)$ transformation gives

$$T(L) = \frac{\Gamma(1+L/2)\Gamma(1-L/2)}{\Gamma(3/2)\Gamma(1/2)} {}_3F_2\left[\begin{matrix}(1-L)/2, (1+L)/2, 1/2 \\ 1+iL/2, 1-iL/2\end{matrix}; 1\right].$$

The series terminates at $(L-1)/2$. The classical Whipple ${}_3F_2$ evaluation, whose upper pair sums to 1 and whose lower pair sums to twice the third upper parameter plus 1, gives

$$T(L) = \frac{2\Gamma(1+L/2)\Gamma(1-L/2)\Gamma(1+iL/2)\Gamma(1-iL/2)}{\prod_{\varepsilon,\eta\in\{\pm1\}} \Gamma((3+\varepsilon L+\eta iL)/4)}. \tag{7.3}$$

These are the classical identities applied in H. Swisher, *On the supercongruence conjectures of van Hamme*, Research in the Mathematical Sciences **2** (2015), article 18, Section 6, equations (38)–(40), https://doi.org/10.1186/s40687-015-0037-6. They are exact terminating identities before any prime-power specialization. Gamma shifts and reflection now yield

$$T(L) = \begin{cases} L \displaystyle\prod_{j=1}^{(L-1)/4} \frac{(4j-2)^2+L^2}{(4j)^2+L^2}, & L \equiv 1 \pmod 4, \\ -L^3 \dfrac{\prod_{j=1}^{(L-3)/4}((4j)^2+L^2)}{\prod_{j=0}^{(L-3)/4}((4j+2)^2+L^2)}, & L \equiv 3 \pmod 4. \end{cases}$$

For the second formula, put $L = 4t+3$ and $z = iL/4$ in (7.3). The shift, reflection and duplication formulas reduce it to $Lz^2(1-z)_t(1+z)_t/((1/2-z)_{t+1}(1/2+z)_{t+1})$, which is exactly the displayed product.

For $p \equiv 3 \pmod 4$ and $L = p^s$, separate the p-divisible factors in either product and let $V(L)$ denote the remaining unit quotient. Direct counting gives

$$T(L)T(L/p) = -p^3(L/p)^4 V(L).$$

For $L \equiv 1 \pmod 4$ the odd-factor numerator has one more divisible factor than the even-factor denominator; for $L \equiv 3 \pmod 4$ the two counts are equal. These are precisely the two cases of the preceding equality. Thus

$$T(p^2) = p^4 \frac{V(p^2)}{V(p)}.$$

The exact unit-product gamma formula in both cases is

$$V(L) = \frac{1}{16}\left(\prod_{\varepsilon,\eta\in\{\pm1\}} \Gamma_p\left(\frac{3}{4} + \frac{\varepsilon+\eta i}{4}L\right)\right)^{-1}.$$

Here $i^2 = -1$ lies in the unramified quadratic extension $K = \mathbb{Q}_p(i)$, with valuation ring $\mathcal{O}_K$. The required extension of Γ_p is justified explicitly by L. Long and R. Ramakrishna, *Some supercongruences occurring in truncated hypergeometric series*, arXiv:1403.5232v2, Proposition 13 and equations (2.2)–(2.3), https://arxiv.org/abs/1403.5232. For rational $a \in \mathbb{Z}_p$, their coefficient bounds are

$$c_k(a) = \frac{\Gamma_p^{(k)}(a)}{k!\Gamma_p(a)}, \qquad c_k(a) \in \mathbb{Z}_p \ (k < p), \qquad v_p(c_k(a)) \geq -k\left(\frac{1}{p} + \frac{1}{p-1}\right).$$

The series $\Gamma_p(a)\sum_{k\geq 0} c_k(a)z^k$ converges for $z \in p\mathcal{O}_K$ and defines the gamma values used here. The step and reflection identities on these extended disks are also equations (2.2)–(2.3) of that source. In particular the half-argument pair has value $(-1)^{(p+1)/2} = 1$, and $\Gamma_p(1+iL/4)\Gamma_p(1-iL/4) = 1$. The factor $1/16$ in $V(L)$ records the single excess quadratic denominator among its unit factors in either residue case; it cancels in $V(p^2)/V(p)$.

For $p \geq 7$ and $z \in p\mathcal{O}_K$, those bounds imply

$$\frac{\Gamma_p(a+z)}{\Gamma_p(a)} \equiv 1 + c_1(a)z + c_2(a)z^2 + c_3(a)z^3 \pmod{p^4\mathcal{O}_K}.$$

Indeed terms $4 \leq k < p$ are integral multiples of p^4; for $k \geq p$ their valuations are at least $k(1-1/p-1/(p-1)) \geq p-2-1/(p-1) > 4$. This argument includes $p = 7$ and does not require the stronger $p \geq 11$ version of Long–Ramakrishna's Theorem 14. The four shifts form one orbit under multiplication by i, so

$$V(p) \equiv V(p^2) \equiv \frac{1}{16}\Gamma_p(3/4)^{-4} \pmod{p^4}.$$

Both are units. Hence $T(p^2) \equiv p^4 \pmod{p^8}$, and (7.2) proves the theorem for $p \geq 7$. Finally, the sole remaining prime $p = 3$ is checked exactly:

$$A(9) - 3^4 = -\frac{419492893}{34359738368}\,3^8.$$

The denominator is a power of 2, completing the proof.

8 Verification and remaining targets

The preceding arguments are mathematical proofs, not inferences from finite numerical tests. The companion code supplies independent checks of the delicate identities, signs, cutoffs, and intermediate valuation bounds. It uses integer arithmetic, exact rational arithmetic, and modular arithmetic with explicit removal of prime factors.

Check	Recorded coverage
A.3	17,752 shifted WZ identities; 101 exact model identities; all intermediate bounds for $p = 7, 11, 19, 23, 31, 43$; the separate exact $p = 3$ calculation.
D.3	20 exact cubic-model identities; all coarse cutoffs $0 \leq C \leq (p-2)/3$ for $p = 5, 11, 17, 23, 29, 41, 47$, including low blocks, model scaling and triangular targets.
G.3	292 normalized two-digit blocks, comprising 275,476 ratio terms, for $p = 7, 11, 19, 23, 31, 43$; 224 cyclotomic prefix and derivative evaluations in exact finite fields; 36 interval partitions with 49,176 leaves.

What is still missing. The A.3 argument removes a fourth-order deformation at the second level. It does not bound its accumulated error at the required p^{5r-2} precision for arbitrary $r \geq 3$. The ordinary D.3 target has precision p^{6r}; the supersingular vanishing method proved here does not address that unit-root recurrence. The even supersingular G.3 target is p^{4r-2}, which is much stronger than the divisibility $p^{3\lfloor r/2 \rfloor}$ proved here. The ordinary G.3 branch also needs a separate residual-sum scaling argument.

For E.3, an exact short-cutoff deformation was derived, but removal of that deformation at the required odd-level precision remains unproved. The companion bundle retains that formula and an ordinary G.3 transformation as explicitly unfinished research checkpoints. Neither checkpoint is used to establish any theorem in this paper.

Reproducibility. The companion archive contains this complete single-file LaTeX source, the PDF, the proof fragments, the exact-check scripts and their recorded JSON results. Its README gives working-directory and dependency instructions. The original statements are taken from Swisher's published Section 7; published inputs are identified at their point of use. Internal checking does not replace external mathematical review.

References

[1] H. Swisher, *On the supercongruence conjectures of van Hamme*, Research in the Mathematical Sciences **2** (2015), Article 18. https://doi.org/10.1186/s40687-015-0037-6.

[2] H.-X. Ni and L.-Y. Wang, *Two q-supercongruences from Watson's transformation*, Revista de la Real Academia de Ciencias Exactas, Fisicas y Naturales. Serie A. Matematicas **116** (2022), Article 30. Theorem 2.3 is also reproduced in [3], equation (2.12).

[3] C. Wei, *q-Supercongruences from Jackson's ${}_8\phi_7$ summation and Watson's ${}_8\phi_7$ transformation*, Journal of Combinatorial Theory, Series A **204** (2024), 105853. https://arxiv.org/abs/2104.07025 (version 4).

[4] L. Long and R. Ramakrishna, *Some supercongruences occurring in truncated hypergeometric series*, Advances in Mathematics **290** (2016), 773–808. https://arxiv.org/abs/1403.5232 (version 2).

[5] Verifiable Mathematical Reasoning Corpus, entries [131] and [132], motivating research material; the present proofs are independent of the results in these entries. https://github.com/QEEDDD/Verifiable-Mathematical-Reasoning-Corpus.

MANUSCRIPT V

Exact Interpolation and Positive Moments for Swisher Supercongruences

Complete A.3 and the remaining D.3 and G.3 branches

Research manuscript 10 November 2026

Abstract

We prove all residue branches and all levels of Swisher's A.3, including the primes 3 and 5. We also prove every ordinary-prime level of D.3 and G.3, with their full respective moduli p^{6r} and p^{4r}, and every even supersingular G.3 level for $p > 3$, with modulus p^{4r-2}. The central tool is an exact interpolation formula for root-of-unity deformations of terminating hypergeometric sums. Every positive interpolation moment admits a polynomial formula whose coefficients have only powers of two in their denominators. These identities remove all unit nodes together and control the entire analytic correction when the remaining nodes are divided by p. A separate termwise interpolation argument proves the supersingular A.3 branch, using an elementary four-factor gamma product that also covers $p = 3$. A companion family with parameter $3/4$ supplies the intermediate step for even supersingular G.3. The odd supersingular E.3 branch is not claimed.

Status. This is a research manuscript with complete proofs of the explicitly stated results, independent internal mathematical checks, and reproducible exact arithmetic. It has not undergone external peer review. No historical-priority assertion is made.

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1 Statements and exact scope

Write $(a)_k = \prod_{j=0}^{k-1}(a+j)$ and $(a)_0 = 1$. We normalize $v_p(p) = 1$ and put $v_p(0) = +\infty$. Congruences are interpreted in $\mathbb{Q}_p$, using the indicated valuation of the difference. Throughout, p denotes an odd prime. The labels refer to Swisher's Section 7, Table 2 [1]. The words ordinary and supersingular name the indicated residue branches; no geometric assertion is implicit in this terminology. For the indicated positive integers define

$$A(N) = \sum_{k=0}^{(N-1)/2} (-1)^k(4k+1)\frac{(1/2)_k^5}{(k!)^5} \qquad (N \text{ odd}),$$

$$D(N) = \sum_{k=0}^{(N-1)/3} (6k+1)\frac{(1/3)_k^6}{(k!)^6} \qquad (N \equiv 1 \pmod 3),$$

$$G(N) = \sum_{k=0}^{(N-1)/4} (8k+1)\frac{(1/4)_k^4}{(k!)^4} \qquad (N \equiv 1 \pmod 4).$$

Theorem 1.1 (All of A.3)**.** *For every prime $p \equiv 1 \pmod 4$ and every $r \geq 1$,*

$$A(p^r) \equiv -p\Gamma_p(1/4)^4 A(p^{r-1}) \pmod{p^{5r}}.$$

For every prime $p \equiv 3 \pmod 4$,

$$A(p) \equiv 0 \pmod{p^3}, \qquad A(p^r) \equiv p^4 A(p^{r-2}) \pmod{p^{5r-2}} \quad (r \geq 2).$$

These statements include $p = 5$ and $p = 3$, respectively.

Theorem 1.2 (All ordinary levels of D.3)**.** *For every prime $p \equiv 1 \pmod 3$ and every $r \geq 1$,*

$$D(p^r) \equiv -p\Gamma_p(1/3)^9 D(p^{r-1}) \pmod{p^{6r}}.$$

Theorem 1.3 (All ordinary levels of G.3)**.** *For every prime $p \equiv 1 \pmod 4$ and every $r \geq 1$,*

$$G(p^r) \equiv p\lambda_p G(p^{r-1}) \pmod{p^{4r}}, \qquad \lambda_p = -(-1)^{(p-1)/4}\Gamma_p(1/2)\Gamma_p(1/4)^2.$$

Theorem 1.4 (All even supersingular levels of G.3)**.** *For every prime $p > 3$ with $p \equiv 3 \pmod 4$ and every $s \geq 1$,*

$$G(p^{2s}) \equiv -p^3 G(p^{2s-2}) \pmod{p^{8s-2}}.$$

Theorem 1.5 (Exact supersingular valuations)**.** *For $p \equiv 3 \pmod 4$ and $r \geq 0$, $v_p(A(p^r)) = 2r + (r \bmod 2)$. For $p > 3$, $p \equiv 3 \pmod 4$, and $s \geq 1$,*

$$v_p(G(p^{2s})) = 3s.$$

Label	Branch	Scope proved here
A.3	Both residue classes modulo 4	Every target level, including $p = 3, 5$.
D.3	$p \equiv 1 \pmod 3$	Every level, modulus p^{6r}.
G.3	$p \equiv 1 \pmod 4$	Every level, modulus p^{4r}.
G.3	$p > 3$, $p \equiv 3 \pmod 4$, even levels	Every target level, modulus p^{4r-2}.
E.3	$p \equiv 2 \pmod 3$, odd $r \geq 3$	Remains unresolved.

The earlier companion manuscript [6] established the $p \equiv 2 \pmod 3$ branch of D.3 and the odd supersingular branch of G.3. Those results are not reused as hypotheses in the new proofs. Combining the ordinary D.3 theorem here with that earlier result completes D.3 within this research series. Similarly, the two new G.3 branches and the earlier odd-level result complete G.3 for $p > 3$. The present document does not repeat the earlier supersingular D and odd-level G proofs. B.3, H.3, I.3 and L.3 are outside the scope of this continuation.

Why moments are needed. Some individual interpolation terms have negative valuations at split primes. The proofs therefore do not infer integrality of a sum from an unjustified termwise bound. Instead every positive moment is evaluated exactly; an Euler-exponent limit removes unit nodes together, and the remaining analytic correction is expanded against the already-controlled moments. The polynomial operator used for these moment identities has unit triangular diagonal over $\mathbb{Z}[1/2]$, so division by a varying odd prime never occurs. The standard terminating evaluations used below are recorded in [2], equations 16.4.7, 16.4.9 and 16.4.10.

2 A.3 in every supersingular level by exact interpolation

Set $b_k = (1/2)_k/k!$ and

$$A(N) = \sum_{k=0}^{(N-1)/2} (-1)^k (4k+1) b_k^5 \qquad (N \text{ positive and odd}).$$

Theorem 2.1. *For every prime $p \equiv 3 \pmod 4$ and every $r \geq 2$,*

$$A(p^r) \equiv p^4 A(p^{r-2}) \pmod{p^{5r-2}}.$$

The same argument also gives $A(p) \equiv 0 \pmod{p^3}$.

2.1 An exact finite interpolation formula

For odd N, write $h = (N-1)/2$ and define

$$S_N(z) = \sum_{k=0}^{h} (-1)^k (4k+1) b_k^5 \frac{\prod_{j=0}^{k-1}(1 - z^4/(2j+1)^4)}{\prod_{j=1}^{k}(1 - z^4/(2j)^4)}.$$

Put $T(H) = S_H(H)$ for every positive odd H. The rational function $S_N(z)$ depends only on $y = z^4$, and multiplication by $D_N(y) = \prod_{j=1}^{h}(1 - y/(2j)^4)$ gives a polynomial of degree at most h. At each of the $h+1$ distinct nodes $y = H^4$, $H = 1, 3, \ldots, N$, the sum terminates at $(H-1)/2$, so $S_N(H) = T(H)$. Lagrange interpolation at zero consequently gives the exact rational identity

$$A(N) = \sum_{\substack{1 \leq H \leq N \\ H \text{ odd}}} W_N(H) T(H), \qquad W_N(H) = \frac{\prod_{\substack{2 \leq E < N \\ E \text{ even}}} (1 - H^4/E^4)}{\prod_{\substack{1 \leq K \leq N \\ K \text{ odd},\ K \neq H}} (1 - H^4/K^4)}. \tag{2.1}$$

All denominators here are nonzero rational numbers. In particular, there is no limiting convention at $H = N$.

Lemma 2.2. *Let $N = p^r$, where p is an odd prime, and let $H \leq N$ be odd with $s = v_p(H) \geq 1$. Put $M = N/p$. Then*

$$\frac{W_N(H)}{W_M(H/p)} \equiv 1 \pmod{p^{r+4s}}.$$

Proof. In the exact product (2.1), separate indices divisible by p. Because p is odd, division by p preserves parity; these factors are exactly $W_M(H/p)$. The other factors give

$$\frac{W_N(H)}{W_M(H/p)} = \frac{\prod_{\substack{2 \leq E < N \\ E \text{ even},\ p \nmid E}} (1 - H^4/E^4)}{\prod_{\substack{1 \leq K < N \\ K \text{ odd},\ p \nmid K}} (1 - H^4/K^4)}.$$

The map $E \mapsto K = N - E$ bijects these two index sets. Each paired factor differs from 1 by

$$\frac{1 - H^4/E^4}{1 - H^4/K^4} - 1 = \frac{H^4(E^4 - K^4)}{E^4(K^4 - H^4)}.$$

Its denominator is a unit. Since $K = N - E$, the numerator has valuation at least $4s + r$. Multiplying proves the assertion, including $H = N$. □

Lemma 2.3. *Suppose $p \equiv 3 \pmod 4$, $N = p^r$, and $H \le N$ is odd. If $s = v_p(H)$, then*

$$v_p(W_N(H)) = 5(r - s).$$

Proof. An elementary factorization of differences of fourth powers gives

$$W_N(H) = 2(N/H)^4 b_{(N-1)/2}^4 b_{(N-H)/2} b_{(N+H)/2} \frac{\prod_{\substack{2 \le E < N \\ E \text{ even}}} (E^2 + H^2)}{\prod_{\substack{1 \le K \le N \\ K \text{ odd}, K \ne H}} (K^2 + H^2)}. \tag{2.2}$$

For clarity, the difference-factor identity used here is

$$\frac{\prod_{E \text{ even}, 2 \le E < N} (E^2 - H^2)}{\prod_{K \text{ odd}, 1 \le K \le N, K \ne H} (K^2 - H^2)} = 2 b_{(N-H)/2} b_{(N+H)/2};$$

the remaining normalization is $(N b_{(N-1)/2}/H)^4$. Since -1 is a nonsquare modulo p, a quadratic factor $X^2 + H^2$ has valuation $2\min(v_p(X), s)$. At every division by p, the numbers of divisible numerator and denominator factors in the final quotient in (2.2) are equal: both are $(p^{r-1} - 1)/2$, because the denominator omits H. Repeating until H/p^s is a unit proves that this quotient is a unit. Also $b_{(N-1)/2}$ and b_N are units, by Lucas's theorem. If $H < N$, put $k = (N - H)/2$, so $v_p(k) = v_p(N - k) = s < r$. The identity

$$\frac{b_k b_{N-k}}{b_N} = \frac{\binom{N}{k}^2}{\binom{2N}{2k}}$$

shows that $v_p(b_k b_{N-k}) = r - s$. Indeed $v_p\binom{N}{k} = r - s$, and $v_p\binom{2N}{2k} = r - s$: in the latter equality use $\binom{2N}{2k} = (N/k)\binom{2N-1}{2k-1}$, whose last factor is a unit by Lucas because $2k - 1 < N$ and all the lower r base-p digits of $2N - 1$ equal $p - 1$. Formula (2.2) now gives $4(r - s) + (r - s)$. At $H = N$ all factors except b_N are units, giving valuation zero as required. □

2.2 The exact Whipple model at arbitrary odd nodes

The classical terminating Whipple transformations give

$$T(L) = \frac{2\Gamma(1 + L/2)\Gamma(1 - L/2)\Gamma(1 + iL/2)\Gamma(1 - iL/2)}{\prod_{\varepsilon, \eta \in \{\pm 1\}} \Gamma((3 + \varepsilon L + \eta i L)/4)}.$$

These are exactly the finite identities in Swisher, Section 6, equations (38)–(40), before specializing L to a prime. Gamma shifts and reflection yield, for every positive odd L,

$$T(L) = \begin{cases} L \displaystyle\prod_{j=1}^{(L-1)/4} \frac{(4j-2)^2 + L^2}{(4j)^2 + L^2}, & L \equiv 1 \pmod 4, \\ -L^3 \dfrac{\prod_{j=1}^{(L-3)/4} ((4j)^2 + L^2)}{\prod_{j=0}^{(L-3)/4} ((4j+2)^2 + L^2)}, & L \equiv 3 \pmod 4. \end{cases} \tag{2.3}$$

For $p \equiv 3 \pmod 4$ and any odd L divisible by p, extraction of the divisible quadratic factors in these two cases gives

$$T(L)T(L/p) = -p^3(L/p)^4 V(L), \qquad V(L) = \frac{1}{16}\left(\prod_{\varepsilon,\eta\in\{\pm1\}} \Gamma_p\left(\frac{3}{4} + \frac{\varepsilon+\eta i}{4}L\right)\right)^{-1}. \tag{2.4}$$

Here $i^2 = -1$ is in the unramified quadratic extension of $\mathbb{Q}_p$. Here is an explicit verification for arbitrary $L = pU$, rather than only for prime powers. Write $p = 4q+3$ and $z = iL/4$. If $L = 4t+1$, then $U = 4v+3$ and $t = pv+3q+2$. The divisible numerator indices in the first line of (2.3) are $j = (p+1)/2 + pw$, $0 \le w \le v$; the divisible denominator indices are $j = pw$, $1 \le w \le v$. Their quotient is $-p^2U^3/T(U)$. If $L = 4t+3$, then $U = 4v+1$ and $t = pv+q$. The divisible numerator indices are $j = pw$, $1 \le w \le v$; the divisible denominator indices, written as $(4j+2)^2 + L^2$, are $j = (p-1)/2 + pw$, $0 \le w < v$. Their quotient is $U/T(U)$. Retaining the unit quotient $V(L)$ and multiplying by the external factor L, respectively $-L^3$, proves the first identity in (2.4).

For the unit quotient, the corresponding Pochhammer pairs are $(1/2 \pm z)_t/(1 \pm z)_t$ in the first case, and $(1 \pm z)_t/(1/2 \pm z)_{t+1}$ in the second. Converting each raw quadratic to these normalized factors contributes 16^{-1}: in the first case the unit numerator count is one smaller, and in the second the total denominator count is one larger while the divisible counts agree. The exact unit-factor formula

$$\prod_{\substack{0\le j<m \\ a+j \text{ a unit}}} (a+j) = (-1)^m \frac{\Gamma_p(a+m)}{\Gamma_p(a)}$$

therefore expresses $V(L)$, in either case, as $1/16$ times

$$\frac{\Gamma_p((L+1)/4+z)\Gamma_p((L+1)/4-z)}{\Gamma_p((L+3)/4+z)\Gamma_p((L+3)/4-z)}.$$

The constant gamma pairs at $1/2 \pm z$ and at $1 \pm z$ both equal 1, because $p \equiv 3 \pmod 4$. Reflect the two numerator arguments to $(3-L)/4 \mp z$. The reflection signs multiply to 1, as the two arguments belong to the same residue disk. This proves the second identity in (2.4).

2.3 An integral four-orbit gamma estimate, including $p = 3$

We use the local analytic extension of Morita's gamma function to $a + p\mathcal{O}_{\mathbb{Q}_p(i)}$. It is available for every odd prime: Dwork, *A note on the p-adic gamma function*, Groupe d'étude d'analyse ultramétrique **9** (1981–1982), no. 3, exposé J5, pp. J1–J10, https://numdam.org/item/GAU_1981-1982___9_3_A6_0.pdf, explicitly assumes $p \neq 2$ and gives the analytic radius $p^{-1/p-1/(p-1)} > p^{-1}$, which contains each of these disks. The step and reflection formulas extend to these disks by the analytic identity theorem: on each disk they agree at the rational p-adic points, which have an accumulation point in that disk. For $p \ge 5$, the same extension and identities are also recorded in Long–Ramakrishna, arXiv:1403.5232v2, Proposition 13 and equations (2.2)–(2.3). We do not use their $p \ge 5$ assumption for the case 3.

Lemma 2.4. *For any odd prime p, $a \in \mathbb{Z}_p$, and $z \in p^s\mathcal{O}$, $s \ge 1$, where $\mathcal{O}$ is the ring of integers of a finite extension containing i, one has*

$$\frac{\prod_{\varepsilon,\eta\in\{\pm1\}} \Gamma_p(a + (\varepsilon+\eta i)z/4)}{\Gamma_p(a)^4} \equiv 1 \pmod{p^{4s}\mathcal{O}}.$$

Proof. Let $\mathcal{C} = \{(\varepsilon + \eta i)/4 : \varepsilon, \eta = \pm 1\}$. The step and reflection identities give $\Gamma_p(x)\Gamma_p(-x) = 1$ for $x \in p\mathcal{O}$. Because $\mathcal{C}$ consists of two opposite pairs, $\prod_{c\in\mathcal{C}} \Gamma_p(cz) = 1$. For every positive integer m, the step formula therefore gives

$$\frac{\prod_{c\in\mathcal{C}} \Gamma_p(m+cz)}{\Gamma_p(m)^4} = \prod_{\substack{1\le t<m \\ p\nmid t}} \prod_{c\in\mathcal{C}} (1 + cz/t) = \prod_{\substack{1\le t<m \\ p\nmid t}} \left(1 + \frac{z^4}{64t^4}\right) \equiv 1 \pmod{p^{4s}\mathcal{O}}.$$

Choose positive integers $m_j \to a$ p-adically. Eventually these centers lie in the same residue disk as a, and $m_j + cz \to a + cz$ inside that one analytic disk. Local analytic continuity consequently allows passage to the limit in the last congruence; the ideal $p^{4s}\mathcal{O}$ is closed. This proves the lemma, including $p = 3$. □

Taking $a = 3/4$ and $z = L$ in this lemma and then using (2.4) proves, whenever $v_p(L) = s \ge 1$,

$$V(L) \equiv \frac{1}{16}\Gamma_p(3/4)^{-4} \pmod{p^{4s}}.$$

If $p \nmid u$, every quadratic factor in $T(u)$ is a unit. Taking valuations in (2.4) and then iterating gives

$$v_p(T(p^s u)) = 2s + (s \bmod 2). \tag{2.5}$$

Dividing two consecutive instances of (2.4) also gives, for $s \ge 2$,

$$\frac{T(p^s u)}{p^4 T(p^{s-2}u)} = \frac{V(p^s u)}{V(p^{s-1}u)} \equiv 1 \pmod{p^{4s-4}}. \tag{2.6}$$

2.4 Transporting the interpolating sum

Apply (2.1) at $N = p^r$. A node H of valuation s contributes a term of valuation

$$v_p(W_N(H)T(H)) = 5(r-s) + 2s + (s \bmod 2) = 5r - 3s + (s \bmod 2).$$

The terms with $s = 0, 1$ therefore vanish modulo p^{5r-2}. Every remaining node has $H = p^2 J$ with $1 \le J \le p^{r-2}$ odd. Applying Lemma 2.2 twice gives

$$\frac{W_{p^r}(H)}{W_{p^{r-2}}(H/p^2)} \equiv 1 \pmod{p^{r+4s-5}}.$$

This precision is at least $4s - 4$, since the difference is $r - 1$. Combining it with (2.6) shows that

$$W_{p^r}(H)T(H) - p^4 W_{p^{r-2}}(H/p^2)T(H/p^2)$$

has valuation at least

$$(5r - 3s + (s \bmod 2)) + (4s - 4) = 5r + s + (s \bmod 2) - 4 \ge 5r - 2,$$

because $s \ge 2$. Summing over the surviving nodes proves the theorem. At $r = 1$, the same valuation formula gives at least 3 for every node, proving the initial vanishing.

Corollary 2.5. *For every prime $p \equiv 3 \pmod 4$ and every $r \ge 0$,*

$$v_p(A(p^r)) = 2r + (r \bmod 2).$$

Proof. The endpoint $H = p^r$ in (2.1) has precisely that valuation. Each other node has $s < r$ and valuation $5r - 3s + (s \bmod 2)$, which is strictly larger: decreasing s by one increases this expression by either 2 or 4. Therefore the endpoint cannot cancel with the other terms. □

Scope of the result. This proves the complete $p \equiv 3 \pmod 4$ branch of A.3, including $p = 3$ and every level. The $p \equiv 1 \pmod 4$ branch is proved in the following section by a different argument: the valuation of the interpolation weights used here relies on -1 being a nonsquare.

3 Positive interpolation moments and the ordinary branch of A.3

Retain the exact interpolation formula

$$A(N) = \sum_{H=1,3,\dots,N} W_N(H)T(H), \qquad b_k = (1/2)_k/k!, \quad h = (N-1)/2.$$

All identities below are exact over $\mathbb{Q}$ before prime-power specialization.

3.1 The positive moments

There are polynomials $P_j(X) \in \mathbb{Z}[1/2][X]$ for $j \geq 0$ such that

$$\sum_{H=1,3,\dots,N} W_N(H)T(H)H^{4j} = (-1)^h N^5 b_h^5 P_{j-1}(N) \qquad (j \geq 1). \tag{3.1}$$

Here $P_0 = 1$, and every P_j is invariant under $X \mapsto -1 - X$. In particular, for every odd prime p, $N = p^r$ and $j \geq 1$, the left side of (3.1) belongs to $p^{5r}\mathbb{Z}_p$.

To prove this, write $y = z^4$ and put

$$D_N(y) = \prod_{a=1}^{h}(1 - y/(2a)^4), \qquad Q_N(y) = \prod_{a=0}^{h}(y - (2a+1)^4), \qquad F_N(y) = D_N(y)S_N(z).$$

The established interpolation says $F_N(H^4) = D_N(H^4)T(H)$. The partial-fraction identity, or Lagrange interpolation of the remainder of $y^j F_N(y)$ modulo $Q_N(y)$, gives

$$\sum_{H} W_N(H)T(H)H^{4j} = -Q_N(0)[y^{-1}]\frac{y^{j-1}F_N(y)}{Q_N(y)}.$$

Cancellation of the first k odd factors in the summand defining S_N gives

$$\frac{F_N(y)}{Q_N(y)} = \frac{(-1)^h}{\prod_{a=1}^{h}(2a)^4}\sum_{k=0}^{h}(-1)^k(4k+1)b_k\frac{\prod_{a=k+1}^{h}(y-(2a)^4)}{\prod_{a=k}^{h}(y-(2a+1)^4)}.$$

Define the formal power series

$$\mathcal{R}_h(x) = \sum_{k=0}^{h}(-1)^k(4k+1)b_k\frac{\prod_{a=k+1}^{h}(1-(2a)^4x)}{\prod_{a=k}^{h}(1-(2a+1)^4x)}.$$

It follows that the moment is $N^4 b_h^4[x^{j-1}]\mathcal{R}_h(x)$. The exact recurrence is

$$(1-(2h+1)^4x)\mathcal{R}_h(x) = (1-(2h)^4x)\mathcal{R}_{h-1}(x) + (-1)^h(4h+1)b_h.$$

The initial series is $\mathcal{R}_0(x) = (1-x)^{-1}$. We now prove

$$\mathcal{R}_h(x) = (-1)^h(2h+1)b_h\sum_{j\geq 0}P_j(2h+1)x^j.$$

The coefficient recurrence required for this assertion is

$$XP_j(X) + (X-1)P_j(X-2) = X^5 P_{j-1}(X) + (X-1)^5 P_{j-1}(X-2) \qquad (j \geq 1). \tag{3.2}$$

For $j = 0$, $P_0 = 1$ gives the inhomogeneous constant coefficient $2X - 1$.

Here is the polynomial existence and integrality argument. Over $R = \mathbb{Z}[1/2]$, the polynomials invariant under $X \mapsto -1 - X$ are exactly $R[X(X+1)]$. The polynomials anti-invariant under $X \mapsto 1 - X$ are exactly $(2X-1)R[X(X-1)]$. The operator

$$\mathcal{L}P = XP(X) + (X-1)P(X-2)$$

maps the first space into the second. In the bases $[X(X+1)]^d$ and $(2X-1)[X(X-1)]^d$, it is triangular with diagonal entries one: both leading terms are $2X^{2d+1}$. It is therefore an isomorphism on each finite degree filtration, over R. If P_{j-1} is invariant, the right side of (3.2) is anti-invariant because the exponent 5 is odd. Thus (3.2) uniquely defines an invariant $P_j \in R[X]$. Moreover $\deg P_j \leq 4j$. At $X = 1$ the same recurrence gives $P_j(1) = P_{j-1}(1) = 1$, which supplies the initial series at $h = 0$. The recurrence for $\mathcal{R}_h$ now proves the asserted coefficient formula by induction on h, and hence (3.1).

3.2 Removing the unit nodes

For $N = p^r$, let $\mu_N(H) = W_N(H)T(H)$. For positive integers $e_v = (p-1)p^v$, one has

$$H^{4e_v} \longrightarrow \begin{cases} 1, & p \nmid H, \\ 0, & p \mid H \end{cases} \quad \text{in } \mathbb{Q}_p.$$

Since the sum has finitely many nodes, (3.1) and closedness of $p^{5r}\mathbb{Z}_p$ imply

$$\sum_{\substack{1 \leq H \leq N \\ H \text{ odd},\ p \nmid H}} \mu_N(H) \in p^{5r}\mathbb{Z}_p. \tag{3.3}$$

Individual unit nodes need not satisfy this bound; their exact moment cancellation is essential.

3.3 Ordinary nodes divisible by p

Let $p \equiv 1 \pmod 4$, so $p \geq 5$, and fix $i \in \mathbb{Z}_p$ with $i^2 = -1$. For every positive odd L divisible by p, the two exact Whipple product forms imply

$$T(L) = -p\Gamma_p(1/4)^4\, T(L/p)\, U(L), \qquad U(L) = \prod_{\varepsilon,\eta \in \{\pm 1\}} \frac{\Gamma_p(1/4 + (\varepsilon + \eta i)L/4)}{\Gamma_p(1/4)}. \tag{3.4}$$

For completeness, write $p = 4q+1$ and $L = pU$. When $L \equiv 1 \pmod 4$, write $U = 4v+1$; the product length is $(L-1)/4 = pv + q$. The divisible numerator indices are $j = (p+1)/2 + pw$, $0 \leq w < v$, and the divisible denominator indices are $j = pw$, $1 \leq w \leq v$. After removing the powers of p, their quotient is exactly $T(U)/U$, and the external factor L contributes p. When $L \equiv 3 \pmod 4$, write $U = 4v+3$; the product length is $(L-3)/4 = pv + 3q$. The divisible numerator indices are $j = pw$, $1 \leq w \leq v$, while the denominator indices are $j = (p-1)/2 + pw$, $0 \leq w \leq v$. Their normalized quotient is $T(U)/(-U^3)$, with one extra denominator power p^2; hence the external factor $-L^3$ again leaves $pT(U)$. In both cases the unit numerator and denominator counts agree, so no factor $1/16$ occurs. The unit quotient is

$$-\left\{\prod_{\varepsilon,\eta} \Gamma_p(3/4 + (\varepsilon + \eta i)L/4)\right\}^{-1}.$$

Indeed the pair at $1/2 \pm iL/4$ has reflection value -1, whereas the pair at $1 \pm iL/4$ has value 1. Reflection of the four arguments changes the displayed inverse product into the product at $1/4$ in (3.4); the four reflection signs multiply to one. These identities hold even when $T(L/p)$ is nonintegral.

The scaled gamma correction has a restricted power series

$$U(pX) = 1 + \sum_{j \geq 1} u_j X^{4j}, \qquad u_j \in p^4 \mathbb{Z}_p, \quad v_p(u_j) \longrightarrow +\infty. \tag{3.5}$$

To justify the entire series, not only its first coefficient, use Long–Ramakrishna's bounds

$$c_k(a) = \frac{\Gamma_p^{(k)}(a)}{k! \Gamma_p(a)}, \qquad c_k(a) \in \mathbb{Z}_p \ (k < p), \qquad v_p(c_k(a)) \geq -k \left(\frac{1}{p} + \frac{1}{p-1} \right).$$

The four shifts form an orbit under multiplication by i, so only degrees divisible by four survive. Degree four has valuation at least four because $4 < p$. A term of degree $4j \geq 8$ in the product of four scaled series has valuation at least

$$4j \left(1 - \frac{1}{p} - \frac{1}{p-1} \right) \geq \frac{11}{20} \, 4j > 4.$$

These lower bounds tend to infinity, proving (3.5), including $p = 5$.

Now let $N = p^r$, $M = p^{r-1}$ and $H = pJ$. The exact weight quotient is

$$R_N(J) = \frac{W_N(pJ)}{W_M(J)} = \frac{\prod_{\substack{2 \leq E < N \\ E \text{ even}, p \nmid E}} (1 - p^4 J^4 / E^4)}{\prod_{\substack{1 \leq K < N \\ K \text{ odd}, p \nmid K}} (1 - p^4 J^4 / K^4)}.$$

As a series in J^4, it has constant term one and its coefficient of degree $j \geq 1$ belongs to $p^{4j} \mathbb{Z}_p$. Thus the product

$$R_N(J) U(pJ) = 1 + \sum_{j \geq 1} e_{N,j} J^{4j}$$

has $e_{N,j} \in p^4 \mathbb{Z}_p$ and $v_p(e_{N,j}) \to +\infty$. Equations (3.4) and the weight quotient give

$$\mu_N(pJ) = -p \Gamma_p(1/4)^4 \mu_M(J) \left(1 + \sum_{j \geq 1} e_{N,j} J^{4j} \right).$$

Each positive moment at scale M is in $p^{5(r-1)} \mathbb{Z}_p$ by (3.1). Summing over the finitely many coarse nodes, then the convergent correction series, proves

$$\sum_{p \mid H} \mu_N(H) \equiv -p \Gamma_p(1/4)^4 A(M) \pmod{p^{5r}}.$$

For $r = 1$ this uses the same identity at $M = 1$, where every positive moment equals one; the coefficient bound still supplies p^5. Finally add (3.3). We have proved

$$\boxed{A(p^r) \equiv -p \Gamma_p(1/4)^4 A(p^{r-1}) \pmod{p^{5r}} \qquad (p \equiv 1 \pmod 4, \ r \geq 1).} \tag{3.6}$$

This is the full ordinary branch of Swisher's A.3, including $p = 5$.

4 Exact interpolation and the ordinary D model

For $N \equiv 1 \pmod 3$, set $n = (N-1)/3$ and

$$b_k = \frac{(1/3)_k}{k!}, \qquad D(N) = \sum_{k=0}^{n} (6k+1) b_k^6.$$

We prove below that for every prime $p \equiv 1 \pmod 3$ and every $r \geq 1$,

$$D(p^r) \equiv -p\Gamma_p(1/3)^9 D(p^{r-1}) \pmod{p^{6r}}.$$

The interpolation formulas first hold over the rational numbers; the subsequent gamma identities are interpreted p-adically.

4.1 Exact fifth-degree interpolation

Define

$$S_N(z) = \sum_{k=0}^{n} (6k+1) b_k^6 \prod_{j=0}^{k-1} \frac{1 - z^5/(3j+1)^5}{1 + z^5/(3j+3)^5}, \qquad T(H) = S_H(H) \quad (H \equiv 1 \pmod 3).$$

In the variable $y = z^5$, multiplication by $\prod_{j=1}^{n}(1 + y/(3j)^5)$ makes S_N a polynomial of degree at most n. At the $n+1$ distinct nodes $y = H^5$, $H = 1, 4, \ldots, N$, the rational function satisfies $S_N(H) = T(H)$; the cleared polynomial has value $\prod_{j=1}^{n}(1 + H^5/(3j)^5) T(H)$. Consequently, Lagrange interpolation at zero gives

$$D(N) = \sum_{\substack{1 \leq H \leq N \\ H \equiv 1 \ (3)}} W_N(H) T(H), \qquad W_N(H) = \frac{\prod_{\substack{1 \leq E < N \\ E \equiv 0 \ (3)}} (1 + H^5/E^5)}{\prod_{\substack{1 \leq K \leq N \\ K \equiv 1 \ (3),\ K \neq H}} (1 - H^5/K^5)}. \tag{4.1}$$

These are exact rational identities with no vanishing denominator.

Lemma 4.1. *Suppose $p \equiv 1 \pmod 3$, $N = p^r$, and $H \equiv 1 \pmod 3$ satisfies $1 \leq H \leq N$, $s = v_p(H) \geq 1$. Then*

$$\frac{W_N(H)}{W_{N/p}(H/p)} \equiv 1 \pmod{p^{r+5s}}.$$

Proof. Division by p preserves both index classes modulo 3, so extraction of the p-divisible indices in (4.1) leaves $W_{N/p}(H/p)$. Pair the remaining numerator index E with the denominator index $K = N - E$. This bijects the unit indices, including when $H = N$, since both omitted nodes H, N are divisible by p. The paired quotient has defect

$$\frac{1 + H^5/E^5}{1 - H^5/K^5} - 1 = \frac{H^5(E^5 + K^5)}{E^5(K^5 - H^5)}.$$

Its denominator is a unit, and $E^5 + (N-E)^5$ is divisible by N. This gives valuation at least $r + 5s$ term by term. □

4.2 The Dougall model at arbitrary nodes

Let ζ be a primitive fifth root of unity in an unramified extension of $\mathbb{Q}_p$, and put $\sigma = \zeta + \zeta^2 + \zeta^3$. For every positive $H \equiv 1 \pmod 3$, with $m = (H-1)/3$, Dougall's terminating identity gives

$$T(H) = H\frac{(1/3)_m \prod_{1\le i<j\le 3}(2/3 + H(\zeta^i+\zeta^j)/3)_m}{\prod_{i=1}^{3}(1+H\zeta^i/3)_m(1/3+H\sigma/3)_m}. \tag{4.2}$$

This is Dougall's classical terminating ${}_7F_6$ evaluation, applied to an arbitrary admissible integer before any prime-power specialization; it is the transformation used for D in Swisher, *On the supercongruence conjectures of van Hamme*, Research in the Mathematical Sciences **2** (2015), article 18, https://doi.org/10.1186/s40687-015-0037-6. For $p \mid H$, write $p = 3q+1$ and $m = pm' + q$, $m' = (H/p-1)/3$. In each Pochhammer symbol $(\alpha + H\beta)_m$, $\alpha \in \{1/3, 2/3, 1\}$, the divisible factors have first indices $q, 2q, p-1$, respectively, and exactly m' terms. After division by p, they are precisely $(\alpha + (H/p)\beta)_{m'}$. The exact unit-factor identity

$$\prod_{\substack{0\le j<m\\ \alpha+j \text{ a unit}}} (\alpha+j) = (-1)^m \frac{\Gamma_p(\alpha+m)}{\Gamma_p(\alpha)}$$

and gamma reflection then give

$$T(H) = C_p T(H/p)\mathcal{U}_p(H), \qquad C_p = -p\Gamma_p(1/3)^9, \tag{4.3}$$

where

$$\mathcal{U}_p(X) = \prod_{a=0}^{4}\Gamma_p(X\zeta^a/3)\prod_{0\le a<b\le 4}\frac{\Gamma_p(1/3 - X(\zeta^a+\zeta^b)/3)}{\Gamma_p(1/3)}.$$

For clarity, the reflection signs and orbit symmetrization are as follows. Since $p = 3q+1$ is odd, q is even and the least positive residue of $1/3$ is $2q+1$. Reflection at $1/3$ has sign -1; the six reciprocal gamma factors centered at $2/3$ give net sign $+1$. The three factors centered at 1 give net sign -1 by $\Gamma_p(1+x) = -\Gamma_p(x)$ near zero. The relation $1+\sigma = -\zeta^4$ and $\Gamma_p(x)\Gamma_p(-x) = 1$ turn the five zero-centered shifts into the complete fifth-root orbit. The other ten shifts are exactly the negative pair sums $-(\zeta^a+\zeta^b)/3$. This proves the displayed product and its constant normalization $\mathcal{U}_p(0) = 1$. The local analytic extension of Γ_p, including the step and reflection formulas at these algebraic arguments, is recorded in Long–Ramakrishna, *Some supercongruences occurring in truncated hypergeometric series*, Adv. Math. **290** (2016), 773–808, Proposition 13 and equations (2.2)–(2.3), https://arxiv.org/abs/1403.5232.

For $s = v_p(H) \ge 1$, this is a unit and

$$\mathcal{U}_p(H) \equiv 1 \pmod{p^{5s}}. \tag{4.4}$$

For completeness, multiplication of the argument by ζ permutes the displayed gamma factors, so the power series of $\mathcal{U}_p(X) - 1$ has only positive degrees divisible by 5. Its degree-five coefficient is integral, because $5 < p$. Long–Ramakrishna's Proposition 13 gives a lower bound $-j(1/p + 1/(p-1))$ for the valuation of the coefficient of degree j in each normalized gamma series, and hence in their finite product. For $j \ge 10$, $p \ge 7$, and $s \ge 1$, this gives $js - j(1/p + 1/(p-1)) \ge 5s$. This proves (4.4), including $p = 7$.

5 Every positive fifth moment of the D interpolating measure

Keep the exact weights $W_N(H)$ and terminating model $T(H)$ from (4.1). No prime hypothesis is needed in the following identity. Let $N = 3n+1$, $n \ge 0$, and $b_n = (1/3)_n/n!$.

Theorem 5.1. *There exist polynomials*

$$P_j(N) \in \mathbb{Z}[1/2][N(N+2)], \qquad P_0 = 1, \quad \deg P_j = 4j, \quad P_j(1) = 1,$$

such that, for every $j \geq 1$,

$$\boxed{\sum_{\substack{1 \leq H \leq N \\ H \equiv 1 \ (3)}} W_N(H) T(H) H^{5j} = (N b_n)^6 P_{j-1}(N).}$$

In particular, if $p \equiv 1 \pmod 3$ and $N = p^r$, every positive fifth moment belongs to $p^{6r}\mathbb{Z}_p$, including when $p \equiv 1 \pmod 5$.

Proof. Put

$$\mathcal{D}_N(y) = \prod_{j=1}^{n} (1 + y/(3j)^5), \quad \mathcal{P}_N(y) = \prod_{h=0}^{n} (1 - y/(3h+1)^5), \quad F_N(y) = \frac{\mathcal{D}_N(y) S_N(y^{1/5})}{\mathcal{P}_N(y)}.$$

The notation $S_N(y^{1/5})$ means the rational function in y; no root is chosen. Partial fractions, or the interpolation identity, give

$$F_N(y) = \sum_{\substack{1 \leq H \leq N \\ H \equiv 1 \ (3)}} \frac{W_N(H) T(H)}{1 - y/H^5}.$$

For $n \geq 1$, separating the last summand of S_N gives the exact rational-function recurrence

$$(1 - y/N^5) F_N(y) = (1 + y/(N-1)^5) F_{N-3}(y) + (2N-1) b_n^6. \tag{5.1}$$

Write the Laurent expansion at infinity as $F_N(y) = -\sum_{j \geq 1} M_j(N) y^{-j}$, where $M_j(N) = \sum_H W_N(H) T(H) H^{5j}$. The constant term of (5.1) gives

$$M_1(N) = N^5 \left((2N-1) b_n^6 - \frac{M_1(N-3)}{(N-1)^5} \right).$$

Since $b_n / b_{n-1} = (N-3)/(N-1)$ and $M_1(1) = 1$, induction proves $M_1(N) = (N b_n)^6$. For $j \geq 1$, comparison of the coefficient of y^{-j} gives

$$M_{j+1}(N) = N^5 (M_j(N) - M_j(N-3)) - \frac{N^5}{(N-1)^5} M_{j+1}(N-3).$$

After substituting the claimed normalization, its polynomial form is

$$N P_j(N) + (N-1) P_j(N-3) = N^6 P_{j-1}(N) - (N-1)^6 P_{j-1}(N-3). \tag{5.2}$$

We now prove that this determines polynomials over $\mathbb{Z}[1/2]$, without dividing by any odd integer.

Let $R = \mathbb{Z}[1/2]$, $X = N(N+2)$, $Y = N(N-1)$, and define $\mathcal{L}(P) = NP(N) + (N-1)P(N-3)$. The invariant polynomials for the involution $N \mapsto -N-2$ are exactly $R[X]$. The anti-invariant polynomials for $N \mapsto 1-N$ are exactly $(2N-1)R[Y]$. Both statements follow by shifting the variable to the center of the corresponding reflection, which is allowed because 2 is invertible in R. For $P \in R[X]$, substitution shows that $\mathcal{L}(P)$ is anti-invariant. Moreover

$$\frac{\mathcal{L}(X^k)}{2N-1}$$

is a polynomial in Y of degree k with leading coefficient 1: the numerator has degree $2k+1$ and leading coefficient 2. Thus $\mathcal{L} : R[X] \to (2N-1)R[Y]$ is an isomorphism of filtered R-modules, with a triangular matrix whose diagonal consists of ones. If $P_{j-1} \in R[X]$, the right side of (5.2) is anti-invariant under $N \mapsto 1-N$. Therefore it lies in $(2N-1)R[Y]$ and has a unique preimage $P_j \in R[X]$. Its degree is $\deg P_{j-1} + 4$; the leading coefficient of its right side before applying $\mathcal{L}^{-1}$ is $(6+3\deg P_{j-1})$ times the nonzero leading coefficient of P_{j-1}, so this degree does not drop. Evaluating at $N=1$ gives $P_j(1) = P_{j-1}(1) = 1$. The moment recurrence with this initial value now proves the exact identity by induction on n and j. Finally, $b_{(p^r-1)/3}$ is a p-adic unit for $p \equiv 1 \pmod 3$: at each power $p^a \le p^r$, the numerator factors $3j+1$, $0 \le j < (p^r-1)/3$, and the denominator factors $3j$, $1 \le j \le (p^r-1)/3$, have the same number of multiples of p^a. Since P_j has no odd prime in its coefficients' denominators, the asserted valuation follows. □

6 The complete ordinary D.3 recursion

Theorem 6.1. *For every prime $p \equiv 1 \pmod 3$ and every $r \ge 1$,*

$$\boxed{D(p^r) \equiv -p\Gamma_p(1/3)^9 D(p^{r-1}) \pmod{p^{6r}}.}$$

Moreover $v_p(D(p^r)) = r$ for all $r \ge 0$.

Proof. Write $N = p^r$, $M = p^{r-1}$ and $C_p = -p\Gamma_p(1/3)^9$. We use the exact interpolation identity (4.1), the arbitrary-node model identity (4.3), and the positive moments of Theorem 5.1.

First remove all unit nodes at once. Put $j_a = (p-1)p^a$. As $a \to \infty$, $H^{5j_a} \to 1$ if $p \nmid H$, and $H^{5j_a} \to 0$ if $p \mid H$. Since the interpolating sum has finitely many rational terms, its limit is

$$\sum_{\substack{1 \le H \le N \\ H \equiv 1 \ (3),\ p \nmid H}} W_N(H)T(H) = \lim_{a\to\infty} \sum_{H} W_N(H)T(H)H^{5j_a} \in p^{6r}\mathbb{Z}_p.$$

The final inclusion follows because every positive moment belongs to the displayed closed ideal. It remains to transport the divisible nodes, which we write $H = pJ$.

Define the unit-index rational function

$$B_N(X) = \frac{\prod_{\substack{1 \le E < N \\ E \equiv 0 \ (3),\ p \nmid E}} (1 + X/E^5)}{\prod_{\substack{1 \le K < N \\ K \equiv 1 \ (3),\ p \nmid K}} (1 - X/K^5)}.$$

Direct extraction of p-divisible factors in the exact weights gives

$$W_N(pJ) = W_M(J)B_N(p^5J^5).$$

Pair E with $K = N - E$. Each paired factor is

$$\frac{1 + X/E^5}{1 - X/K^5} = 1 + X\frac{E^5 + K^5}{E^5K^5}\frac{1}{1 - X/K^5}.$$

Since $E^5 + K^5$ is divisible by N, this proves the coefficientwise statement

$$B_N(X) \in 1 + p^r X\mathbb{Z}_p[[X]]. \tag{6.1}$$

In particular $B_N(p^5J^5)$ is a convergent power series in J^5, whose nonconstant coefficients have valuation at least $r+5$ and tend to infinity with their index.

The exact gamma factor $\mathcal{U}_p$ in (4.3) has a series

$$\mathcal{U}_p(X) = 1 + \sum_{j\ge 1} u_j X^{5j}.$$

For $p \ge 7$, its coefficient u_1 is integral, and the general coefficient bound is $v_p(u_j) \ge -5j(1/p+1/(p-1))$. Thus

$$\mathcal{U}_p(pJ) = 1 + \sum_{j\ge 1} p^{5j} u_j J^{5j},$$

where all nonconstant coefficients belong to $p^5\mathcal{O}$ and tend to zero p-adically. At $j = 1$ use the integrality of u_1; for $j \ge 2$ use $5j(1 - 1/p - 1/(p-1)) \ge 10(1 - 13/42) > 5$. Here $\mathcal{O}$ is the ring of integers of the unramified extension containing the fifth roots of unity. Combining this fact with (6.1) gives a convergent series

$$B_N(p^5J^5)\mathcal{U}_p(pJ) - 1 = \sum_{j\ge 1} c_j J^{5j}, \qquad c_j \in p^5\mathcal{O}, \quad c_j \longrightarrow 0. \tag{6.2}$$

This convergence is uniform for $J \in \mathbb{Z}_p$.

The difference of the divisible-node contributions from the two levels is therefore exactly

$$\begin{aligned}
&\sum_{\substack{1\le H\le N\\ H\equiv 1\ (3),\ p\mid H}} W_N(H)T(H) - C_pD(M)\\
&\quad = C_p \sum_{\substack{1\le J\le M\\ J\equiv 1\ (3)}} W_M(J)T(J) \sum_{j\ge 1} c_j J^{5j}\\
&\quad = C_p \sum_{j\ge 1} c_j \sum_{\substack{1\le J\le M\\ J\equiv 1\ (3)}} W_M(J)T(J)J^{5j}.
\end{aligned}$$

Interchanging the finite sum with the convergent series is valid even if individual interpolation weights have negative valuations. Every inner positive moment belongs to $p^{6(r-1)}\mathbb{Z}_p$. Since $v_p(C_p) = 1$ and $c_j \in p^5\mathcal{O}$, the last expression belongs to $p^{1+5+6(r-1)}\mathcal{O} = p^{6r}\mathcal{O}$. It is rational over $\mathbb{Q}_p$, so it belongs to $p^{6r}\mathbb{Z}_p$. Together with the unit-node limit this proves the asserted congruence. Starting with $D(1) = 1$, its valuation statement follows inductively because $6r > r$. □

7 The entire ordinary branch of G.3 by positive cubic moments

For $N = 4n + 1$ put

$$b_k = \frac{(1/4)_k}{k!}, \qquad G(N) = \sum_{k=0}^{n} (8k+1) b_k^4.$$

We prove, for every prime $p \equiv 1 \pmod 4$ and every $r \ge 1$,

$$\boxed{G(p^r) \equiv p\lambda_p G(p^{r-1}) \pmod{p^{4r}}, \qquad \lambda_p = -(-1)^{(p-1)/4}\Gamma_p(1/2)\Gamma_p(1/4)^2.} \tag{7.1}$$

This includes primes $p \equiv 1 \pmod 3$; no estimate of individual interpolation weights is required.

7.1 Exact interpolation and its terminating model

Define the rational function in $y = z^3$

$$S_N(z) = \sum_{k=0}^{n} (8k+1) b_k^4 \prod_{j=0}^{k-1} \frac{1 - z^3/(4j+1)^3}{1 + z^3/(4j+4)^3}.$$

Multiplication by $\mathcal{D}_N(y) = \prod_{j=1}^{n} (1 + y/(4j)^3)$ gives a polynomial of degree at most n. At the distinct nodes $y = H^3$, $H = 1, 5, \ldots, N$, the sum terminates at $(H-1)/4$. Writing $T(H) = S_H(H)$, interpolation at zero gives

$$G(N) = \sum_{H=1,5,\ldots,N} W_N(H) T(H), \qquad W_N(H) = \frac{\prod_{\substack{1 \le E < N \\ E \equiv 0 \ (4)}} (1 + H^3/E^3)}{\prod_{\substack{1 \le K \le N \\ K \equiv 1 \ (4),\ K \ne H}} (1 - H^3/K^3)}. \tag{7.2}$$

All these are exact finite identities. Their rational denominators are nonzero, regardless of their p-adic valuations.

Let ζ be a primitive cube root of unity. For $H = 4m+1$, Dougall's terminating very-well-poised ${}_5F_4$ identity gives

$$T(H) = (-1)^m H \frac{(1/4)_m (1/2)_m}{(1 + \zeta H/4)_m (1 + \zeta^2 H/4)_m}. \tag{7.3}$$

Indeed its parameters are $a = 1/4$, $b = (1 - \zeta H)/4$, $c = (1 - \zeta^2 H)/4$, and $-m = (1-H)/4$. The terminating evaluation is

$$\frac{(1+a)_m (1 + a - b - c)_m}{(1 + a - b)_m (1 + a - c)_m} = \frac{(5/4)_m (1/2 - m)_m}{(1 + \zeta H/4)_m (1 + \zeta^2 H/4)_m},$$

which proves (7.3). The denominator product is rational and nonzero.

7.2 Every positive cubic moment

There are polynomials $P_j(X) \in \mathbb{Z}[1/2][X(X+3)]$, $P_0 = 1$, $P_j(1) = 1$, with degree at most $2j$, such that

$$M_j(N) := \sum_{H=1,5,\ldots,N} W_N(H) T(H) H^{3j} = (N b_n)^4 P_{j-1}(N) \qquad (j \ge 1). \tag{7.4}$$

To prove this, put

$$\mathcal{P}_N(y) = \prod_{h=0}^{n} (1 - y/(4h+1)^3), \qquad F_N(y) = \frac{\mathcal{D}_N(y) S_N(y^{1/3})}{\mathcal{P}_N(y)}.$$

Here $S_N(y^{1/3})$ denotes the rational function in y. The partial-fraction decomposition of the interpolation formula is

$$F_N(y) = \sum_{H=1,5,\ldots,N} \frac{W_N(H) T(H)}{1 - y/H^3}.$$

Separating the last summand of S_N yields, for $n \ge 1$,

$$(1 - y/N^3) F_N(y) = (1 + y/(N-1)^3) F_{N-4}(y) + (2N-1) b_n^4. \tag{7.5}$$

Expand at infinity: $F_N(y) = -\sum_{j \ge 1} M_j(N) y^{-j}$. The constant term of (7.5) gives

$$M_1(N) = N^3 \left((2N-1) b_n^4 - \frac{M_1(N-4)}{(N-1)^3} \right).$$

Since $b_n/b_{n-1} = (N-4)/(N-1)$ and $M_1(1) = 1$, this proves $M_1(N) = (Nb_n)^4$. The remaining coefficients give

$$M_{j+1}(N) = N^3\big(M_j(N) - M_j(N-4)\big) - \frac{N^3}{(N-1)^3}M_{j+1}(N-4).$$

The claimed normalization therefore reduces to

$$XP_j(X) + (X-1)P_j(X-4) = X^4P_{j-1}(X) - (X-1)^4P_{j-1}(X-4). \tag{7.6}$$

Here polynomial existence requires no division by an odd prime. Set $R = \mathbb{Z}[1/2]$ and $\mathcal{L}P = XP(X) + (X-1)P(X-4)$. The invariant polynomials under $X \mapsto -X-3$ are exactly $R[X(X+3)]$. The anti-invariant polynomials under $X \mapsto 1-X$ are exactly $(2X-1)R[X(X-1)]$. Substitution shows that $\mathcal{L}$ maps the former module into the latter. In the corresponding bases $[X(X+3)]^d$ and $(2X-1)[X(X-1)]^d$ its matrix is triangular with diagonal entries one, since both leading terms are $2X^{2d+1}$. It is consequently an isomorphism on every finite degree filtration. If P_{j-1} is invariant, the right side of (7.6) is anti-invariant and has degree at most $\deg P_{j-1} + 3$. Thus the equation uniquely defines an invariant $P_j \in R[X]$ of degree at most $2j$. At $X = 1$ it gives $P_j(1) = P_{j-1}(1) = 1$. The moment recurrence now proves (7.4) by induction.

For $N = p^r$ with $p \equiv 1 \pmod 4$, b_n is a p-adic unit: at each power $p^a \le N$, its numerator and denominator have the same number $(p^{r-a}-1)/4$ of divisible factors. Hence

$$M_j(p^r) \in p^{4r}\mathbb{Z}_p \qquad (j \ge 1). \tag{7.7}$$

This also holds at $r = 0$, where every moment equals one.

7.3 Discarding the unit nodes by their moments

Write $\mu_N(H) = W_N(H)T(H)$. For $e_v = (p-1)p^v$,

$$H^{3e_v} \longrightarrow \begin{cases} 1, & p \nmid H, \\ 0, & p \mid H \end{cases} \quad \text{in } \mathbb{Q}_p.$$

There are finitely many nodes, so (7.7) implies

$$\sum_{\substack{H=1,5,\dots,N \\ p\nmid H}} \mu_N(H) \in p^{4r}\mathbb{Z}_p \qquad (N = p^r). \tag{7.8}$$

This step uses the sum of unit nodes; their individual valuations need not satisfy the displayed bound.

7.4 Exact scaling of the model

Now let $p = 4q+1$, $p \mid H$, $J = H/p$, and $m = (H-1)/4 = pm' + q$, $m' = (J-1)/4$. In $(1/4)_m$, $(1/2)_m$, and $(1+\zeta^i H/4)_m$, the indices of the first p-divisible factors are respectively q, $2q$, and $p-1$. Each list has exactly m' divisible factors; after division by p they give the identical factors for the model at J. Using the functional equation for the remaining unit factors gives

$$\frac{T(H)}{pT(J)} = (-1)^q \frac{\Gamma_p(H/4)\Gamma_p(1/4+H/4)\Gamma_p(1+\zeta H/4)\Gamma_p(1+\zeta^2 H/4)}{\Gamma_p(1/4)\Gamma_p(1/2)\Gamma_p(3/4-\zeta H/4)\Gamma_p(3/4-\zeta^2 H/4)}.$$

The two functional equations $\Gamma_p(1+u) = -\Gamma_p(u)$ apply because $u \in p\mathcal{O}_K$. Reflection at $1/4+\zeta^i H/4$ cancels the two equal signs. Finally $\Gamma_p(1/2)^2 = -1$ for $p \equiv 1 \pmod 4$. Therefore the exact model scaling is

$$T(H) = p\lambda_p T(J)U(H), \qquad U(X) = \prod_{i=0}^{2}\Gamma_p(\zeta^i X/4)\prod_{i=0}^{2}\frac{\Gamma_p(1/4+\zeta^i X/4)}{\Gamma_p(1/4)}. \tag{7.9}$$

These gamma values lie in the unramified extension containing ζ, and the identities are valid on the disks $p\mathcal{O}_K$. No integrality assertion for an individual $T(J)$ is needed.

The scaled correction has a restricted series

$$U(pX) = 1 + \sum_{j\geq 1} u_j X^{3j}, \qquad u_j \in p^3\mathbb{Z}_p, \quad v_p(u_j) \longrightarrow +\infty. \tag{7.10}$$

Indeed multiplication of X by ζ permutes each triple, so only degrees divisible by three occur. Galois invariance puts the coefficients in $\mathbb{Q}_p$. For the normalized Taylor coefficients of each gamma factor, Long–Ramakrishna's bounds give integrality in degrees $k < p$ and valuation at least $-k(1/p+1/(p-1))$ in all degrees. The degree-three coefficient of the scaled product is thus in $p^3\mathbb{Z}_p$, as $p \geq 5$. For total degree $k \geq 6$ its valuation is at least

$$k\left(1 - \frac{1}{p} - \frac{1}{p-1}\right) \geq \frac{11k}{20} > 3.$$

These bounds also prove convergence on $X \in \mathbb{Z}_p$ and the coefficient limit in (7.10).

7.5 Transport using all positive moments

Put $N = p^r$, $M = N/p$. Stripping the p-divisible indices from (7.2), using $p \equiv 1 \pmod 4$, gives exactly

$$R_N(J) := \frac{W_N(pJ)}{W_M(J)} = \frac{\prod_{\substack{1\leq E<N \\ E\equiv 0\ (4),\ p\nmid E}} (1 + p^3J^3/E^3)}{\prod_{\substack{1\leq K<N \\ K\equiv 1\ (4),\ p\nmid K}} (1 - p^3J^3/K^3)}.$$

Its constant coefficient is one and its coefficient of J^{3j} belongs to $p^{3j}\mathbb{Z}_p$. Thus

$$R_N(J)U(pJ) = 1 + \sum_{j\geq 1} e_{N,j}J^{3j}, \qquad e_{N,j} \in p^3\mathbb{Z}_p, \quad v_p(e_{N,j}) \to +\infty.$$

Equations (7.9) and the weight quotient show

$$\sum_{p|H} \mu_N(H) = p\lambda_p \left(G(M) + \sum_{j\geq 1} e_{N,j} M_j(M) \right).$$

The interchange is legitimate: there are finitely many coarse nodes, and the restricted series converges at each of them. Every correction term has valuation at least $1 + 3 + 4(r-1) = 4r$ by (7.7). The ideal $p^{4r}\mathbb{Z}_p$ is closed, so the entire correction belongs to it. Adding (7.8) proves (7.1), including $r = 1$.

8 The even supersingular branch of G.3 through an alternate family

Let $p > 3$ be prime with $p \equiv 3 \pmod 4$. For $a \in \{1, 3\}$ and $N \equiv a \pmod 4$, put $n = (N-a)/4$ and define

$$b_{a,k} = \frac{(a/4)_k}{k!}, \qquad G_a(N) = \sum_{k=0}^{n} (8k + a) b_{a,k}^4.$$

In particular $G_1(p^{2s})$ is the original even-level G.3 sum. We prove

$$\boxed{G_1(p^{2s}) \equiv -p^3 G_1(p^{2s-2}) \pmod{p^{8s-2}} \qquad (s \geq 1).} \tag{8.1}$$

The alternate family uses the weight $8k + 3$, rather than $(8k + 3)/3$; thus $G_3(3) = 3$.

8.1 Interpolation and moments for both node classes

Define

$$S_{a,N}(z)=\sum_{k=0}^{n}(8k+a)b_{a,k}^{4}\prod_{j=0}^{k-1}\frac{1-z^{3}/(4j+a)^{3}}{1+z^{3}/(4j+4)^{3}},\qquad T_{a}(H)=S_{a,H}(H).$$

Multiplication by $\prod_{j=1}^{n}(1+y/(4j)^{3})$ makes $S_{a,N}(y^{1/3})$ a polynomial of degree at most n. At the nodes $H=a,a+4,\dots,N$, the undeclared root is unnecessary: the rational function in y takes the value $T_{a}(H)$ at $y=H^{3}$. Lagrange interpolation therefore gives the exact identity

$$G_{a}(N)=\sum_{\substack{1\le H\le N\\ H\equiv a\ (4)}}W_{a,N}(H)T_{a}(H),\tag{8.2}$$

where

$$W_{a,N}(H)=\frac{\prod_{j=1}^{n}(1+H^{3}/(4j)^{3})}{\prod_{\substack{0\le j\le n\\ 4j+a\neq H}}(1-H^{3}/(4j+a)^{3})}.$$

All denominators are nonzero rational numbers.

For every $j\ge 1$, the positive moments have the exact form

$$\sum_{H}W_{a,N}(H)T_{a}(H)H^{3j}=(Nb_{a,n})^{4}P_{j-1}^{(a)}(N),\tag{8.3}$$

where

$$P_{j}^{(a)}\in\mathbb{Z}[1/2][N(N+4-a)],\quad P_{0}^{(a)}=1,\quad \deg P_{j}^{(a)}=2j,\quad P_{j}^{(a)}(a)=a^{3j}.$$

Here is a proof including the alternate normalization. Let $F_{a,N}(y)$ be the proper rational function obtained by multiplying $S_{a,N}(y^{1/3})$ by

$$\frac{\prod_{j=1}^{n}(1+y/(4j)^{3})}{\prod_{j=0}^{n}(1-y/(4j+a)^{3})}.$$

Its partial fractions are $F_{a,N}(y)=\sum_{H}W_{a,N}(H)T_{a}(H)/(1-y/H^{3})$. For $n\ge 1$, separating the last summand gives

$$(1-y/N^{3})F_{a,N}(y)=(1+y/(N-a)^{3})F_{a,N-4}(y)+(2N-a)b_{a,n}^{4}.$$

Write $F_{a,N}(y)=-\sum_{j\ge1}M_{j}(N)y^{-j}$ at infinity. Since $b_{a,n}/b_{a,n-1}=(N-4)/(N-a)$, comparison of constant terms proves $M_{1}(N)=(Nb_{a,n})^{4}$, starting from $F_{a,a}(y)=a/(1-y/a^{3})$. The higher coefficients give the recurrence

$$NP_{j}^{(a)}(N)+(N-a)P_{j}^{(a)}(N-4)=N^{4}P_{j-1}^{(a)}(N)-(N-a)^{4}P_{j-1}^{(a)}(N-4).\tag{8.4}$$

Over $R=\mathbb{Z}[1/2]$, the operator $\mathcal{L}_{a}P=NP(N)+(N-a)P(N-4)$ maps

$$R[N(N+4-a)]\longrightarrow(2N-a)R[N(N-a)].$$

The domain is invariant under $N\mapsto a-4-N$, and the codomain is anti-invariant under $N\mapsto a-N$. In the displayed power bases this map is triangular with diagonal one, because both leading terms are $2N^{2d+1}$. It is therefore an isomorphism over R on every finite degree filtration. The right side of (8.4) is anti-invariant, so it defines a unique $P_{j}^{(a)}$ over R. If the preceding degree is d with leading coefficient c, the right side has leading term $4(a+d)cN^{d+3}$, so its degree increases by two after

applying $\mathcal{L}_a^{-1}$. Evaluating at $N = a$ gives $P_j^{(a)}(a) = a^3 P_{j-1}^{(a)}(a)$. Thus the initial values and the rational-function recurrence prove (8.3).

If $N = p^r \equiv a \pmod 4$, the factor $b_{a,n}$ is a p-adic unit. Indeed, for every $p^\ell \mid N$, the first numerator index k_ℓ satisfying $p^\ell \mid 4k_\ell + a$ is precisely the residue of n modulo p^ℓ. Since the numerator product stops at $n-1$, its number of divisible factors equals $\lfloor n/p^\ell \rfloor$, the same as the denominator's count. Consequently every positive moment in (8.3) belongs to $p^{4r}\mathbb{Z}_p$. For $e_v = (p-1)p^v$, the limit of H^{3e_v} is one at unit nodes and zero at nodes divisible by p. Applying this to the finite moment sum shows

$$\sum_{p\nmid H} W_{a,p^r}(H) T_a(H) \in p^{4r}\mathbb{Z}_p. \tag{8.5}$$

This argument does not require individual nodes to be integral.

8.2 The two terminating models and their exact scaling

Let ζ be a primitive cube root in an unramified extension of $\mathbb{Q}_p$. Dougall's terminating ${}_5F_4$ identity, with central parameter $a/4$ and other parameters $(a-H)/4, (a-\zeta H)/4, (a-\zeta^2 H)/4$, gives

$$T_a(H) = (-1)^m H \frac{(a/4)_m (a/2)_m}{(1+\zeta H/4)_m (1+\zeta^2 H/4)_m}, \qquad m = (H-a)/4. \tag{8.6}$$

The factor a converting the very-well-poised weight to $8k+a$ is included here. Indeed, $a(1+a/4)_m = H(a/4)_m$, and the other numerator is $(1-a/2-m)_m = (-1)^m (a/2)_m$.

Put $q = (p-3)/4$, and write Γ for Morita's Γ_p. Set

$$C_1 = \frac{(-1)^q p^2 \Gamma(1/4)^2}{2\Gamma(1/2)}, \qquad C_3 = -\frac{2(-1)^q p \Gamma(3/4)^2}{\Gamma(1/2)},$$

$$U_a(X) = \prod_{j=0}^{2} \Gamma(X\zeta^j/4) \prod_{j=0}^{2} \frac{\Gamma(a/4 + X\zeta^j/4)}{\Gamma(a/4)}.$$

For every admissible H divisible by p, exact factor extraction gives

$$T_1(H) = C_1 T_3(H/p) U_1(H), \qquad T_3(H) = C_3 T_1(H/p) U_3(H). \tag{8.7}$$

To check counts and constants, let $H/p = 4v+3$ in the first case. Then $m = pv + 3q + 2$. The divisible counts in $(1/4)_m, (1/2)_m$ are $v, v+1$, while each denominator count is v. The extracted numerator pair is $(3/4)_v (1/2)_{v+1} = (1/2)(3/4)_v (3/2)_v$. The external H therefore contributes $p^2/2$ relative to T_3. In the second case $H/p = 4v+1$ and $m = pv + q$; every count is v, and the numerator parameters become $1/4, 1/2$, giving the external factor p. In both cases $(-1)^{m-v} = (-1)^q$.

The remaining gamma quotient in case a is

$$\frac{\Gamma(H/4)\Gamma(a/4+H/4)\Gamma(1+\zeta H/4)\Gamma(1+\zeta^2 H/4)}{\Gamma(a/4)\Gamma(a/2)\Gamma(1-a/4-\zeta H/4)\Gamma(1-a/4-\zeta^2 H/4)}.$$

The step formula at the two arguments centered at one and reflection at the two arguments centered at $1-a/4$ convert this to $\Gamma(a/4)^2 U_a(H)/\Gamma(a/2)$; the paired signs cancel. Finally $\Gamma(3/2) = -\Gamma(1/2)/2$, proving (8.7). The same identities show

$$v_p(C_1) = 2, \qquad v_p(C_3) = 1, \qquad C_1 C_3 = -p^3. \tag{8.8}$$

For the last equality use $(\Gamma(1/4)\Gamma(3/4))^2 = 1$ and $\Gamma(1/2)^2 = 1$ for $p \equiv 3 \pmod 4$.

We need the whole correction series, not merely its value at a single node. The normalized analytic gamma series and their functional identities are valid on the displayed residue disks. Long–Ramakrishna's Proposition 13 gives integral coefficients in degrees less than p, and the general coefficient bound $-k(1/p+1/(p-1))$. Multiplication of X by ζ permutes all factors in U_a, so its positive degrees are multiples of three. Since $p \geq 7$, the degree-three coefficient is integral. Every scaled coefficient in $U_a(pX)$ of degree $d \geq 6$ has valuation at least

$$d\left(1-\frac{1}{p}-\frac{1}{p-1}\right) \geq \frac{29d}{42} > 3.$$

These bounds tend to infinity. Hence

$$U_a(pX) = 1 + \sum_{j\geq 1} u_{a,j} X^{3j}, \qquad u_{a,j} \in p^3\mathbb{Z}_p, \qquad v_p(u_{a,j}) \longrightarrow \infty. \tag{8.9}$$

The coefficients lie in $\mathbb{Q}_p$ by Galois symmetry of the complete cube-root products. All estimates can equivalently be made in the unramified extension and then restricted to $\mathbb{Q}_p$.

8.3 Transport between the two families

Let $N = p^r \equiv a \pmod 4$, $r \geq 1$, and put $M = N/p$ and $b = 4-a$, so $M \equiv b \pmod 4$. Dividing the p-divisible indices in the exact interpolation weight by p exchanges the two residue classes and gives

$$\frac{W_{a,N}(pJ)}{W_{b,M}(J)} = \frac{\prod_{\substack{1\leq E<N \\ E\equiv 0\ (4),\ p\nmid E}} (1+p^3J^3/E^3)}{\prod_{\substack{1\leq K\leq N \\ K\equiv a\ (4),\ p\nmid K}} (1-p^3J^3/K^3)} =: R_{a,N}(J).$$

The numerator range is precisely $E = 4, 8, \ldots, N-a$; the displayed congruence description has that same endpoint for $a \in \{1,3\}$. This restricted series has constant term one and its coefficient of J^{3j} lies in $p^{3j}\mathbb{Z}_p$. By (8.9),

$$R_{a,N}(J)U_a(pJ) = 1 + \sum_{j\geq 1} e_{a,N,j} J^{3j}, \qquad e_{a,N,j} \in p^3\mathbb{Z}_p, \quad v_p(e_{a,N,j}) \longrightarrow \infty.$$

Writing $\mu_{a,N}(H) = W_{a,N}(H)T_a(H)$, the exact model steps imply

$$\mu_{a,N}(pJ) = C_a \mu_{b,M}(J)\left(1 + \sum_{j\geq 1} e_{a,N,j} J^{3j}\right).$$

Each positive coarse moment belongs to $p^{4(r-1)}\mathbb{Z}_p$. The aggregated correction therefore has valuation at least $v_p(C_a) + 3 + 4(r-1) \geq 4r$. The infinite series may be summed after the finite node sum because its coefficients tend to zero and there are only finitely many rational node values. Adding the unit-node contribution (8.5) proves

$$G_a(p^r) \equiv C_a G_{4-a}(p^{r-1}) \pmod{p^{4r}}. \tag{8.10}$$

For the smallest odd level $r = 1$, $a = 3$, the coarse level is $M = 1$, $b = 1$; its positive moments equal one, so the same bound still applies.

Apply (8.10) first at $r = 2s$, $a = 1$, and then at $r = 2s-1$, $a = 3$. The second error has valuation at least $8s-4$; after multiplication by C_1 it has valuation at least $8s-2$. The first error belongs to $p^{8s}\mathbb{Z}_p$. Equation (8.8) now gives (8.1). The exclusion of $p = 3$ is necessary for this even-level statement; no assertion for that prime is made here.

Corollary 8.1. *For $p > 3$, $p \equiv 3 \pmod 4$ and $s \geq 0$, $v_p(G(p^{2s})) = 3s$.*

Proof. The base case is $G(1) = 1$. If the preceding value has valuation $3s-3$, its product with $-p^3$ has valuation $3s$. The error in (8.1) has valuation at least $8s-2 > 3s$ for $s \geq 1$, which completes the induction. □

9 What remains and how the results were checked

The new results remove all remaining A.3 levels, all ordinary D.3 levels, all ordinary G.3 levels, and every even supersingular G.3 level for $p > 3$. Together with the earlier companion results, A.3, D.3 and G.3 (with the stated prime domain) are complete within the research series. The remaining target in the active scope is E.3 at $p \equiv 2 \pmod 3$, odd $r \geq 3$, with modulus p^{3r-1}. Explicitly, for

$$E_r = \sum_{k=0}^{(p^r-2)/3} (-1)^k (6k+1) \frac{(1/3)_k^3}{(k!)^3},$$

the unresolved assertion is $E_r \equiv p^2 E_{r-2} \pmod{p^{3r-1}}$. Exact parameter reflections and vanishing deformations are recorded in the archive, but removal of those deformations at the target precision has not been proved. The prime 3 requires a domain distinction for the even G formula. Direct calculation gives

$$G(9) + 27 = \frac{29407617}{1048576} = \frac{3^5 \cdot 121019}{2^{20}},$$

so the literal modulus-p^6 even-level assertion fails at $p = 3, r = 2$. All even supersingular G claims in this manuscript therefore explicitly assume $p > 3$. Swisher's Table 2 does not display this exception in its G.3 row; our completion claim for G.3 is for the corrected prime domain $p > 3$.

Reproducibility. The companion archive contains this single-file source, the PDF, the proof fragments, exact-arithmetic programs and their recorded outputs. The checks include rational interpolation identities, polynomial moment recurrences and evaluations, node-model formulas, weight valuations, and original prime-power congruences. The earlier second-level ordinary G calculation also checked all 38 primes $p \leq 401$ in its residue class: 1,862 centered blocks and 496,238 fine terms. It is retained as an independent finite-level check, although the all-level proof no longer relies on it. The supersingular A recurrence was checked in 26 cases, including $p = 3$ through $r = 8$, $p = 7$ through $r = 5$, and $p = 11$ through $r = 4$. There are also 136 exact A interpolation and factorization checks. Finite calculations verify the formulas; the algebraic and p-adic proofs establish the universal assertions.

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MANUSCRIPT VI

Exact Reductions for the Remaining Odd Branch of Swisher's E.3 Supercongruence

Research checkpoint 11 November 2026

Abstract

We study only the unresolved branch with $p > 3$, $p \equiv 2 \pmod 3$ and odd level $r \geq 3$. We prove an exact symmetric companion transformation, an interpolation formula on three grids adapted to the short cutoff, a rational-polynomial formula for every positive interpolation moment, and an exact two-step residual identity. The first two moments vanish; the third moment contains a factor of valuation $(7r-3)/2$. We also exhibit an explicit failure of a proposed coefficient bound and verify the cancellation between two contributions at the first lifting level. These are proved structural reductions and finite checks. The full E.3 congruence remains unproved, and no new infinite range of its prime-power instances is claimed in this note.

Status. The exact identities were independently checked internally. The note has not undergone external peer review. The distinction between identities, numerical evidence, and the remaining conjectural divisibility is explicit throughout.

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1 The single remaining target

The Pochhammer symbol is $(a)_k = \prod_{j=0}^{k-1}(a+j)$, with $(a)_0 = 1$. Write $v_p(p) = 1$ and $v_p(0) = +\infty$. For every integer $L \equiv 2 \pmod 3$, $L \geq 2$, define

$$E(L) = \sum_{k=0}^{(L-2)/3} (-1)^k (6k+1) \frac{(1/3)_k^3}{(k!)^3}.$$

The target inherited from Swisher's Table 2, label E.3 [1], is

$$E(p^r) - p^2 E(p^{r-2}) \in p^{3r-1}\mathbb{Z}_p, \qquad p > 3,\ p \equiv 2 \pmod 3,\ r \geq 3 \text{ odd}. \tag{1.1}$$

All other labels and all previously completed branches are outside this continuation. In particular, no proof from the earlier A, D, or G manuscripts is used as a black-box solution of (1.1).

What is proved. The symmetric finite expression below removes the alternating weight by an exact transformation. The three-grid formula supplies explicitly evaluated nodes at the actual short cutoff. Its moments satisfy $M_1 = M_2 = 0$ and

$$M_3(L) = (-1)^{m+1} L^2 (L-1)^3 \left(\frac{(1/3)_m}{m!} \right)^3, \qquad m = (L-2)/3.$$

For $L = p^r$ in the target branch this gives $v_p(M_3(L)) = (7r-3)/2$. This valuation concerns a moment of the interpolation weights, not the original E.3 recurrence difference. The final section gives an exact equation identifying the cancellation still needed to establish (1.1).

2 An exact symmetric companion for the odd E.3 cutoff

The identities in this section are exact. They do not, by themselves, prove the two-step prime-power recurrence in E.3.

For any nonnegative integer n, set

$$S_\alpha(n) = \sum_{k=0}^{n} (-1)^k \frac{2k+\alpha}{\alpha} \frac{(\alpha)_k^3}{(k!)^3}, \qquad W_\alpha(n) = \sum_{k=0}^{n} \frac{(\alpha)_k (-n-\alpha)_k}{(k!)^2}.$$

The finite transformation is

$$S_\alpha(n) = \frac{(1+\alpha)_n}{n!} W_\alpha(n). \tag{2.1}$$

Here is a short self-contained verification. For $n \geq 1$, put

$$q_n(k) = \frac{(-n-\alpha)_k (\alpha)_k}{(k!)^2}, \qquad R_n(k) = -\frac{k^2}{(n+\alpha)^2}.$$

Cancellation of consecutive factors verifies

$$q_n(k+1) R_n(k+1) - q_n(k) R_n(k) = q_n(k) - \frac{n}{n+\alpha} q_{n-1}(k).$$

Sum over $0 \leq k \leq n$. The upper boundary and the extra term $q_{n-1}(n)$ both equal $(-1)^n((\alpha)_n/n!)^2$. Consequently

$$W_\alpha(n) - \frac{n}{n+\alpha} W_\alpha(n-1) = (-1)^n \frac{2n+\alpha}{n+\alpha} \left(\frac{(\alpha)_n}{n!} \right)^2.$$

Multiplying by $(1+\alpha)_n/n!$ proves that the right side of (2.1) has exactly the defining increment of $S_\alpha(n)$. Both sides equal one at $n=0$, proving the identity. These calculations hold first away from zero denominators and then as rational-function identities in α. There is a second exact transformation

$$W_\alpha(n) = (-1)^n \sum_{k=0}^{n} \frac{(-n)_k (n+\alpha+1)_k (1-\alpha)_k}{(k!)^3}. \tag{2.2}$$

Here is a formal power-series proof, which avoids convergence issues. The left side is the coefficient of z^n in

$$(1-z)^{-1}\,{}_2F_1(\alpha, -n-\alpha; 1; z).$$

Euler's transformation, valid as a formal identity, rewrites this as

$$(1-z)^n\,{}_2F_1(1-\alpha, n+\alpha+1; 1; z).$$

The coefficient of z^{n-k} in $(1-z)^n$ is $(-1)^{n-k}\binom{n}{k} = (-1)^n(-n)_k/k!$. Taking the coefficient of z^n gives (2.2).

Now let $L \equiv 5 \pmod 6$, $n=(L-2)/3$ and $t=L/3$. Then n is odd and $-n = 2/3 - t$. Put

$$P_n = \frac{(4/3)_n}{n!}, \qquad H_n(t) = \sum_{k=0}^{n} \frac{(2/3-t)_k (2/3)_k (2/3+t)_k}{(k!)^3}.$$

The original E.3 sum is exactly

$$\boxed{E(L) = -P_n H_n(L/3).} \tag{2.3}$$

In particular the new finite hypergeometric expression has a genuine quadratic symmetry in its deformation parameter. With $c_k = (2/3)_k/k!$, it is

$$H_n(L/3) = \sum_{k=0}^{n} c_k^3 \prod_{j=0}^{k-1} \left(1 - \frac{L^2}{(3j+2)^2}\right). \tag{2.4}$$

No denominator deformation occurs in this expression.

2.1 Exact prefactor valuation and a sharp termwise obstruction

Let $p>3$, $p \equiv 2 \pmod 3$, $r = 2s+1 \geq 3$, $L=p^r$ and $n=(p^r-2)/3$. Then

$$v_p(P_n) = s. \tag{2.5}$$

Indeed $P_n = (3n+1)(1/3)_n/n! = (p^r-1)(1/3)_n/n!$. For each $\ell \leq r$, the denominator contributes $\lfloor n/p^\ell \rfloor$ factors divisible by p^ℓ. If ℓ is odd, the first numerator index is $(2p^\ell-1)/3$, while $n \bmod p^\ell = (p^\ell-2)/3$; there is no extra numerator factor. If ℓ is even, the first index is $(p^\ell-1)/3$, while $n \bmod p^\ell = 2(p^\ell-1)/3$; there is exactly one extra factor. There are s positive even indices below r. No higher level contributes, proving (2.5).

Every c_k is p-integral (equivalently, $(2/3)_k/k! = (-1)^k\binom{-2/3}{k}$ is a p-adic binomial coefficient). The factors $3j+2$ in (2.4) satisfy $0<3j+2<L$ for $j<n$, so each displayed deformation factor is in $1+p^2\mathbb{Z}_p$. Consequently, with $U_n = \sum_{k=0}^{n} c_k^3$,

$$H_n(L/3) - U_n \in p^2\mathbb{Z}_p, \qquad E(L) + P_n U_n \in p^{s+2}\mathbb{Z}_p. \tag{2.6}$$

The first bound cannot be improved by bounding each summand separately. Take $k=p^{r-1}$, which lies below n since $p \geq 5$. The coefficient c_k is a p-adic unit: at every level $\ell \leq r-1$, the numerator

and denominator counts both equal $p^{r-1-\ell}$, and at level r neither product has a divisible factor. Among the integers $3j+2$, $0 \le j < k$, the only one of valuation $r-1$ is $2p^{r-1}$. All others have valuation at most $r-2$. Therefore

$$\prod_{j=0}^{k-1}\left(1-\frac{p^{2r}}{(3j+2)^2}\right)-1 \equiv -\frac{p^2}{4} \pmod{p^4},$$

and this single summand of $H_n(L/3)-U_n$ has valuation exactly 2. Thus any improvement needed for E.3 must use cancellation between different values of k.

2.2 A related exactly vanishing Dixon deformation

For the same n and $t = n+2/3$, define

$$D_n(t) = \sum_{k=0}^{n} \frac{(-n)_k(2/3)_k(2/3+t)_k}{k!(1-t)_k(1-2t)_k}.$$

This sum vanishes exactly. More generally,

$$\sum_{k=0}^{n} \frac{(-n)_k(b)_k(c)_k}{k!(1-n-b)_k(1-n-c)_k} = 0 \qquad (n \text{ odd}),$$

whenever the finite denominators are nonzero. Reversing k to $n-k$ multiplies every term by $(-1)^n$, proving the identity. Here $b = 2/3$, $c = 2/3+t$ give precisely the displayed denominators. Their linear shifts are not symmetric about one; consequently this zero identity does not justify removing the denominator shifts at the E.3 target precision.

Precise remaining target in this representation. Let $m = (p^{r-2}-2)/3$ and $t' = p^{r-2}/3$. The unsolved E.3 recurrence is equivalent, by (2.3), to

$$P_n H_n(p^r/3) - p^2 P_m H_m(t') \in p^{3r-1}\mathbb{Z}_p.$$

For $r \ge 3$, $v_p(P_n) = s$ and $v_p(P_m) = s-1$. Thus after dividing by P_n one needs the aggregate congruence

$$H_n(p^r/3) - p^2\frac{P_m}{P_n}H_m(t') \in p^{5s+2}\mathbb{Z}_p,$$

where the multiplier p^2P_m/P_n has valuation one. This is an exact reformulation with a symmetric deformation, not a proof of the required aggregate congruence.

2.3 Exact two-step stripping of the prefactor

There is also an exact symmetric gamma description of the prefactor ratio. Write $L = p^r$, $M = L/p^2$, $n = (L-2)/3$, $m = (M-2)/3$, and $n_1 = (L/p-1)/3$. Let $\Gamma = \Gamma_p$ and put, for $X \in p\mathbb{Z}_p$,

$$R_p(X) = \frac{1}{\Gamma(1/3-X/3)\Gamma(1/3+X/3)}.$$

Then

$$P_n = -pP_m\frac{R_p(L)}{R_p(L/p)}. \tag{2.7}$$

To check both the constant and the sign, first extract the factors divisible by p from $(4/3)_n/n!$. The numerator and denominator have n_1 such factors, and their normalized ratio is $(2/3)_{n_1}/n_1!$. The remaining unit ratio is

$$-\frac{\Gamma(2/3+L/3)}{\Gamma(4/3)\Gamma(1/3+L/3)}=\frac{3\Gamma(2/3+L/3)}{\Gamma(1/3)\Gamma(1/3+L/3)}.$$

In the second extraction, the numerator has $m+1$ factors divisible by p and the denominator has m. Their ratio is

$$p\frac{(1/3)_{m+1}}{m!}=\frac{p}{3}P_m.$$

The unit ratio in this step is

$$-\frac{\Gamma(1/3+L/(3p))}{\Gamma(2/3)\Gamma(2/3+L/(3p))}.$$

For $p\equiv 2 \pmod 3$, the least positive residue of $1/3$ modulo p is $(p+1)/3$, which is even. Thus reflection gives $\Gamma(1/3)\Gamma(2/3)=1$ and, for $X\in p\mathbb{Z}_p$,

$$\frac{\Gamma(2/3+X/3)}{\Gamma(1/3+X/3)}=R_p(X).$$

Combining the two extractions proves (2.7). In particular R_p is even, so this second deformation has no linear term either.

Using (2.7), the exact remaining aggregate congruence can be written

$$H_n(L/3)+p\frac{R_p(L/p)}{R_p(L)}H_m(M/3)\in p^{5s+2}\mathbb{Z}_p,\qquad r=2s+1.$$

The identities above do not establish this final congruence.

3 Three exact interpolation grids at the short E cutoff

For an integer $m\ge 0$ put $N=3m+2$ and

$$\mathcal{S}_m(z)=\sum_{k=0}^{m}(-1)^k(6k+1)\frac{(1/3)_k(1/3+z/6)_k(1/3-z/3)_k(2/3-z/3)_k}{k!(1-z/6)_k(1+z/3)_k(2/3+z/3)_k}.$$

At $z=0$ this is the original E sum with cutoff m. Define

$$D_m(z)=\prod_{j=1}^{m}(1-z/(6j))(1+z/(3j))(1+z/(3j-1)),$$

$$Q_m(z)=\prod_{h=0}^{m}(1-z/(3h+1))(1-z/(3h+2))(1+z/(6h+2)).$$

The cleared expression $P_m(z)=D_m(z)\mathcal{S}_m(z)$ is a polynomial of degree at most $3m$. Its $3m+3$ distinct interpolation nodes are

$$C_h=3h+1,\qquad F_h=3h+2,\qquad B_h=-6h-2\quad(0\le h\le m).$$

At these nodes the exact cleared values are

$$P_m(H)=D_m(H)T(H),\qquad \begin{cases}T(C_h)=C_h, & h \text{ even},\\ T(C_h)=0, & h \text{ odd},\\ T(F_h)=(-1)^hF_h/2,\\ T(B_h)=-B_h/2.\end{cases}\tag{3.1}$$

3.1 The two positive grids

For complex z with positive real part and away from denominator poles, apply the nonterminating Whipple ${}_6F_5(-1)$ transformation [2] with

$$a = 1/3, \quad b = 1/3 + z/6, \quad c = 1/3 - z/3, \quad e = 2/3, \quad f = 2/3 - z/3.$$

The equal factors at $e = 1 + a - e = 2/3$ cancel. The transformed Dixon ${}_3F_2(1)$ has upper parameters $2/3, 2/3 + z/6, 2/3 - z/3$ and lower parameters $1 - z/6, 1 + z/3$. The required convergence conditions reduce to $\mathrm{Re}(z/3) > 0$ and $\mathrm{Re}(z/6) > 0$. Whipple followed by Dixon and gamma reflection evaluates the series as

$$\frac{z}{2}\,\frac{\sin(\pi(1/3 + z/6))}{\sin(\pi z/6)}.$$

At $z = C_h$ or F_h, the original series terminates at h; its lower parameters are never nonpositive integers at these positive nodes. The displayed expression is respectively C_h or zero according to the parity of h, and $(-1)^h F_h/2$. This proves the first three lines of (3.1). It also explains why the same sine expression must not be substituted on the negative grid: there a lower parameter becomes a nonpositive integer, and analytic continuation does not represent the specified finite truncation.

3.2 A finite WZ certificate for the negative grid

At $z = B_h$, the finite terminating sum through h is

$$\sum_{k=0}^{h} t(h,k), \qquad t(h,k) = (-1)^k(6k+1)\frac{(1/3)_k(-h)_k(2h+1)_k(2h+4/3)_k}{k!(h+4/3)_k(1/3-2h)_k(-2h)_k}.$$

For $0 \le k \le h$ the displayed denominator has no zero; at $h = 0$ only $k = 0$ is used. We prove

$$\sum_{k=0}^{h} t(h,k) = 3h + 1. \tag{3.2}$$

Put $u(h,k) = t(h,k)/(3h+1)$ and define

$$\begin{aligned} A(h,k) &= 864h^4 + 3024h^3 - 144h^2k^2 + 96h^2k + 3912h^2 \\ &\quad - 264hk^2 + 176hk + 2212h + 9k^4 - 12k^3 - 119k^2 + 82k + 460, \end{aligned}$$

$$R(h,k) = -\frac{k(3k-1)(2h-k+1)(6h-3k+2)A(h,k)}{16(h+1)(2h+1)^2(3h+2)(6h+5)(6h+7)(6k+1)(h-k+1)}.$$

Direct rational simplification gives the certificate

$$u(h+1,k) - u(h,k) = R(h,k+1)u(h,k+1) - R(h,k)u(h,k). \tag{3.3}$$

To interpret the upper boundary, cancel the apparent pole using

$$\frac{(-h)_k}{h-k+1} = \frac{(-h-1)_k}{h+1}.$$

Thus the product $R(h,k)u(h,k)$ has a well-defined value at $k = h+1$. Direct cancellation gives

$$R(h,0)u(h,0) = 0, \qquad R(h,h+1)u(h,h+1) = -u(h+1,h+1).$$

Summing (3.3) for $0 \le k \le h$ and adding the last term $u(h+1,h+1)$ proves that the sum of $u(h,k)$ over $0 \le k \le h$ is independent of h. Its value at $h = 0$ is one, proving (3.2).

3.3 The apparent poles in the cleared interpolation

At $z = B_h$, the only possible zero in D_m is its factor indexed by $j = 2h+1$ in $1 + z/(3j-1)$. If $2h \geq m$, this index is outside D_m; the original finite sum terminates at h, and (3.2) gives $P_m(B_h) = D_m(B_h)(3h+1)$. If $2h < m$, then $D_m(B_h) = 0$. In each cleared summand with $k \leq h$ the same zero remains in the denominator-clearing tail. In each summand with $k \geq h+1$ the numerator contains the factor $(1/3 + z/6 + h)$, which vanishes at B_h. Every cleared summand is therefore zero, so $P_m(B_h) = 0$. This is again exactly $D_m(B_h)(3h+1)$ and proves (3.1) at every negative node, without evaluating any $0/0$ summand.

Consequently interpolation of P_m at all the listed nodes gives the exact identity

$$\mathcal{S}_m(0) = \sum_H W_m(H)T(H), \qquad W_m(H) = \frac{D_m(H)}{\prod_{K\neq H}(1 - H/K)},$$

with K, H ranging over the three grids. The weights may be zero; their rational denominators are never zero. Equivalently,

$$\frac{D_m(z)\mathcal{S}_m(z)}{Q_m(z)} = \sum_H \frac{W_m(H)T(H)}{1 - z/H}.$$

The left side is $O(z^{-3})$ at infinity, proving the exact vanishing of its first and second positive moments.

4 Exact moments for the three-grid short E deformation

The statements in this section are exact. The uniform p-integrality assertion needed by the proposed moment argument is not assumed or proved here. Let $L = 3m+2$, $m \geq 0$, and $b_k = (1/3)_k/k!$. Define

$$S_m(z) = \sum_{k=0}^{m} (-1)^k (6k+1) b_k^3 \prod_{h=0}^{k-1} \frac{(1 - z/(3h+1))(1 - z/(3h+2))(1 + z/(6h+2))}{(1 - z/(6h+6))(1 + z/(3h+3))(1 + z/(3h+2))}.$$

Put

$$q_h(z) = (1 - z/(3h+1))(1 - z/(3h+2))(1 + z/(6h+2)),$$
$$d_h(z) = (1 - z/(6h+6))(1 + z/(3h+3))(1 + z/(3h+2)),$$
$$Q_m(z) = \prod_{h=0}^{m} q_h(z), \qquad D_m(z) = \prod_{h=0}^{m-1} d_h(z), \qquad F_m(z) = \frac{D_m(z)S_m(z)}{Q_m(z)}.$$

The numerator $D_m S_m$ is a polynomial of degree at most $3m$, whereas Q_m has degree $3m+3$ and distinct roots. Thus F_m is proper and has a Laurent expansion at infinity

$$F_m(z) = -\sum_{j\geq 1} M_j(L) z^{-j}, \qquad M_1(L) = M_2(L) = 0.$$

Equivalently, if $F_m(z) = \sum_H \mu_m(H)/(1 - z/H)$ is its partial fraction expansion at its three node grids, then $M_j(L) = \sum_H \mu_m(H) H^j$. This definition allows a cancelled pole to have residue weight zero; it does not assume that all values of S_m at lower-parameter collisions are terminating evaluations.

Proposition 4.1. *Set*

$$C(L) = (-1)^{m+1} L^2 (L-1)^3 b_m^3.$$

There are uniquely determined even polynomials $P_j(L) \in \mathbb{Q}[L^2]$, $j \geq 0$, with $P_0 = 1$ and $\deg P_j \leq 2\lfloor 2j/3 \rfloor$, such that

$$M_{j+3}(L) = C(L) P_j(L).$$

In particular,

$$P_1 = 1, \qquad P_2 = 2L^2 - 3, \qquad P_3 = -\frac{L^4 - 13L^2 + 16}{4}.$$

Proof. At infinity the ratio of the leading terms of q_h and d_h is $-(3h+3)^2/(3h+1)^2$. Hence

$$S_m(\infty) = \sum_{k=0}^{m} (6k+1)b_k = \frac{L(L-1)b_m}{2}.$$

The last equality telescopes: the difference between $(3k+1)(3k+2)b_k/2$ and its value at $k-1$ is $(6k+1)b_k$. The leading coefficient of D_m/Q_m is $(-1)^m 2L(L-1)^2 b_m^2 z^{-3}$. This proves $M_3 = C(L)$.

Separating the last summand of S_m gives

$$q_m(z)F_m(z) = d_{m-1}(z)F_{m-1}(z) + (-1)^m(2L-3)b_m^3. \tag{4.1}$$

Write

$$A = L(L-1), \quad B = (L-2)(L-3), \quad a = (L-1)(3L-2), \quad c = (L-2)(3L-7).$$

The two cubic factors have expansions

$$q_m(z) = \frac{z^3 - z^2 - az + 2L(L-1)^2}{2L(L-1)^2},$$

$$d_{m-1}(z) = -\frac{z^3 - z^2 - cz - 2(L-3)(L-2)^2}{2(L-3)(L-2)^2}.$$

Since $b_m/b_{m-1} = (L-4)/(L-2)$, comparison of coefficients in (4.1), after dividing by $C(L)$, yields

$$\begin{aligned} AP_j(L) - BP_j(L-3) = {} & AP_{j-1}(L) - BP_{j-1}(L-3) \\ & + AaP_{j-2}(L) - BcP_{j-2}(L-3) \\ & - 2AL(L-1)^2 P_{j-3}(L) - 2B(L-3)(L-2)^2 P_{j-3}(L-3), \end{aligned}$$

for $j \geq 1$, where $P_{-1} = P_{-2} = P_{-3} = 0$ and $P_0 = 1$. This also provides an independent recursive definition of the polynomials. Indeed the operator

$$\mathcal{L}(P) = L(L-1)P(L) - (L-2)(L-3)P(L-3)$$

maps $\mathbb{Q}[L^2]$ to $(2L-3)\mathbb{Q}[L(L-3)]$. The first space consists of even polynomials; the second consists of polynomials anti-invariant under $L \mapsto 3-L$. In the displayed power bases, $\mathcal{L}(L^{2k})/(2L-3)$ has degree k and leading coefficient $3k+2$. Consequently this operator is an isomorphism of the filtered vector spaces over $\mathbb{Q}$. For even preceding polynomials, the right side of the recurrence is anti-invariant under $L \mapsto 3-L$; the two cubic terms exchange with opposite signs under this reflection. Thus the recurrence uniquely defines an even P_j at every step. It gives the degree bound

$$\deg P_j \leq \max\{\deg P_{j-1},\ \deg P_{j-2} + 2, \deg P_{j-3} + 4\} \leq 2\lfloor 2j/3 \rfloor.$$

At $L = 2$, evaluation of the polynomial recurrence gives the coefficients of

$$\sum_{j \geq 0} P_j(2)t^j = \frac{1}{1 - t - 4t^2 + 4t^3}.$$

This is exactly the normalized expansion of $F_0 = 1/q_0$. The recurrence in m now proves the moment identities for every integer $m \geq 0$. □

Prime-power valuation of the first moments. For $p > 3$, $p \equiv 2 \pmod 3$, and odd r, take $L = p^r$. Then

$$v_p(b_{(p^r-2)/3}) = \frac{r-1}{2}, \qquad v_p(C(p^r)) = \frac{7r-3}{2}.$$

For the first equality count multiples of p^ℓ in the numerator and denominator of b_m. If ℓ is even, the numerator acquires one additional multiple, while for odd ℓ the counts agree; there are $(r-1)/2$ positive even levels at most r. Therefore $M_1 = M_2 = 0$, and M_3, M_4, M_5, M_6 all have exactly the valuation $(7r-3)/2$ for these primes. This exceeds or equals the required $3r-1$.

The remaining arithmetic issue. The polynomial operator is not an isomorphism over $\mathbb{Z}[1/2]$: its diagonal is $3k+2$, not a unit there. For example

$$P_6(L) = \frac{L^8 + 290L^6 - 3325L^4 + 16666L^2 - 29900}{28},$$

and P_9 has denominator 280. Thus the rational-polynomial existence proof alone does not establish $P_j(p^r) \in \mathbb{Z}_p$ for all j at primes $p \equiv 2 \pmod 3$. Such an integrality theorem, or an alternative sufficient uniform bound for the full moments, remains to be proved before using an Euler-power limit to discard all unit-node contributions. The recurrence and the first moment valuations above do not resolve that remaining issue.

5 Exact two-step transport and the remaining cancellation

Let $p > 3$, $p \equiv 2 \pmod 3$, and $r \geq 3$ be odd. Set $L = p^r$, $M = L/p^2$, and $m = (L-2)/3$. Use the node and denominator root sets $\mathcal{H}_L$ and $\mathcal{D}_L$ of the three-grid construction, and write $\mu_L(H) = W_L(H)T(H)$. Define the restricted rational power series

$$R_{L,p}(X) = \frac{\prod_{d \in \mathcal{D}_L,\, p^2 \nmid d}(1 - p^2X/d)}{\prod_{h \in \mathcal{H}_L,\, p^2 \nmid h}(1 - p^2X/h)} = 1 + \sum_{j \geq 1} c_j X^j.$$

Every retained root has valuation zero or one. Hence each linear coefficient p^2/d or p^2/h belongs to $p\mathbb{Z}_p$. Expanding the finitely many inverse factors proves

$$c_j \in p^j\mathbb{Z}_p, \qquad v_p(c_j) \longrightarrow +\infty.$$

The series converges at every integral coarse node J.

Proposition 5.1 (Exact residual identity)**.** *Put*

$$U_{L,p} = \sum_{\substack{H \in \mathcal{H}_L \\ p^2 \nmid H}} \mu_L(H).$$

Then

$$\boxed{E(L) - p^2E(M) = U_{L,p} + p^2 \sum_{j \geq 3} c_j M_j(M).} \tag{5.1}$$

No assertion that either term on the right is separately divisible by the target modulus is made.

Proof. Since $p^2 \equiv 1 \pmod 6$, dividing all p^2-divisible nodes by p^2 gives exactly $\mathcal{H}_M$. Dividing the corresponding denominator roots gives exactly $\mathcal{D}_M$. For example the positive grids end at $L-1$ and L, and become the positive grids ending at $M-1$ and M; the negative grid is -2 times the first

positive grid. The positive denominator roots $6i$ and the negative roots $-3i$ both have coarse index range $1 \le i \le (M-2)/3$. The third denominator grid ends at $-(L-3)$ and becomes the grid ending at $-(M-3)$.

The exact model values satisfy $T(p^2J) = p^2T(J)$, because multiplication by p^2 preserves the residues modulo six and the sign of each node. Separating divisible factors in the weights therefore gives the product identities

$$W_L(p^2J) = W_M(J)R_{L,p}(J), \qquad \mu_L(p^2J) = p^2\mu_M(J)R_{L,p}(J).$$

These are identities of products, including at zero weights; no division by a vanishing weight is needed. Sum over coarse nodes and add the nodes not divisible by p^2. Interchanging the finite node sum with the convergent series is legitimate. The identities $M_1(M) = M_2(M) = 0$ remove its first two positive terms and give (5.1). □

For $r = 2s+1$, the previously proved moment formula rewrites the remaining assertion as the single explicit divisibility requirement

$$U_{L,p} + p^2C(M)\sum_{j\ge 3} c_j P_{j-3}(M) \in p^{6s+2}\mathbb{Z}_p.$$

This is equivalent to the original conjecture, not a completed proof of it. The polynomial identities do not yet control the whole series at this precision. Moreover $U_{L,p}$ contains both the unit nodes and the nodes divisible by p but not by p^2. Eliminating only the first class does not eliminate the second.

A verified cancellation that must be proved uniformly. Exact modular arithmetic at $p = 11$, $r = 3$ gives

$$v_{11}\left(\sum_{11\nmid H} \mu_{11^3}(H)\right) = 9, \qquad v_{11}\left(\sum_{v_{11}(H)=1} \mu_{11^3}(H)\right) = 7,$$

while the second term on the right of (5.1) also has valuation seven. Dividing the last two contributions by 11^7 gives residues 8 and 3 modulo 11. They cancel, and the original recurrence difference has valuation eight. These are finite checks, not a uniform proof. They show why separately discarding both contributions would give a false proof.

Audit of the cubic coefficient in the three-grid transport

Let $p > 3$ satisfy $p \equiv 2 \pmod 3$, put $L = p^3$ and $m = (L-2)/3$. With multiplicities, let

$$\mathcal{H} = \{3j+1, 3j+2, -2(3j+1) : 0 \le j \le m\},$$
$$\mathcal{D} = \{6j, -3j : 1 \le j \le m\} \cup \{-(3j+2) : 0 \le j < m\}.$$

The exact factor left after stripping the p^2-divisible nodes and denominator roots is

$$R(X) = \frac{\prod_{d\in\mathcal{D},\, p^2\nmid d}(1 - p^2X/d)}{\prod_{h\in\mathcal{H},\, p^2\nmid h}(1 - p^2X/h)}.$$

Write $R(X) = \sum c_j X^j$. The elementary estimate $c_j \in p^j\mathbb{Z}_p$ holds, but the proposed improvement $c_3 \in p^4\mathbb{Z}_p$ is false in general.

For a direct exact test, define

$$S_i = \sum_{\substack{h\in\mathcal{H}\\ v_p(h)=1}} (h/p)^{-i} - \sum_{\substack{d\in\mathcal{D}\\ v_p(d)=1}} (d/p)^{-i} \quad \text{in } \mathbb{F}_p.$$

The formal logarithm of R through degree three gives

$$\frac{c_3}{p^3} \equiv \frac{S_1^3 + 3S_1S_2 + 2S_3}{6} \pmod{p}.$$

Roots of valuation zero cannot affect this congruence, since their degree-i logarithmic contribution is divisible by p^{2i}. At $p = 11$, the differences of multiplicities at the nonzero residues $1, \ldots, 10$ are

$$(-1, 0, -1, 0, 2, 0, -1, 0, -1, 0).$$

Thus $(S_1, S_2, S_3) = (0, 1, 8)$ and $c_3/p^3 \equiv -1 \pmod{11}$. At $p = 17$ the residue is also -1, and at $p = 23$ it is 3. The companion script computes these coefficients directly modulo p^4 as well as by the displayed Newton-sum formula.

This is an obstruction to the proposed transport argument, not a counterexample to the original E.3 congruence. A unit-node Euler projector removes the nodes $p \nmid H$, but it does not remove nodes with $v_p(H) = 1$. A p^2-strip argument must retain those nodes or give an additional estimate for their combined contribution.

There is also a removable-specialization detail at the negative nodes. For $H = -2(3j + 1)$ and $2j < m$, a denominator root equals H. The cleared numerator polynomial vanishes there: terms with $k \leq j$ retain that clearing factor, whereas terms with $k > j$ contain a zero upper factor. These nodes therefore have interpolation weight zero. Any stripping statement should be written as the exact product identity $W_L(p^2J) = W_{L/p^2}(J)R(J)$, rather than as a quotient of weights which can both vanish.

6 Verification and unresolved obligations

The archive contains the single-file source and reproducible exact-arithmetic programs. Independent checks include 123 terminating node evaluations, 693 cleared-node values (including denominator collisions), 21 complete interpolation identities, 84 low-moment identities, and a second implementation of 90 moment identities from direct Laurent division. The negative-grid WZ identity and its boundary term were also checked symbolically. Direct modular evaluations at $p = 5, 11, 17$, $r = 3$ cross-check the three-grid decomposition against the original truncated sums. Their recurrence-difference valuations are respectively $8, 8, 9$. These checks support the formulas at the specified inputs; they are not substitutes for an all-prime, all-level proof.

The remaining work is arithmetic: a sufficient uniform estimate for all the moments, together with control of the combined valuation-one-node contribution and the analytic transport correction. The fact that the first few moments have large valuation does not justify an estimate for all of them. The explicit $p = 11$ cubic-coefficient calculation also rules out discarding the correction terms separately.

Thus (1.1) remains open within this research effort. The note records exact reductions and the precise surviving divisibility requirement, without labeling it a completed proof.

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MANUSCRIPT VII

Finite Polynomial Separation and Higher Odd Levels of Swisher's E.3

Keio research notes 11 November 2026

Abstract

For primes $p > 3$ with $p \equiv 2 \pmod 3$, we prove the odd-level Swisher E.3 recurrence with its original cutoff and original modulus at every odd level $r \geq 5$. The mechanism combines two exact finite interpolation formulas, polynomial moment recurrences, a uniform logarithmic bound on denominator loss, and a finite polynomial separating unit nodes. An additional finite-field reflection produces a full extra factor of p in the companion numerator. The leakage of the unit polynomial onto divisible nodes is estimated through the other moment family, rather than discarded termwise. All identities and valuation estimates needed for these ranges are included, with explicit finite certificates for the companion interpolation. An explicitly exhausted finite set of small-prime cases is settled by exact integer certificates. The third level $r = 3$ remains unresolved. Historical priority is not asserted.

Scope and verification. This is a proof manuscript with internal independent checks and exact symbolic certificates; it has not undergone external peer review. Finite computations are identified separately from uniform theorems. In particular, this manuscript does not claim the whole of Swisher's E.3.

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1 The unchanged problem and the proved range

Write $(a)_k = \prod_{j=0}^{k-1}(a+j)$, $(a)_0 = 1$, and $v_p(p) = 1$, $v_p(0) = +\infty$. For $L \equiv 2 \pmod 3$, $L \geq 2$, set

$$E(L) = \sum_{k=0}^{(L-2)/3} (-1)^k (6k+1) \frac{(1/3)_k^3}{k!^3}.$$

The odd branch of Swisher's Table 2, E.3 [1], asks for

$$E(p^r) - p^2 E(p^{r-2}) \in p^{3r-1}\mathbb{Z}_p, \qquad p > 3, \quad p \equiv 2 \pmod 3, \quad r \geq 3 \text{ odd}. \tag{1.1}$$

Only this previously remaining branch is studied here. No result from a different Swisher label is assumed as a proof of this one.

Theorem 1.1 (Range established in this manuscript)**.** *Congruence* (1.1) *holds for every prime $p > 3$ with $p \equiv 2 \pmod 3$ and every odd integer $r \geq 5$:*

$$\boxed{E(p^r) \equiv p^2 E(p^{r-2}) \pmod{p^{3r-1}}, \qquad r \geq 5 \text{ odd}.}$$

The range is established by uniform arguments plus exact finite certificates for the explicitly enumerated exceptions.

The proof is supplied below after the exact interpolation and valuation lemmas. Its central estimate is stronger than an initial-moment calculation: every normalized moment loses at most $O(\log j)$ powers of p at moment order j. A polynomial of degree $O(pr)$ can therefore separate the unit nodes at the target precision. An exact companion formula controls its contribution on the other nodes. This is the step that turns the previous exact reductions into a proof on an infinite subrange.

2 Three exact interpolation grids at the short E cutoff

For an integer $m \geq 0$ put $N = 3m + 2$ and

$$\mathcal{S}_m(z) = \sum_{k=0}^{m} (-1)^k (6k+1) \frac{(1/3)_k (1/3 + z/6)_k (1/3 - z/3)_k (2/3 - z/3)_k}{k!(1 - z/6)_k (1 + z/3)_k (2/3 + z/3)_k}.$$

At $z = 0$ this is the original E sum with cutoff m. Define

$$D_m(z) = \prod_{j=1}^{m} (1 - z/(6j))(1 + z/(3j))(1 + z/(3j-1)),$$

$$Q_m(z) = \prod_{h=0}^{m} (1 - z/(3h+1))(1 - z/(3h+2))(1 + z/(6h+2)).$$

The cleared expression $P_m(z) = D_m(z)\mathcal{S}_m(z)$ is a polynomial of degree at most $3m$. Its $3m + 3$ distinct interpolation nodes are

$$C_h = 3h + 1, \qquad F_h = 3h + 2, \qquad B_h = -6h - 2 \quad (0 \leq h \leq m).$$

At these nodes the exact cleared values are

$$P_m(H) = D_m(H)T(H), \qquad \begin{cases} T(C_h) = C_h, & h \text{ even}, \\ T(C_h) = 0, & h \text{ odd}, \\ T(F_h) = (-1)^h F_h/2, & \\ T(B_h) = -B_h/2. & \end{cases} \tag{2.1}$$

2.1 The two positive grids

For complex z with positive real part and away from denominator poles, apply the nonterminating Whipple ${}_6F_5(-1)$ transformation [2] with

$$a = 1/3, \quad b = 1/3 + z/6, \quad c = 1/3 - z/3, \quad e = 2/3, \quad f = 2/3 - z/3.$$

The equal factors at $e = 1 + a - e = 2/3$ cancel. The transformed Dixon ${}_3F_2(1)$ has upper parameters $2/3, 2/3 + z/6, 2/3 - z/3$ and lower parameters $1 - z/6, 1 + z/3$. The required convergence conditions reduce to $\mathrm{Re}(z/3) > 0$ and $\mathrm{Re}(z/6) > 0$. Whipple followed by Dixon and gamma reflection evaluates the series as

$$\frac{z}{2}\,\frac{\sin(\pi(1/3 + z/6))}{\sin(\pi z/6)}.$$

At $z = C_h$ or F_h, the original series terminates at h; its lower parameters are never nonpositive integers at these positive nodes. The displayed expression is respectively C_h or zero according to the parity of h, and $(-1)^h F_h/2$. This proves the first three lines of (2.1). It also explains why the same sine expression must not be substituted on the negative grid: there a lower parameter becomes a nonpositive integer, and analytic continuation does not represent the specified finite truncation.

2.2 A finite WZ certificate for the negative grid

At $z = B_h$, the finite terminating sum through h is

$$\sum_{k=0}^{h} t(h,k), \qquad t(h,k) = (-1)^k(6k+1)\frac{(1/3)_k(-h)_k(2h+1)_k(2h+4/3)_k}{k!(h+4/3)_k(1/3-2h)_k(-2h)_k}.$$

For $0 \le k \le h$ the displayed denominator has no zero; at $h = 0$ only $k = 0$ is used. We prove

$$\sum_{k=0}^{h} t(h,k) = 3h + 1. \tag{2.2}$$

Put $u(h,k) = t(h,k)/(3h+1)$ and define

$$\begin{aligned} A(h,k) &= 864h^4 + 3024h^3 - 144h^2k^2 + 96h^2k + 3912h^2 \\ &\quad - 264hk^2 + 176hk + 2212h + 9k^4 - 12k^3 - 119k^2 + 82k + 460, \end{aligned}$$

$$R(h,k) = -\frac{k(3k-1)(2h-k+1)(6h-3k+2)A(h,k)}{16(h+1)(2h+1)^2(3h+2)(6h+5)(6h+7)(6k+1)(h-k+1)}.$$

Direct rational simplification gives the certificate

$$u(h+1,k) - u(h,k) = R(h,k+1)u(h,k+1) - R(h,k)u(h,k). \tag{2.3}$$

To interpret the upper boundary, cancel the apparent pole using

$$\frac{(-h)_k}{h-k+1} = \frac{(-h-1)_k}{h+1}.$$

Thus the product $R(h,k)u(h,k)$ has a well-defined value at $k = h+1$. Direct cancellation gives

$$R(h,0)u(h,0) = 0, \qquad R(h,h+1)u(h,h+1) = -u(h+1,h+1).$$

Summing (2.3) for $0 \le k \le h$ and adding the last term $u(h+1,h+1)$ proves that the sum of $u(h,k)$ over $0 \le k \le h$ is independent of h. Its value at $h = 0$ is one, proving (2.2).

2.3 The apparent poles in the cleared interpolation

At $z = B_h$, the only possible zero in D_m is its factor indexed by $j = 2h + 1$ in $1 + z/(3j - 1)$. If $2h \geq m$, this index is outside D_m; the original finite sum terminates at h, and (2.2) gives $P_m(B_h) = D_m(B_h)(3h + 1)$. If $2h < m$, then $D_m(B_h) = 0$. In each cleared summand with $k \leq h$ the same zero remains in the denominator-clearing tail. In each summand with $k \geq h + 1$ the numerator contains the factor $(1/3 + z/6 + h)$, which vanishes at B_h. Every cleared summand is therefore zero, so $P_m(B_h) = 0$. This is again exactly $D_m(B_h)(3h + 1)$ and proves (2.1) at every negative node, without evaluating any $0/0$ summand.

Consequently interpolation of P_m at all the listed nodes gives the exact identity

$$\mathcal{S}_m(0) = \sum_H W_m(H)T(H), \qquad W_m(H) = \frac{D_m(H)}{\prod_{K\neq H}(1 - H/K)},$$

with K, H ranging over the three grids. The weights may be zero; their rational denominators are never zero. Equivalently,

$$\frac{D_m(z)\mathcal{S}_m(z)}{Q_m(z)} = \sum_H \frac{W_m(H)T(H)}{1 - z/H}.$$

The left side is $O(z^{-3})$ at infinity, proving the exact vanishing of its first and second positive moments.

3 Exact moments for the three-grid short E deformation

The statements in this section are exact. The uniform p-integrality assertion needed by the proposed moment argument is not assumed or proved here. Let $L = 3m+2$, $m \geq 0$, and $b_k = (1/3)_k/k!$. Define

$$S_m(z) = \sum_{k=0}^{m}(-1)^k(6k + 1)b_k^3 \prod_{h=0}^{k-1} \frac{(1 - z/(3h + 1))(1 - z/(3h + 2))(1 + z/(6h + 2))}{(1 - z/(6h + 6))(1 + z/(3h + 3))(1 + z/(3h + 2))}.$$

Put

$$q_h(z) = (1 - z/(3h + 1))(1 - z/(3h + 2))(1 + z/(6h + 2)),$$
$$d_h(z) = (1 - z/(6h + 6))(1 + z/(3h + 3))(1 + z/(3h + 2)),$$
$$Q_m(z) = \prod_{h=0}^{m} q_h(z), \qquad D_m(z) = \prod_{h=0}^{m-1} d_h(z), \qquad F_m(z) = \frac{D_m(z)S_m(z)}{Q_m(z)}.$$

The numerator D_mS_m is a polynomial of degree at most $3m$, whereas Q_m has degree $3m + 3$ and distinct roots. Thus F_m is proper and has a Laurent expansion at infinity

$$F_m(z) = -\sum_{j\geq 1} M_j(L)z^{-j}, \qquad M_1(L) = M_2(L) = 0.$$

Equivalently, if $F_m(z) = \sum_H \mu_m(H)/(1 - z/H)$ is its partial fraction expansion at its three node grids, then $M_j(L) = \sum_H \mu_m(H)H^j$. This definition allows a cancelled pole to have residue weight zero; it does not assume that all values of S_m at lower-parameter collisions are terminating evaluations.

Proposition 3.1. *Set*

$$C(L) = (-1)^{m+1}L^2(L - 1)^3b_m^3.$$

There are uniquely determined even polynomials $P_j(L) \in \mathbb{Q}[L^2]$, $j \geq 0$, with $P_0 = 1$ and $\deg P_j \leq 2\lfloor 2j/3\rfloor$, such that

$$M_{j+3}(L) = C(L)P_j(L).$$

In particular,

$$P_1 = 1, \qquad P_2 = 2L^2 - 3, \qquad P_3 = -\frac{L^4 - 13L^2 + 16}{4}.$$

Proof. At infinity the ratio of the leading terms of q_h and d_h is $-(3h+3)^2/(3h+1)^2$. Hence

$$S_m(\infty) = \sum_{k=0}^{m} (6k+1)b_k = \frac{L(L-1)b_m}{2}.$$

The last equality telescopes: the difference between $(3k+1)(3k+2)b_k/2$ and its value at $k-1$ is $(6k+1)b_k$. The leading coefficient of D_m/Q_m is $(-1)^m 2L(L-1)^2 b_m^2 z^{-3}$. This proves $M_3 = C(L)$.

Separating the last summand of S_m gives

$$q_m(z)F_m(z) = d_{m-1}(z)F_{m-1}(z) + (-1)^m(2L-3)b_m^3. \tag{3.1}$$

Write

$$A = L(L-1), \quad B = (L-2)(L-3), \quad a = (L-1)(3L-2), \quad c = (L-2)(3L-7).$$

The two cubic factors have expansions

$$q_m(z) = \frac{z^3 - z^2 - az + 2L(L-1)^2}{2L(L-1)^2},$$

$$d_{m-1}(z) = -\frac{z^3 - z^2 - cz - 2(L-3)(L-2)^2}{2(L-3)(L-2)^2}.$$

Since $b_m/b_{m-1} = (L-4)/(L-2)$, comparison of coefficients in (3.1), after dividing by $C(L)$, yields

$$\begin{aligned} AP_j(L) - BP_j(L-3) = {} & AP_{j-1}(L) - BP_{j-1}(L-3) \\ & + AaP_{j-2}(L) - BcP_{j-2}(L-3) \\ & - 2AL(L-1)^2 P_{j-3}(L) - 2B(L-3)(L-2)^2 P_{j-3}(L-3), \end{aligned}$$

for $j \geq 1$, where $P_{-1} = P_{-2} = P_{-3} = 0$ and $P_0 = 1$. This also provides an independent recursive definition of the polynomials. Indeed the operator

$$\mathcal{L}(P) = L(L-1)P(L) - (L-2)(L-3)P(L-3)$$

maps $\mathbb{Q}[L^2]$ to $(2L-3)\mathbb{Q}[L(L-3)]$. The first space consists of even polynomials; the second consists of polynomials anti-invariant under $L \mapsto 3-L$. In the displayed power bases, $\mathcal{L}(L^{2k})/(2L-3)$ has degree k and leading coefficient $3k+2$. Consequently this operator is an isomorphism of the filtered vector spaces over $\mathbb{Q}$. For even preceding polynomials, the right side of the recurrence is anti-invariant under $L \mapsto 3-L$; the two cubic terms exchange with opposite signs under this reflection. Thus the recurrence uniquely defines an even P_j at every step. It gives the degree bound

$$\deg P_j \leq \max\{\deg P_{j-1},\ \deg P_{j-2} + 2, \deg P_{j-3} + 4\} \leq 2\lfloor 2j/3 \rfloor.$$

At $L = 2$, evaluation of the polynomial recurrence gives the coefficients of

$$\sum_{j \geq 0} P_j(2)t^j = \frac{1}{1 - t - 4t^2 + 4t^3}.$$

This is exactly the normalized expansion of $F_0 = 1/q_0$. The recurrence in m now proves the moment identities for every integer $m \geq 0$. □

Prime-power valuation of the first moments. For $p > 3$, $p \equiv 2 \pmod 3$, and odd r, take $L = p^r$. Then

$$v_p(b_{(p^r-2)/3}) = \frac{r-1}{2}, \qquad v_p(C(p^r)) = \frac{7r-3}{2}.$$

For the first equality count multiples of p^ℓ in the numerator and denominator of b_m. If ℓ is even, the numerator acquires one additional multiple, while for odd ℓ the counts agree; there are $(r-1)/2$ positive even levels at most r. Therefore $M_1 = M_2 = 0$, and M_3, M_4, M_5, M_6 all have exactly the valuation $(7r-3)/2$ for these primes. This exceeds or equals the required $3r-1$.

Why a uniform denominator estimate is needed. The polynomial operator is not an isomorphism over $\mathbb{Z}[1/2]$: its diagonal is $3k+2$, not a unit there. For example

$$P_6(L) = \frac{L^8 + 290L^6 - 3325L^4 + 16666L^2 - 29900}{28},$$

and P_9 has denominator 280. Thus the rational-polynomial existence proof alone does not establish $P_j(p^r) \in \mathbb{Z}_p$ for all j at primes $p \equiv 2 \pmod 3$. The logarithmic bounds proved below supply the uniform estimate needed in this manuscript. They are used with finite polynomial separation, so no Euler-power limiting argument is required.

4 An exact negative-parameter companion and its first three moments

This section proves the exact finite companion identity used in the higher-level argument. The identity alone is not a supercongruence.

Let $M = 3n+1$, $n \geq 0$, and put

$$\mathcal{H}_n = \{3h+1 : 0 \leq h \leq n\} \cup \{3h+2 : 0 \leq h < n\} \cup \{-2(3h+2) : 0 \leq h < n\},$$

$$\mathcal{D}_n = \{6j, -3j : 1 \leq j \leq n\} \cup \{-3h-1 : 0 \leq h < n\}.$$

All $3n+1$ nodes in $\mathcal{H}_n$ are distinct and nonzero. Define

$$U(3h+1) = -(-1)^h(3h+1)/2, \qquad U(3h+2) = \begin{cases} 3h+2, & h \text{ odd}, \\ 0, & h \text{ even}, \end{cases} \qquad U(-2(3h+2)) = 3h+2,$$

$$D_n(z) = \prod_{D \in \mathcal{D}_n} (1 - z/D), \qquad Q_n(z) = \prod_{H \in \mathcal{H}_n} (1 - z/H), \qquad \widetilde{W}_n(H) = \frac{D_n(H)}{\prod_{K \in \mathcal{H}_n, K \neq H}(1 - H/K)}.$$

The companion and its positive moments are

$$\mathcal{A}(M) = \sum_H \widetilde{W}_n(H) U(H), \qquad \widetilde{M}_j(M) = \sum_H \widetilde{W}_n(H) U(H) H^j.$$

Set $d_k = (-1/3)_k/k!$ and $c_k = (2/3)_k/k!$.

Theorem 4.1. *For every integer $n \geq 0$,*

$$\begin{aligned} \mathcal{A}(3n+1) &= -\frac{1}{2}\sum_{k=0}^{n} (-1)^k (6k-1)(9k^2-3k-1) d_k^3 \\ &= \sum_{k=0}^{n} (-1)^k (6k-1) d_k^3 + \frac{(-1)^n}{2} c_n^3. \end{aligned}$$

Moreover, with $C_n = (-1)^{n+1} M^2 c_n^3/2$,

$$\widetilde{M}_1(M) = C_n, \qquad \widetilde{M}_2(M) = C_n(M^2+1)/2, \qquad \widetilde{M}_3(M) = C_n(2M^2-1).$$

4.1 A rational deformation with exactly the companion node values

Put

$$K_k(z)=\prod_{j=0}^{k-1}\frac{(1+z/[2(3j-1)])(1-z/(3j-1))(1-z/(3j+1))}{(1-z/[2(3j+3)])(1+z/(3j+3))(1+z/(3j+1))},$$

$$V_n(z)=-\frac{1}{2(2-z)}\sum_{k=0}^{n}(-1)^k(6k-1)(z+18k^2-6k-2)d_k^3K_k(z). \tag{4.1}$$

Although this expression displays $2-z$ in its denominator, its $k=0$ term is the constant $-1/2$. Every $k\geq 1$ term has the factor $1-z/2$ in its numerator, cancelling this additional denominator. Consequently $D_n(z)V_n(z)$ is a polynomial of degree at most $3n$. At $z=2$, the terms with $k\geq 2$ vanish after cancellation; the $k=1$ term simplifies to $-5(z-1)(z+10)/(6(z-6)(z+3))$ and has value $1/2$. Thus $V_n(2)=0$ for every $n\geq 1$.

The three finite certificate proofs supplied below establish

$$[D_n(z)V_n(z)]_{z=H}=D_n(H)U(H)\qquad(H\in\mathcal{H}_n). \tag{4.2}$$

On the first positive grid the sum terminates at h; on the second it terminates at $h+1$, with $z=2$ handled above. The lower parameters are regular on both positive grids. On the negative grid the finite certificate gives the sum through $h+1$; where a later denominator would vanish, the cleared identity is proved term by term. Thus (4.2) never assigns a value to a raw $0/0$ summand.

Interpolation of the degree-$3n$ polynomial D_nV_n at its $3n+1$ distinct nodes now proves

$$V_n(0)=\sum_H\widetilde{W}_n(H)U(H)=\mathcal{A}(M),\qquad F_n(z):=\frac{D_n(z)V_n(z)}{Q_n(z)}=\sum_H\frac{\widetilde{W}_n(H)U(H)}{1-z/H}.$$

Evaluation of (4.1) at zero proves the first equality in Theorem 4.1.

4.2 An elementary telescoping simplification

Let $u_k=(-1)^kd_k^3$ and $B_k=k^3u_k$. Since $B_{k+1}=-(k-1/3)^3u_k$, direct expansion gives

$$(6k-1)(9k^2-3k-1)u_k=-27(B_{k+1}-B_k)-2(6k-1)u_k.$$

Summing uses $B_0=0$ and $27B_{n+1}=(-1)^nc_n^3$ and proves the second equality of the theorem, without any asymptotic argument.

4.3 The adjacent-cutoff recurrence and the three moments

For $n\geq 1$, write

$$A_n(z)=(1-z/M)(1-z/(M-2))(1+z/[2(M-2)]),$$

$$B_n(z)=(1-z/[2(M-1)])(1+z/(M-1))(1+z/(M-3)).$$

These are precisely Q_n/Q_{n-1} and D_n/D_{n-1}. The numerator of $K_n(z)$ is $(1+z)(1-z/2)Q_{n-1}(z)$. Separating the last summand of (4.1) therefore gives the exact polynomial recurrence

$$A_n(z)F_n(z)-B_n(z)F_{n-1}(z)=G_n(z), \tag{4.3}$$

$$G_n(z)=-\frac{(-1)^n(6n-1)d_n^3}{4}(z+1)(z+18n^2-6n-2).$$

This derivation uses the summand formula, not the moment identities being proved.

At infinity, write $F_n(z) = -\sum_{j\ge1} \widetilde{M}_j(M) z^{-j}$. The coefficients of z^2, z^1, z^0 in (4.3) determine its first three moments successively. For complete verification, write $A_n = \sum a_i z^i$, $B_n = \sum b_i z^i$, and $G_n = \sum g_i z^i$. Those three equations are

$$-a_3\widetilde{M}_1 + b_3\widetilde{M}_1' = g_2,$$

$$-a_2\widetilde{M}_1 - a_3\widetilde{M}_2 + b_2\widetilde{M}_1' + b_3\widetilde{M}_2' = g_1,$$

$$-a_1\widetilde{M}_1 - a_2\widetilde{M}_2 - a_3\widetilde{M}_3 + b_1\widetilde{M}_1' + b_2\widetilde{M}_2' + b_3\widetilde{M}_3' = g_0,$$

where primes mean the preceding cutoff $M-3$. Here

$$a_3 = \frac{1}{2M(M-2)^2}, \quad a_2 = -2a_3, \quad a_1 = -\frac{3M-4}{2M(M-2)},$$

$$b_3 = -\frac{1}{2(M-1)^2(M-3)}, \quad b_2 = -2b_3, \quad b_1 = \frac{3M-5}{2(M-1)(M-3)}.$$

Substitution of the three claimed formulas satisfies these rational identities, using

$$\frac{C_{n-1}}{C_n} = -\frac{(M-3)^2(M-1)^3}{M^2(M-2)^3}.$$

At $n=0$, $F_0(z) = -1/[2(1-z)]$, so every positive moment is $-1/2$, exactly the claimed initial values. Since $a_3 \neq 0$ for $n \ge 1$, induction proves all three formulas. This completes the theorem.

5 A uniform logarithmic valuation bound for the primary E moments

Let $P_j(L) \in \mathbb{Q}[L^2]$ be the primary moment polynomials defined by the exact three-grid recurrence, normalized by $P_0 = 1$. They satisfy

$$\deg P_j \le d_j := 2\left\lfloor \frac{2j}{3} \right\rfloor, \qquad M_{j+3}(L) = C(L)P_j(L),$$

where, for $L = 3m+2$,

$$C(L) = (-1)^{m+1}L^2(L-1)^3 b_m^3, \qquad b_m = \frac{(1/3)_m}{m!}.$$

For a prime $p > 3$, $p \equiv 2 \pmod 3$, define

$$a_j = \left\lfloor \log_p(3d_j+2) \right\rfloor, \qquad e_j = a_j + \left\lfloor \frac{a_j}{2} \right\rfloor.$$

Here the logarithm only denotes the integer characterized by $p^{a_j} \le 3d_j + 2 < p^{a_j+1}$.

Theorem 5.1. *For every $j \ge 0$ and every $x \in \mathbb{Z}_p$,*

$$\boxed{v_p(P_j(x)) \ge -e_j.}$$

Equivalently, $p^{e_j}P_j$ is integer-valued on $\mathbb{Z}_p$. Moreover $e_0 = 0$, and $e_j \le j-1$ for every $j \ge 1$.

This is an all-degree theorem with a logarithmic loss. It does not assert the stronger, still unproved statement $P_j \in \mathrm{Int}(\mathbb{Z}_p)$ for all j.

5.1 An integral-coefficient formula before normalization

For $h \geq 0$, put

$$\widehat{q}_h(t) = (1-(3h+1)t)(1-(3h+2)t)(1+(6h+2)t),$$
$$\widehat{d}_h(t) = (1-(6h+6)t)(1+(3h+3)t)(1+(3h+2)t).$$

For $0 \leq k \leq m$, define integers $\phi_j(k,m)$ by

$$\frac{\prod_{h=k}^{m-1} \widehat{d}_h(t)}{\prod_{h=k}^{m} \widehat{q}_h(t)} = \sum_{j \geq 0} \phi_j(k,m) t^j.$$

All these coefficients are integers because every denominator polynomial has constant term one. The exact normalized moment formula is

$$P_j(3m+2) = \frac{2}{L(L-1)b_m} \sum_{k=0}^{m} (6k+1) b_k \phi_j(k,m), \qquad L = 3m+2. \tag{5.1}$$

To verify it directly, expand the proper rational function $F_m = D_m S_m / Q_m$ term by term at $z = \infty$, using $t = 1/z$. The kth summand is

$$(-1)^k (6k+1) b_k^3 \frac{\prod_{h=k}^{m-1} d_h(z)}{\prod_{h=k}^{m} q_h(z)}.$$

Its leading coefficient is $(-1)^m 2L(L-1)^2 b_m^2 (6k+1) b_k z^{-3}$. Dividing the reversed numerator and denominator polynomials by their leading coefficients gives exactly the series defining $\phi_j(k,m)$. Comparison with $F_m = -C(L) \sum_{j \geq 0} P_j(L) z^{-j-3}$ proves (5.1). In particular this argument holds for every coefficient, not only for any prescribed finite range.

5.2 Valuation of the normalizing denominator

Lemma 5.2. *Let $m \geq 0$, $L = 3m+2$, and $a = \lfloor \log_p L \rfloor$. Then*

$$b_k \in \mathbb{Z}_p \quad (k \geq 0), \qquad v_p(L(L-1)b_m) \leq a + \lfloor a/2 \rfloor.$$

Proof. At the level $P = p^\ell$, let u_ℓ be the least nonnegative solution of $3u_\ell + 1 \equiv 0 \pmod{P}$. Explicitly,

$$u_\ell = \begin{cases} (P-1)/3, & \ell \text{ even}, \\ (2P-1)/3, & \ell \text{ odd}. \end{cases}$$

The difference between the counts of multiples of P in $\prod_{h=0}^{m-1}(3h+1)$ and in $m!$ is

$$\delta_\ell = \begin{cases} 1, & m \bmod P > u_\ell, \\ 0, & m \bmod P \leq u_\ell. \end{cases}$$

The factors of 3 in b_m are units. Thus $v_p(b_m) = \sum_{\ell \geq 1} \delta_\ell \geq 0$, proving integrality. Also $\delta_\ell = 0$ for $p^\ell > L$: all positive numerator factors are at most $L-4$ and all denominator indices are at most m.

If $t = v_p(L) > 0$, then for $\ell \leq t$ the congruence $3m \equiv -2 \pmod{p^\ell}$ gives

$$m \bmod p^\ell = \begin{cases} 2(p^\ell - 1)/3, & \ell \text{ even}, \\ (p^\ell - 2)/3, & \ell \text{ odd}. \end{cases}$$

Consequently $\delta_\ell = 1$ precisely at the even levels among $1, \dots, t$. Since $L-1$ is a unit,

$$v_p(L(L-1)b_m) \le t + \lfloor t/2 \rfloor + (a-t) = a + \lfloor t/2 \rfloor \le a + \lfloor a/2 \rfloor.$$

If instead $u = v_p(L-1) > 0$, then $m \bmod p^\ell = u_\ell$ for $\ell \le u$. All those δ_ℓ are zero, and $v_p(L(L-1)b_m) \le u + (a-u) = a$. If both L and $L-1$ are units, the same bound follows from $\sum \delta_\ell \le a$. These cases include $m = 0$, where $L = 2$ and $a = 0$. □

Proof of Theorem 5.1. Formula (5.1) and the preceding lemma imply, for every nonnegative integer m,

$$v_p(P_j(3m+2)) \ge -\left(\lfloor \log_p(3m+2) \rfloor + \left\lfloor \frac{\lfloor \log_p(3m+2) \rfloor}{2} \right\rfloor \right).$$

Use this for $m = 0, 1, \dots, d_j$. Every one of these values lies in $p^{-e_j}\mathbb{Z}_p$. Set $R_j(X) = P_j(3X+2)$, whose degree is at most d_j. Its exact Newton expansion is

$$R_j(X) = \sum_{k=0}^{d_j} \Delta^k R_j(0) \binom{X}{k}.$$

Each finite difference is an integer linear combination of the sampled values, so it too belongs to $p^{-e_j}\mathbb{Z}_p$. For every $X \in \mathbb{Z}_p$, the binomial polynomial $\binom{X}{k}$ is p-integral, by continuity from the nonnegative integers. Taking $X = (x-2)/3 \in \mathbb{Z}_p$ proves the uniform bound. No division by $k!$ is being applied to the finite-difference coefficient: that factorial belongs to the integer-valued basis.

Finally, for $j \ge 1$ the elementary inequality

$$6\lfloor 2j/3 \rfloor + 2 < 5^{\lfloor (2j-1)/3 \rfloor + 1}$$

follows by checking $j = 1, 2, 3$ and then increasing j by three; the left side increases by 12, while the right side is multiplied by 25. Since $p \ge 5$, this implies $a_j \le \lfloor (2j-1)/3 \rfloor$, and hence $a_j + \lfloor a_j/2 \rfloor \le j-1$. Also $d_0 = 0$, so $e_0 = 0$. □

5.3 The double-strip transport is controlled at every higher odd level

Corollary 5.3. *Let $r \ge 3$ be odd, $M = p^{r-2}$, and let $c_j \in \mathbb{Q}_p$ satisfy $v_p(c_j) \ge j$ for every $j \ge 1$. Then the series*

$$V_r = p^2 \sum_{j \ge 1} c_j M_j(M)$$

converges. For every odd $r \ge 5$,

$$V_r \in p^{3r-1}\mathbb{Z}_p.$$

At the remaining level $r = 3$, one has the precise reduction

$$V_3 \equiv p^2 c_3 M_3(p) \pmod{p^8}.$$

Proof. The first two moments vanish exactly. For $j \ge 3$, put $n = j-3$. The exact primary moment normalization and the proven valuation $v_p(C(p^{r-2})) = (7r-17)/2$ give

$$v_p(p^2 c_j M_j(M)) \ge 2 + j + \frac{7r-17}{2} - e_n = \frac{7r-7}{2} + n - e_n.$$

The right side tends to infinity, because $e_n = O(\log n)$, so convergence follows. For $n = 0$ it is at least $(7r-7)/2$; for $n \ge 1$ it is at least $(7r-5)/2$. When $r \ge 5$, the former bound is at least $3r-1$. When $r = 3$, all terms with $n \ge 1$ belong to $p^8\mathbb{Z}_p$, leaving only the displayed cubic term. □

6 Polynomial structure of every companion moment

Let $\widetilde{M}_j(M)$ be the companion moments from the proved negative-parameter companion identity, and put

$$C_n = \frac{(-1)^{n+1}M^2}{2}\left(\frac{(2/3)_n}{n!}\right)^3, \qquad M = 3n+1.$$

Proposition 6.1. *There exist even polynomials $R_j(M) \in \mathbb{Q}[M^2]$, $j \geq 0$, with*

$$\deg R_j \leq 2\left\lfloor \frac{2j+1}{3} \right\rfloor, \qquad R_j(1) = 1,$$

such that, for every integer $n \geq 0$,

$$\widetilde{M}_{j+1}(3n+1) = C_n R_j(3n+1).$$

Their first values are $R_0 = 1$, $R_1 = (M^2+1)/2$, and $R_2 = 2M^2 - 1$.

Proof. Start with the already proved rational-function recurrence $A_n(z)F_n(z) - B_n(z)F_{n-1}(z) = G_n(z)$, where G_n has degree two. Its first three Laurent equations give the three initial moments just stated. Set

$$A = M(M-2), \quad B = (M-3)(M-1), \quad a = (M-2)(3M-4), \quad c = (M-1)(3M-5).$$

The monic forms of its cubic factors are

$$A_n(z) = \frac{z^3 - 2z^2 - az + 2M(M-2)^2}{2M(M-2)^2},$$

$$B_n(z) = -\frac{z^3 - 2z^2 - cz - 2(M-3)(M-1)^2}{2(M-3)(M-1)^2}.$$

Use the exact ratio

$$\frac{C_{n-1}}{C_n} = -\frac{(M-3)^2(M-1)^3}{M^2(M-2)^3}.$$

The homogeneous Laurent equations, starting at the coefficient of z^{-1}, yield, for every $j \geq 3$,

$$\begin{aligned} AR_j(M) - BR_j(M-3) = {} & 2\big[AR_{j-1}(M) - BR_{j-1}(M-3)\big] \\ & + AaR_{j-2}(M) - BcR_{j-2}(M-3) \\ & - 2AM(M-2)^2R_{j-3}(M) - 2B(M-3)(M-1)^2R_{j-3}(M-3). \end{aligned}$$

We show this recurrence uniquely defines even rational polynomials. The operator

$$\mathcal{L}'(P) = M(M-2)P(M) - (M-3)(M-1)P(M-3)$$

maps $\mathbb{Q}[M^2]$ to $(2M-3)\mathbb{Q}[M(M-3)]$. Indeed its image is anti-invariant under $M \mapsto 3-M$. For the basis element M^{2k}, division of its image by $2M-3$ gives a polynomial of degree k in $M(M-3)$, with leading coefficient $3k+1$. Thus $\mathcal{L}'$ is an isomorphism of the filtered vector spaces over $\mathbb{Q}$. The right side of the displayed recurrence is anti-invariant when the preceding polynomials are even. It therefore has a unique even preimage. Its degree is bounded by

$$\deg R_j \leq \max\{\deg R_{j-1}, \deg R_{j-2} + 2, \deg R_{j-3} + 4\}.$$

The given initial degrees imply the asserted bound. At $M = 1$ the recurrence reduces to

$$R_j(1) = 2R_{j-1}(1) + R_{j-2}(1) - 2R_{j-3}(1),$$

so all these values are one. This agrees with the exact base $F_0(z) = -1/[2(1-z)]$, whose every positive moment equals $-1/2$. Finally the Laurent recurrence determines the moments uniquely successively in the cutoff and the moment index, since its leading cubic coefficient is nonzero for $n \geq 1$. Hence the constructed polynomials give the actual moments for all n and j. □

The diagonal $3k+1$ need not be a p-adic unit, so this proposition asserts rational-polynomial structure and a uniform degree bound; it does not assert coefficient integrality or integer-valuedness.

7 A uniform logarithmic loss for all companion moments

Use the polynomials R_j of Proposition 6.1, so that

$$\widetilde{M}_{j+1}(3n+1) = C_n R_j(3n+1), \qquad d_j := 2\left\lfloor \frac{2j+1}{3} \right\rfloor \geq \deg R_j.$$

For $p > 3$, $p \equiv 2 \pmod 3$, define

$$a_j = \left\lfloor \log_p(3d_j+1) \right\rfloor, \qquad \eta_j = a_j + \left\lceil \frac{a_j}{2} \right\rceil = \left\lceil \frac{3a_j}{2} \right\rceil.$$

Theorem 7.1. *For every $j \geq 0$ and every $x \in \mathbb{Z}_p$,*

$$v_p(R_j(x)) \geq -\eta_j.$$

In particular, if $M = p^{2s}$, then for every $j \geq 1$,

$$v_p(\widetilde{M}_j(M)) \geq 7s - e_p(j), \qquad e_p(j) = \left\lceil \frac{3}{2}\left\lfloor \log_p(4j+1) \right\rfloor \right\rceil.$$

Proof. We first prove a finite-value formula with integral coefficients before normalization. Write $w = 1/z$. For each integer root a, reversing the factor $1 - z/a$ gives its leading monomial times $1 - aw$. Thus the reversed products in D_n/Q_n and in each K_k have integer coefficients as formal series in w: their denominator factors all have constant term one and integral coefficients. The additional factor $(z + 18k^2 - 6k - 2)/(z-2)$ becomes

$$\frac{1 + (18k^2 - 6k - 2)w}{1 - 2w} \in \mathbb{Z}[[w]].$$

The leading coefficient of K_k is $(-1)^k/d_k^2$, where $d_k = (-1/3)_k/k!$. Also

$$\frac{D_n(z)}{Q_n(z)} = -\nu_n w\,(1 + O(w)), \qquad \nu_n = (-1)^n M c_n^2, \quad M = 3n+1, \quad c_n = (2/3)_n/n!.$$

It follows from the proved summand formula for V_n that there exist integers $\phi_j(k,n)$, for every $j \geq 0$, such that

$$F_n(z) = -\frac{\nu_n}{2} \sum_{j \geq 0} \left(\sum_{k=0}^{n} (6k-1) d_k \phi_j(k,n) \right) w^{j+1}.$$

Here ϕ_j is simply the coefficient of the product of the reversed factors just described, so integrality holds for every degree, not only for a finite initial range. Since $C_n = -\nu_n M c_n/2$, comparison of Laurent coefficients proves the exact formula

$$R_j(3n+1) = -\frac{1}{Mc_n}\sum_{k=0}^{n}(6k-1)d_k\phi_j(k,n). \tag{7.1}$$

Both d_k and c_k belong to $\mathbb{Z}_p$: for any $\alpha \in \mathbb{Z}_p$, $(\alpha)_k/k! = (-1)^k\binom{-\alpha}{k}$ is integral. Put $b = \lfloor \log_p M \rfloor$ and $t = v_p(M)$. At the level $q = p^\ell$, the excess numerator count in c_n is zero or one; it is zero when $q > M$. For $\ell \le t$, the residue of $n = (M-1)/3$ is $(q-1)/3$ when ℓ is even, and $(2q-1)/3$ when ℓ is odd. The first numerator index for $3k+2 \equiv 0 \pmod q$ is respectively $(2q-2)/3$ and $(q-2)/3$. Hence exactly the odd levels among $1,\dots,t$ contribute an excess. The remaining $b-t$ levels each contribute at most one. Consequently

$$v_p(Mc_n) \le t + \lceil t/2 \rceil + (b-t) \le b + \lceil b/2 \rceil.$$

Equation (7.1) therefore bounds every finite value $R_j(3n+1)$ by the negative of this quantity.

Apply this bound at $n = 0, 1, \dots, d_j$. All these values belong to $p^{-\eta_j}\mathbb{Z}_p$. The polynomial $T_j(X) = R_j(3X+1)$ has exact Newton expansion

$$T_j(X) = \sum_{k=0}^{d_j} \Delta^k T_j(0)\binom{X}{k}.$$

Each finite difference is an integer linear combination of the sampled values. Each $\binom{X}{k}$ is integral on $\mathbb{Z}_p$, by continuity from the nonnegative integers. Taking $X = (x-1)/3$ proves the uniform bound for $R_j(x)$.

Finally, at $M = p^{2s}$ the excess-count argument gives $v_p(c_n) = s$ exactly, so $v_p(C_n) = 7s$. For the moment with index j, its polynomial is R_{j-1}, and

$$3d_{j-1} + 1 = 6\left\lfloor \frac{2j-1}{3} \right\rfloor + 1 \le 4j - 1 \le 4j + 1.$$

Thus $\eta_{j-1} \le e_p(j)$, proving the last assertion. □

8 A uniform lower bound for every companion interpolation mass

Let $p > 3$, $p \equiv 2 \pmod 3$, $s \ge 1$, $M = p^{2s}$ and $n = (M-1)/3$. Retain the companion grids, denominator roots, and models from Theorem 4.1, and put

$$\mu_M(H) = \widetilde{W}_n(H)U(H).$$

Proposition 8.1. *For every companion node H, with $t = v_p(H)$,*

$$\boxed{v_p(\mu_M(H)) \ge 3s - \lceil t/2 \rceil \ge 2s.}$$

Zero masses have valuation $+\infty$. In particular every unit-node mass belongs to $p^{3s}\mathbb{Z}_p$.

Proof. We may suppose $U(H) \ne 0$ and $H \notin \mathcal{D}_n$; otherwise the mass is zero. At an active node, $U(H)/H$ is one of $1, 1/2, -1/2$, hence is a unit. Direct multiplication of the grids gives

$$\frac{\prod_{K\in\mathcal{H}_n} K}{\prod_{D\in\mathcal{D}_n} D} = (-1)^n M\left(\frac{(2/3)_n}{n!}\right)^2.$$

The Pochhammer ratio has valuation s: at each odd level p^ℓ, $1 \le \ell \le 2s$, its numerator has one extra divisible factor, and at each even level its numerator and denominator counts agree. Thus the displayed normalization has valuation $4s$. The exact mass, up to an irrelevant sign, is this normalization times

$$\frac{U(H)}{H}\frac{\prod_{D\in\mathcal{D}_n}(D-H)}{\prod_{K\in\mathcal{H}_n,K\neq H}(K-H)}.$$

All nonzero differences here have absolute value less than $4M$. Since $p \ge 5$, no power p^ℓ with $\ell > 2s$ divides any of them.

For $q = p^\ell$ and $x = H \bmod q$, define

$$\delta_q(x) = 1 + \#\{D \in \mathcal{D}_n : D \equiv x \pmod q\} - \#\{K \in \mathcal{H}_n : K \equiv x \pmod q\}.$$

The initial one removes the node $K = H$ from the denominator count. Consequently

$$v_p(\mu_M(H)) = 4s + \sum_{\ell=1}^{2s} \delta_{p^\ell}(H \bmod p^\ell). \tag{8.1}$$

All three arithmetic progressions have steps prime to q. Their complete blocks of q consecutive indices contribute equally to the numerator and denominator counts and cancel. If ℓ is even, the remaining cutoff is $b = (q-1)/3$; if ℓ is odd, it is $b = (2q-1)/3$. The residual grids and denominators are exactly the companion sets with $n = b$, respectively $M' = q$ or $M' = 2q$.

Here are the complete residual counts for nonzero residues $1 \le x < q$, listed by x mod 6:

$x \bmod 6$	0	1	2	3	4	5
$\#K$ $(q \equiv 1 \pmod 6)$	0	1	1	1	2	1
$\#D$ $(q \equiv 1 \pmod 6)$	2	1	0	1	1	1
δ_q $(q \equiv 1 \pmod 6)$	3	1	0	1	0	1
$\#K$ $(q \equiv 5 \pmod 6)$	2	2	2	1	2	3
$\#D$ $(q \equiv 5 \pmod 6)$	2	3	2	2	2	1
δ_q $(q \equiv 5 \pmod 6)$	1	2	1	2	1	−1

These counts follow directly by solving the three progression congruences in the residual ranges. For example, in the even case $M' = q$, the negative node grid contributes once for residues $x \equiv 3, 4 \pmod 6$, while the positive denominator grid contributes once for $x \equiv 0, 5 \pmod 6$. In the odd case $M' = 2q$, those same contributions are respectively $(1,1,0,0,1,1)$ and $(1,1,1,1,0,0)$ across the six residue classes. The other two grids have step three, yielding the displayed totals. At the zero residue, the even case has one node and no denominator root, so $\delta_q(0) = 0$; the odd case has three nodes and no denominator root, so $\delta_q(0) = -2$.

Thus every even level contributes a nonnegative amount. Each odd level contributes at least -1, with a possible extra loss of one precisely when $p^\ell \mid H$. There are s odd levels altogether and $\lceil t/2 \rceil$ of them divide H; here $t \le 2s$ because $0 < |H| < 2M < p^{2s+1}$. Equation (8.1) proves

$$v_p(\mu_M(H)) \ge 4s - s - \lceil t/2 \rceil,$$

as asserted. □

This is an individual-mass estimate. It is independent of the unresolved cancellation in E.3 and can be combined with exact moment formulas or polynomial projectors without assuming any positive-moment integrality.

8.1 The corresponding primary mass bound

Let $L = p^{2s+1}$, $m = (L-2)/3$, and let $\mu_L(H)$ be the primary three-grid mass. Then

$$\boxed{v_p(\mu_L(H)) \geq 3s + 1 - \lfloor v_p(H)/2 \rfloor \geq 2s + 1.} \tag{8.2}$$

Again zero masses are harmless. We give the full analogous count, so this bound does not rely on numerical interpolation tests. The primary normalization is

$$\frac{\prod K}{\prod D} = (-1)^{m+1} 2L(L-1)^2 \left(\frac{(1/3)_m}{m!} \right)^2 .$$

Its valuation is $4s+1$, since the Pochhammer ratio has valuation s. At an active node $T(H)/H$ is a unit, and all nonzero differences have absolute value less than $4L < pL$. The same formula $1 + \#D_x - \#K_x$ gives each level's correction. For odd ℓ, the residual cutoff is $(q-2)/3$ with $q = p^\ell$; for even ℓ, it is $(2q-2)/3$. The corresponding residual primary endpoint is q or $2q$ respectively. Complete blocks cancel as before. For $1 \leq x < q$, the exact tables are

$x \bmod 6$	0	1	2	3	4	5
$\#K$ (ℓ odd)	0	1	2	1	1	1
$\#D$ (ℓ odd)	2	1	1	1	0	1
δ_q (ℓ odd)	3	1	0	1	0	1
$\#K$ (ℓ even)	2	3	2	1	2	2
$\#D$ (ℓ even)	2	1	2	2	2	3
δ_q (ℓ even)	1	-1	1	2	1	2

For example, at an odd level the negative node grid contributes in residue classes $2, 3 \pmod 6$, while the positive denominator grid contributes in classes $0, 1 \pmod 6$. At an even level their respective six-class contributions are $(1,1,1,0,0,1)$ and $(1,0,0,1,1,1)$. The remaining step-three grids give the totals shown above. At $x = 0$ the correction is zero for odd ℓ and -2 for even ℓ: there are respectively one or three nodes and no denominator root at that residue. Thus odd levels cost nothing, while each of the s even levels costs at most one, plus one more when it divides H. There are $\lfloor v_p(H)/2 \rfloor$ such divisible even levels. Subtracting these losses from $4s+1$ proves (8.2). In particular unit-node masses belong to $p^{3s+1}\mathbb{Z}_p$.

8.2 The unit-node contribution at level three

Corollary 8.2. *For every prime $p > 3$ with $p \equiv 2 \pmod 3$,*

$$\sum_{H \in \mathcal{H}_{(p^3-2)/3},\ p \nmid H} \mu_{p^3}(H) \in p^8 \mathbb{Z}_p.$$

Proof. First assume $p \geq 41$. Let $g(Y) = 3Y^2 - 2Y^3$ and $\Pi(X) = g(g(X^{p-1}))$. This polynomial has integer coefficients, no terms of degree below $4(p-1)$, and degree $9(p-1)$. For a unit integer H, Fermat's congruence and $g(1+u) = 1 - 3u^2 - 2u^3$ give $\Pi(H) \equiv 1 \pmod{p^4}$. For $p \mid H$, one has $\Pi(H) \in p^{4(p-1)}\mathbb{Z}_p$. The preceding proposition, with $s = 1$, therefore gives

$$\sum_{p \nmid H} \mu_{p^3}(H) \equiv \sum_{H} \mu_{p^3}(H)\Pi(H) \pmod{p^8}.$$

The right-hand side is an integer linear combination of the moments $M_j(p^3)$ with $4(p-1) \leq j \leq 9(p-1)$. Their common normalization $C(p^3)$ has valuation nine. For every such j, the logarithmic moment bound in Theorem 5.1 has loss at most one: indeed, putting $n = j - 3$,

$$3d_n + 2 = 6\lfloor 2n/3 \rfloor + 2 \leq 4j - 10 \leq 36(p-1) - 10 < p^2 \qquad (p \geq 41).$$

Consequently $M_j(p^3) \in p^8\mathbb{Z}_p$.

The remaining primes are exactly $5, 11, 17, 23, 29$. They can be checked by finite integer arithmetic in the exact mass formula above. Here is an explicit modular certificate, computed modulo p^{10}:

p	5	11	17	23	29
$p^{-9}\sum_{p\nmid H}\mu_{p^3}(H) \pmod p$	1	6	6	18	10.

In particular every one of those sums lies in $p^9\mathbb{Z}_p$. The reproducible certificate is fast_primary_masses.py: it evaluates the displayed product formula modulo p^{10}, removing the exact power of p from every factor before taking a modular inverse. Prefix products in each residue class modulo three and six evaluate every arithmetic progression in the formula. A zero factor is omitted only at the unique denominator occurrence $K = H$; numerator zeroes are excluded by testing $H \in \mathcal{D}_m$ first. Thus all modular inverses are applied to units and the finite checks are exact. They complete the five exceptional cases. □

The corollary controls only the unit-node contribution. The shell $v_p(H) = 1$ and its cancellation with the cubic double-strip term remain separate obligations.

9 One-step transport of the valuation-one stratum

Let $L = p^{2s+1}$, $M = L/p = p^{2s}$, $p > 3$, $p \equiv 2 \pmod 3$. Write $\mu_L(H)$ for the primary interpolation masses and $\widetilde{\mu}_M(J)$ for the proved companion masses. Let $\mathcal{H}_L, \mathcal{D}_L$ denote the primary root sets and set

$$R^{(1)}_{L,p}(X) = \frac{\prod_{d\in\mathcal{D}_L,\, p\nmid d}(1 - pX/d)}{\prod_{h\in\mathcal{H}_L,\, p\nmid h}(1 - pX/h)} = \sum_{j\ge 0} a_j X^j.$$

Every retained root is a unit, so $a_0 = 1$ and $a_j \in p^j\mathbb{Z}_p$. The series is restricted and converges at every integral node.

The exact one-step product identity is

$$\mu_L(pJ) = p\widetilde{\mu}_M(J)R^{(1)}_{L,p}(J). \tag{9.1}$$

Indeed dividing the divisible primary nodes by p gives respectively

$$\{2, 5, \dots, M-2\}, \quad \{1, 4, \dots, M\}, \quad -2\{2, 5, \dots, M-2\},$$

exactly the companion nodes. Dividing the divisible denominator roots gives $6j, -3j$ for $1 \le j \le (M-1)/3$ and $-1, -4, \dots, -(M-3)$, exactly the companion denominator roots. For the first two denominator grids this follows from $\lfloor (L-2)/(3p)\rfloor = (M-1)/3$; the last grid has fine endpoint $-(L-3)$, so its largest divisible quotient is $-(M-3)$. Since $p \equiv 5 \pmod 6$, the primary model formulas give $T(pJ) = pU(J)$ on each companion grid, including zero model values. Separating the divisible and unit factors in the interpolation products proves (9.1); no division by a zero weight is used.

In particular, the omitted part of the two-step transport has the exact split

$$U_{L,p} = \sum_{p\nmid H} \mu_L(H) + p\sum_{p\nmid J} \widetilde{\mu}_M(J)R^{(1)}_{L,p}(J). \tag{9.2}$$

This identity places the formerly correlated valuation-one stratum inside a weighted unit sum of the exact companion family.

10 The original odd E.3 congruence at all sufficiently high levels

Let $p > 3$ be a prime with $p \equiv 2 \pmod 3$, and define

$$E(L) = \sum_{k=0}^{(L-2)/3} (-1)^k (6k+1) \left(\frac{(1/3)_k}{k!} \right)^3 \qquad (L \equiv 2 \pmod 3).$$

All interpolation identities and moment bounds used below are exact or have been proved in the preceding sections. In particular, no assertion of integral normalized moments of every degree is required.

Theorem 10.1. *Put $r = 2s+1$, with $s \geq 2$, and suppose*

$$\left\lceil \frac{3}{2} \left\lfloor \log_p(4(3s+1)(p-1)+1) \right\rfloor \right\rceil \leq s-1. \tag{10.1}$$

Then the original E.3 congruence holds:

$$\boxed{E(p^{2s+1}) \equiv p^2 E(p^{2s-1}) \pmod{p^{6s+2}}.}$$

In particular it holds in each of the following ranges:

1. *every $p > 3$, $p \equiv 2 \pmod 3$, and every odd $r \geq 13$;*
2. *every such $p \geq 11$ and every odd $r \geq 9$;*
3. *every such $p \geq 41$ and every odd $r \geq 7$.*

10.1 The two exact one-step identities

Set $N = p^{2s+1}$, $M = p^{2s}$, and $K = p^{2s-1}$. Write $\mu_N(H)$ and $\mu_K(H)$ for the primary three-grid masses, and $\widetilde{\mu}_M(J)$ for the companion masses. Their sums are $E(N)$, $E(K)$, and $\mathcal{A}(M)$, respectively. Their positive moments are denoted by M_j and $\widetilde{M}_j$.

For either grid family, form the quotient of unit denominator-root factors by unit node factors, evaluated at pX:

$$R(X) = \frac{\prod_{d \in \mathcal{D},\ p \nmid d} (1 - pX/d)}{\prod_{h \in \mathcal{H},\ p \nmid h} (1 - pX/h)}.$$

Let R_0 use the primary family at N, and R_1 the companion family at M. Both have expansions

$$R_i(X) = 1 + \sum_{j \geq 1} a_{i,j} X^j, \qquad a_{i,j} \in p^j \mathbb{Z}_p \quad (i = 0, 1). \tag{10.2}$$

These are restricted power series and converge at every $X \in \mathbb{Z}_p$. Separating divisible and unit roots in the interpolation weights gives

$$\mu_N(pJ) = p\widetilde{\mu}_M(J) R_0(J), \qquad \widetilde{\mu}_M(pH) = p\mu_K(H) R_1(H). \tag{10.3}$$

Here the correspondence between divisible nodes and the lower family is exact, including endpoints and zero model values.

For completeness, the first correspondence is the one established in (9.1). For the reverse correspondence, divide the divisible companion nodes by p. The positive grids become $\{2, 5, \ldots, K\}$ and $\{1, 4, \ldots, K-1\}$, and the negative grid becomes $-2\{1, 4, \ldots, K-1\}$, precisely the primary grids at K. The divided denominator roots are

$$\{6j, -3j : 1 \leq j \leq (K-2)/3\} \ \cup \ \{-2, -5, \ldots, -(K-3)\},$$

also precisely the primary denominator roots. The explicit models satisfy $U(pH) = pT(H)$ on these grids because $p \equiv 5 \pmod 6$. Thus the same product separation proves the second identity in (10.3), without dividing by a model value that might vanish.

10.2 Moment estimates in the forms needed for transport

The proved primary normalizations and logarithmic bounds give

$$M_1(N) = M_2(N) = M_1(K) = M_2(K) = 0, \tag{10.4}$$

$$v_p(M_j(N)) \geq 7s + 2 - e_{j-3} \qquad (j \geq 3), \tag{10.5}$$

$$v_p(M_j(K)) \geq 7s - 5 - e_{j-3} \qquad (j \geq 3), \tag{10.6}$$

where $e_0 = 0$, $e_h \leq h - 1$ for $h \geq 1$, and $e_h = O(\log(h+2))$. The companion logarithmic bound gives

$$v_p(\widetilde{M}_j(M)) \geq 7s - \eta_{j-1}.$$

In particular,

$$v_p(\widetilde{M}_j(M)) \geq 7s - (j-1) \qquad (j \geq 1). \tag{10.7}$$

For $j = 1, 2$ this follows from the exact normalized polynomials $R_0^{\mathrm{mom}} = 1$ and $R_1^{\mathrm{mom}} = (M^2+1)/2$. For the other indices use $\eta_h \leq h$ for $h \geq 2$. To verify this last elementary inequality, the companion definition is

$$\eta_h = \left\lceil \frac{3}{2} \left\lfloor \log_p \left(6 \left\lfloor \frac{2h+1}{3} \right\rfloor + 1 \right) \right\rfloor \right\rceil.$$

The inequality

$$6 \left\lfloor \frac{2h+1}{3} \right\rfloor + 1 < 5^{\lfloor 2h/3 \rfloor + 1}$$

holds at $h = 2, 3, 4$ and is preserved by increasing h by three: the left side increases by 12, whereas the right side is multiplied by 25. Since $p \geq 5$, it implies $\eta_h \leq h$.

A common nondecreasing logarithmic bound is

$$\epsilon_p(j) = \left\lceil \frac{3}{2} \left\lfloor \log_p(4j+1) \right\rfloor \right\rceil. \tag{10.8}$$

For $j \geq 3$ it bounds e_{j-3}, and for $j \geq 1$ it bounds η_{j-1}. Consequently the positive moments of the two fine families satisfy

$$v_p(M_j(N)) \geq 7s + 2 - \epsilon_p(j), \qquad v_p(\widetilde{M}_j(M)) \geq 7s - \epsilon_p(j).$$

The first inequality also holds for $j = 1, 2$, whose moments vanish. The proved individual unit-node estimates are

$$v_p(\mu_N(H)) \geq 3s + 1 \ (p \nmid H), \qquad v_p(\widetilde{\mu}_M(J)) \geq 3s \ (p \nmid J). \tag{10.9}$$

10.3 A low-degree unit projector and its leakage

Put

$$t = 3s + 1, \qquad D = t(p-1), \qquad \chi_t(X) = 1 - (1 - X^{p-1})^t \in \mathbb{Z}[X].$$

Its nonzero monomials have degrees $i(p-1)$ for $1 \leq i \leq t$, so its smallest degree is $p - 1 \geq 4$ and its degree is D. For every integral unit x, Fermat's congruence gives $1 - \chi_t(x) \in p^t \mathbb{Z}_p$. Let $B = \epsilon_p(D)$; the theorem assumes $B \leq s - 1$.

First consider the primary unit sum $U_N = \sum_{p \nmid H} \mu_N(H)$. Replacing 1 by $\chi_t(H)$ on its unit nodes changes it by an element of $p^{6s+2}\mathbb{Z}_p$, by (10.9). The full-node polynomial sum satisfies

$$\sum_H \mu_N(H)\chi_t(H) \in p^{7s+2-B}\mathbb{Z}_p \subseteq p^{6s+3}\mathbb{Z}_p.$$

Its contribution from nonunit nodes is, by the first exact strip,

$$\sum_{p|H} \mu_N(H)\chi_t(H) = p\sum_J \widetilde{\mu}_M(J)R_0(J)\chi_t(pJ).$$

Write $R_0(X)\chi_t(pX) = \sum_{j\geq p-1} b_j X^j$. Every b_j belongs to $p^j\mathbb{Z}_p$: both factors have this coefficient property, and their convolution preserves it. Hence (10.7) shows that each term of its moment expansion has valuation at least

$$1 + j + 7s - (j-1) = 7s + 2.$$

The expansion converges, since the sharper logarithmic bound makes these valuations tend to infinity. We have proved

$$U_N \in p^{6s+2}\mathbb{Z}_p. \tag{10.10}$$

Apply the same projector to the companion unit sum $\widetilde{U}_M = \sum_{p\nmid J} \widetilde{\mu}_M(J)$. The approximation error on unit nodes belongs to $p^{3s+t}\mathbb{Z}_p = p^{6s+1}\mathbb{Z}_p$. The full-node polynomial sum has valuation at least $7s - B \geq 6s + 1$. Its nonunit leakage is

$$p\sum_H \mu_K(H)R_1(H)\chi_t(pH).$$

Again the product series starts in degree $p - 1 \geq 4$ and its degree-j coefficient has valuation at least j. For all such j, $e_{j-3} \leq j - 4$. Therefore each leakage term has valuation at least

$$1 + j + (7s - 5) - (j - 4) = 7s \geq 6s + 1.$$

Convergence follows from the logarithmic loss as before. Thus

$$\widetilde{U}_M \in p^{6s+1}\mathbb{Z}_p. \tag{10.11}$$

This calculation explicitly controls the nonunit leakage; the projector is not being treated as a literal indicator at nonunit nodes.

10.4 Completion of the two one-step congruences

Sum the first identity in (10.3) and add the primary unit nodes. It gives exactly

$$E(N) = U_N + p\mathcal{A}(M) + p\sum_{j\geq 1} a_{0,j}\widetilde{M}_j(M).$$

Every term of the last series has valuation at least $7s + 2$ by (10.2) and (10.7). Thus

$$E(N) \equiv p\mathcal{A}(M) \pmod{p^{6s+2}}. \tag{10.12}$$

The reverse strip similarly gives

$$\mathcal{A}(M) = \widetilde{U}_M + pE(K) + p\sum_{j\geq 3} a_{1,j}M_j(K),$$

where the first two moments have vanished exactly. The $j = 3$ term has valuation at least $7s - 1$; every $j \geq 4$ term has valuation at least $7s$. Since $s \geq 2$, these are at least $6s + 1$. Together with (10.11), this proves

$$\mathcal{A}(M) \equiv pE(K) \pmod{p^{6s+1}}. \tag{10.13}$$

All series manipulations are legitimate: there are finitely many nodes, the root-product series converge on $\mathbb{Z}_p$, and the logarithmic moment estimates also give convergence after rearranging by degree. Combining the two displayed congruences proves the assertion of Theorem 10.1.

10.5 Verification of the explicit ranges

For integer $s \geq 2$, condition (10.1) is equivalent to the strict inequality

$$(12s+4)(p-1)+1 < p^{\lfloor 2(s-1)/3 \rfloor + 1}. \tag{10.14}$$

For fixed s let $A = 12s+4$ and $a = \lfloor 2(s-1)/3 \rfloor + 1$. In each range considered below, $a \geq 2$. The function $(Ap - A + 1)/p^a$ is decreasing for real $p \geq 5$, because its derivative has numerator $Ap(1-a) + a(A-1) < 0$. It therefore suffices to check the smallest value of p in each claimed range.

For $p \geq 5$, the left side at $p = 5$ is $48s + 17$. At $s = 6, 7, 8$ the required comparisons are

$$305 < 625, \qquad 353 < 3125, \qquad 401 < 3125.$$

Increasing s by three adds 144 to the left side and multiplies the right side by 25, preserving the inequality. Thus it holds for all $s \geq 6$.

For $p \geq 11$, the left side at $p = 11$ is $120s + 41$. At $s = 4, 5, 6$ the comparisons are

$$521 < 1331, \qquad 641 < 1331, \qquad 761 < 14641.$$

The step $s \mapsto s+3$ adds 360 to the left side and multiplies the right side by 121. Thus the inequality holds for all $s \geq 4$.

For $p \geq 41$, the left side at $p = 41$ is $480s + 161$. At $s = 3, 4, 5$ the comparisons are

$$1601 < 1681, \qquad 2081 < 68921, \qquad 2561 < 68921.$$

The step $s \mapsto s+3$ adds 1440 to the left side and multiplies the right side by 1681. This proves the final stated range.

11 A reflection gain for companion interpolation samples

Retain $d_k = (-1/3)_k/k!$, $c_n = (2/3)_n/n!$, $M = 3n+1$, and the exact normalized companion moments $R_j(M)$.

Lemma 11.1. *Let $p > 3$, $p \equiv 2 \pmod 3$. For every $n \geq 0$ with $p \mid 3n+1$, every coefficient of the integral numerator series*

$$\mathcal{N}_n(w) = \sum_{k=0}^{n} (6k-1) d_k \phi_k(w), \qquad \phi_k(w) = \sum_{j \geq 0} \phi_j(k,n) w^j,$$

from (7.1) belongs to $p\mathbb{Z}_p$.

Proof. Use the monic reversed product of the companion summand kernel:

$$\widehat{K}_k(w) = \prod_{j=0}^{k-1} f_j(w), \qquad f_j(w) = \frac{(1+(6j-2)w)(1-(3j-1)w)(1-(3j+1)w)}{(1-(6j+6)w)(1+(3j+3)w)(1+(3j+1)w)}.$$

The coefficient series ϕ_k has the form

$$\phi_k(w) = A_n(w) \widehat{K}_k(w) \frac{1+(18k^2-6k-2)w}{1-2w}, \qquad A_n(w) \in \mathbb{Z}[[w]], \quad A_n(0) = 1.$$

Here A_n is the common reversed denominator-root/node-root quotient. Its precise roots are immaterial for this argument; every root is integral, so A_n is an integral unit series. It suffices to prove that

$$\sum_{k=0}^{n} (6k-1) d_k \widehat{K}_k(w) (1+(18k^2-6k-2)w)$$

vanishes in $\mathbb{F}_p[[w]]$.

Put $a = (p+1)/3$, which is an even integer. The binomial identity $d_k = (-1)^k\binom{1/3}{k}$ and the p-adic Lucas congruence give

$$d_{pb+c} \equiv \frac{(1/3)_b}{b!}\,(-1)^c\binom{a}{c} \pmod p \qquad (0 \le c < p). \tag{11.1}$$

Indeed $1/3 = a + p(-1/3)$ in $\mathbb{Z}_p$; the ordinary Lucas identity extends to binomial coefficients with p-adic top argument by continuity from nonnegative integers. This also proves that the left side vanishes when $c > a$.

For any consecutive p indices, each of the three linear root progressions in the numerator and denominator of f_j runs through all residues of $\mathbb{F}_p$. Thus their products agree, and

$$\prod_{j=bp}^{(b+1)p-1} f_j(w) = 1 \quad \text{in } \mathbb{F}_p[[w]], \qquad \widehat{K}_{pb+c}(w) = \widehat{K}_c(w).$$

The remaining linear and quadratic factors in k depend only on k mod p. Therefore every complete block reduces, up to the scalar $(1/3)_b/b!$, to

$$B(w) = \sum_{c=0}^{a}(6c-1)(-1)^c\binom{a}{c}\widehat{K}_c(w)(1 + (18c^2 - 6c - 2)w).$$

The same holds for the final partial block: since $3n+1 \equiv 0$ modulo p, its last residue is $(2p-1)/3 = 2a - 1 \ge a$ and therefore includes the entire nonzero Lucas support.

It remains to show $B(w) = 0$. The exact root lists defining f_j give, using $3a = 1$ in $\mathbb{F}_p$,

$$f_{a-1-j}(w) = f_j(w)^{-1}, \qquad \widehat{K}_a(w) = 1, \qquad \widehat{K}_{a-c}(w) = \widehat{K}_c(w).$$

The first equality pairs the numerator roots $2-6(a-1-j), 3(a-1-j)-1, 3(a-1-j)+1$ with the denominator roots $6j+6, -3j-3, -3j-1$, and conversely. Because a is even, $(-1)^c\binom{a}{c}$ is unchanged under $c \mapsto a-c$. Also

$$6(a-c) - 1 = -(6c-1), \qquad 18(a-c)^2 - 6(a-c) - 2 = 18c^2 - 6c - 2 \quad \text{in } \mathbb{F}_p.$$

Consequently the summands cancel in reflected pairs. If there is a middle summand, its factor $6c-1$ is zero. Hence $B(w) = 0$, proving the claimed coefficientwise divisibility. □

Corollary 11.2. *With the notation of Theorem 7.1, the bound improves to*

$$v_p(R_j(x)) \ge -\max\{0, \eta_j - 1\} \qquad (j \ge 0,\ x \in \mathbb{Z}_p).$$

Proof. At a sample $M = 3n+1$, put $b = \lfloor \log_p M \rfloor$. If $p \nmid M$, its normalizing denominator Mc_n has valuation at most b. If $p \mid M$, the previous count and the new numerator gain give a loss of at most $b + \lceil b/2 \rceil - 1$. For $b \ge 1$ the latter dominates b; for $b = 0$ there is no loss. Apply these bounds to the same Newton samples as in Theorem 7.1. □

Corollary 11.3. *For every integer $n \ge 0$ with $1 \le M = 3n+1 < p^2$, and every $j \ge 0$,*

$$v_p(R_j(M)) \ge -1.$$

Consequently, if $3d_j + 1 < p^2$, where $d_j = 2\lfloor (2j+1)/3 \rfloor$, then

$$pR_j(x) \in \mathbb{Z}_p \qquad \textit{for every } x \in \mathbb{Z}_p.$$

Proof. Use the exact formula $R_j(M) = -(Mc_n)^{-1}[w^j]\mathcal{N}_n(w)$. If $p \nmid M$, the excess-count proof gives $v_p(c_n) \le 1$ since $M < p^2$. Thus its normalization costs at most one power of p. If $p \mid M$, then $v_p(M) = v_p(c_n) = 1$, so the normalizing denominator costs two powers, but Lemma 11.1 restores one. This proves the first assertion, including $M = 1$. For the second, apply it at the Newton interpolation samples $M = 3n + 1$, $0 \le n \le d_j$. Their maximum is below p^2. All their finite differences belong to $p^{-1}\mathbb{Z}_p$, and the binomial basis is integral on $\mathbb{Z}_p$. □

Theorem 11.4. *For every prime $p \ge 29$ with $p \equiv 2 \pmod 3$,*

$$\boxed{E(p^5) \equiv p^2 E(p^3) \pmod{p^{14}}.}$$

Proof. Repeat the proof of Theorem 10.1 with $s = 2$, $t = 7$, and $D = 7(p - 1)$. The leakage and one-step transport estimates in that proof are valid for every $s \ge 2$ and require no logarithmic threshold. Only its two full-node projector sums need the following improved estimates.

For each companion moment index $1 \le j \le D$, its polynomial is R_{j-1} and its largest interpolation sample satisfies

$$3d_{j-1} + 1 = 6\left\lfloor \frac{2j-1}{3} \right\rfloor + 1 \le 4j - 1 \le 28(p-1) - 1 < p^2.$$

The last inequality holds for $p \ge 29$. The preceding corollary therefore gives $v_p(\widetilde{M}_j(p^4)) \ge 14 - 1 = 13$. The full-node companion projector sum belongs to $p^{13}\mathbb{Z}_p$, as required.

The primary moments have their first two terms zero. For $3 \le j \le D$, the largest sample for P_{j-3} is at most $4(j-3) + 2 \le 28(p-1) - 10 < p^2$. The primary logarithmic bound is therefore at most one, giving $v_p(M_j(p^5)) \ge 16 - 1 = 15$. This exceeds the required primary projector precision 14. The unchanged leakage calculations now give $E(p^5) \equiv p\mathcal{A}(p^4) \pmod{p^{14}}$ and $\mathcal{A}(p^4) \equiv pE(p^3) \pmod{p^{13}}$. Their combination proves the theorem. □

12 Exact finite certificates for the remaining higher-level cases

The analytic theorem leaves only the following pairs (s, p) at odd levels $r = 2s + 1 \ge 7$:

$$(3,5), (3,11), (3,17), (3,23), (3,29), (4,5), (5,5).$$

This is a finite exhaustive list, not a numerical sample of the remaining higher-level range. We give exact polynomial-interpolation certificates for the moments required by the proved projector argument.

Put $t = 3s + 1$. The projector $1 - (1 - X^{p-1})^t$ uses precisely the moment indices $j = i(p-1)$, $1 \le i \le t$. Its proof only needs

$$v_p(R_{i(p-1)-1}(p^{2s})) \ge -(s-1), \qquad v_p(P_{i(p-1)-3}(p^{2s+1})) \ge -s. \tag{12.1}$$

The first bound controls the full companion projector sum, and the second controls the full primary projector sum. The bounds on unit-node approximation, nonunit leakage, and the infinite transport corrections remain the previously proved uniform estimates; none of those is replaced by a finite truncation.

12.1 The integer recurrences used for certification

For the companion define a formal series

$$Y_n(w) = -Mc_n \sum_{j \ge 0} R_j(M) w^j, \qquad M = 3n + 1.$$

The proved rational-function recurrence gives, for $n \geq 1$,

$$\begin{aligned}&(1-2w-(M-2)(3M-4)w^2+2M(M-2)^2w^3)Y_n(w)\\&\quad=(1-2w-(M-1)(3M-5)w^2-2(M-3)(M-1)^2w^3)Y_{n-1}(w)\\&\qquad+(6n-1)d_n(1+w)(1+(18n^2-6n-2)w),\end{aligned}$$

with $Y_0(w) = -1/(1-w)$, $d_n = (-1/3)_n/n!$, and $c_n = (2/3)_n/n!$. All polynomial coefficients are integers and the left constant term is one. Thus each coefficient is determined using only addition and multiplication and already known lower coefficients.

For the primary family let $L = 3n+2$, $b_n = (1/3)_n/n!$, and

$$Z_n(w) = \frac{L(L-1)b_n}{2}\sum_{j\geq 0} P_j(L)w^j.$$

Its integer recurrence is

$$\begin{aligned}&(1-w-(L-1)(3L-2)w^2+2L(L-1)^2w^3)Z_n(w)\\&\quad=(1-w-(L-2)(3L-7)w^2-2(L-2)^2(L-3)w^3)Z_{n-1}(w)+(6n+1)b_n.\end{aligned}$$

At $n = 0$ take the preceding series as zero, giving $Z_0(w) = 1/(1-w-4w^2+4w^3)$. This recurrence follows equally from the integral reversed-factor formula for the primary normalized moments.

To compute a Pochhammer ratio modulo a power of p, each numerator and denominator factor is written as a power of p times a unit. Only the unit denominator is inverted; the accumulated valuation is tracked as an integer. This computes b_n, c_n, d_n without inverting p modulo its own power.

12.2 Degree bounds, precision, and exact Newton evaluation

For any selected normalized index j, the proved degree bounds are

$$d_j^R = 2\lfloor (2j+1)/3 \rfloor, \qquad d_j^P = 2\lfloor 2j/3 \rfloor.$$

Therefore its values at the small cutoffs $n = 0, \ldots, d_j$ determine the desired polynomial exactly. Let $d_{\max}$ be the largest degree needed for a fixed family and pair (s,p). Choose the common scale

$$A_R = \left\lceil \frac{3}{2}\lfloor \log_p(3d_{\max}+1)\rfloor \right\rceil, \qquad A_P = \left\lfloor \frac{3}{2}\lfloor \log_p(3d_{\max}+2)\rfloor \right\rfloor,$$

and compute modulo p^{A+2} in the corresponding family. The previously proved normalization bounds ensure $v_p(Mc_n) \leq A_R$ and $v_p(L(L-1)b_n) \leq A_P$ throughout those samples. If a normalization is $p^v u$ with u a unit, the scaled sample is computed as p^{A-v} times the unnormalized coefficient times u^{-1}, with the indicated factor -1 or 2. No modular division by p is performed and no precision is lost: the exponent $A - v$ is nonnegative.

For either polynomial, let $f(n)$ denote its scaled value at the appropriate affine cutoff. Its exact Newton formula is

$$f(X) = \sum_{k=0}^{d_j} \Delta^k f(0)\binom{X}{k}.$$

Here $X = (p^{2s}-1)/3$ for the companion and $X = (p^{2s+1}-2)/3$ for the primary. Both are nonnegative integers. The binomial coefficients are computed as exact integers before reduction modulo p^{A+2}. All finite differences and final sums are then integer operations in that explicitly specified residue ring. This evaluates the actual polynomial value, not an approximation obtained by guessing a polynomial from too few samples.

If the scaled residue is nonzero with valuation $v < A+2$, its unscaled value has exactly valuation $v - A$. If the residue is zero, the unscaled value has valuation at least two. Thus the residue arithmetic certifies every inequality in (12.1).

12.3 Certificate output and exhaustion of the exceptional set

The complete reproducible program is work_e7_companion/finite_projector_certificates.py; its integer outputs are in the adjacent JSON file. It verifies 158 selected moment indices across the seven pairs and both families. The following table gives the largest small sample index and the minimum valuation of the scaled residues in each family; the scale A is displayed alongside, so the minimum unscaled valuation is their difference.

s	p	$d^R_{\max}$	A_R	$\min v(p^{A_R}R)$	$d^P_{\max}$	A_P	$\min v(p^{A_P}P)$
3	5	52	5	5	48	4	4
3	11	132	3	2	128	3	3
3	17	212	3	2	208	3	3
3	23	292	3	2	288	3	3
3	29	372	3	2	368	3	3
4	5	68	5	5	64	4	4
5	5	84	5	5	80	4	4

The moduli used are respectively p^{A_R+2} and p^{A_P+2}. In particular every companion minimum is at least -1 after removing the scale, and every primary minimum is at least zero. These are stronger than the requirements in (12.1).

Applying the already proved projector-and-leakage argument with these 158 certified moment bounds proves the unchanged E.3 congruence for all seven exceptional pairs. Combining them with the analytic ranges therefore gives the unconditional computationally certified conclusion

$$\boxed{E(p^r) \equiv p^2 E(p^{r-2}) \pmod{p^{3r-1}} \quad \text{for every } p > 3,\ p \equiv 2 \pmod 3, \text{ and every odd } r \geq 7.}$$

The finite computations establish the finite exceptional set only; the assertion for every other prime and level comes from the uniform analytic theorem. The fifth level is supplied by the reflection argument and the additional finite certificates in the next section.

13 Exact finite completion of the level $r = 5$

The uniform reflection-gain argument proves the original $r = 5$ congruence for every prime $p \geq 29$ with $p \equiv 2 \pmod 3$. The only primes satisfying $3 < p < 29$ and $p \equiv 2 \pmod 3$ are $5, 11, 17, 23$. This section certifies that exhaustive finite exception set using the already proved polynomial-moment method.

Here $s = 2$, so the projector $1 - (1 - X^{p-1})^7$ uses exactly the seven positive moment indices $j = i(p-1)$, $1 \leq i \leq 7$. The sufficient bounds from the proved projector-and-leakage argument are

$$v_p\big(R_{i(p-1)-1}(p^4)\big) \geq -1, \qquad v_p\big(P_{i(p-1)-3}(p^5)\big) \geq -2.$$

All nonunit leakage and all infinite transport terms continue to use the uniform estimates; the calculation below only supplies these fourteen polynomial values per prime.

Use exactly the integer Laurent recurrences, proved degree bounds, and scaled Newton interpolation described in Section 12. In all eight family-prime cases, the common scaling exponent is $A = 3$ and the calculation is modulo $p^5 = p^{A+2}$. The normalization divisors at every sampled small cutoff have valuation at most three. Thus multiplication by p^{A-v} and inversion of the unit part supplies each scaled sample without losing precision. The Newton evaluation uses exact integer binomial coefficients at $(p^4 - 1)/3$ or $(p^5 - 2)/3$.

The complete reproducible invocation, from the archive root, is

```
python work_e7_companion/r5_finite_closure_driver.py
```

It imports the shared, independently audited function `certify` from finite_projector_certificates.py and writes r5_finite_closure.json. It does not overwrite the earlier 158-index higher-level certificate. Only the Python standard library is required.

The output for all 56 required values is summarized below. The minimum scaled valuation is the valuation of the exact residue of p^3R or p^3P modulo p^5; subtracting three gives the certified unscaled valuation. All minima shown are below the precision cap, so their listed valuations are exact.

p	$d^R_{\max}$	$\min v_p(p^3R)$	$d^P_{\max}$	$\min v_p(p^3P)$
5	36	3	32	3
11	92	2	88	3
17	148	2	144	3
23	204	2	200	3

Thus every companion value has valuation at least -1, and every primary value has valuation at least zero, which is stronger than the required primary bound. The finite sufficient conditions hold for all four primes. The uniform leakage argument therefore gives

$$E(p^5) \equiv p\mathcal{A}(p^4) \pmod{p^{14}}, \qquad \mathcal{A}(p^4) \equiv pE(p^3) \pmod{p^{13}}.$$

Combining them proves the unchanged $r = 5$ target for this exhaustive finite exception set. Together with the separate uniform argument for $p \geq 29$, it yields

$$\boxed{E(p^5) \equiv p^2E(p^3) \pmod{p^{14}} \qquad \text{for every prime } p > 3,\ p \equiv 2 \pmod 3.}$$

The finite output is not being extrapolated to other primes: those primes are covered by the uniform reflection-gain proof. This section makes no claim about the still separate $r = 3$ case.

14 A uniform first-step congruence at the third level

Corollary 14.1. *For every prime $p > 3$, $p \equiv 2 \pmod 3$, the exact companion sum of Theorem 4.1 satisfies*

$$\boxed{E(p^3) \equiv p\mathcal{A}(p^2) \pmod{p^8}.}$$

In fact the positive-moment transport part of this difference belongs to $p^9\mathbb{Z}_p$.

Proof. Write $R^{(1)}_{p^3,p}(X) = 1+\sum_{j\geq1} a_jX^j$ for the exact one-step kernel. Its coefficients satisfy $v_p(a_j) \geq j$. Summing the proved identity $\mu_{p^3}(pJ) = p\widetilde{\mu}_{p^2}(J)R^{(1)}_{p^3,p}(J)$ over the companion nodes gives

$$E(p^3) - p\mathcal{A}(p^2) = \sum_{p\nmid H} \mu_{p^3}(H) + p\sum_{j\geq1} a_j\widetilde{M}_j(p^2).$$

The first sum belongs to $p^8\mathbb{Z}_p$ by Corollary 8.2.

The common companion moment normalization has valuation seven. The first two explicit moments thus have valuation at least seven, so their transport terms have valuation at least nine. For $j \geq 3$, Theorem 7.1 gives

$$v_p(\widetilde{M}_j(p^2)) \geq 7 - \eta_{j-1}, \qquad \eta_{j-1} = \left\lceil \frac{3}{2}\left\lfloor \log_p(6\lfloor(2j-1)/3\rfloor + 1)\right\rfloor\right\rceil.$$

Here $\eta_{j-1} \le j-1$. Indeed $6\lfloor(2j-1)/3\rfloor + 1 \le 4j-1$, and

$$4j - 1 < 5^{\lfloor 2(j-1)/3\rfloor + 1} \qquad (j \ge 3).$$

The latter inequality is checked for $j = 3, 4, 5$; increasing j by three adds twelve on the left and multiplies the right by twenty-five. It follows that each remaining transport term has valuation at least $1 + j + 7 - (j-1) = 9$. The logarithmic growth of η_{j-1} proves convergence of the series. This proves the corollary. □

Consequently the still-unproved third-level E.3 congruence is exactly equivalent to the shorter companion estimate

$$\mathcal{A}(p^2) - pE(p) \in p^7\mathbb{Z}_p.$$

This equivalence includes an independently proved precision bound for one entire transport step. It does not assert the remaining companion estimate.

15 Exact two-step transport and the remaining cancellation

Let $p > 3$, $p \equiv 2 \pmod 3$, and $r \ge 3$ be odd. Set $L = p^r$, $M = L/p^2$, and $m = (L-2)/3$. Use the node and denominator root sets $\mathcal{H}_L$ and $\mathcal{D}_L$ of the three-grid construction, and write $\mu_L(H) = W_L(H)T(H)$. Define the restricted rational power series

$$R_{L,p}(X) = \frac{\prod_{d\in\mathcal{D}_L,\, p^2\nmid d}(1 - p^2X/d)}{\prod_{h\in\mathcal{H}_L,\, p^2\nmid h}(1 - p^2X/h)} = 1 + \sum_{j\ge1} c_j X^j.$$

Every retained root has valuation zero or one. Hence each linear coefficient p^2/d or p^2/h belongs to $p\mathbb{Z}_p$. Expanding the finitely many inverse factors proves

$$c_j \in p^j\mathbb{Z}_p, \qquad v_p(c_j) \longrightarrow +\infty.$$

The series converges at every integral coarse node J.

Proposition 15.1 (Exact residual identity)**.** *Put*

$$U_{L,p} = \sum_{\substack{H\in\mathcal{H}_L \\ p^2\nmid H}} \mu_L(H).$$

Then

$$\boxed{E(L) - p^2E(M) = U_{L,p} + p^2\sum_{j\ge3} c_jM_j(M).} \tag{15.1}$$

No assertion that either term on the right is separately divisible by the target modulus is made.

Proof. Since $p^2 \equiv 1 \pmod 6$, dividing all p^2-divisible nodes by p^2 gives exactly $\mathcal{H}_M$. Dividing the corresponding denominator roots gives exactly $\mathcal{D}_M$. For example the positive grids end at $L-1$ and L, and become the positive grids ending at $M-1$ and M; the negative grid is -2 times the first positive grid. The positive denominator roots $6i$ and the negative roots $-3i$ both have coarse index range $1 \le i \le (M-2)/3$. The third denominator grid ends at $-(L-3)$ and becomes the grid ending at $-(M-3)$.

The exact model values satisfy $T(p^2J) = p^2T(J)$, because multiplication by p^2 preserves the residues modulo six and the sign of each node. Separating divisible factors in the weights therefore gives the product identities

$$W_L(p^2J) = W_M(J)R_{L,p}(J), \qquad \mu_L(p^2J) = p^2\mu_M(J)R_{L,p}(J).$$

These are identities of products, including at zero weights; no division by a vanishing weight is needed. Sum over coarse nodes and add the nodes not divisible by p^2. Interchanging the finite node sum with the convergent series is legitimate. The identities $M_1(M) = M_2(M) = 0$ remove its first two positive terms and give (15.1). □

16 The leading kernel coefficients from three short harmonic sums

Let $p > 3$ be prime with $p \equiv 2 \pmod 3$, let $r \geq 3$ be odd, and put $L = p^r$, $q = (p-2)/3$. Set

$$H_i = \sum_{a=1}^{q} a^{-i} \in \mathbb{F}_p \quad (1 \leq i \leq 3).$$

Use the exact two-step kernel

$$R_{L,p}(X) = \frac{\prod_{d \in \mathcal{D}_L,\ p^2 \nmid d}(1 - p^2X/d)}{\prod_{h \in \mathcal{H}_L,\ p^2 \nmid h}(1 - p^2X/h)} = 1 + \sum_{j \geq 1} c_j X^j.$$

All products here count multiplicities. In particular, an overlap between the node and denominator lists may be canceled but need not be removed in defining the products.

Proposition 16.1. *For every such p, r,*

$$\begin{aligned} c_1/p &\equiv -H_1/2 \pmod p, \\ c_2/p^2 &\equiv H_1^2/8 + 5H_2/24 \pmod p, \\ \kappa_p := c_3/p^3 &\equiv -\frac{3H_1^3 + 15H_1H_2 + 2H_3}{144} \pmod p. \end{aligned}$$

Thus these leading coefficients are independent of the odd level r.

Proof. For $i \leq 3$ define the normalized root-power difference

$$s_i = p^{-i}\left(\sum_{h \in \mathcal{H}_L,\ p^2 \nmid h} (p^2/h)^i - \sum_{d \in \mathcal{D}_L,\ p^2 \nmid d} (p^2/d)^i \right).$$

Every term is integral at p. Modulo p, the unit roots contribute zero, leaving exactly the roots of valuation one. After dividing those roots by p, their lists are the unit roots of the companion grids at $M = p^{r-1} = 3n + 1$:

$$\begin{aligned} \mathcal{H}'_M &= \{3j+1 : 0 \leq j \leq n\} \cup \{3j+2, -2(3j+2) : 0 \leq j < n\}, \\ \mathcal{D}'_M &= \{6j, -3j : 1 \leq j \leq n\} \cup \{-(3j+1) : 0 \leq j < n\}. \end{aligned}$$

These identifications follow directly by solving the original linear root congruences modulo p; they respect multiplicities.

Put $t = (2p-1)/3 = 2q + 1$. Since $n \equiv t \pmod p$, complete blocks of p consecutive indices contribute zero modulo p to each inverse-power sum under consideration. Indeed, $\sum_{u \in \mathbb{F}_p^\times} u^{-i} = 0$ for $1 \leq i \leq 3 < p - 1$. Let

$$T_i = \sum_{j=1}^{t} j^{-i}, \qquad A_i = \sum_{j=0}^{t-1} (3j+1)^{-i}, \qquad B_i = \sum_{\substack{0 \leq j < t \\ j \neq q}} (3j+2)^{-i}.$$

The extra endpoint $3t+1 = 2p$ of the first node list is excluded as a nonunit. The involution $j \mapsto t-1-j$ gives $A_i = (-3)^{-i}T_i$. In the B list, the residues are $3a$ for $a = -q, \dots, -1, 1, \dots, q$. Consequently

$$B_1 = B_3 = 0, \qquad B_2 = 2H_2/9.$$

Since $t = p-1-q$, the full nonzero-residue sum also gives $T_i = (-1)^{i+1}H_i$. Taking the difference of the companion node and denominator power sums yields

$$s_i \equiv (1-(-1)^i)A_i + (1+(-2)^{-i})B_i - (6^{-i}+(-3)^{-i})T_i \pmod p.$$

Substitution gives

$$s_1 = -H_1/2, \qquad s_2 = 5H_2/12, \qquad s_3 = -H_3/24 \quad \text{in } \mathbb{F}_p.$$

Finally the logarithmic product expansion, or equivalently the first three Newton identities, gives

$$\frac{c_1}{p} = s_1, \quad \frac{c_2}{p^2} = \frac{s_1^2+s_2}{2}, \quad \frac{c_3}{p^3} = \frac{s_1^3+3s_1s_2+2s_3}{6}.$$

Here these identities are exact before the reductions of s_i. The stated congruences follow; all displayed denominators are units because $p > 3$. □

This proposition computes one contribution to the residual and does not prove its cancellation with the other contribution. For example $p = 11$ gives $(H_1, H_2, H_3) = (0, 9, 6)$ modulo 11, so $\kappa_{11} = -1$. In particular the cubic kernel coefficient is not always divisible by p^4.

17 A uniform coarse-moment estimate and a rigorous transport-tail bound

Let $p > 3$ be prime, $p \equiv 2 \pmod 3$, and put $q = (p-2)/3$. Then q is odd. Use the original three-grid interpolation at $N = p$, and denote its positive moments by $M_j(p)$.

Proposition 17.1. *For every positive integer j,*

$$\boxed{M_j(p) \in p^2\mathbb{Z}_p.}$$

This bound is uniform in j; it is not inferred from finitely many polynomial moment formulas.

Proof. Write $P_q(z) = D_q(z)\mathcal{S}_q(z)$ for the cleared polynomial of the three-grid construction. All its coefficients belong to $\mathbb{Z}_p$: the fixed denominators $k!$, $(2/3)_k$, and the normalizing roots of D_q are units for $0 \le k \le q$. We prove the stronger coefficientwise assertion

$$P_q(z) \in p\mathbb{Z}_p[z]. \tag{17.1}$$

Work first in $\mathbb{F}_p(z)$. Insert the canceled numerator and denominator parameter $e = 2/3$ in $\mathcal{S}_q$ to exhibit the very-well-poised ${}_6F_5(-1)$ form. Modulo p its central parameter satisfies $a = 1/3 = -2q-1$, and the inserted parameter satisfies $e = 2/3 = -q$. The finite canceled factors $(-q)_k$ are nonzero for $k \le q$, so this insertion is legitimate. Apply Whipple's terminating transformation using this $-q$ as the terminating upper parameter. The other parameters may be written

$$b = a + z/6, \quad c = a - z/3, \quad e' = -q - z/3, \quad f = -q.$$

The transformed terminating ${}_3F_2$ is

$$ {}_3F_2\left[\begin{matrix}-q,\ -q+z/6,\ -q-z/3\\ 1-z/6,\ 1+z/3\end{matrix};1\right]_q.$$

Its prefactor is $(-2q)_q/(-q+z/3)_q$, a well-defined element of $\mathbb{F}_p(z)$. The displayed sum is a terminating Dixon sum with odd central parameter $-q$: the two lower parameters equal $1-q-(-q+z/6)$ and $1-q-(-q-z/3)$. Reversing the summation index $k\mapsto q-k$ negates every term. The sum is therefore zero. All transformations are finite rational identities whose denominators are nonzero in $\mathbb{F}_p(z)$; no analytic continuation in characteristic p is used. It follows that the reduction of $\mathcal{S}_q$ is zero as a rational function. Since D_q is a nonzero polynomial modulo p, this proves (17.1).

Let $\widehat{Q}_q(z)$ be the monic polynomial whose roots are the three interpolation grids. These roots are integers. Exactly one root is divisible by p, namely the positive node p itself; every other root is a p-adic unit. Thus $Q_q(z)=\ell_q\widehat{Q}_q(z)$ with $v_p(\ell_q)=-1$. By (17.1),

$$\frac{D_q(z)\mathcal{S}_q(z)}{Q_q(z)}=\frac{A_q(z)}{\widehat{Q}_q(z)},\qquad A_q(z)=\ell_q^{-1}P_q(z)\in p^2\mathbb{Z}_p[z].$$

The inverse Laurent series at infinity of a monic polynomial in $\mathbb{Z}_p[z]$ has integral coefficients after its leading power of z is removed. Hence every Laurent coefficient of this proper rational function belongs to $p^2\mathbb{Z}_p$. Those coefficients are $-M_j(p)$, which proves the proposition. □

Corollary 17.2. *At $L=p^3$, let $R_{L,p}(X)=1+\sum_{j\ge1}c_jX^j$ be the exact two-step weight kernel. Then*

$$p^2\sum_{j\ge4}c_jM_j(p)\in p^8\mathbb{Z}_p.$$

In particular the entire transport error, including its infinite tail, satisfies

$$p^2\sum_{j\ge3}c_jM_j(p)\equiv p^2c_3M_3(p)\pmod{p^8}.$$

Proof. The established coefficient estimate is $c_j\in p^j\mathbb{Z}_p$. Proposition 17.1 bounds every summand for $j\ge4$ by p^{2+j+2}, hence by p^8. These bounds tend to infinity, so the entire convergent tail belongs to that closed ideal. □

For an explicit leading term put

$$b_q=\frac{(1/3)_q}{q!},\qquad \kappa_p\equiv c_3/p^3\pmod p.$$

Since q is odd, the exact first nonzero moment is $M_3(p)=p^2(p-1)^3b_q^3$. The corollary therefore gives

$$\boxed{p^2\sum_{j\ge3}c_jM_j(p)\equiv-p^7\kappa_pb_q^3\pmod{p^8}.}$$

Together with the separately derived formula

$$\kappa_p\equiv-\frac{3H_1^3+15H_1H_2+2H_3}{144}\pmod p,\qquad H_i=\sum_{a=1}^{q}a^{-i},$$

this determines the full transport contribution modulo the target modulus by three harmonic sums of length $(p-2)/3$.

The remaining unit and valuation-one node contribution has not been bounded here. In particular this proposition and corollary do not claim the uniform $r=3$ recurrence: they prove that all orders beyond the cubic transport term are already harmless, leaving one explicit finite-node cancellation to establish.

18 The single remaining odd level

The preceding uniform proofs and exact finite certificates establish every odd level $r \geq 5$. The sole level not settled uniformly in this manuscript is $r = 3$, for primes $p > 3$, $p \equiv 2 \pmod 3$. The independently proved first-step congruence reduces it exactly to

$$\boxed{\mathcal{A}(p^2) - pE(p) \in p^7\mathbb{Z}_p.}$$

This is a smaller finite sum but is still a conjectural uniform estimate; its statement is not counted as its proof.

At $r = 3$, let $q = (p-2)/3$, $b_q = (1/3)_q/q!$, and use the coefficient κ_p evaluated in Section 16. The proved exact residual and the uniform tail bound reduce (1.1) to

$$U_{p^3,p} \equiv p^7\kappa_p b_q^3 \pmod{p^8}.$$

This congruence remains to be proved uniformly. The cubic coefficient need not vanish: $\kappa_{11} = -1$ modulo 11. Thus it would be incorrect to eliminate the two contributions separately.

Novelty and success criteria. The named problem, cutoff, recurrence factor and modulus have been preserved. The proof is unconditional in its stated ranges and contains a uniform working estimate rather than merely an equivalent formulation. It does not meet the requirement of a complete proof of the whole odd branch. No claim that the technique or subrange result has historical priority is made; that requires a separate comparison with the full relevant literature.

Verification. The symbolic WZ certificates, boundary identities, polynomial recurrences, root-count tables, normalized moment formulas and valuation budgets were checked independently. The accompanying archive contains the actual derivation and validation scripts. The higher-level finite reduction certifies 158 specified moments, and the fifth-level finite reduction certifies 56 further moments. Both batches use proved degree bounds and exact integer Newton interpolation, with modular normalization performed without division by p. The complete certificate evaluator is reproduced below. Finite modular checks of the harmonic kernel coefficients used $p = 5, 11, 17, 23, 29, 41$ at level three and $p = 5$ at level five. Those checks do not replace the proofs for arbitrary parameters.

A Finite certificates on the two positive companion grids

A.1 A finite certificate on the grid $H = 3h + 1$

Retain the rational deformation $V_n(z)$ defined in the companion construction. For every $h \geq 0$ and $n \geq h$,

$$V_n(3h+1) = -\frac{(-1)^h(3h+1)}{2}. \tag{A.1}$$

All lower parameters at $z = 3h + 1$ are regular: in particular $(5 - 3h)/6$ is never an integer. The factor $(-h)_k$ makes the sum terminate at $k = h$, so it suffices to prove the identity for $n = h$.

Set

$$P(h,k) = 18k^2 - 6k + 3h - 1,$$

$$u(h,k) = \frac{(-1)^{k-h}(6k-1)P(h,k)}{(1-3h)(3h+1)} \frac{(-1/3)_k((3h-1)/6)_k(-h-2/3)_k(-h)_k}{k!((5-3h)/6)_k(h+4/3)_k(h+2/3)_k}.$$

This is exactly the summand of $V_h(3h+1)$ divided by the right side of (A.1). We prove $\sum_{k=0}^{h} u(h,k) = 1$ by a recurrence with step two in h.

Define the polynomial

$$\begin{aligned}A(h,k) = {} & 486h^5 + (1458k^2 - 1944k + 3564)h^4 \\ & + (7857k^2 - 10476k + 10233)h^3 \\ & + (-972k^4 + 2592k^3 + 13149k^2 - 19836k + 14229)h^2 \\ & + (-2511k^4 + 6696k^3 + 7137k^2 - 15468k + 9486)h \\ & + 243k^6 - 972k^5 - 189k^4 + 3384k^3 + 414k^2 - 4072k + 2392.\end{aligned}$$

The rational certificate is

$$R(h,k) = \frac{-k(3k-2)(3h-6k+1)A(h,k)}{D_+(h,k)},$$

where

$$\begin{aligned}D_+(h,k) = {} & (6k-1)(h-k+1)(h-k+2)(3h-3k+5)(3h-3k+8) \\ & \cdot (3h+3k+2)(3h+3k+4)P(h,k).\end{aligned}$$

Cancellation of consecutive rising factors and expansion of the remaining polynomials verifies the rational identity

$$u(h+2,k) - u(h,k) = R(h,k+1)u(h,k+1) - R(h,k)u(h,k). \tag{A.2}$$

For an independently checkable form, if $q(h,k) = u(h,k+1)/u(h,k)$ and $r(h,k) = u(h+2,k)/u(h,k)$, then the certificate is the polynomial equality obtained by clearing all denominators in $q(h,k)R(h,k+1) - R(h,k) = r(h,k) - 1$. The ratio q is explicitly

$$q(h,k) = -\frac{(6k+5)P(h,k+1)}{(6k-1)P(h,k)}\,\frac{(k-1/3)(k+(3h-1)/6)(k-h-2/3)(k-h)}{(k+1)(k+(5-3h)/6)(k+h+4/3)(k+h+2/3)}.$$

Here are the upper-boundary details. Put $g(h,k) = R(h,k)u(h,k)$. The apparent poles at $k = h+1, h+2$ cancel because

$$\frac{(-h)_k}{(h-k+1)(h-k+2)} = \frac{(-h-2)_k}{(h+1)(h+2)}.$$

Thus g has a finite value at both boundary indices. Also $g(h,0) = 0$. Direct cancellation gives

$$\frac{R(h,k)}{r(h,k)}\Big|_{k=h+1} = -\frac{4(3h+1)(9h^3+39h^2+46h+15)}{(h+2)(6h+5)(18h^2+33h+17)},$$

$$q(h+2,h+1) = -\frac{5(3h+2)(9h+11)}{(h+2)(6h+5)(18h^2+33h+17)}.$$

The first displayed quotient plus $1 + q(h+2,h+1)$ is identically zero. Multiplication by $u(h+2,h+1)$ therefore gives

$$g(h,h+1) = -u(h+2,h+1) - u(h+2,h+2).$$

Summing (A.2) for $0 \le k \le h$ and adding these two final terms proves

$$\sum_{k=0}^{h+2} u(h+2,k) = \sum_{k=0}^{h} u(h,k).$$

At $h = 0$ the sum is 1; at $h = 1$ its terms are $-1/4$ and $5/4$, again summing to one. Induction separately on the two parity classes proves (A.1) for every h.

This proves one exact node family of the companion interpolation. It does not, by itself, prove any prime-power lifting congruence.

A.2 The second positive grid: a complete finite certificate

For $h \geq 1$, let $t_h(k)$ be the summand of $V_{h+1}(3h+2)$ and normalize it by $u_h(k) = t_h(k)/(3h+2)$. Explicitly,

$$u_h(k) = \frac{(-1)^k(6k-1)(h+6k^2-2k)}{2h(3h+2)} \frac{(-1/3, h/2, -h-1, -h-1/3)_k}{k!(2/3-h/2, h+5/3, h+1)_k}.$$

The sum terminates at $h+1$. The desired exact assertion is

$$\sum_{k=0}^{h+1} u_h(k) = \begin{cases} 1, & h \text{ odd}, \\ 0, & h \text{ even}. \end{cases}$$

All denominator Pochhammers are nonzero on this range.

Put

$$\begin{aligned} J(h,k) = {} & 54h^5 + 162h^4k^2 - 216h^4k + 486h^4 + 1089h^3k^2 - 1452h^3k + 1725h^3 \\ & - 108h^2k^4 + 288h^2k^3 + 2442h^2k^2 - 3512h^2k + 3002h^2 \\ & - 351hk^4 + 936hk^3 + 2082hk^2 - 3608hk + 2549h \\ & + 27k^6 - 108k^5 - 126k^4 + 656k^3 + 507k^2 - 1316k + 840, \end{aligned}$$

$$R(h,k) = -\frac{k(3k-2)(3h-6k+2)J(h,k)}{D_2(h,k)},$$

where

$$\begin{aligned} D_2(h,k) = {} & (6k-1)(h-k+2)(h-k+3)(h+k+1)(h+6k^2-2k) \\ & \cdot (3h-3k+4)(3h-3k+7)(3h+3k+5). \end{aligned}$$

Direct rational cancellation proves the finite WZ certificate

$$u_{h+2}(k) - u_h(k) = R(h,k+1)u_h(k+1) - R(h,k)u_h(k). \tag{A.3}$$

For a compact independently reproducible verification, the consecutive ratios are

$$q(h,k) = \frac{u_h(k+1)}{u_h(k)} = \frac{(h+2k)(3k-1)(6k+5)(h-k+1)(3h-3k+1)(h+6k^2+10k+4)}{(k+1)(6k-1)(h+k+1)(h+6k^2-2k)(3h-6k-4)(3h+3k+5)},$$

$$\begin{aligned} r(h,k) = {} & \frac{u_{h+2}(k)}{u_h(k)} \\ = {} & \frac{(h+1)(h+2)(h+3)(h+2k)(3h+4)(3h+5)(3h+7)}{(h-k+2)(h-k+3)(h+k+1)(h+k+2)(h+6k^2-2k)} \\ & \times \frac{(3h-6k+2)(h+6k^2-2k+2)}{(3h-3k+4)(3h-3k+7)(3h+3k+5)(3h+3k+8)}. \end{aligned}$$

Substitution in $q(h,k)R(h,k+1) - R(h,k) = r(h,k) - 1$ leaves identically zero after clearing denominators.

At the lower boundary $R(h,0)u_h(0) = 0$. At the upper boundary, cancel the apparent zeros and poles before evaluating. For instance,

$$\frac{(-h-1)_k}{(h-k+2)(h-k+3)} = \frac{(-h-3)_k}{(h+2)(h+3)}.$$

The needed endpoint can also be checked entirely from rational ratios:

$$\frac{R(h,k)}{r(h,k)}\Big|_{k=h+2} = -\frac{4(9h^3+60h^2+127h+87)}{(h+3)(6h+11)^2},$$

$$\frac{u_{h+2}(h+3)}{u_{h+2}(h+2)} = -\frac{3(3h+5)}{(h+3)(6h+11)^2}.$$

The first displayed ratio plus one plus the second ratio is exactly zero. Hence the upper boundary is $-u_{h+2}(h+2)-u_{h+2}(h+3)$. Summing (A.3) over $0 \le k \le h+1$ therefore proves that the normalized sum is unchanged when h increases by two. The initial normalized terms are

$$h=1: \quad -\frac{1}{10},\ \frac{5}{4},\ -\frac{3}{20}, \qquad h=2: \quad -\frac{1}{16},\ -\frac{105}{176},\ \frac{11}{16},\ -\frac{5}{176}.$$

Their sums are respectively one and zero, proving both parity cases. At $h=0$ the removable value $V_n(2)=0$ for $n \ge 1$ supplies the remaining case. For an ambient truncation $n \ge h+1$, all subsequent terms vanish and their denominators remain nonzero; the same node value therefore holds for every admissible ambient cutoff.

B A finite certificate for the negative companion grid

The following establishes the negative grid of the proposed companion interpolation. It is an exact finite identity and makes no claim about the remaining prime-power error estimate.

For $h \ge 0$, set

$$t(h,k) = -\frac{(-1)^k(6k-1)(3k^2-k-h-1)}{2(h+1)}\,\frac{(-1/3)_k(-h-1)_k(2h+1)_k(2h+5/3)_k}{k!(h+5/3)_k(-2h-1/3)_k(-2h-1)_k}.$$

All denominator factors are nonzero for $0 \le k \le h+1$. We prove

$$\sum_{k=0}^{h+1} t(h,k) = 3h+2. \tag{B.1}$$

Put $u(h,k)=t(h,k)/(3h+2)$ and define the polynomial

$$\begin{aligned} A(h,k) = {} & 1728h^5 - 2592h^4k^2 + 3456h^4k + 9504h^4 \\ & - 12024h^3k^2 + 16032h^3k + 20712h^3 \\ & + 432h^2k^4 - 1152h^2k^3 - 19704h^2k^2 + 27296h^2k + 22408h^2 \\ & + 1026hk^4 - 2736hk^3 - 13308hk^2 + 20176hk + 12074h \\ & - 27k^6 + 108k^5 + 450k^4 - 1520k^3 - 3027k^2 + 5444k + 2604. \end{aligned}$$

The certificate is

$$R(h,k) = -\frac{k(3k-2)(2h-k+2)(6h-3k+4)A(h,k)}{D_-(h,k)},$$

where

$$\begin{aligned} D_-(h,k) = {} & 16(h+1)(2h+1)(2h+3)(3h+4)(6h+5)(6h+7) \\ & \cdot (6k-1)(h-k+2)(h-3k^2+k+1). \end{aligned}$$

Direct rational cancellation gives

$$u(h+1,k)-u(h,k)=R(h,k+1)u(h,k+1)-R(h,k)u(h,k). \tag{B.2}$$

For reproducibility, this is the polynomial identity obtained after substituting the displayed R and the consecutive-factor ratios of u; the exact residual is zero. The quadratic factor $h-3k^2+k+1$ cancels from the product Ru, so roots of that factor cause no singularity.

We specify the upper boundary without interpreting an undefined terminating summand. Write $r(h,k)=u(h+1,k)/u(h,k)$ as its rational consecutive-parameter ratio. Cancellation gives

$$\left.\frac{R(h,k)}{r(h,k)}\right|_{k=h+2}=-1.$$

The canceled quotient has no pole at this substitution for any integer $h\geq 0$. Thus the continued product at the upper boundary is

$$[R(h,k)u(h,k)]_{k=h+2}=-u(h+1,h+2).$$

This can equivalently be obtained by cancelling $(-h-1)_k/(h-k+2)=(-h-2)_k/(h+2)$ before specializing; the quotient R/r also covers the coincident zero factors at $h=0$. At the lower boundary $R(h,0)u(h,0)=0$. Sum (B.2) over $0\leq k\leq h+1$ and add the new endpoint $u(h+1,h+2)$. The sums of u at successive h are equal. For $h=0$, the unnormalized terms are $-1/2$ and $5/2$, whose sum is two. Hence the normalized sum equals one, proving (B.1).

B.1 Embedding in the cleared interpolation

At the negative node $H=-2(3h+2)$, the candidate $V_n(z)$ has exactly the summand $t(h,k)$ above. Its numerator parameter $-h-1$ terminates the finite sum at $k=h+1$. For this node to belong to the companion grid one has $n\geq h+1$. The companion denominator is

$$D_n(z)=\prod_{j=1}^{n}(1-z/(6j))(1+z/(3j))\prod_{j=0}^{n-1}(1+z/(3j+1)).$$

The only possible zero of $D_n(H)$ is the last product's factor at $j=2h+1$. It occurs precisely when $n\geq 2h+2$. If $h+1\leq n\leq 2h+1$, all original summand denominators are nonzero, the sum terminates at $h+1$, and (B.1) gives $V_n(H)=3h+2$.

If $n\geq 2h+2$, treat $D_n(z)V_n(z)$ term by term after clearing its finite denominator factors. In a summand $k\leq h+1$, the factor indexed by $j=2h+1$ remains in the denominator-clearing tail and vanishes at H. In a summand $k\geq h+2$, the numerator contains the factor at $j=h+1$ in $1+z/[2(3j-1)]$, and that factor vanishes at H. The additional factor $2-z$ in V_n is nonzero at H. Thus every cleared summand is zero, giving

$$[D_n(z)V_n(z)]_{z=H}=0=D_n(H)(3h+2).$$

Combining the two ranges proves the full cleared negative-grid identity

$$\boxed{[D_n(z)V_n(z)]_{z=-2(3h+2)}=D_n(-2(3h+2))(3h+2)\qquad(0\leq h<n).} \tag{B.3}$$

No raw 0/0 hypergeometric summand is used in this argument.

C Reproducible integer certificate evaluator

The following complete program is the evaluator used for the 158 higher-level moment certificates. For the 56 fifth-level certificates, the exhaustive case list is instead $[(5,2),(11,2),(17,2),(23,2)]$; all other calculations are identical. These lists supplement the uniform proofs and do not extrapolate from a finite sample to an infinite family.

```
"""Exact finite certificates for exceptional polynomial-projector parameters.

The proved degree and logarithmic denominator bounds reduce each required
prime-power moment to Newton interpolation from small integer cutoffs.
All calculations use Python integers modulo explicit prime powers; no
floating point, symbolic heuristic, or original enormous truncated sum.
"""
from pathlib import Path
from math import comb
import json


def strip(x,p):
    assert x
    v=0
    while x%p==0:x//=p;v+=1
    return x,v


def ilog(x,p):
    k=0
    while x>=p:x//=p;k+=1
    return k


def vmod(x,p,cap):
    return cap if x==0 else strip(x,p)[1]


def poch_step(unit,val,num,den,p,mod):
    nu,nv=strip(num,p);du,dv=strip(den,p)
    return unit*nu*pow(du,-1,mod)%mod,val+nv-dv


def solve_series(a,b,previous,source,J,mod):
    out=[0]*(J+1)
    for j in range(J+1):
        rhs=sum(b[i]*previous[j-i] for i in range(min(3,j)+1))
        if j<len(source):rhs+=source[j]
        rhs-=sum(a[i]*out[j-i] for i in range(1,min(3,j)+1))
        out[j]=rhs%mod
    return out


def newton_value(values,X,mod):
    delta=values[:];ans=0;binom=1
    for k in range(len(values)):
        if k:binom=binom*(X-k+1)//k
        ans=(ans+delta[0]*(binom%mod))%mod
        delta=[(delta[i+1]-delta[i])%mod for i in range(len(delta)-1)]
    return ans


def certify(p,s,family):
    indices=[i*(p-1)-(1 if family=='companion' else 3) for i in range(1,3*s+2)]
    degrees={j:2*((2*j+1)//3) if family=='companion' else 2*((2*j)//3) for j in indices}
    J=max(indices);nmax=max(degrees.values())
    log=ilog(3*nmax+(1 if family=='companion' else 2),p)
    A=(3*log+1)//2 if family=='companion' else (3*log)//2
    cap=A+2;mod=p**cap
    sample={j:[] for j in indices}
```

```
    previous=[0]*(J+1)
    du=cu=bu=1;dv=cv=bv=0
    max_norm_v=0
    for n in range(nmax+1):
        if n:
            du,dv=poch_step(du,dv,3*n-4,3*n,p,mod)
            cu,cv=poch_step(cu,cv,3*n-1,3*n,p,mod)
            bu,bv=poch_step(bu,bv,3*n-2,3*n,p,mod)
        assert min(dv,cv,bv)>=0
        if family=='companion':
            M=3*n+1
            if n==0:
                current=[mod-1]*(J+1)
            else:
                a=[1,-2,-(M-2)*(3*M-4),2*M*(M-2)**2]
                b=[1,-2,-(M-1)*(3*M-5),-2*(M-3)*(M-1)**2]
                t=(6*n-1)*du*pow(p,dv,mod)%mod
                alpha=18*n*n-6*n-2
                current=solve_series(a,b,previous,[t,t*(alpha+1),t*alpha],J,mod)
            mu,mv=strip(M,p);norm_v=mv+cv;norm_unit=mu*cu%mod
            sign=-1
        else:
            L=3*n+2
            a=[1,-1,-(L-1)*(3*L-2),2*L*(L-1)**2]
            b=[1,-1,-(L-2)*(3*L-7),-2*(L-2)**2*(L-3)]
            t=(6*n+1)*bu*pow(p,bv,mod)%mod
            current=solve_series(a,b,previous,[t],J,mod)
            lu,lv=strip(L,p);vu,vv=strip(L-1,p)
            norm_v=lv+vv+bv;norm_unit=lu*vu*bu%mod;sign=2
        assert norm_v<=A,(family,p,s,n,norm_v,A)
        max_norm_v=max(max_norm_v,norm_v)
        scale=sign*pow(p,A-norm_v,mod)*pow(norm_unit,-1,mod)%mod
        for j in indices:
            if n<=degrees[j]:sample[j].append(current[j]*scale%mod)
        previous=current
    X=(p**(2*s)-1)//3 if family=='companion' else (p**(2*s+1)-2)//3
    rows=[]
    for j in indices:
        result=newton_value(sample[j],X,mod)
        valuation=vmod(result,p,cap)-A
        target=-(s-1) if family=='companion' else -s
        assert valuation>=target,(p,s,family,j,valuation,target)
        rows.append({'normalized_index':j,'polynomial_degree_bound':degrees[j],
            'scaled_value_residue':result,'valuation_lower_bound':valuation,
            'required_lower_bound':target,'passed':True})
    return {'p':p,'s':s,'r':2*s+1,'family':family,'precision_exponent':cap,
        'common_scale_exponent':A,'max_sample_cutoff':nmax,
        'max_normalizer_valuation':max_norm_v,'number_of_moments':len(rows),
        'checks':rows}


if __name__=='__main__':
    cases=[(5,3),(11,3),(17,3),(23,3),(29,3),(5,4),(5,5)]
    records=[]
    for p,s in cases:
        for family in ['companion','primary']:
            row=certify(p,s,family);records.append(row)
            print(json.dumps({k:v for k,v in row.items() if k!='checks'}),flush=True)
    result={'status':'All exact finite projector-moment certificates passed.',
        'pair_count':len(cases),'family_pair_count':len(records),
        'moment_count':sum(r['number_of_moments'] for r in records),
        'method':'Proved small-cutoff Laurent recurrence, common p-power scaling, exact Newton interpolation at
      the prime-power cutoff.',
        'records':records}
    Path(__file__).with_suffix('.json').write_text(json.dumps(result,indent=2)+'\n')
```

D Reproducible third-level unit-mass evaluator

This evaluator supplies the five finite exceptions in the proof of the third-level first-step congruence. The final companion step at that level is not asserted by this computation.

```
"""Exact modular interpolation masses, using interval products in residue classes.
No floating arithmetic; products remove powers of p before inversion.
"""
from pathlib import Path
from collections import Counter
import json,time

class Products:
    def __init__(self,p,L,N):
        self.p=p;self.mod=p**L;self.tables={}
        B=4*N
        for step in (3,6):
            for residue in range(step):
                low=(-B-residue+step-1)//step;high=(B-residue)//step
                vs=[0];us=[1]; factors=[]
                for j in range(low,high+1):
                    x=residue+step*j;v=0
                    if x==0:x=1
                    else:
                        while x%p==0:x//=p;v+=1
                    factors.append(x%self.mod)
                    vs.append(vs[-1]+v);us.append(us[-1]*x%self.mod)
                inv=[0]*len(us);inv[-1]=pow(us[-1],-1,self.mod)
                for j in range(len(factors)-1,-1,-1):
                    inv[j]=inv[j+1]*factors[j]%self.mod
                self.tables[(step,residue)]=(low,vs,us,inv)
    def product(self,lo,hi,step):
        assert (hi-lo)%step==0 and hi>=lo
        residue=lo%step;base,vs,us,inv=self.tables[(step,residue)]
        a=(lo-residue)//step-base;b=(hi-residue)//step-base+1
        return vs[b]-vs[a],us[b]*inv[a]%self.mod

def analyze(p,r,L=10):
    N=p**r;m=(N-2)//3;mod=p**L;pr=Products(p,L,N)
    def prods(H,numer):
        ranges=([(6-H,6*m-H,6),(-3*m-H,-3-H,3),(-3*m+1-H,-2-H,3)] if numer
                else [(1-H,3*m+1-H,3),(2-H,3*m+2-H,3),(-6*m-2-H,-2-H,6)])
        v=0;u=1
        for lo,hi,step in ranges:
            a,b=pr.product(lo,hi,step);v+=a;u=u*b%mod
        return v,u
    vq,uq=prods(0,False);vd,ud=prods(0,True)
    cv=vq-vd;cu=uq*pow(ud,-1,mod)%mod
    denominator_roots=set(z for j in range(1,m+1) for z in (6*j,-3*j,-3*j+1))
    mass=[0]*(r+1);counts=[0]*(r+1);minimum=[None]*(r+1);dist=[Counter() for _ in range(r+1)]
    def vp(x):
        n=0
        if not x:return L
        while x%p==0:n+=1;x//=p
        return n
    half=pow(2,-1,mod)
    for h in range(m+1):
        for H,c in [(3*h+1,1 if h%2==0 else 0),(3*h+2,(-1)**h*half),(-6*h-2,-half)]:
            if not c or H in denominator_roots:continue
            a,b=prods(H,True);d,e=prods(H,False)
            v=cv+a-d;assert v>=0
            u=cu*c*b*pow(e,-1,mod)%mod
            term=u*p**v%mod;t=vp(H)
            mass[t]=(mass[t]+term)%mod;counts[t]+=1;dist[t][v]+=1
            minimum[t]=v if minimum[t] is None else min(minimum[t],v)
    return {'p':p,'r':r,'modulus_exponent':L,'mass_mod':mass,
            'mass_valuations':[vp(z) for z in mass],'counts':counts,
            'minimum_individual_valuations':minimum,
            'individual_valuation_counts':[dict(z) for z in dist]}
```

```
if __name__=='__main__':
    rows=[]
    for p in (5,11,17,23,29):
        start=time.time();row=analyze(p,3);rows.append(row)
        print(p,row['mass_valuations'],row['minimum_individual_valuations'],'seconds',round(time.time()-start,2)
     ,flush=True)
    Path(__file__).with_suffix('.json').write_text(json.dumps(rows,indent=2))
```

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MANUSCRIPT VIII

Reflection, Rational Rigidity, and the Odd Branch of Swisher's E.3

Keio research notes 11 November 2026

Abstract

For every prime $p > 3$ with $p \equiv 2 \pmod 3$ and every odd integer $r \geq 3$, we prove Swisher's E.3 recurrence with its original cutoff and modulus p^{3r-1}. The higher levels follow from exact finite interpolation, polynomial moments, a logarithmic denominator bound, and an exhaustive finite set of integer certificates. The third level is settled uniformly by a finite-field rational identity. A reflected block derivative has a zero of order $p-1$ at the origin, only simple finite poles, a specified regular point, and a known leading coefficient at infinity. These properties determine the entire rational function. Its unit-pole residue cancels exactly the cubic transport correction, without evaluating either separately. All proof ingredients and finite certificates are included in this single source. The third-level argument includes $p = 5$ and $p = 11$. Historical priority is not asserted.

Scope and verification. This manuscript supplies a proof of the complete odd branch $p \equiv 2 \pmod 3$, $r \geq 3$. It incorporates the earlier higher-level proof and the new uniform third-level completion. It has received internal independent checks, but has not undergone external peer review. Other branches and other Swisher labels are not re-proved here. Finite computations are identified separately from proofs valid for arbitrary parameters.

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1 The unchanged problem and the proved range

Write $(a)_k = \prod_{j=0}^{k-1}(a+j)$, $(a)_0 = 1$, and $v_p(p) = 1$, $v_p(0) = +\infty$. For $L \equiv 2 \pmod 3$, $L \geq 2$, set

$$E(L) = \sum_{k=0}^{(L-2)/3} (-1)^k (6k+1) \frac{(1/3)_k^3}{k!^3}.$$

The odd branch of Swisher's Table 2, E.3 [1], asks for

$$E(p^r) - p^2 E(p^{r-2}) \in p^{3r-1}\mathbb{Z}_p, \qquad p > 3, \quad p \equiv 2 \pmod 3, \quad r \geq 3 \text{ odd}. \tag{1.1}$$

Only this previously remaining branch is studied here. No result from a different Swisher label is assumed as a proof of this one.

Theorem 1.1 (Range established in this manuscript)**.** *Congruence* (1.1) *holds for every prime $p > 3$ with $p \equiv 2 \pmod 3$ and every odd integer $r \geq 3$:*

$$\boxed{E(p^r) \equiv p^2 E(p^{r-2}) \pmod{p^{3r-1}}, \qquad r \geq 3 \text{ odd}.}$$

The higher levels use uniform arguments and exact finite certificates for explicitly enumerated exceptions. The third level is proved uniformly in Theorem 24.1, with no exceptional prime.

The proof is supplied below after the exact interpolation and valuation lemmas. Its central estimate is stronger than an initial-moment calculation: every normalized moment loses at most $O(\log j)$ powers of p at moment order j. A polynomial of degree $O(pr)$ can therefore separate the unit nodes at the target precision. An exact companion formula controls its contribution on the other nodes. This is the step that turns the previous exact reductions into a proof on an infinite subrange.

2 Three exact interpolation grids at the short E cutoff

For an integer $m \geq 0$ put $N = 3m + 2$ and

$$\mathcal{S}_m(z) = \sum_{k=0}^{m} (-1)^k (6k+1) \frac{(1/3)_k (1/3 + z/6)_k (1/3 - z/3)_k (2/3 - z/3)_k}{k!(1 - z/6)_k (1 + z/3)_k (2/3 + z/3)_k}.$$

At $z = 0$ this is the original E sum with cutoff m. Define

$$D_m(z) = \prod_{j=1}^{m} (1 - z/(6j))(1 + z/(3j))(1 + z/(3j-1)),$$

$$Q_m(z) = \prod_{h=0}^{m} (1 - z/(3h+1))(1 - z/(3h+2))(1 + z/(6h+2)).$$

The cleared expression $P_m(z) = D_m(z)\mathcal{S}_m(z)$ is a polynomial of degree at most $3m$. Its $3m + 3$ distinct interpolation nodes are

$$C_h = 3h + 1, \qquad F_h = 3h + 2, \qquad B_h = -6h - 2 \quad (0 \leq h \leq m).$$

At these nodes the exact cleared values are

$$P_m(H) = D_m(H)T(H), \qquad \begin{cases} T(C_h) = C_h, & h \text{ even}, \\ T(C_h) = 0, & h \text{ odd}, \\ T(F_h) = (-1)^h F_h/2, \\ T(B_h) = -B_h/2. \end{cases} \tag{2.1}$$

2.1 The two positive grids

For complex z with positive real part and away from denominator poles, apply the nonterminating Whipple ${}_6F_5(-1)$ transformation [2] with

$$a = 1/3, \quad b = 1/3 + z/6, \quad c = 1/3 - z/3, \quad e = 2/3, \quad f = 2/3 - z/3.$$

The equal factors at $e = 1 + a - e = 2/3$ cancel. The transformed Dixon ${}_3F_2(1)$ has upper parameters $2/3, 2/3 + z/6, 2/3 - z/3$ and lower parameters $1 - z/6, 1 + z/3$. The required convergence conditions reduce to $\mathrm{Re}(z/3) > 0$ and $\mathrm{Re}(z/6) > 0$. Whipple followed by Dixon and gamma reflection evaluates the series as

$$\frac{z}{2} \frac{\sin(\pi(1/3 + z/6))}{\sin(\pi z/6)}.$$

At $z = C_h$ or F_h, the original series terminates at h; its lower parameters are never nonpositive integers at these positive nodes. The displayed expression is respectively C_h or zero according to the parity of h, and $(-1)^h F_h/2$. This proves the first three lines of (2.1). It also explains why the same sine expression must not be substituted on the negative grid: there a lower parameter becomes a nonpositive integer, and analytic continuation does not represent the specified finite truncation.

2.2 A finite WZ certificate for the negative grid

At $z = B_h$, the finite terminating sum through h is

$$\sum_{k=0}^{h} t(h,k), \qquad t(h,k) = (-1)^k (6k+1) \frac{(1/3)_k (-h)_k (2h+1)_k (2h+4/3)_k}{k!(h+4/3)_k (1/3-2h)_k (-2h)_k}.$$

For $0 \le k \le h$ the displayed denominator has no zero; at $h = 0$ only $k = 0$ is used. We prove

$$\sum_{k=0}^{h} t(h,k) = 3h + 1. \tag{2.2}$$

Put $u(h,k) = t(h,k)/(3h+1)$ and define

$$\begin{aligned} A(h,k) &= 864h^4 + 3024h^3 - 144h^2k^2 + 96h^2k + 3912h^2 \\ &\quad - 264hk^2 + 176hk + 2212h + 9k^4 - 12k^3 - 119k^2 + 82k + 460, \end{aligned}$$

$$R(h,k) = -\frac{k(3k-1)(2h-k+1)(6h-3k+2)A(h,k)}{16(h+1)(2h+1)^2(3h+2)(6h+5)(6h+7)(6k+1)(h-k+1)}.$$

Direct rational simplification gives the certificate

$$u(h+1,k) - u(h,k) = R(h,k+1)u(h,k+1) - R(h,k)u(h,k). \tag{2.3}$$

To interpret the upper boundary, cancel the apparent pole using

$$\frac{(-h)_k}{h-k+1} = \frac{(-h-1)_k}{h+1}.$$

Thus the product $R(h,k)u(h,k)$ has a well-defined value at $k = h+1$. Direct cancellation gives

$$R(h,0)u(h,0) = 0, \qquad R(h,h+1)u(h,h+1) = -u(h+1,h+1).$$

Summing (2.3) for $0 \le k \le h$ and adding the last term $u(h+1,h+1)$ proves that the sum of $u(h,k)$ over $0 \le k \le h$ is independent of h. Its value at $h = 0$ is one, proving (2.2).

2.3 The apparent poles in the cleared interpolation

At $z = B_h$, the only possible zero in D_m is its factor indexed by $j = 2h+1$ in $1 + z/(3j-1)$. If $2h \ge m$, this index is outside D_m; the original finite sum terminates at h, and (2.2) gives $P_m(B_h) = D_m(B_h)(3h+1)$. If $2h < m$, then $D_m(B_h) = 0$. In each cleared summand with $k \le h$ the same zero remains in the denominator-clearing tail. In each summand with $k \ge h+1$ the numerator contains the factor $(1/3 + z/6 + h)$, which vanishes at B_h. Every cleared summand is therefore zero, so $P_m(B_h) = 0$. This is again exactly $D_m(B_h)(3h+1)$ and proves (2.1) at every negative node, without evaluating any $0/0$ summand.

Consequently interpolation of P_m at all the listed nodes gives the exact identity

$$\mathcal{S}_m(0) = \sum_H W_m(H)T(H), \qquad W_m(H) = \frac{D_m(H)}{\prod_{K\neq H}(1 - H/K)},$$

with K, H ranging over the three grids. The weights may be zero; their rational denominators are never zero. Equivalently,

$$\frac{D_m(z)\mathcal{S}_m(z)}{Q_m(z)} = \sum_H \frac{W_m(H)T(H)}{1 - z/H}.$$

The left side is $O(z^{-3})$ at infinity, proving the exact vanishing of its first and second positive moments.

3 Exact moments for the three-grid short E deformation

The statements in this section are exact. The uniform p-integrality assertion needed by the proposed moment argument is not assumed or proved here. Let $L = 3m+2$, $m \ge 0$, and $b_k = (1/3)_k/k!$. Define

$$S_m(z) = \sum_{k=0}^{m} (-1)^k (6k+1) b_k^3 \prod_{h=0}^{k-1} \frac{(1 - z/(3h+1))(1 - z/(3h+2))(1 + z/(6h+2))}{(1 - z/(6h+6))(1 + z/(3h+3))(1 + z/(3h+2))}.$$

Put

$$q_h(z) = (1 - z/(3h+1))(1 - z/(3h+2))(1 + z/(6h+2)),$$
$$d_h(z) = (1 - z/(6h+6))(1 + z/(3h+3))(1 + z/(3h+2)),$$
$$Q_m(z) = \prod_{h=0}^{m} q_h(z), \qquad D_m(z) = \prod_{h=0}^{m-1} d_h(z), \qquad F_m(z) = \frac{D_m(z)S_m(z)}{Q_m(z)}.$$

The numerator $D_m S_m$ is a polynomial of degree at most $3m$, whereas Q_m has degree $3m+3$ and distinct roots. Thus F_m is proper and has a Laurent expansion at infinity

$$F_m(z) = -\sum_{j\ge 1} M_j(L) z^{-j}, \qquad M_1(L) = M_2(L) = 0.$$

Equivalently, if $F_m(z) = \sum_H \mu_m(H)/(1 - z/H)$ is its partial fraction expansion at its three node grids, then $M_j(L) = \sum_H \mu_m(H) H^j$. This definition allows a cancelled pole to have residue weight zero; it does not assume that all values of S_m at lower-parameter collisions are terminating evaluations.

Proposition 3.1. *Set*

$$C(L) = (-1)^{m+1} L^2 (L-1)^3 b_m^3.$$

There are uniquely determined even polynomials $P_j(L) \in \mathbb{Q}[L^2]$, $j \geq 0$, with $P_0 = 1$ and $\deg P_j \leq 2\lfloor 2j/3 \rfloor$, such that

$$M_{j+3}(L) = C(L) P_j(L).$$

In particular,

$$P_1 = 1, \qquad P_2 = 2L^2 - 3, \qquad P_3 = -\frac{L^4 - 13L^2 + 16}{4}.$$

Proof. At infinity the ratio of the leading terms of q_h and d_h is $-(3h+3)^2/(3h+1)^2$. Hence

$$S_m(\infty) = \sum_{k=0}^{m} (6k+1) b_k = \frac{L(L-1) b_m}{2}.$$

The last equality telescopes: the difference between $(3k+1)(3k+2)b_k/2$ and its value at $k-1$ is $(6k+1)b_k$. The leading coefficient of D_m/Q_m is $(-1)^m 2L(L-1)^2 b_m^2 z^{-3}$. This proves $M_3 = C(L)$.

Separating the last summand of S_m gives

$$q_m(z) F_m(z) = d_{m-1}(z) F_{m-1}(z) + (-1)^m (2L-3) b_m^3. \tag{3.1}$$

Write

$$A = L(L-1), \quad B = (L-2)(L-3), \quad a = (L-1)(3L-2), \quad c = (L-2)(3L-7).$$

The two cubic factors have expansions

$$q_m(z) = \frac{z^3 - z^2 - az + 2L(L-1)^2}{2L(L-1)^2},$$

$$d_{m-1}(z) = -\frac{z^3 - z^2 - cz - 2(L-3)(L-2)^2}{2(L-3)(L-2)^2}.$$

Since $b_m/b_{m-1} = (L-4)/(L-2)$, comparison of coefficients in (3.1), after dividing by $C(L)$, yields

$$\begin{aligned} AP_j(L) - BP_j(L-3) = {} & AP_{j-1}(L) - BP_{j-1}(L-3) \\ & + AaP_{j-2}(L) - BcP_{j-2}(L-3) \\ & - 2AL(L-1)^2 P_{j-3}(L) - 2B(L-3)(L-2)^2 P_{j-3}(L-3), \end{aligned}$$

for $j \geq 1$, where $P_{-1} = P_{-2} = P_{-3} = 0$ and $P_0 = 1$. This also provides an independent recursive definition of the polynomials. Indeed the operator

$$\mathcal{L}(P) = L(L-1)P(L) - (L-2)(L-3)P(L-3)$$

maps $\mathbb{Q}[L^2]$ to $(2L-3)\mathbb{Q}[L(L-3)]$. The first space consists of even polynomials; the second consists of polynomials anti-invariant under $L \mapsto 3-L$. In the displayed power bases, $\mathcal{L}(L^{2k})/(2L-3)$ has degree k and leading coefficient $3k+2$. Consequently this operator is an isomorphism of the filtered vector spaces over $\mathbb{Q}$. For even preceding polynomials, the right side of the recurrence is anti-invariant under $L \mapsto 3-L$; the two cubic terms exchange with opposite signs under this reflection. Thus the recurrence uniquely defines an even P_j at every step. It gives the degree bound

$$\deg P_j \leq \max\{\deg P_{j-1},\ \deg P_{j-2} + 2, \deg P_{j-3} + 4\} \leq 2\lfloor 2j/3 \rfloor.$$

At $L = 2$, evaluation of the polynomial recurrence gives the coefficients of

$$\sum_{j \geq 0} P_j(2) t^j = \frac{1}{1 - t - 4t^2 + 4t^3}.$$

This is exactly the normalized expansion of $F_0 = 1/q_0$. The recurrence in m now proves the moment identities for every integer $m \geq 0$. □

Prime-power valuation of the first moments. For $p > 3$, $p \equiv 2 \pmod 3$, and odd r, take $L = p^r$. Then

$$v_p(b_{(p^r-2)/3}) = \frac{r-1}{2}, \qquad v_p(C(p^r)) = \frac{7r-3}{2}.$$

For the first equality count multiples of p^ℓ in the numerator and denominator of b_m. If ℓ is even, the numerator acquires one additional multiple, while for odd ℓ the counts agree; there are $(r-1)/2$ positive even levels at most r. Therefore $M_1 = M_2 = 0$, and M_3, M_4, M_5, M_6 all have exactly the valuation $(7r-3)/2$ for these primes. This exceeds or equals the required $3r-1$.

Why a uniform denominator estimate is needed. The polynomial operator is not an isomorphism over $\mathbb{Z}[1/2]$: its diagonal is $3k+2$, not a unit there. For example

$$P_6(L) = \frac{L^8 + 290L^6 - 3325L^4 + 16666L^2 - 29900}{28},$$

and P_9 has denominator 280. Thus the rational-polynomial existence proof alone does not establish $P_j(p^r) \in \mathbb{Z}_p$ for all j at primes $p \equiv 2 \pmod 3$. The logarithmic bounds proved below supply the uniform estimate needed in this manuscript. They are used with finite polynomial separation, so no Euler-power limiting argument is required.

4 An exact negative-parameter companion and its first three moments

This section proves the exact finite companion identity used in the higher-level argument. The identity alone is not a supercongruence.

Let $M = 3n+1$, $n \geq 0$, and put

$$\mathcal{H}_n = \{3h+1 : 0 \leq h \leq n\} \cup \{3h+2 : 0 \leq h < n\} \cup \{-2(3h+2) : 0 \leq h < n\},$$

$$\mathcal{D}_n = \{6j, -3j : 1 \leq j \leq n\} \cup \{-3h-1 : 0 \leq h < n\}.$$

All $3n+1$ nodes in $\mathcal{H}_n$ are distinct and nonzero. Define

$$U(3h+1) = -(-1)^h(3h+1)/2, \qquad U(3h+2) = \begin{cases} 3h+2, & h \text{ odd}, \\ 0, & h \text{ even}, \end{cases} \qquad U(-2(3h+2)) = 3h+2,$$

$$D_n(z) = \prod_{D \in \mathcal{D}_n} (1 - z/D), \qquad Q_n(z) = \prod_{H \in \mathcal{H}_n} (1 - z/H), \qquad \widetilde{W}_n(H) = \frac{D_n(H)}{\prod_{K \in \mathcal{H}_n, K \neq H}(1 - H/K)}.$$

The companion and its positive moments are

$$\mathcal{A}(M) = \sum_H \widetilde{W}_n(H) U(H), \qquad \widetilde{M}_j(M) = \sum_H \widetilde{W}_n(H) U(H) H^j.$$

Set $d_k = (-1/3)_k/k!$ and $c_k = (2/3)_k/k!$.

Theorem 4.1. *For every integer $n \geq 0$,*

$$\begin{aligned} \mathcal{A}(3n+1) &= -\frac{1}{2}\sum_{k=0}^{n} (-1)^k (6k-1)(9k^2-3k-1) d_k^3 \\ &= \sum_{k=0}^{n} (-1)^k (6k-1) d_k^3 + \frac{(-1)^n}{2} c_n^3. \end{aligned}$$

Moreover, with $C_n = (-1)^{n+1} M^2 c_n^3/2$,

$$\widetilde{M}_1(M) = C_n, \qquad \widetilde{M}_2(M) = C_n(M^2+1)/2, \qquad \widetilde{M}_3(M) = C_n(2M^2-1).$$

4.1 A rational deformation with exactly the companion node values

Put

$$K_k(z)=\prod_{j=0}^{k-1}\frac{(1+z/[2(3j-1)])(1-z/(3j-1))(1-z/(3j+1))}{(1-z/[2(3j+3)])(1+z/(3j+3))(1+z/(3j+1))},$$

$$V_n(z)=-\frac{1}{2(2-z)}\sum_{k=0}^{n}(-1)^k(6k-1)(z+18k^2-6k-2)d_k^3K_k(z). \tag{4.1}$$

Although this expression displays $2-z$ in its denominator, its $k=0$ term is the constant $-1/2$. Every $k\geq 1$ term has the factor $1-z/2$ in its numerator, cancelling this additional denominator. Consequently $D_n(z)V_n(z)$ is a polynomial of degree at most $3n$. At $z=2$, the terms with $k\geq 2$ vanish after cancellation; the $k=1$ term simplifies to $-5(z-1)(z+10)/(6(z-6)(z+3))$ and has value $1/2$. Thus $V_n(2)=0$ for every $n\geq 1$.

The three finite certificate proofs supplied below establish

$$[D_n(z)V_n(z)]_{z=H}=D_n(H)U(H)\qquad (H\in\mathcal{H}_n). \tag{4.2}$$

On the first positive grid the sum terminates at h; on the second it terminates at $h+1$, with $z=2$ handled above. The lower parameters are regular on both positive grids. On the negative grid the finite certificate gives the sum through $h+1$; where a later denominator would vanish, the cleared identity is proved term by term. Thus (4.2) never assigns a value to a raw $0/0$ summand.

Interpolation of the degree-$3n$ polynomial D_nV_n at its $3n+1$ distinct nodes now proves

$$V_n(0)=\sum_H\widetilde{W}_n(H)U(H)=\mathcal{A}(M),\qquad F_n(z):=\frac{D_n(z)V_n(z)}{Q_n(z)}=\sum_H\frac{\widetilde{W}_n(H)U(H)}{1-z/H}.$$

Evaluation of (4.1) at zero proves the first equality in Theorem 4.1.

4.2 An elementary telescoping simplification

Let $u_k=(-1)^kd_k^3$ and $B_k=k^3u_k$. Since $B_{k+1}=-(k-1/3)^3u_k$, direct expansion gives

$$(6k-1)(9k^2-3k-1)u_k=-27(B_{k+1}-B_k)-2(6k-1)u_k.$$

Summing uses $B_0=0$ and $27B_{n+1}=(-1)^nc_n^3$ and proves the second equality of the theorem, without any asymptotic argument.

4.3 The adjacent-cutoff recurrence and the three moments

For $n\geq 1$, write

$$A_n(z)=(1-z/M)(1-z/(M-2))(1+z/[2(M-2)]),$$

$$B_n(z)=(1-z/[2(M-1)])(1+z/(M-1))(1+z/(M-3)).$$

These are precisely Q_n/Q_{n-1} and D_n/D_{n-1}. The numerator of $K_n(z)$ is $(1+z)(1-z/2)Q_{n-1}(z)$. Separating the last summand of (4.1) therefore gives the exact polynomial recurrence

$$A_n(z)F_n(z)-B_n(z)F_{n-1}(z)=G_n(z), \tag{4.3}$$

$$G_n(z)=-\frac{(-1)^n(6n-1)d_n^3}{4}(z+1)(z+18n^2-6n-2).$$

This derivation uses the summand formula, not the moment identities being proved.

At infinity, write $F_n(z) = -\sum_{j\ge1} \widetilde{M}_j(M) z^{-j}$. The coefficients of z^2, z^1, z^0 in (4.3) determine its first three moments successively. For complete verification, write $A_n = \sum a_i z^i$, $B_n = \sum b_i z^i$, and $G_n = \sum g_i z^i$. Those three equations are

$$-a_3\widetilde{M}_1 + b_3\widetilde{M}_1' = g_2,$$
$$-a_2\widetilde{M}_1 - a_3\widetilde{M}_2 + b_2\widetilde{M}_1' + b_3\widetilde{M}_2' = g_1,$$
$$-a_1\widetilde{M}_1 - a_2\widetilde{M}_2 - a_3\widetilde{M}_3 + b_1\widetilde{M}_1' + b_2\widetilde{M}_2' + b_3\widetilde{M}_3' = g_0,$$

where primes mean the preceding cutoff $M-3$. Here

$$a_3 = \frac{1}{2M(M-2)^2}, \quad a_2 = -2a_3, \quad a_1 = -\frac{3M-4}{2M(M-2)},$$

$$b_3 = -\frac{1}{2(M-1)^2(M-3)}, \quad b_2 = -2b_3, \quad b_1 = \frac{3M-5}{2(M-1)(M-3)}.$$

Substitution of the three claimed formulas satisfies these rational identities, using

$$\frac{C_{n-1}}{C_n} = -\frac{(M-3)^2(M-1)^3}{M^2(M-2)^3}.$$

At $n=0$, $F_0(z) = -1/[2(1-z)]$, so every positive moment is $-1/2$, exactly the claimed initial values. Since $a_3 \neq 0$ for $n \ge 1$, induction proves all three formulas. This completes the theorem.

5 A uniform logarithmic valuation bound for the primary E moments

Let $P_j(L) \in \mathbb{Q}[L^2]$ be the primary moment polynomials defined by the exact three-grid recurrence, normalized by $P_0 = 1$. They satisfy

$$\deg P_j \le d_j := 2\left\lfloor \frac{2j}{3} \right\rfloor, \qquad M_{j+3}(L) = C(L)P_j(L),$$

where, for $L = 3m+2$,

$$C(L) = (-1)^{m+1}L^2(L-1)^3 b_m^3, \qquad b_m = \frac{(1/3)_m}{m!}.$$

For a prime $p > 3$, $p \equiv 2 \pmod 3$, define

$$a_j = \left\lfloor \log_p(3d_j+2) \right\rfloor, \qquad e_j = a_j + \left\lfloor \frac{a_j}{2} \right\rfloor.$$

Here the logarithm only denotes the integer characterized by $p^{a_j} \le 3d_j + 2 < p^{a_j+1}$.

Theorem 5.1. *For every $j \ge 0$ and every $x \in \mathbb{Z}_p$,*

$$\boxed{v_p(P_j(x)) \ge -e_j.}$$

Equivalently, $p^{e_j}P_j$ is integer-valued on $\mathbb{Z}_p$. Moreover $e_0 = 0$, and $e_j \le j-1$ for every $j \ge 1$.

This is an all-degree theorem with a logarithmic loss. It does not assert the stronger, still unproved statement $P_j \in \mathrm{Int}(\mathbb{Z}_p)$ for all j.

5.1 An integral-coefficient formula before normalization

For $h \geq 0$, put

$$\widehat{q}_h(t) = (1-(3h+1)t)(1-(3h+2)t)(1+(6h+2)t),$$
$$\widehat{d}_h(t) = (1-(6h+6)t)(1+(3h+3)t)(1+(3h+2)t).$$

For $0 \leq k \leq m$, define integers $\phi_j(k,m)$ by

$$\frac{\prod_{h=k}^{m-1} \widehat{d}_h(t)}{\prod_{h=k}^{m} \widehat{q}_h(t)} = \sum_{j\geq 0} \phi_j(k,m) t^j.$$

All these coefficients are integers because every denominator polynomial has constant term one. The exact normalized moment formula is

$$P_j(3m+2) = \frac{2}{L(L-1)b_m} \sum_{k=0}^{m} (6k+1) b_k \phi_j(k,m), \qquad L = 3m+2. \tag{5.1}$$

To verify it directly, expand the proper rational function $F_m = D_m S_m / Q_m$ term by term at $z = \infty$, using $t = 1/z$. The kth summand is

$$(-1)^k (6k+1) b_k^3 \frac{\prod_{h=k}^{m-1} d_h(z)}{\prod_{h=k}^{m} q_h(z)}.$$

Its leading coefficient is $(-1)^m 2L(L-1)^2 b_m^2 (6k+1) b_k z^{-3}$. Dividing the reversed numerator and denominator polynomials by their leading coefficients gives exactly the series defining $\phi_j(k,m)$. Comparison with $F_m = -C(L)\sum_{j\geq 0} P_j(L) z^{-j-3}$ proves (5.1). In particular this argument holds for every coefficient, not only for any prescribed finite range.

5.2 Valuation of the normalizing denominator

Lemma 5.2. *Let $m \geq 0$, $L = 3m+2$, and $a = \lfloor \log_p L \rfloor$. Then*

$$b_k \in \mathbb{Z}_p \quad (k \geq 0), \qquad v_p(L(L-1)b_m) \leq a + \lfloor a/2 \rfloor.$$

Proof. At the level $P = p^\ell$, let u_ℓ be the least nonnegative solution of $3u_\ell + 1 \equiv 0 \pmod{P}$. Explicitly,

$$u_\ell = \begin{cases} (P-1)/3, & \ell \text{ even}, \\ (2P-1)/3, & \ell \text{ odd}. \end{cases}$$

The difference between the counts of multiples of P in $\prod_{h=0}^{m-1}(3h+1)$ and in $m!$ is

$$\delta_\ell = \begin{cases} 1, & m \bmod P > u_\ell, \\ 0, & m \bmod P \leq u_\ell. \end{cases}$$

The factors of 3 in b_m are units. Thus $v_p(b_m) = \sum_{\ell \geq 1} \delta_\ell \geq 0$, proving integrality. Also $\delta_\ell = 0$ for $p^\ell > L$: all positive numerator factors are at most $L-4$ and all denominator indices are at most m.

If $t = v_p(L) > 0$, then for $\ell \leq t$ the congruence $3m \equiv -2 \pmod{p^\ell}$ gives

$$m \bmod p^\ell = \begin{cases} 2(p^\ell - 1)/3, & \ell \text{ even}, \\ (p^\ell - 2)/3, & \ell \text{ odd}. \end{cases}$$

Consequently $\delta_\ell = 1$ precisely at the even levels among $1, \dots, t$. Since $L-1$ is a unit,

$$v_p(L(L-1)b_m) \le t + \lfloor t/2 \rfloor + (a-t) = a + \lfloor t/2 \rfloor \le a + \lfloor a/2 \rfloor.$$

If instead $u = v_p(L-1) > 0$, then $m \bmod p^\ell = u_\ell$ for $\ell \le u$. All those δ_ℓ are zero, and $v_p(L(L-1)b_m) \le u + (a-u) = a$. If both L and $L-1$ are units, the same bound follows from $\sum \delta_\ell \le a$. These cases include $m = 0$, where $L = 2$ and $a = 0$. □

Proof of Theorem 5.1. Formula (5.1) and the preceding lemma imply, for every nonnegative integer m,

$$v_p(P_j(3m+2)) \ge -\left(\lfloor \log_p(3m+2) \rfloor + \left\lfloor \frac{\lfloor \log_p(3m+2) \rfloor}{2} \right\rfloor \right).$$

Use this for $m = 0, 1, \dots, d_j$. Every one of these values lies in $p^{-e_j}\mathbb{Z}_p$. Set $R_j(X) = P_j(3X+2)$, whose degree is at most d_j. Its exact Newton expansion is

$$R_j(X) = \sum_{k=0}^{d_j} \Delta^k R_j(0) \binom{X}{k}.$$

Each finite difference is an integer linear combination of the sampled values, so it too belongs to $p^{-e_j}\mathbb{Z}_p$. For every $X \in \mathbb{Z}_p$, the binomial polynomial $\binom{X}{k}$ is p-integral, by continuity from the nonnegative integers. Taking $X = (x-2)/3 \in \mathbb{Z}_p$ proves the uniform bound. No division by $k!$ is being applied to the finite-difference coefficient: that factorial belongs to the integer-valued basis.

Finally, for $j \ge 1$ the elementary inequality

$$6\lfloor 2j/3 \rfloor + 2 < 5^{\lfloor (2j-1)/3 \rfloor + 1}$$

follows by checking $j = 1, 2, 3$ and then increasing j by three; the left side increases by 12, while the right side is multiplied by 25. Since $p \ge 5$, this implies $a_j \le \lfloor (2j-1)/3 \rfloor$, and hence $a_j + \lfloor a_j/2 \rfloor \le j-1$. Also $d_0 = 0$, so $e_0 = 0$. □

5.3 The double-strip transport is controlled at every higher odd level

Corollary 5.3. *Let $r \ge 3$ be odd, $M = p^{r-2}$, and let $c_j \in \mathbb{Q}_p$ satisfy $v_p(c_j) \ge j$ for every $j \ge 1$. Then the series*

$$V_r = p^2 \sum_{j \ge 1} c_j M_j(M)$$

converges. For every odd $r \ge 5$,

$$V_r \in p^{3r-1}\mathbb{Z}_p.$$

At the remaining level $r = 3$, one has the precise reduction

$$V_3 \equiv p^2 c_3 M_3(p) \pmod{p^8}.$$

Proof. The first two moments vanish exactly. For $j \ge 3$, put $n = j-3$. The exact primary moment normalization and the proven valuation $v_p(C(p^{r-2})) = (7r-17)/2$ give

$$v_p(p^2 c_j M_j(M)) \ge 2 + j + \frac{7r-17}{2} - e_n = \frac{7r-7}{2} + n - e_n.$$

The right side tends to infinity, because $e_n = O(\log n)$, so convergence follows. For $n = 0$ it is at least $(7r-7)/2$; for $n \ge 1$ it is at least $(7r-5)/2$. When $r \ge 5$, the former bound is at least $3r-1$. When $r = 3$, all terms with $n \ge 1$ belong to $p^8\mathbb{Z}_p$, leaving only the displayed cubic term. □

6 Polynomial structure of every companion moment

Let $\widetilde{M}_j(M)$ be the companion moments from the proved negative-parameter companion identity, and put

$$C_n = \frac{(-1)^{n+1}M^2}{2}\left(\frac{(2/3)_n}{n!}\right)^3, \qquad M = 3n+1.$$

Proposition 6.1. *There exist even polynomials $R_j(M) \in \mathbb{Q}[M^2]$, $j \geq 0$, with*

$$\deg R_j \leq 2\left\lfloor \frac{2j+1}{3} \right\rfloor, \qquad R_j(1) = 1,$$

such that, for every integer $n \geq 0$,

$$\widetilde{M}_{j+1}(3n+1) = C_n R_j(3n+1).$$

Their first values are $R_0 = 1$, $R_1 = (M^2+1)/2$, and $R_2 = 2M^2 - 1$.

Proof. Start with the already proved rational-function recurrence $A_n(z)F_n(z) - B_n(z)F_{n-1}(z) = G_n(z)$, where G_n has degree two. Its first three Laurent equations give the three initial moments just stated. Set

$$A = M(M-2), \quad B = (M-3)(M-1), \quad a = (M-2)(3M-4), \quad c = (M-1)(3M-5).$$

The monic forms of its cubic factors are

$$A_n(z) = \frac{z^3 - 2z^2 - az + 2M(M-2)^2}{2M(M-2)^2},$$

$$B_n(z) = -\frac{z^3 - 2z^2 - cz - 2(M-3)(M-1)^2}{2(M-3)(M-1)^2}.$$

Use the exact ratio

$$\frac{C_{n-1}}{C_n} = -\frac{(M-3)^2(M-1)^3}{M^2(M-2)^3}.$$

The homogeneous Laurent equations, starting at the coefficient of z^{-1}, yield, for every $j \geq 3$,

$$\begin{aligned} AR_j(M) - BR_j(M-3) = {} & 2\big[AR_{j-1}(M) - BR_{j-1}(M-3)\big] \\ & + AaR_{j-2}(M) - BcR_{j-2}(M-3) \\ & - 2AM(M-2)^2R_{j-3}(M) - 2B(M-3)(M-1)^2R_{j-3}(M-3). \end{aligned}$$

We show this recurrence uniquely defines even rational polynomials. The operator

$$\mathcal{L}'(P) = M(M-2)P(M) - (M-3)(M-1)P(M-3)$$

maps $\mathbb{Q}[M^2]$ to $(2M-3)\mathbb{Q}[M(M-3)]$. Indeed its image is anti-invariant under $M \mapsto 3-M$. For the basis element M^{2k}, division of its image by $2M-3$ gives a polynomial of degree k in $M(M-3)$, with leading coefficient $3k+1$. Thus $\mathcal{L}'$ is an isomorphism of the filtered vector spaces over $\mathbb{Q}$. The right side of the displayed recurrence is anti-invariant when the preceding polynomials are even. It therefore has a unique even preimage. Its degree is bounded by

$$\deg R_j \leq \max\{\deg R_{j-1}, \deg R_{j-2}+2, \deg R_{j-3}+4\}.$$

The given initial degrees imply the asserted bound. At $M = 1$ the recurrence reduces to

$$R_j(1) = 2R_{j-1}(1) + R_{j-2}(1) - 2R_{j-3}(1),$$

so all these values are one. This agrees with the exact base $F_0(z) = -1/[2(1-z)]$, whose every positive moment equals $-1/2$. Finally the Laurent recurrence determines the moments uniquely successively in the cutoff and the moment index, since its leading cubic coefficient is nonzero for $n \geq 1$. Hence the constructed polynomials give the actual moments for all n and j. □

The diagonal $3k+1$ need not be a p-adic unit, so this proposition asserts rational-polynomial structure and a uniform degree bound; it does not assert coefficient integrality or integer-valuedness.

7 A uniform logarithmic loss for all companion moments

Use the polynomials R_j of Proposition 6.1, so that

$$\widetilde{M}_{j+1}(3n+1) = C_n R_j(3n+1), \qquad d_j := 2\left\lfloor \frac{2j+1}{3} \right\rfloor \geq \deg R_j.$$

For $p > 3$, $p \equiv 2 \pmod 3$, define

$$a_j = \left\lfloor \log_p(3d_j+1) \right\rfloor, \qquad \eta_j = a_j + \left\lceil \frac{a_j}{2} \right\rceil = \left\lceil \frac{3a_j}{2} \right\rceil.$$

Theorem 7.1. *For every $j \geq 0$ and every $x \in \mathbb{Z}_p$,*

$$v_p(R_j(x)) \geq -\eta_j.$$

In particular, if $M = p^{2s}$, then for every $j \geq 1$,

$$v_p(\widetilde{M}_j(M)) \geq 7s - e_p(j), \qquad e_p(j) = \left\lceil \frac{3}{2}\left\lfloor \log_p(4j+1) \right\rfloor \right\rceil.$$

Proof. We first prove a finite-value formula with integral coefficients before normalization. Write $w = 1/z$. For each integer root a, reversing the factor $1 - z/a$ gives its leading monomial times $1 - aw$. Thus the reversed products in D_n/Q_n and in each K_k have integer coefficients as formal series in w: their denominator factors all have constant term one and integral coefficients. The additional factor $(z + 18k^2 - 6k - 2)/(z-2)$ becomes

$$\frac{1 + (18k^2 - 6k - 2)w}{1 - 2w} \in \mathbb{Z}[[w]].$$

The leading coefficient of K_k is $(-1)^k/d_k^2$, where $d_k = (-1/3)_k/k!$. Also

$$\frac{D_n(z)}{Q_n(z)} = -\nu_n w\,(1 + O(w)), \qquad \nu_n = (-1)^n M c_n^2, \quad M = 3n+1, \quad c_n = (2/3)_n/n!.$$

It follows from the proved summand formula for V_n that there exist integers $\phi_j(k,n)$, for every $j \geq 0$, such that

$$F_n(z) = -\frac{\nu_n}{2} \sum_{j \geq 0} \left(\sum_{k=0}^{n} (6k-1) d_k \phi_j(k,n) \right) w^{j+1}.$$

Here ϕ_j is simply the coefficient of the product of the reversed factors just described, so integrality holds for every degree, not only for a finite initial range. Since $C_n = -\nu_n M c_n/2$, comparison of Laurent coefficients proves the exact formula

$$R_j(3n+1) = -\frac{1}{Mc_n}\sum_{k=0}^{n}(6k-1)d_k\phi_j(k,n). \tag{7.1}$$

Both d_k and c_k belong to $\mathbb{Z}_p$: for any $\alpha \in \mathbb{Z}_p$, $(\alpha)_k/k! = (-1)^k\binom{-\alpha}{k}$ is integral. Put $b = \lfloor \log_p M \rfloor$ and $t = v_p(M)$. At the level $q = p^\ell$, the excess numerator count in c_n is zero or one; it is zero when $q > M$. For $\ell \le t$, the residue of $n = (M-1)/3$ is $(q-1)/3$ when ℓ is even, and $(2q-1)/3$ when ℓ is odd. The first numerator index for $3k+2 \equiv 0 \pmod q$ is respectively $(2q-2)/3$ and $(q-2)/3$. Hence exactly the odd levels among $1,\dots,t$ contribute an excess. The remaining $b-t$ levels each contribute at most one. Consequently

$$v_p(Mc_n) \le t + \lceil t/2 \rceil + (b-t) \le b + \lceil b/2 \rceil.$$

Equation (7.1) therefore bounds every finite value $R_j(3n+1)$ by the negative of this quantity.

Apply this bound at $n = 0, 1, \dots, d_j$. All these values belong to $p^{-\eta_j}\mathbb{Z}_p$. The polynomial $T_j(X) = R_j(3X+1)$ has exact Newton expansion

$$T_j(X) = \sum_{k=0}^{d_j} \Delta^k T_j(0)\binom{X}{k}.$$

Each finite difference is an integer linear combination of the sampled values. Each $\binom{X}{k}$ is integral on $\mathbb{Z}_p$, by continuity from the nonnegative integers. Taking $X = (x-1)/3$ proves the uniform bound for $R_j(x)$.

Finally, at $M = p^{2s}$ the excess-count argument gives $v_p(c_n) = s$ exactly, so $v_p(C_n) = 7s$. For the moment with index j, its polynomial is R_{j-1}, and

$$3d_{j-1} + 1 = 6\left\lfloor \frac{2j-1}{3} \right\rfloor + 1 \le 4j - 1 \le 4j + 1.$$

Thus $\eta_{j-1} \le e_p(j)$, proving the last assertion. □

8 A uniform lower bound for every companion interpolation mass

Let $p > 3$, $p \equiv 2 \pmod 3$, $s \ge 1$, $M = p^{2s}$ and $n = (M-1)/3$. Retain the companion grids, denominator roots, and models from Theorem 4.1, and put

$$\mu_M(H) = \widetilde{W}_n(H)U(H).$$

Proposition 8.1. *For every companion node H, with $t = v_p(H)$,*

$$\boxed{v_p(\mu_M(H)) \ge 3s - \lceil t/2 \rceil \ge 2s.}$$

Zero masses have valuation $+\infty$. In particular every unit-node mass belongs to $p^{3s}\mathbb{Z}_p$.

Proof. We may suppose $U(H) \ne 0$ and $H \notin \mathcal{D}_n$; otherwise the mass is zero. At an active node, $U(H)/H$ is one of $1, 1/2, -1/2$, hence is a unit. Direct multiplication of the grids gives

$$\frac{\prod_{K\in\mathcal{H}_n} K}{\prod_{D\in\mathcal{D}_n} D} = (-1)^n M\left(\frac{(2/3)_n}{n!}\right)^2.$$

The Pochhammer ratio has valuation s: at each odd level p^ℓ, $1 \le \ell \le 2s$, its numerator has one extra divisible factor, and at each even level its numerator and denominator counts agree. Thus the displayed normalization has valuation $4s$. The exact mass, up to an irrelevant sign, is this normalization times

$$\frac{U(H)}{H}\frac{\prod_{D\in\mathcal{D}_n}(D-H)}{\prod_{K\in\mathcal{H}_n,K\neq H}(K-H)}.$$

All nonzero differences here have absolute value less than $4M$. Since $p \ge 5$, no power p^ℓ with $\ell > 2s$ divides any of them.

For $q = p^\ell$ and $x = H \bmod q$, define

$$\delta_q(x) = 1 + \#\{D \in \mathcal{D}_n : D \equiv x \pmod q\} - \#\{K \in \mathcal{H}_n : K \equiv x \pmod q\}.$$

The initial one removes the node $K = H$ from the denominator count. Consequently

$$v_p(\mu_M(H)) = 4s + \sum_{\ell=1}^{2s} \delta_{p^\ell}(H \bmod p^\ell). \tag{8.1}$$

All three arithmetic progressions have steps prime to q. Their complete blocks of q consecutive indices contribute equally to the numerator and denominator counts and cancel. If ℓ is even, the remaining cutoff is $b = (q-1)/3$; if ℓ is odd, it is $b = (2q-1)/3$. The residual grids and denominators are exactly the companion sets with $n = b$, respectively $M' = q$ or $M' = 2q$.

Here are the complete residual counts for nonzero residues $1 \le x < q$, listed by x mod 6:

$x \bmod 6$	0	1	2	3	4	5
$\#K$ $(q \equiv 1 \pmod 6)$	0	1	1	1	2	1
$\#D$ $(q \equiv 1 \pmod 6)$	2	1	0	1	1	1
δ_q $(q \equiv 1 \pmod 6)$	3	1	0	1	0	1
$\#K$ $(q \equiv 5 \pmod 6)$	2	2	2	1	2	3
$\#D$ $(q \equiv 5 \pmod 6)$	2	3	2	2	2	1
δ_q $(q \equiv 5 \pmod 6)$	1	2	1	2	1	−1

These counts follow directly by solving the three progression congruences in the residual ranges. For example, in the even case $M' = q$, the negative node grid contributes once for residues $x \equiv 3, 4 \pmod 6$, while the positive denominator grid contributes once for $x \equiv 0, 5 \pmod 6$. In the odd case $M' = 2q$, those same contributions are respectively $(1,1,0,0,1,1)$ and $(1,1,1,1,0,0)$ across the six residue classes. The other two grids have step three, yielding the displayed totals. At the zero residue, the even case has one node and no denominator root, so $\delta_q(0) = 0$; the odd case has three nodes and no denominator root, so $\delta_q(0) = -2$.

Thus every even level contributes a nonnegative amount. Each odd level contributes at least -1, with a possible extra loss of one precisely when $p^\ell \mid H$. There are s odd levels altogether and $\lceil t/2 \rceil$ of them divide H; here $t \le 2s$ because $0 < |H| < 2M < p^{2s+1}$. Equation (8.1) proves

$$v_p(\mu_M(H)) \ge 4s - s - \lceil t/2 \rceil,$$

as asserted. □

This is an individual-mass estimate. It is independent of the later cancellation argument for E.3 and can be combined with exact moment formulas or polynomial projectors without assuming any positive-moment integrality.

8.1 The corresponding primary mass bound

Let $L = p^{2s+1}$, $m = (L-2)/3$, and let $\mu_L(H)$ be the primary three-grid mass. Then

$$\boxed{v_p(\mu_L(H)) \geq 3s + 1 - \lfloor v_p(H)/2 \rfloor \geq 2s + 1.} \tag{8.2}$$

Again zero masses are harmless. We give the full analogous count, so this bound does not rely on numerical interpolation tests. The primary normalization is

$$\frac{\prod K}{\prod D} = (-1)^{m+1} 2L(L-1)^2 \left(\frac{(1/3)_m}{m!} \right)^2 .$$

Its valuation is $4s+1$, since the Pochhammer ratio has valuation s. At an active node $T(H)/H$ is a unit, and all nonzero differences have absolute value less than $4L < pL$. The same formula $1 + \#D_x - \#K_x$ gives each level's correction. For odd ℓ, the residual cutoff is $(q-2)/3$ with $q = p^\ell$; for even ℓ, it is $(2q-2)/3$. The corresponding residual primary endpoint is q or $2q$ respectively. Complete blocks cancel as before. For $1 \leq x < q$, the exact tables are

$x \bmod 6$	0	1	2	3	4	5
$\#K$ (ℓ odd)	0	1	2	1	1	1
$\#D$ (ℓ odd)	2	1	1	1	0	1
δ_q (ℓ odd)	3	1	0	1	0	1
$\#K$ (ℓ even)	2	3	2	1	2	2
$\#D$ (ℓ even)	2	1	2	2	2	3
δ_q (ℓ even)	1	-1	1	2	1	2

For example, at an odd level the negative node grid contributes in residue classes $2, 3 \pmod 6$, while the positive denominator grid contributes in classes $0, 1 \pmod 6$. At an even level their respective six-class contributions are $(1,1,1,0,0,1)$ and $(1,0,0,1,1,1)$. The remaining step-three grids give the totals shown above. At $x = 0$ the correction is zero for odd ℓ and -2 for even ℓ: there are respectively one or three nodes and no denominator root at that residue. Thus odd levels cost nothing, while each of the s even levels costs at most one, plus one more when it divides H. There are $\lfloor v_p(H)/2 \rfloor$ such divisible even levels. Subtracting these losses from $4s+1$ proves (8.2). In particular unit-node masses belong to $p^{3s+1}\mathbb{Z}_p$.

8.2 The unit-node contribution at level three

Corollary 8.2. *For every prime $p > 3$ with $p \equiv 2 \pmod 3$,*

$$\sum_{H \in \mathcal{H}_{(p^3-2)/3},\ p \nmid H} \mu_{p^3}(H) \in p^8 \mathbb{Z}_p.$$

Proof. First assume $p \geq 41$. Let $g(Y) = 3Y^2 - 2Y^3$ and $\Pi(X) = g(g(X^{p-1}))$. This polynomial has integer coefficients, no terms of degree below $4(p-1)$, and degree $9(p-1)$. For a unit integer H, Fermat's congruence and $g(1+u) = 1 - 3u^2 - 2u^3$ give $\Pi(H) \equiv 1 \pmod{p^4}$. For $p \mid H$, one has $\Pi(H) \in p^{4(p-1)}\mathbb{Z}_p$. The preceding proposition, with $s = 1$, therefore gives

$$\sum_{p \nmid H} \mu_{p^3}(H) \equiv \sum_{H} \mu_{p^3}(H)\Pi(H) \pmod{p^8}.$$

The right-hand side is an integer linear combination of the moments $M_j(p^3)$ with $4(p-1) \leq j \leq 9(p-1)$. Their common normalization $C(p^3)$ has valuation nine. For every such j, the logarithmic moment bound in Theorem 5.1 has loss at most one: indeed, putting $n = j - 3$,

$$3d_n + 2 = 6\lfloor 2n/3 \rfloor + 2 \leq 4j - 10 \leq 36(p-1) - 10 < p^2 \qquad (p \geq 41).$$

Consequently $M_j(p^3) \in p^8\mathbb{Z}_p$.

The remaining primes are exactly $5, 11, 17, 23, 29$. They can be checked by finite integer arithmetic in the exact mass formula above. Here is an explicit modular certificate, computed modulo p^{10}:

p	5	11	17	23	29
$p^{-9}\sum_{p\nmid H}\mu_{p^3}(H) \pmod p$	1	6	6	18	10.

In particular every one of those sums lies in $p^9\mathbb{Z}_p$. The reproducible certificate is fast_primary_masses.py: it evaluates the displayed product formula modulo p^{10}, removing the exact power of p from every factor before taking a modular inverse. Prefix products in each residue class modulo three and six evaluate every arithmetic progression in the formula. A zero factor is omitted only at the unique denominator occurrence $K = H$; numerator zeroes are excluded by testing $H \in \mathcal{D}_m$ first. Thus all modular inverses are applied to units and the finite checks are exact. They complete the five exceptional cases. □

The corollary controls only the unit-node contribution. The shell $v_p(H) = 1$ and its cancellation with the cubic double-strip term remain separate obligations.

9 One-step transport of the valuation-one stratum

Let $L = p^{2s+1}$, $M = L/p = p^{2s}$, $p > 3$, $p \equiv 2 \pmod 3$. Write $\mu_L(H)$ for the primary interpolation masses and $\widetilde{\mu}_M(J)$ for the proved companion masses. Let $\mathcal{H}_L, \mathcal{D}_L$ denote the primary root sets and set

$$R^{(1)}_{L,p}(X) = \frac{\prod_{d\in\mathcal{D}_L,\, p\nmid d}(1 - pX/d)}{\prod_{h\in\mathcal{H}_L,\, p\nmid h}(1 - pX/h)} = \sum_{j\ge 0} a_j X^j.$$

Every retained root is a unit, so $a_0 = 1$ and $a_j \in p^j\mathbb{Z}_p$. The series is restricted and converges at every integral node.

The exact one-step product identity is

$$\mu_L(pJ) = p\widetilde{\mu}_M(J)R^{(1)}_{L,p}(J). \tag{9.1}$$

Indeed dividing the divisible primary nodes by p gives respectively

$$\{2, 5, \dots, M-2\}, \quad \{1, 4, \dots, M\}, \quad -2\{2, 5, \dots, M-2\},$$

exactly the companion nodes. Dividing the divisible denominator roots gives $6j, -3j$ for $1 \le j \le (M-1)/3$ and $-1, -4, \dots, -(M-3)$, exactly the companion denominator roots. For the first two denominator grids this follows from $\lfloor (L-2)/(3p)\rfloor = (M-1)/3$; the last grid has fine endpoint $-(L-3)$, so its largest divisible quotient is $-(M-3)$. Since $p \equiv 5 \pmod 6$, the primary model formulas give $T(pJ) = pU(J)$ on each companion grid, including zero model values. Separating the divisible and unit factors in the interpolation products proves (9.1); no division by a zero weight is used.

In particular, the omitted part of the two-step transport has the exact split

$$U_{L,p} = \sum_{p\nmid H}\mu_L(H) + p\sum_{p\nmid J}\widetilde{\mu}_M(J)R^{(1)}_{L,p}(J). \tag{9.2}$$

This identity places the formerly correlated valuation-one stratum inside a weighted unit sum of the exact companion family.

10 The original odd E.3 congruence at all sufficiently high levels

Let $p > 3$ be a prime with $p \equiv 2 \pmod 3$, and define

$$E(L) = \sum_{k=0}^{(L-2)/3} (-1)^k (6k+1) \left(\frac{(1/3)_k}{k!} \right)^3 \qquad (L \equiv 2 \pmod 3).$$

All interpolation identities and moment bounds used below are exact or have been proved in the preceding sections. In particular, no assertion of integral normalized moments of every degree is required.

Theorem 10.1. *Put $r = 2s+1$, with $s \geq 2$, and suppose*

$$\left\lceil \frac{3}{2} \left\lfloor \log_p(4(3s+1)(p-1)+1) \right\rfloor \right\rceil \leq s-1. \tag{10.1}$$

Then the original E.3 congruence holds:

$$\boxed{E(p^{2s+1}) \equiv p^2 E(p^{2s-1}) \pmod{p^{6s+2}}.}$$

In particular it holds in each of the following ranges:

1. *every $p > 3$, $p \equiv 2 \pmod 3$, and every odd $r \geq 13$;*
2. *every such $p \geq 11$ and every odd $r \geq 9$;*
3. *every such $p \geq 41$ and every odd $r \geq 7$.*

10.1 The two exact one-step identities

Set $N = p^{2s+1}$, $M = p^{2s}$, and $K = p^{2s-1}$. Write $\mu_N(H)$ and $\mu_K(H)$ for the primary three-grid masses, and $\widetilde{\mu}_M(J)$ for the companion masses. Their sums are $E(N)$, $E(K)$, and $\mathcal{A}(M)$, respectively. Their positive moments are denoted by M_j and $\widetilde{M}_j$.

For either grid family, form the quotient of unit denominator-root factors by unit node factors, evaluated at pX:

$$R(X) = \frac{\prod_{d \in \mathcal{D},\ p \nmid d} (1 - pX/d)}{\prod_{h \in \mathcal{H},\ p \nmid h} (1 - pX/h)}.$$

Let R_0 use the primary family at N, and R_1 the companion family at M. Both have expansions

$$R_i(X) = 1 + \sum_{j \geq 1} a_{i,j} X^j, \qquad a_{i,j} \in p^j \mathbb{Z}_p \quad (i = 0, 1). \tag{10.2}$$

These are restricted power series and converge at every $X \in \mathbb{Z}_p$. Separating divisible and unit roots in the interpolation weights gives

$$\mu_N(pJ) = p\widetilde{\mu}_M(J) R_0(J), \qquad \widetilde{\mu}_M(pH) = p\mu_K(H) R_1(H). \tag{10.3}$$

Here the correspondence between divisible nodes and the lower family is exact, including endpoints and zero model values.

For completeness, the first correspondence is the one established in (9.1). For the reverse correspondence, divide the divisible companion nodes by p. The positive grids become $\{2, 5, \ldots, K\}$ and $\{1, 4, \ldots, K-1\}$, and the negative grid becomes $-2\{1, 4, \ldots, K-1\}$, precisely the primary grids at K. The divided denominator roots are

$$\{6j, -3j : 1 \leq j \leq (K-2)/3\} \ \cup \ \{-2, -5, \ldots, -(K-3)\},$$

also precisely the primary denominator roots. The explicit models satisfy $U(pH) = pT(H)$ on these grids because $p \equiv 5 \pmod 6$. Thus the same product separation proves the second identity in (10.3), without dividing by a model value that might vanish.

10.2 Moment estimates in the forms needed for transport

The proved primary normalizations and logarithmic bounds give

$$M_1(N) = M_2(N) = M_1(K) = M_2(K) = 0, \tag{10.4}$$

$$v_p(M_j(N)) \geq 7s + 2 - e_{j-3} \qquad (j \geq 3), \tag{10.5}$$

$$v_p(M_j(K)) \geq 7s - 5 - e_{j-3} \qquad (j \geq 3), \tag{10.6}$$

where $e_0 = 0$, $e_h \leq h - 1$ for $h \geq 1$, and $e_h = O(\log(h+2))$. The companion logarithmic bound gives

$$v_p(\widetilde{M}_j(M)) \geq 7s - \eta_{j-1}.$$

In particular,

$$v_p(\widetilde{M}_j(M)) \geq 7s - (j-1) \qquad (j \geq 1). \tag{10.7}$$

For $j = 1, 2$ this follows from the exact normalized polynomials $R_0^{\mathrm{mom}} = 1$ and $R_1^{\mathrm{mom}} = (M^2+1)/2$. For the other indices use $\eta_h \leq h$ for $h \geq 2$. To verify this last elementary inequality, the companion definition is

$$\eta_h = \left\lceil \frac{3}{2} \left\lfloor \log_p \left(6 \left\lfloor \frac{2h+1}{3} \right\rfloor + 1 \right) \right\rfloor \right\rceil.$$

The inequality

$$6 \left\lfloor \frac{2h+1}{3} \right\rfloor + 1 < 5^{\lfloor 2h/3 \rfloor + 1}$$

holds at $h = 2, 3, 4$ and is preserved by increasing h by three: the left side increases by 12, whereas the right side is multiplied by 25. Since $p \geq 5$, it implies $\eta_h \leq h$.

A common nondecreasing logarithmic bound is

$$\epsilon_p(j) = \left\lceil \frac{3}{2} \left\lfloor \log_p(4j+1) \right\rfloor \right\rceil. \tag{10.8}$$

For $j \geq 3$ it bounds e_{j-3}, and for $j \geq 1$ it bounds η_{j-1}. Consequently the positive moments of the two fine families satisfy

$$v_p(M_j(N)) \geq 7s + 2 - \epsilon_p(j), \qquad v_p(\widetilde{M}_j(M)) \geq 7s - \epsilon_p(j).$$

The first inequality also holds for $j = 1, 2$, whose moments vanish. The proved individual unit-node estimates are

$$v_p(\mu_N(H)) \geq 3s + 1 \ (p \nmid H), \qquad v_p(\widetilde{\mu}_M(J)) \geq 3s \ (p \nmid J). \tag{10.9}$$

10.3 A low-degree unit projector and its leakage

Put

$$t = 3s + 1, \qquad D = t(p-1), \qquad \chi_t(X) = 1 - (1 - X^{p-1})^t \in \mathbb{Z}[X].$$

Its nonzero monomials have degrees $i(p-1)$ for $1 \leq i \leq t$, so its smallest degree is $p - 1 \geq 4$ and its degree is D. For every integral unit x, Fermat's congruence gives $1 - \chi_t(x) \in p^t \mathbb{Z}_p$. Let $B = \epsilon_p(D)$; the theorem assumes $B \leq s - 1$.

First consider the primary unit sum $U_N = \sum_{p \nmid H} \mu_N(H)$. Replacing 1 by $\chi_t(H)$ on its unit nodes changes it by an element of $p^{6s+2}\mathbb{Z}_p$, by (10.9). The full-node polynomial sum satisfies

$$\sum_H \mu_N(H)\chi_t(H) \in p^{7s+2-B}\mathbb{Z}_p \subseteq p^{6s+3}\mathbb{Z}_p.$$

Its contribution from nonunit nodes is, by the first exact strip,

$$\sum_{p|H} \mu_N(H)\chi_t(H) = p\sum_J \widetilde{\mu}_M(J)R_0(J)\chi_t(pJ).$$

Write $R_0(X)\chi_t(pX) = \sum_{j\geq p-1} b_j X^j$. Every b_j belongs to $p^j\mathbb{Z}_p$: both factors have this coefficient property, and their convolution preserves it. Hence (10.7) shows that each term of its moment expansion has valuation at least

$$1 + j + 7s - (j-1) = 7s + 2.$$

The expansion converges, since the sharper logarithmic bound makes these valuations tend to infinity. We have proved

$$U_N \in p^{6s+2}\mathbb{Z}_p. \tag{10.10}$$

Apply the same projector to the companion unit sum $\widetilde{U}_M = \sum_{p\nmid J} \widetilde{\mu}_M(J)$. The approximation error on unit nodes belongs to $p^{3s+t}\mathbb{Z}_p = p^{6s+1}\mathbb{Z}_p$. The full-node polynomial sum has valuation at least $7s - B \geq 6s + 1$. Its nonunit leakage is

$$p\sum_H \mu_K(H)R_1(H)\chi_t(pH).$$

Again the product series starts in degree $p - 1 \geq 4$ and its degree-j coefficient has valuation at least j. For all such j, $e_{j-3} \leq j - 4$. Therefore each leakage term has valuation at least

$$1 + j + (7s - 5) - (j - 4) = 7s \geq 6s + 1.$$

Convergence follows from the logarithmic loss as before. Thus

$$\widetilde{U}_M \in p^{6s+1}\mathbb{Z}_p. \tag{10.11}$$

This calculation explicitly controls the nonunit leakage; the projector is not being treated as a literal indicator at nonunit nodes.

10.4 Completion of the two one-step congruences

Sum the first identity in (10.3) and add the primary unit nodes. It gives exactly

$$E(N) = U_N + p\mathcal{A}(M) + p\sum_{j\geq 1} a_{0,j}\widetilde{M}_j(M).$$

Every term of the last series has valuation at least $7s + 2$ by (10.2) and (10.7). Thus

$$E(N) \equiv p\mathcal{A}(M) \pmod{p^{6s+2}}. \tag{10.12}$$

The reverse strip similarly gives

$$\mathcal{A}(M) = \widetilde{U}_M + pE(K) + p\sum_{j\geq 3} a_{1,j}M_j(K),$$

where the first two moments have vanished exactly. The $j = 3$ term has valuation at least $7s - 1$; every $j \geq 4$ term has valuation at least $7s$. Since $s \geq 2$, these are at least $6s + 1$. Together with (10.11), this proves

$$\mathcal{A}(M) \equiv pE(K) \pmod{p^{6s+1}}. \tag{10.13}$$

All series manipulations are legitimate: there are finitely many nodes, the root-product series converge on $\mathbb{Z}_p$, and the logarithmic moment estimates also give convergence after rearranging by degree. Combining the two displayed congruences proves the assertion of Theorem 10.1.

10.5 Verification of the explicit ranges

For integer $s \geq 2$, condition (10.1) is equivalent to the strict inequality

$$(12s+4)(p-1)+1 < p^{\lfloor 2(s-1)/3 \rfloor + 1}. \tag{10.14}$$

For fixed s let $A = 12s+4$ and $a = \lfloor 2(s-1)/3 \rfloor + 1$. In each range considered below, $a \geq 2$. The function $(Ap - A + 1)/p^a$ is decreasing for real $p \geq 5$, because its derivative has numerator $Ap(1-a) + a(A-1) < 0$. It therefore suffices to check the smallest value of p in each claimed range.

For $p \geq 5$, the left side at $p = 5$ is $48s + 17$. At $s = 6, 7, 8$ the required comparisons are

$$305 < 625, \qquad 353 < 3125, \qquad 401 < 3125.$$

Increasing s by three adds 144 to the left side and multiplies the right side by 25, preserving the inequality. Thus it holds for all $s \geq 6$.

For $p \geq 11$, the left side at $p = 11$ is $120s + 41$. At $s = 4, 5, 6$ the comparisons are

$$521 < 1331, \qquad 641 < 1331, \qquad 761 < 14641.$$

The step $s \mapsto s+3$ adds 360 to the left side and multiplies the right side by 121. Thus the inequality holds for all $s \geq 4$.

For $p \geq 41$, the left side at $p = 41$ is $480s + 161$. At $s = 3, 4, 5$ the comparisons are

$$1601 < 1681, \qquad 2081 < 68921, \qquad 2561 < 68921.$$

The step $s \mapsto s+3$ adds 1440 to the left side and multiplies the right side by 1681. This proves the final stated range.

11 A reflection gain for companion interpolation samples

Retain $d_k = (-1/3)_k/k!$, $c_n = (2/3)_n/n!$, $M = 3n+1$, and the exact normalized companion moments $R_j(M)$.

Lemma 11.1. *Let $p > 3$, $p \equiv 2 \pmod 3$. For every $n \geq 0$ with $p \mid 3n+1$, every coefficient of the integral numerator series*

$$\mathcal{N}_n(w) = \sum_{k=0}^{n} (6k-1) d_k \phi_k(w), \qquad \phi_k(w) = \sum_{j \geq 0} \phi_j(k,n) w^j,$$

from (7.1) belongs to $p\mathbb{Z}_p$.

Proof. Use the monic reversed product of the companion summand kernel:

$$\widehat{K}_k(w) = \prod_{j=0}^{k-1} f_j(w), \qquad f_j(w) = \frac{(1+(6j-2)w)(1-(3j-1)w)(1-(3j+1)w)}{(1-(6j+6)w)(1+(3j+3)w)(1+(3j+1)w)}.$$

The coefficient series ϕ_k has the form

$$\phi_k(w) = A_n(w) \widehat{K}_k(w) \frac{1 + (18k^2 - 6k - 2)w}{1 - 2w}, \qquad A_n(w) \in \mathbb{Z}[[w]], \quad A_n(0) = 1.$$

Here A_n is the common reversed denominator-root/node-root quotient. Its precise roots are immaterial for this argument; every root is integral, so A_n is an integral unit series. It suffices to prove that

$$\sum_{k=0}^{n} (6k-1) d_k \widehat{K}_k(w) (1 + (18k^2 - 6k - 2)w)$$

vanishes in $\mathbb{F}_p[[w]]$.

Put $a = (p+1)/3$, which is an even integer. The binomial identity $d_k = (-1)^k\binom{1/3}{k}$ and the p-adic Lucas congruence give

$$d_{pb+c} \equiv \frac{(1/3)_b}{b!}\,(-1)^c\binom{a}{c} \pmod p \qquad (0 \le c < p). \tag{11.1}$$

Indeed $1/3 = a + p(-1/3)$ in $\mathbb{Z}_p$; the ordinary Lucas identity extends to binomial coefficients with p-adic top argument by continuity from nonnegative integers. This also proves that the left side vanishes when $c > a$.

For any consecutive p indices, each of the three linear root progressions in the numerator and denominator of f_j runs through all residues of $\mathbb{F}_p$. Thus their products agree, and

$$\prod_{j=bp}^{(b+1)p-1} f_j(w) = 1 \quad \text{in } \mathbb{F}_p[[w]], \qquad \widehat{K}_{pb+c}(w) = \widehat{K}_c(w).$$

The remaining linear and quadratic factors in k depend only on $k \bmod p$. Therefore every complete block reduces, up to the scalar $(1/3)_b/b!$, to

$$B(w) = \sum_{c=0}^{a}(6c-1)(-1)^c\binom{a}{c}\widehat{K}_c(w)(1 + (18c^2 - 6c - 2)w).$$

The same holds for the final partial block: since $3n+1 \equiv 0$ modulo p, its last residue is $(2p-1)/3 = 2a - 1 \ge a$ and therefore includes the entire nonzero Lucas support.

It remains to show $B(w) = 0$. The exact root lists defining f_j give, using $3a = 1$ in $\mathbb{F}_p$,

$$f_{a-1-j}(w) = f_j(w)^{-1}, \qquad \widehat{K}_a(w) = 1, \qquad \widehat{K}_{a-c}(w) = \widehat{K}_c(w).$$

The first equality pairs the numerator roots $2-6(a-1-j), 3(a-1-j)-1, 3(a-1-j)+1$ with the denominator roots $6j+6, -3j-3, -3j-1$, and conversely. Because a is even, $(-1)^c\binom{a}{c}$ is unchanged under $c \mapsto a-c$. Also

$$6(a-c) - 1 = -(6c-1), \qquad 18(a-c)^2 - 6(a-c) - 2 = 18c^2 - 6c - 2 \quad \text{in } \mathbb{F}_p.$$

Consequently the summands cancel in reflected pairs. If there is a middle summand, its factor $6c-1$ is zero. Hence $B(w) = 0$, proving the claimed coefficientwise divisibility. □

Corollary 11.2. *With the notation of Theorem 7.1, the bound improves to*

$$v_p(R_j(x)) \ge -\max\{0, \eta_j - 1\} \qquad (j \ge 0,\ x \in \mathbb{Z}_p).$$

Proof. At a sample $M = 3n+1$, put $b = \lfloor \log_p M \rfloor$. If $p \nmid M$, its normalizing denominator Mc_n has valuation at most b. If $p \mid M$, the previous count and the new numerator gain give a loss of at most $b + \lceil b/2 \rceil - 1$. For $b \ge 1$ the latter dominates b; for $b = 0$ there is no loss. Apply these bounds to the same Newton samples as in Theorem 7.1. □

Corollary 11.3. *For every integer $n \ge 0$ with $1 \le M = 3n+1 < p^2$, and every $j \ge 0$,*

$$v_p(R_j(M)) \ge -1.$$

Consequently, if $3d_j + 1 < p^2$, where $d_j = 2\lfloor (2j+1)/3 \rfloor$, then

$$pR_j(x) \in \mathbb{Z}_p \qquad \textit{for every } x \in \mathbb{Z}_p.$$

Proof. Use the exact formula $R_j(M) = -(Mc_n)^{-1}[w^j]\mathcal{N}_n(w)$. If $p \nmid M$, the excess-count proof gives $v_p(c_n) \le 1$ since $M < p^2$. Thus its normalization costs at most one power of p. If $p \mid M$, then $v_p(M) = v_p(c_n) = 1$, so the normalizing denominator costs two powers, but Lemma 11.1 restores one. This proves the first assertion, including $M = 1$. For the second, apply it at the Newton interpolation samples $M = 3n + 1$, $0 \le n \le d_j$. Their maximum is below p^2. All their finite differences belong to $p^{-1}\mathbb{Z}_p$, and the binomial basis is integral on $\mathbb{Z}_p$. □

Theorem 11.4. *For every prime $p \ge 29$ with $p \equiv 2 \pmod 3$,*

$$\boxed{E(p^5) \equiv p^2 E(p^3) \pmod{p^{14}}.}$$

Proof. Repeat the proof of Theorem 10.1 with $s = 2$, $t = 7$, and $D = 7(p-1)$. The leakage and one-step transport estimates in that proof are valid for every $s \ge 2$ and require no logarithmic threshold. Only its two full-node projector sums need the following improved estimates.

For each companion moment index $1 \le j \le D$, its polynomial is R_{j-1} and its largest interpolation sample satisfies

$$3d_{j-1} + 1 = 6\left\lfloor \frac{2j-1}{3} \right\rfloor + 1 \le 4j - 1 \le 28(p-1) - 1 < p^2.$$

The last inequality holds for $p \ge 29$. The preceding corollary therefore gives $v_p(\widetilde{M}_j(p^4)) \ge 14 - 1 = 13$. The full-node companion projector sum belongs to $p^{13}\mathbb{Z}_p$, as required.

The primary moments have their first two terms zero. For $3 \le j \le D$, the largest sample for P_{j-3} is at most $4(j-3) + 2 \le 28(p-1) - 10 < p^2$. The primary logarithmic bound is therefore at most one, giving $v_p(M_j(p^5)) \ge 16 - 1 = 15$. This exceeds the required primary projector precision 14. The unchanged leakage calculations now give $E(p^5) \equiv p\mathcal{A}(p^4) \pmod{p^{14}}$ and $\mathcal{A}(p^4) \equiv pE(p^3) \pmod{p^{13}}$. Their combination proves the theorem. □

12 Exact finite certificates for the remaining higher-level cases

The analytic theorem leaves only the following pairs (s, p) at odd levels $r = 2s + 1 \ge 7$:

$$(3,5), (3,11), (3,17), (3,23), (3,29), (4,5), (5,5).$$

This is a finite exhaustive list, not a numerical sample of the remaining higher-level range. We give exact polynomial-interpolation certificates for the moments required by the proved projector argument.

Put $t = 3s + 1$. The projector $1 - (1 - X^{p-1})^t$ uses precisely the moment indices $j = i(p-1)$, $1 \le i \le t$. Its proof only needs

$$v_p(R_{i(p-1)-1}(p^{2s})) \ge -(s-1), \qquad v_p(P_{i(p-1)-3}(p^{2s+1})) \ge -s. \tag{12.1}$$

The first bound controls the full companion projector sum, and the second controls the full primary projector sum. The bounds on unit-node approximation, nonunit leakage, and the infinite transport corrections remain the previously proved uniform estimates; none of those is replaced by a finite truncation.

12.1 The integer recurrences used for certification

For the companion define a formal series

$$Y_n(w) = -Mc_n \sum_{j \ge 0} R_j(M) w^j, \qquad M = 3n + 1.$$

The proved rational-function recurrence gives, for $n \geq 1$,

$$\begin{aligned}&(1-2w-(M-2)(3M-4)w^2+2M(M-2)^2w^3)Y_n(w)\\&\quad=(1-2w-(M-1)(3M-5)w^2-2(M-3)(M-1)^2w^3)Y_{n-1}(w)\\&\qquad+(6n-1)d_n(1+w)(1+(18n^2-6n-2)w),\end{aligned}$$

with $Y_0(w) = -1/(1-w)$, $d_n = (-1/3)_n/n!$, and $c_n = (2/3)_n/n!$. All polynomial coefficients are integers and the left constant term is one. Thus each coefficient is determined using only addition and multiplication and already known lower coefficients.

For the primary family let $L = 3n+2$, $b_n = (1/3)_n/n!$, and

$$Z_n(w) = \frac{L(L-1)b_n}{2}\sum_{j\geq 0} P_j(L)w^j.$$

Its integer recurrence is

$$\begin{aligned}&(1-w-(L-1)(3L-2)w^2+2L(L-1)^2w^3)Z_n(w)\\&\quad=(1-w-(L-2)(3L-7)w^2-2(L-2)^2(L-3)w^3)Z_{n-1}(w)+(6n+1)b_n.\end{aligned}$$

At $n = 0$ take the preceding series as zero, giving $Z_0(w) = 1/(1-w-4w^2+4w^3)$. This recurrence follows equally from the integral reversed-factor formula for the primary normalized moments.

To compute a Pochhammer ratio modulo a power of p, each numerator and denominator factor is written as a power of p times a unit. Only the unit denominator is inverted; the accumulated valuation is tracked as an integer. This computes b_n, c_n, d_n without inverting p modulo its own power.

12.2 Degree bounds, precision, and exact Newton evaluation

For any selected normalized index j, the proved degree bounds are

$$d_j^R = 2\lfloor (2j+1)/3\rfloor, \qquad d_j^P = 2\lfloor 2j/3\rfloor.$$

Therefore its values at the small cutoffs $n = 0, \ldots, d_j$ determine the desired polynomial exactly. Let $d_{\max}$ be the largest degree needed for a fixed family and pair (s,p). Choose the common scale

$$A_R = \left\lceil \frac{3}{2}\lfloor \log_p(3d_{\max}+1)\rfloor \right\rceil, \qquad A_P = \left\lfloor \frac{3}{2}\lfloor \log_p(3d_{\max}+2)\rfloor \right\rfloor,$$

and compute modulo p^{A+2} in the corresponding family. The previously proved normalization bounds ensure $v_p(Mc_n) \leq A_R$ and $v_p(L(L-1)b_n) \leq A_P$ throughout those samples. If a normalization is $p^v u$ with u a unit, the scaled sample is computed as p^{A-v} times the unnormalized coefficient times u^{-1}, with the indicated factor -1 or 2. No modular division by p is performed and no precision is lost: the exponent $A - v$ is nonnegative.

For either polynomial, let $f(n)$ denote its scaled value at the appropriate affine cutoff. Its exact Newton formula is

$$f(X) = \sum_{k=0}^{d_j} \Delta^k f(0)\binom{X}{k}.$$

Here $X = (p^{2s}-1)/3$ for the companion and $X = (p^{2s+1}-2)/3$ for the primary. Both are nonnegative integers. The binomial coefficients are computed as exact integers before reduction modulo p^{A+2}. All finite differences and final sums are then integer operations in that explicitly specified residue ring. This evaluates the actual polynomial value, not an approximation obtained by guessing a polynomial from too few samples.

If the scaled residue is nonzero with valuation $v < A+2$, its unscaled value has exactly valuation $v - A$. If the residue is zero, the unscaled value has valuation at least two. Thus the residue arithmetic certifies every inequality in (12.1).

12.3 Certificate output and exhaustion of the exceptional set

The complete reproducible program is work_e7_companion/finite_projector_certificates.py; its integer outputs are in the adjacent JSON file. It verifies 158 selected moment indices across the seven pairs and both families. The following table gives the largest small sample index and the minimum valuation of the scaled residues in each family; the scale A is displayed alongside, so the minimum unscaled valuation is their difference.

s	p	$d^R_{\max}$	A_R	$\min v(p^{A_R}R)$	$d^P_{\max}$	A_P	$\min v(p^{A_P}P)$
3	5	52	5	5	48	4	4
3	11	132	3	2	128	3	3
3	17	212	3	2	208	3	3
3	23	292	3	2	288	3	3
3	29	372	3	2	368	3	3
4	5	68	5	5	64	4	4
5	5	84	5	5	80	4	4

The moduli used are respectively p^{A_R+2} and p^{A_P+2}. In particular every companion minimum is at least -1 after removing the scale, and every primary minimum is at least zero. These are stronger than the requirements in (12.1).

Applying the already proved projector-and-leakage argument with these 158 certified moment bounds proves the unchanged E.3 congruence for all seven exceptional pairs. Combining them with the analytic ranges therefore gives the unconditional computationally certified conclusion

$$\boxed{E(p^r) \equiv p^2 E(p^{r-2}) \pmod{p^{3r-1}} \quad \text{for every } p>3,\ p\equiv 2 \pmod 3, \text{ and every odd } r \geq 7.}$$

The finite computations establish the finite exceptional set only; the assertion for every other prime and level comes from the uniform analytic theorem. The fifth level is supplied by the reflection argument and the additional finite certificates in the next section.

13 Exact finite completion of the level $r = 5$

The uniform reflection-gain argument proves the original $r = 5$ congruence for every prime $p \geq 29$ with $p \equiv 2 \pmod 3$. The only primes satisfying $3 < p < 29$ and $p \equiv 2 \pmod 3$ are $5, 11, 17, 23$. This section certifies that exhaustive finite exception set using the already proved polynomial-moment method.

Here $s = 2$, so the projector $1 - (1 - X^{p-1})^7$ uses exactly the seven positive moment indices $j = i(p-1)$, $1 \leq i \leq 7$. The sufficient bounds from the proved projector-and-leakage argument are

$$v_p\big(R_{i(p-1)-1}(p^4)\big) \geq -1, \qquad v_p\big(P_{i(p-1)-3}(p^5)\big) \geq -2.$$

All nonunit leakage and all infinite transport terms continue to use the uniform estimates; the calculation below only supplies these fourteen polynomial values per prime.

Use exactly the integer Laurent recurrences, proved degree bounds, and scaled Newton interpolation described in Section 12. In all eight family-prime cases, the common scaling exponent is $A = 3$ and the calculation is modulo $p^5 = p^{A+2}$. The normalization divisors at every sampled small cutoff have valuation at most three. Thus multiplication by p^{A-v} and inversion of the unit part supplies each scaled sample without losing precision. The Newton evaluation uses exact integer binomial coefficients at $(p^4 - 1)/3$ or $(p^5 - 2)/3$.

The complete reproducible invocation, from the archive root, is

```
python work_e7_companion/r5_finite_closure_driver.py
```

It imports the shared, independently audited function `certify` from finite_projector_certificates.py and writes r5_finite_closure.json. It does not overwrite the earlier 158-index higher-level certificate. Only the Python standard library is required.

The output for all 56 required values is summarized below. The minimum scaled valuation is the valuation of the exact residue of p^3R or p^3P modulo p^5; subtracting three gives the certified unscaled valuation. All minima shown are below the precision cap, so their listed valuations are exact.

p	$d^R_{\max}$	$\min v_p(p^3R)$	$d^P_{\max}$	$\min v_p(p^3P)$
5	36	3	32	3
11	92	2	88	3
17	148	2	144	3
23	204	2	200	3

Thus every companion value has valuation at least -1, and every primary value has valuation at least zero, which is stronger than the required primary bound. The finite sufficient conditions hold for all four primes. The uniform leakage argument therefore gives

$$E(p^5) \equiv p\mathcal{A}(p^4) \pmod{p^{14}}, \qquad \mathcal{A}(p^4) \equiv pE(p^3) \pmod{p^{13}}.$$

Combining them proves the unchanged $r = 5$ target for this exhaustive finite exception set. Together with the separate uniform argument for $p \geq 29$, it yields

$$\boxed{E(p^5) \equiv p^2E(p^3) \pmod{p^{14}} \qquad \text{for every prime } p > 3,\ p \equiv 2 \pmod 3.}$$

The finite output is not being extrapolated to other primes: those primes are covered by the uniform reflection-gain proof. This section settles the fifth level. The third level is completed below.

14 A uniform first-step congruence at the third level

Corollary 14.1. *For every prime $p > 3$, $p \equiv 2 \pmod 3$, the exact companion sum of Theorem 4.1 satisfies*

$$\boxed{E(p^3) \equiv p\mathcal{A}(p^2) \pmod{p^8}.}$$

In fact the positive-moment transport part of this difference belongs to $p^9\mathbb{Z}_p$.

Proof. Write $R^{(1)}_{p^3,p}(X) = 1+\sum_{j\geq 1} a_jX^j$ for the exact one-step kernel. Its coefficients satisfy $v_p(a_j) \geq j$. Summing the proved identity $\mu_{p^3}(pJ) = p\widetilde{\mu}_{p^2}(J)R^{(1)}_{p^3,p}(J)$ over the companion nodes gives

$$E(p^3) - p\mathcal{A}(p^2) = \sum_{p\nmid H} \mu_{p^3}(H) + p\sum_{j\geq 1} a_j\widetilde{M}_j(p^2).$$

The first sum belongs to $p^8\mathbb{Z}_p$ by Corollary 8.2.

The common companion moment normalization has valuation seven. The first two explicit moments thus have valuation at least seven, so their transport terms have valuation at least nine. For $j \geq 3$, Theorem 7.1 gives

$$v_p(\widetilde{M}_j(p^2)) \geq 7 - \eta_{j-1}, \qquad \eta_{j-1} = \left\lceil \frac{3}{2}\left\lfloor \log_p(6\lfloor(2j-1)/3\rfloor + 1)\right\rfloor\right\rceil.$$

Here $\eta_{j-1} \le j-1$. Indeed $6\lfloor(2j-1)/3\rfloor + 1 \le 4j-1$, and

$$4j - 1 < 5^{\lfloor 2(j-1)/3\rfloor + 1} \qquad (j \ge 3).$$

The latter inequality is checked for $j = 3, 4, 5$; increasing j by three adds twelve on the left and multiplies the right by twenty-five. It follows that each remaining transport term has valuation at least $1 + j + 7 - (j-1) = 9$. The logarithmic growth of η_{j-1} proves convergence of the series. This proves the corollary. □

Consequently the third-level E.3 congruence is exactly equivalent to the shorter companion estimate

$$\mathcal{A}(p^2) - pE(p) \in p^7\mathbb{Z}_p.$$

This equivalence includes an independently proved precision bound for one entire transport step. The companion estimate is proved below in Theorem 24.1.

15 Exact two-step transport and the remaining cancellation

Let $p > 3$, $p \equiv 2 \pmod 3$, and $r \ge 3$ be odd. Set $L = p^r$, $M = L/p^2$, and $m = (L-2)/3$. Use the node and denominator root sets $\mathcal{H}_L$ and $\mathcal{D}_L$ of the three-grid construction, and write $\mu_L(H) = W_L(H)T(H)$. Define the restricted rational power series

$$R_{L,p}(X) = \frac{\prod_{d\in\mathcal{D}_L,\, p^2\nmid d}(1 - p^2X/d)}{\prod_{h\in\mathcal{H}_L,\, p^2\nmid h}(1 - p^2X/h)} = 1 + \sum_{j\ge1} c_j X^j.$$

Every retained root has valuation zero or one. Hence each linear coefficient p^2/d or p^2/h belongs to $p\mathbb{Z}_p$. Expanding the finitely many inverse factors proves

$$c_j \in p^j\mathbb{Z}_p, \qquad v_p(c_j) \longrightarrow +\infty.$$

The series converges at every integral coarse node J.

Proposition 15.1 (Exact residual identity)**.** *Put*

$$U_{L,p} = \sum_{\substack{H\in\mathcal{H}_L \\ p^2\nmid H}} \mu_L(H).$$

Then

$$\boxed{E(L) - p^2E(M) = U_{L,p} + p^2\sum_{j\ge3} c_j M_j(M).} \tag{15.1}$$

No assertion that either term on the right is separately divisible by the target modulus is made.

Proof. Since $p^2 \equiv 1 \pmod 6$, dividing all p^2-divisible nodes by p^2 gives exactly $\mathcal{H}_M$. Dividing the corresponding denominator roots gives exactly $\mathcal{D}_M$. For example the positive grids end at $L-1$ and L, and become the positive grids ending at $M-1$ and M; the negative grid is -2 times the first positive grid. The positive denominator roots $6i$ and the negative roots $-3i$ both have coarse index range $1 \le i \le (M-2)/3$. The third denominator grid ends at $-(L-3)$ and becomes the grid ending at $-(M-3)$.

The exact model values satisfy $T(p^2J) = p^2T(J)$, because multiplication by p^2 preserves the residues modulo six and the sign of each node. Separating divisible factors in the weights therefore gives the product identities

$$W_L(p^2J) = W_M(J)R_{L,p}(J), \qquad \mu_L(p^2J) = p^2\mu_M(J)R_{L,p}(J).$$

These are identities of products, including at zero weights; no division by a vanishing weight is needed. Sum over coarse nodes and add the nodes not divisible by p^2. Interchanging the finite node sum with the convergent series is legitimate. The identities $M_1(M) = M_2(M) = 0$ remove its first two positive terms and give (15.1). □

16 The leading kernel coefficients from three short harmonic sums

Let $p > 3$ be prime with $p \equiv 2 \pmod 3$, let $r \geq 3$ be odd, and put $L = p^r$, $q = (p-2)/3$. Set

$$H_i = \sum_{a=1}^{q} a^{-i} \in \mathbb{F}_p \quad (1 \leq i \leq 3).$$

Use the exact two-step kernel

$$R_{L,p}(X) = \frac{\prod_{d \in \mathcal{D}_L,\ p^2 \nmid d}(1 - p^2X/d)}{\prod_{h \in \mathcal{H}_L,\ p^2 \nmid h}(1 - p^2X/h)} = 1 + \sum_{j \geq 1} c_j X^j.$$

All products here count multiplicities. In particular, an overlap between the node and denominator lists may be canceled but need not be removed in defining the products.

Proposition 16.1. *For every such p, r,*

$$\begin{aligned} c_1/p &\equiv -H_1/2 \pmod p, \\ c_2/p^2 &\equiv H_1^2/8 + 5H_2/24 \pmod p, \\ \kappa_p := c_3/p^3 &\equiv -\frac{3H_1^3 + 15H_1H_2 + 2H_3}{144} \pmod p. \end{aligned}$$

Thus these leading coefficients are independent of the odd level r.

Proof. For $i \leq 3$ define the normalized root-power difference

$$s_i = p^{-i}\left(\sum_{h \in \mathcal{H}_L,\ p^2 \nmid h} (p^2/h)^i - \sum_{d \in \mathcal{D}_L,\ p^2 \nmid d} (p^2/d)^i \right).$$

Every term is integral at p. Modulo p, the unit roots contribute zero, leaving exactly the roots of valuation one. After dividing those roots by p, their lists are the unit roots of the companion grids at $M = p^{r-1} = 3n + 1$:

$$\begin{aligned} \mathcal{H}'_M &= \{3j+1 : 0 \leq j \leq n\} \cup \{3j+2, -2(3j+2) : 0 \leq j < n\}, \\ \mathcal{D}'_M &= \{6j, -3j : 1 \leq j \leq n\} \cup \{-(3j+1) : 0 \leq j < n\}. \end{aligned}$$

These identifications follow directly by solving the original linear root congruences modulo p; they respect multiplicities.

Put $t = (2p-1)/3 = 2q + 1$. Since $n \equiv t \pmod p$, complete blocks of p consecutive indices contribute zero modulo p to each inverse-power sum under consideration. Indeed, $\sum_{u \in \mathbb{F}_p^\times} u^{-i} = 0$ for $1 \leq i \leq 3 < p - 1$. Let

$$T_i = \sum_{j=1}^{t} j^{-i}, \qquad A_i = \sum_{j=0}^{t-1} (3j+1)^{-i}, \qquad B_i = \sum_{\substack{0 \leq j < t \\ j \neq q}} (3j+2)^{-i}.$$

The extra endpoint $3t+1 = 2p$ of the first node list is excluded as a nonunit. The involution $j \mapsto t-1-j$ gives $A_i = (-3)^{-i}T_i$. In the B list, the residues are $3a$ for $a = -q, \dots, -1, 1, \dots, q$. Consequently

$$B_1 = B_3 = 0, \qquad B_2 = 2H_2/9.$$

Since $t = p-1-q$, the full nonzero-residue sum also gives $T_i = (-1)^{i+1}H_i$. Taking the difference of the companion node and denominator power sums yields

$$s_i \equiv (1-(-1)^i)A_i + (1+(-2)^{-i})B_i - (6^{-i}+(-3)^{-i})T_i \pmod p.$$

Substitution gives

$$s_1 = -H_1/2, \qquad s_2 = 5H_2/12, \qquad s_3 = -H_3/24 \quad \text{in } \mathbb{F}_p.$$

Finally the logarithmic product expansion, or equivalently the first three Newton identities, gives

$$\frac{c_1}{p} = s_1, \quad \frac{c_2}{p^2} = \frac{s_1^2+s_2}{2}, \quad \frac{c_3}{p^3} = \frac{s_1^3+3s_1s_2+2s_3}{6}.$$

Here these identities are exact before the reductions of s_i. The stated congruences follow; all displayed denominators are units because $p > 3$. □

This proposition computes one contribution to the residual and does not prove its cancellation with the other contribution. For example $p = 11$ gives $(H_1, H_2, H_3) = (0, 9, 6)$ modulo 11, so $\kappa_{11} = -1$. In particular the cubic kernel coefficient is not always divisible by p^4.

17 A uniform coarse-moment estimate and a rigorous transport-tail bound

Let $p > 3$ be prime, $p \equiv 2 \pmod 3$, and put $q = (p-2)/3$. Then q is odd. Use the original three-grid interpolation at $N = p$, and denote its positive moments by $M_j(p)$.

Proposition 17.1. *For every positive integer j,*

$$\boxed{M_j(p) \in p^2\mathbb{Z}_p.}$$

This bound is uniform in j; it is not inferred from finitely many polynomial moment formulas.

Proof. Write $P_q(z) = D_q(z)\mathcal{S}_q(z)$ for the cleared polynomial of the three-grid construction. All its coefficients belong to $\mathbb{Z}_p$: the fixed denominators $k!$, $(2/3)_k$, and the normalizing roots of D_q are units for $0 \le k \le q$. We prove the stronger coefficientwise assertion

$$P_q(z) \in p\mathbb{Z}_p[z]. \tag{17.1}$$

Work first in $\mathbb{F}_p(z)$. Insert the canceled numerator and denominator parameter $e = 2/3$ in $\mathcal{S}_q$ to exhibit the very-well-poised ${}_6F_5(-1)$ form. Modulo p its central parameter satisfies $a = 1/3 = -2q-1$, and the inserted parameter satisfies $e = 2/3 = -q$. The finite canceled factors $(-q)_k$ are nonzero for $k \le q$, so this insertion is legitimate. Apply Whipple's terminating transformation using this $-q$ as the terminating upper parameter. The other parameters may be written

$$b = a + z/6, \quad c = a - z/3, \quad e' = -q - z/3, \quad f = -q.$$

The transformed terminating ${}_3F_2$ is

$$ {}_3F_2\left[\begin{matrix}-q,\ -q+z/6,\ -q-z/3\\ 1-z/6,\ 1+z/3\end{matrix};1\right]_q .$$

Its prefactor is $(-2q)_q/(-q+z/3)_q$, a well-defined element of $\mathbb{F}_p(z)$. The displayed sum is a terminating Dixon sum with odd central parameter $-q$: the two lower parameters equal $1-q-(-q+z/6)$ and $1-q-(-q-z/3)$. Reversing the summation index $k \mapsto q-k$ negates every term. The sum is therefore zero. All transformations are finite rational identities whose denominators are nonzero in $\mathbb{F}_p(z)$; no analytic continuation in characteristic p is used. It follows that the reduction of $\mathcal{S}_q$ is zero as a rational function. Since D_q is a nonzero polynomial modulo p, this proves (17.1).

Let $\widehat{Q}_q(z)$ be the monic polynomial whose roots are the three interpolation grids. These roots are integers. Exactly one root is divisible by p, namely the positive node p itself; every other root is a p-adic unit. Thus $Q_q(z) = \ell_q \widehat{Q}_q(z)$ with $v_p(\ell_q) = -1$. By (17.1),

$$\frac{D_q(z)\mathcal{S}_q(z)}{Q_q(z)} = \frac{A_q(z)}{\widehat{Q}_q(z)}, \qquad A_q(z) = \ell_q^{-1} P_q(z) \in p^2\mathbb{Z}_p[z].$$

The inverse Laurent series at infinity of a monic polynomial in $\mathbb{Z}_p[z]$ has integral coefficients after its leading power of z is removed. Hence every Laurent coefficient of this proper rational function belongs to $p^2\mathbb{Z}_p$. Those coefficients are $-M_j(p)$, which proves the proposition. □

Corollary 17.2. *At $L = p^3$, let $R_{L,p}(X) = 1 + \sum_{j\ge1} c_j X^j$ be the exact two-step weight kernel. Then*

$$p^2 \sum_{j\ge4} c_j M_j(p) \in p^8\mathbb{Z}_p.$$

In particular the entire transport error, including its infinite tail, satisfies

$$p^2 \sum_{j\ge3} c_j M_j(p) \equiv p^2 c_3 M_3(p) \pmod{p^8}.$$

Proof. The established coefficient estimate is $c_j \in p^j\mathbb{Z}_p$. Proposition 17.1 bounds every summand for $j \ge 4$ by p^{2+j+2}, hence by p^8. These bounds tend to infinity, so the entire convergent tail belongs to that closed ideal. □

For an explicit leading term put

$$b_q = \frac{(1/3)_q}{q!}, \qquad \kappa_p \equiv c_3/p^3 \pmod{p}.$$

Since q is odd, the exact first nonzero moment is $M_3(p) = p^2(p-1)^3 b_q^3$. The corollary therefore gives

$$\boxed{p^2 \sum_{j\ge3} c_j M_j(p) \equiv -p^7 \kappa_p b_q^3 \pmod{p^8}.}$$

Together with the separately derived formula

$$\kappa_p \equiv -\frac{3H_1^3 + 15H_1H_2 + 2H_3}{144} \pmod{p}, \qquad H_i = \sum_{a=1}^{q} a^{-i},$$

this determines the full transport contribution modulo the target modulus by three harmonic sums of length $(p-2)/3$.

These estimates show that all orders beyond the cubic transport term are harmless. The following sections evaluate the unit-node contribution and prove its exact cancellation with the cubic term.

18 The first resonance determines the companion pole residues

Let $R_j(X)$ be the normalized companion moment polynomials, and set $\mathcal{R}(X,t)=\sum_{j\ge0}R_j(X)t^j$. The following statements are unconditional finite algebraic results. Their low-order integrality consequence will force a high-order zero of the block derivative at the origin.

18.1 An exactly diagonal factorial basis

For $k\ge0$, define monic polynomials

$$P_k(X)=\prod_{h=0}^{k-1}(X^2-(3h+1)^2),\qquad Q_k(Y)=\prod_{h=0}^{k-1}(Y-(3h+1)(3h-2)).$$

These are auxiliary basis polynomials, not the primary moment polynomials denoted by P_j elsewhere. Put $Y=X(X-3)$ and

$$\mathcal{L}'f=X(X-2)f(X)-(X-3)(X-1)f(X-3).$$

Then

$$\mathcal{L}'P_k=(3k+1)(2X-3)Q_k(Y). \tag{18.1}$$

Indeed the two products on the left have the common factor $Q_k(Y)$, and their remaining factors are respectively $X(X+3k-2)$ and $(X-3)(X-3k-1)$. Their difference is $(3k+1)(2X-3)$. This proves the identity over $\mathbb{Z}[X]$.

Define

$$g_X(t)=(1-Xt)(1-(X-2)t)(1+2(X-2)t).$$

The exact companion recurrence, including its first three initial moments, is equivalent to

$$\begin{aligned}X(X-2)g_X(t)\mathcal{R}(X,t)-(X-3)(X-1)g_{3-X}(t)\mathcal{R}(X-3,t)\\ =(2X-3)\{1+(2Y+3)t+(2Y+2)t^2\}.\end{aligned} \tag{18.2}$$

For example, the t^0,t^1,t^2 equations use $R_0=1$, $R_1=(X^2+1)/2$, and $R_2=2X^2-1$; all later equations are precisely the proved homogeneous companion recurrence.

Expand in the factorial basis:

$$\mathcal{R}(X,t)=\sum_{k\ge0}C_k(t)P_k(X).$$

Each coefficient of t^j is a finite sum, since $\deg R_j\le 2\lfloor(2j+1)/3\rfloor$. Direct multiplication using $YQ_k=Q_{k+1}+(3k+1)(3k-2)Q_k$ gives

$$\begin{aligned}&\frac{X(X-2)g_XP_k-(X-3)(X-1)g_{3-X}P_k(X-3)}{2X-3}\\ &\quad=(3k+1)g_{3k+1}(t)Q_k(Y)\\ &\qquad+t^2\{-9k-2+2(18k^2+18k+5)t\}Q_{k+1}(Y)+2t^3Q_{k+2}(Y).\end{aligned}$$

Consequently the coefficient functions are the following explicit rational functions:

$$\begin{aligned}C_0(t)&=\frac{1}{1-t},\\ C_1(t)&=\frac{t(1+3t)}{2(1-t)(1-16t^2)},\\ C_k(t)&=\frac{t^2\{9k-7-2(18k^2-18k+5)t\}C_{k-1}(t)-2t^3C_{k-2}(t)}{(3k+1)g_{3k+1}(t)}\qquad(k\ge2).\end{aligned} \tag{18.3}$$

Here all g_{3k+1} have constant term one. The recurrence also gives $\operatorname{ord}_t C_k\ge\lfloor3k/2\rfloor$, so it independently justifies the finite coefficient sums.

18.2 A rigorous first-resonance and local-constancy theorem

Theorem 18.1. *Let $p > 3$, $p \equiv 2 \pmod 3$, and put $k_0 = (2p-1)/3$. If $0 \le j \le (p^2-3)/2$, then for every $x \in p\mathbb{Z}_p$, $pR_j(x) \in \mathbb{Z}_p$ and*

$$pR_j(x) \equiv P_{k_0}(0)\,[t^j](pC_{k_0}(t)) \pmod p. \tag{18.4}$$

In particular $pR_j(x)$ modulo p is constant on $p\mathbb{Z}_p$. For every $p \ge 11$ in this residue class, this applies to all $j \le 4p-5$, and hence

$$pR_j(0) \equiv pR_j(p^2) \equiv pR_j(2p) \pmod p \qquad (0 \le j \le 4p-5).$$

Also $pR_j(x) \equiv 0 \pmod p$ for $j < p-1$.

Proof. For a fixed k, let

$$m_k = \sum_{i=1}^{k} v_p(3i+1).$$

Because the nonconstant denominators in (18.3) are unit series, induction gives the coefficientwise bound

$$C_k(t) \in p^{-m_k}\mathbb{Z}_p[[t]].$$

For a moment index in the stated range, only $k \le \lfloor (2j+1)/3 \rfloor \le (p^2-4)/3$ can occur. Therefore $3k+1 < p^2$; every positive valuation appearing in m_k is exactly one. Thus m_k simply counts the indices $i \le k$ for which $p \mid 3i+1$.

If $k < k_0$, then $m_k = 0$, and the corresponding term in $pR_j(x)$ is divisible by p. At $k = k_0$, there is exactly one such divisor, namely $3k_0+1 = 2p$. Hence pC_{k_0} is coefficientwise integral and $P_{k_0}(x) \equiv P_{k_0}(0)$ modulo p.

Suppose $k > k_0$, and set $\delta_k = 1$ if $p \mid 3k+1$ and $\delta_k = 0$ otherwise. Among the factors of $P_k(x)$ there are exactly $m_k - \delta_k$ indices $h < k$ with $p \mid 3h+1$. Each of those factors is divisible by p^2 because $x \in p\mathbb{Z}_p$. Consequently every coefficient contribution has valuation at least

$$1 - m_k + 2(m_k - \delta_k) = 1 + m_k - 2\delta_k.$$

If $\delta_k = 0$, this is at least two. If $\delta_k = 1$, then $k > k_0$ implies $m_k \ge 2$, so the same expression is at least one. All these terms vanish modulo p, proving (18.4). The argument also proves the asserted integrality.

For $p \ge 11$, $4p-5 \le (p^2-3)/2$. Finally $\operatorname{ord}_t C_{k_0} \ge \lfloor 3k_0/2 \rfloor = p-1$, proving the last assertion. □

19 Exact reflection and second-order unit-support reduction

This section proves an exact reflection identity and a valuation-two reduction of the unit-support part of the level-three numerator. The next sections combine this identity with the carry-support contribution and rational-function rigidity.

Let $p > 3$ be prime with $p \equiv 2 \pmod 3$, and put

$$a = (p+1)/3, \qquad q = a-1, \qquad B_b = (1/3)_b/b!.$$

Then a is even. Work in $\mathcal{R} = \mathbb{Z}_p[[w]]$. Define

$$f_x(w) = \frac{(1+(6x-2)w)(1-(3x-1)w)(1-(3x+1)w)}{(1-(6x+6)w)(1+(3x+3)w)(1+(3x+1)w)}, \qquad v_j(t,w) = \frac{j-1/3+t}{j+1+t} f_{j+t}(w),$$

and, for $0 \le c \le a$,

$$T_c(t,w) = \prod_{j=0}^{c-1} v_j(t,w), \qquad F(x,w) = (6x-1)(1+(18x^2-6x-2)w).$$

$$\mathcal{B}(t,w) = \sum_{c=0}^{a} T_c(t,w)F(c+t,w).$$

Every numerical factor in $v_j(t,w)$ is a p-adic unit for $0 \le j < a$ and $t \in p\mathbb{Z}_p$. In fact these rational functions have integral Taylor coefficients in t on this residue disk, coefficientwise in w. All formal denominators have constant term one in w.

Lemma 19.1 (Exact reflection)**.** *As rational functions of t, w, with $p = 3a-1$,*

$$v_{a-1-j}(t,w) = v_j(-p/3-t,w)^{-1}, \qquad 0 \le j < a.$$

Consequently, if $u = -p/3 - t$,

$$T_{a-c}(t,w) = T_a(t,w)T_c(u,w), \qquad T_a(t,w)T_a(u,w) = 1,$$

and

$$\boxed{\mathcal{B}(t,w) = -T_a(t,w)\mathcal{B}(u,w).}$$

At $t_0 = -p/6$ one has the exact identities

$$T_a(t_0,w) = 1, \qquad \boxed{\mathcal{B}(t_0,w) = 0}.$$

Proof. First,

$$\frac{a-1-j-1/3+t}{a-j+t} = \frac{j+1-p/3-t}{j-1/3-p/3-t}.$$

For the kernel, the three numerator factors of $f_{a-1-j+t}$ are

$$1-(6j+6)w+2pw+6tw, \quad 1+(3j+3)w-pw-3tw, \quad 1+(3j+1)w-pw-3tw.$$

They are precisely the denominator factors of f_{j+u}; its three numerator factors similarly equal the reflected denominator factors. This proves the single-factor identity. Reflecting the final c factors of $T_a(t)$ gives the identity for $T_{a-c}(t)$, and taking $c = a$ proves self-reciprocity.

A direct polynomial calculation gives $F(1/3-x,w) = -F(x,w)$. Since $a-c+t = 1/3-(c+u)$, replacing c by $a-c$ in the finite sum proves the displayed reflection of $\mathcal{B}$. Finally $u = t = t_0$ and the a factors of $T_a(t_0)$ pair into reciprocal pairs. There is no unpaired factor because a is even. Thus $T_a(t_0) = 1$ exactly, and $\mathcal{B}(t_0) = -\mathcal{B}(t_0)$ gives zero. □

Write

$$g = \partial_t \log T_a(t,w)|_{t=0}, \qquad B_i = \partial_t^i \mathcal{B}(t,w)\Big|_{t=0} \quad (i = 0,1,2).$$

Logarithmic derivatives mean rational derivatives and require no choice of a logarithm of the constant factor. The quantities g, B_i are in $\mathcal{R}$.

Corollary 19.2 (Reflection jets)**.** *One has*

$$B_2 \equiv gB_1 \pmod{p\mathcal{R}}, \qquad B_0 \equiv \frac{p}{6}B_1 - \frac{p^2}{72}B_2 \pmod{p^3\mathcal{R}}.$$

For every integer b,

$$\mathcal{B}(pb,w) \equiv p\left(b+\frac{1}{6}\right)B_1 + p^2\left(\frac{b^2}{2}-\frac{1}{72}\right)gB_1 \pmod{p^3\mathcal{R}}. \tag{19.1}$$

Proof. Differentiate the exact reflection twice, using $u' = -1$. At $t = t_0$, its value is zero and $T_a(t_0) = 1$, so

$$\mathcal{B}''(t_0, w) = (\partial_t \log T_a)(t_0, w)\mathcal{B}'(t_0, w).$$

Reducing modulo p yields the first congruence, because $t_0 \in p\mathbb{Z}_p$. Taylor expansion of the exact equality $\mathcal{B}(-p/6, w) = 0$ gives

$$0 \equiv B_0 - \frac{p}{6}B_1 + \frac{p^2}{72}B_2 \pmod{p^3}.$$

Taylor expansion at pb and substitution prove (19.1). The Taylor remainders are integral multiples of p^3, since all rational factors have unit numerical denominators on the residue disk. □

For comparison with the original length-p block, let

$$\widehat{K}_k(w) = \prod_{j=0}^{k-1} f_j(w), \qquad d_k = (-1/3)_k/k!.$$

Let $Z(t, w)$ be the complete block after the numerical factors divisible by p have been stripped:

$$Z(t, w) = \frac{\prod_{0\le j<p,\ j\neq a}(j - 1/3 + t)}{\prod_{1\le j\le p,\ j\neq p}(j + t)} \prod_{j=0}^{p-1} f_{j+t}(w).$$

The complete-block residue calculation proves

$$Z(0, w) \equiv 1 \pmod{p}, \qquad \partial_t \log Z(0, w) \in p\mathcal{R}.$$

For completeness, the numerical numerator and denominator residues both cover $\mathbb{F}_p^\times$ once. Their reciprocal first-power sums vanish. Each affine root progression in the kernel covers $\mathbb{F}_p$ once, so its complete product is the same for all three numerator and denominator progressions. In the logarithmic derivative, the numerator slope sum is $6 - 3 - 3 = 0$ and the denominator slope sum is $-6 + 3 + 3 = 0$. Thus both assertions hold coefficientwise. Put

$$\beta = \frac{Z(0, w) - 1}{p} \in \mathcal{R}.$$

Proposition 19.3 (The whole unit-support block has valuation two)**.** *Define*

$$\mathcal{U}_p(w) = \sum_{b=0}^{q}\sum_{c=0}^{a} d_{pb+c}\widehat{K}_{pb+c}(w)F(pb + c, w).$$

Then $\mathcal{U}_p \in p^2\mathcal{R}$, with the explicit residue

$$\boxed{\frac{\mathcal{U}_p(w)}{p^2} \equiv B_q B_1(w)\left(-\frac{1}{12} + \frac{\beta(w)}{28} + \frac{g(w)}{56}\right) \pmod{p\mathcal{R}}.}$$

Proof. For $c \le a$, extraction of the numerical factors divisible by p gives the exact factorization

$$d_{pb+c}\widehat{K}_{pb+c} = B_b\left(\prod_{\ell=0}^{b-1} Z(p\ell, w)\right) T_c(pb, w).$$

No partial-block numerator factor divisible by p has yet appeared; the first such index is $c = a + 1$. Since $Z(p\ell, w) \equiv Z(0, w) \pmod{p^2}$,

$$\prod_{\ell=0}^{b-1} Z(p\ell, w) \equiv 1 + pb\beta \pmod{p^2}.$$

Equation (19.1) puts $\mathcal{B}(pb, w)$ in $p\mathcal{R}$. Therefore multiplying the last approximation by the entire fine-support sum loses only a multiple of p^3. We obtain

$$\begin{aligned}\mathcal{U}_p \equiv\ & pB_1 \sum_{b=0}^{q} B_b(b + 1/6) \\ & + p^2 B_1 \left\{ \beta \sum_{b=0}^{q} B_b b(b + 1/6) + g \sum_{b=0}^{q} B_b(b^2/2 - 1/72) \right\} \pmod{p^3}.\end{aligned}$$

The generalized hockey-stick identity gives

$$\sum_{b=0}^{q} \binom{b}{j} B_b = \frac{(p-1)\binom{q}{j}}{3j+1} B_q \qquad (j = 0, 1, 2).$$

Using $q = (p - 2)/3$ yields the exact equalities

$$\begin{aligned}\sum_{b=0}^{q} B_b(b + 1/6) &= \frac{p(p-1)}{12} B_q, \\ \sum_{b=0}^{q} B_b b(b + 1/6) &= \frac{(p-2)(p-1)(8p+9)}{504} B_q, \\ \sum_{b=0}^{q} B_b(b^2/2 - 1/72) &= \frac{(p-1)(4p^2 - 7p - 9)}{504} B_q.\end{aligned}$$

The last two coefficients reduce to $1/28$ and $1/56$, respectively. All their denominators are units for the primes at issue: $p > 3$ and $p \equiv 2 \pmod 3$ exclude $p = 7$. This proves the proposition. □

The fine first derivative in this result is explicit:

$$B_1(w) = \sum_{c=0}^{a} T_c(0, w) \left(F'(c, w) + F(c, w)\, \partial_t \log T_c(t, w)|_{t=0} \right).$$

The proposition removes the coarse-index sums from the unit-support contribution. The following identity simplifies its correction terms.

Lemma 19.4 (Cancellation of the complete-block correction)**.** *The complete-block correction satisfies*

$$\beta \equiv -g/2 \pmod{p\mathcal{R}}.$$

Consequently the unit-support formula simplifies to

$$\boxed{\mathcal{U}_p(w)/p^2 \equiv -B_q B_1(w)/12 \pmod{p}.}$$

Proof. Set $K = p + a = 4a - 1$, and extend $T_c(t, w)$ by the same original product when $c > a$. Define its unitized value

$$V(t, w) = T_K(t, w) \frac{p + t}{p/3 + t}.$$

The product up to K contains just one numerical numerator factor divisible by p, at index a, and just one denominator factor divisible by p, at index p. Thus V is analytic and integral on $t \in p\mathbb{Z}_p$ with unit value. Splitting the product at p gives the exact equality

$$V(t,w) = Z(t,w)T_a(t+p,w).$$

The single-factor reflection for length K is

$$v_{K-1-j}(t,w) = v_j(-4p/3 - t, w)^{-1}.$$

At $t_* = -2p/3$, all factors pair into reciprocals except the central one, because K is odd. Its argument is $(K-1)/2 + t_* = -1/3$, for which $f_{-1/3} = 1$ and the numerical ratio is -1. Hence $T_K(t_*, w) = -1$. Since $(p + t_*)/(p/3 + t_*) = -1$, it follows that $V(t_*, w) = 1$.

The logarithmic derivative of V at zero reduces to g modulo p: this follows from $V = ZT_a(t+p)$ and the complete-block derivative $\partial_t \log Z(0) \in p\mathcal{R}$. Therefore

$$V(0,w) \equiv 1 + \frac{2p}{3}g \pmod{p^2}.$$

On the other hand $T_a(-p/6, w) = 1$, so

$$T_a(p,w) \equiv 1 + \frac{7p}{6}g \pmod{p^2}, \qquad V(0,w) = Z(0,w)T_a(p,w) \equiv 1 + p\beta + \frac{7p}{6}g \pmod{p^2}.$$

Comparison gives $\beta \equiv -g/2 \pmod{p}$. Substitution into Proposition 19.3 cancels its last two terms. □

20 A coefficientwise valuation-two theorem and a full-block derivative

This coefficientwise identity is the bridge from the companion numerator to a rational function over the finite field.

Let $p > 3$, $p \equiv 2 \pmod 3$, $a = (p+1)/3$, $q = a - 1$, and $n = (p^2-1)/3$. Use the functions f_x, F, v_j from Section 19, and write

$$P_c(t,w) = \prod_{j=0}^{c-1} v_j(t,w), \qquad Z_p(t,w) = \sum_{c=0}^{p-1} P_c(t,w)F(c+t,w).$$

Here P_c has integral Taylor coefficients at $t = 0$ whenever $c < p$; its numerator may contain a nonunit factor, but its denominators do not. Put $B_b = (1/3)_b/b!$, $d_k = (-1/3)_k/k!$, and

$$\mathcal{S}_p(w) = \sum_{k=0}^{n} d_k \widehat{K}_k(w)F(k,w), \qquad \widehat{K}_k(w) = \prod_{j=0}^{k-1} f_j(w).$$

Theorem 20.1. *Coefficientwise in $\mathbb{Z}_p[[w]]$,*

$$\mathcal{S}_p(w) \in p^2\mathbb{Z}_p[[w]], \qquad \boxed{\frac{\mathcal{S}_p(w)}{p^2} \equiv -\frac{B_q}{12}\,\partial_t Z_p(t,w)|_{t=0} \pmod{p}.}$$

Consequently the integral companion numerator satisfies

$$\boxed{\frac{\mathcal{N}_n(w)}{p^2} \equiv -\frac{B_q A_n(w)}{12(1-2w)}\,\partial_t Z_p(t,w)|_{t=0} \pmod{p},}$$

where $\mathcal{N}_n = A_n\mathcal{S}_p/(1-2w)$ is its exact reversed-root representation.

Proof. All congruences below are coefficientwise. Let $Z(t,w)$ denote the complete length-p product after deleting the numerical numerator factor at $j=a$ and the numerical denominator factor at $j=p-1$. Thus

$$P_p(t,w)=\frac{p/3+t}{p+t}Z(t,w),\qquad \beta=(Z(0,w)-1)/p.$$

The complete-block calculation gives $Z(0,w)\equiv 1 \pmod p$ and $\partial_t \log Z(0,w)\equiv 0 \pmod p$. Write $T_a=P_a$, $g=\partial_t \log T_a(0,w)$, and

$$\mathcal{B}(t,w)=\sum_{c=0}^{a} P_c(t,w)F(c+t,w),\qquad B_1(w)=\partial_t\mathcal{B}(0,w).$$

The exact short reflection proved in the accompanying lemma gives $T_a(-p/6,w)=1$ and

$$\frac{\mathcal{U}_p}{p^2}\equiv B_qB_1\left(-\frac{1}{12}+\frac{\beta}{28}+\frac{g}{56}\right)\pmod p, \tag{20.1}$$

where $\mathcal{U}_p$ is the part with fine index $0\le c\le a$. In particular $\mathcal{U}_p\in p^2\mathbb{Z}_p[[w]]$.

We now treat every remaining term. Set $K=p+a=4a-1$ and define

$$U_c(t,w)=\frac{P_c(t,w)}{p/3+t}\quad (a<c<p),\qquad U_K(t,w)=\frac{p+t}{p/3+t}P_K(t,w).$$

$$W_c(t,w)=U_c(t,w)F(c+t,w).$$

These unitized rational functions have integral Taylor coefficients on $p\mathbb{Z}_p$. The general single-factor reflection is

$$v_{K-1-j}(t,w)=v_j(-4p/3-t,w)^{-1}.$$

Writing $t'=-4p/3-t$, canceling the single deleted numerator factor on each side, and using $F(1/3-x,w)=-F(x,w)$ yields the exact identity

$$W_{K-c}(t,w)=U_K(t,w)W_c(t',w)\qquad (a<c<p). \tag{20.2}$$

At $t=-2p/3$, the K factors of P_K pair reciprocally except the central factor, whose value is -1. Hence $P_K=-1$ there; the removed ratio $(p/3+t)/(p+t)$ is also -1. Therefore $U_K(-2p/3,w)=1$ exactly.

Put $g_K=\partial_t\log U_K(0,w)$. Taylor expansion of the last identity and (20.2) gives, with W_c, W_c' evaluated at zero,

$$\begin{aligned} W_{K-c}(0,w)&\equiv W_c+\frac{2p}{3}(g_KW_c-2W_c')\pmod{p^2},\\ W_{K-c}'(0,w)&\equiv g_KW_c-W_c'\pmod p.\end{aligned}$$

The exact product factorization

$$U_K(t,w)=Z(t,w)T_a(t+p,w)$$

first gives $g_K\equiv g \pmod p$. It also gives a useful cancellation. Indeed $T_a(-p/6)=1$ implies $T_a(p)\equiv 1+7pg/6 \pmod{p^2}$, whereas $U_K(-2p/3)=1$ implies $U_K(0)\equiv 1+2pg/3 \pmod{p^2}$. Comparing the product at zero yields

$$\boxed{\beta\equiv -g/2\pmod p.} \tag{20.3}$$

Pair each $c=a+1,\dots,2a-1$ with $K-c=2a,\dots,p-1$. The first member has coarse range $0\le b\le q$, and the second has range $0\le b\le q-1$. Exact extraction of the nonunit Pochhammer

factors makes their common outside factor $pB_b(b+1/3)$; their full-block product is $1+pb\beta \pmod{p^2}$. Define

$$H_i^+ = \sum_{b=0}^{q} b^i(b+1/3)B_b, \qquad H_i^- = \sum_{b=0}^{q-1} b^i(b+1/3)B_b \quad (i=0,1).$$

The elementary finite binomial sums give exactly

$$H_0^+/B_q = (p-1)(p+2)/12, \qquad H_0^-/B_q = (p-2)(p-1)/12,$$
$$H_1^+/B_q = (p-2)(p-1)(p+2)/63, \qquad H_1^-/B_q = (p-5)(p-2)(p-1)/63.$$

In particular $H_0^+ + H_0^- = p(p-1)B_q/6$. The paired contribution, divided by p, is congruent modulo p^2 to

$$H_0^+W_c + H_0^-W_{K-c} + pH_1^+(W_c' + \beta W_c) + pH_1^-(W_{K-c}' + \beta W_{K-c}).$$

Its constant term is divisible by p. On substitution, the coefficient of W_c' after division by the second p is $B_q(-2/9+14/63)=0$. The coefficient of W_c is

$$B_q\left(-\frac{1}{6} - \frac{g_K}{21} - \frac{2\beta}{21}\right) \equiv -\frac{B_q}{6} \pmod{p}$$

by (20.3). Thus the entire carry-support contribution $\mathcal{V}_p$ satisfies

$$\mathcal{V}_p \in p^2\mathbb{Z}_p[[w]], \qquad \frac{\mathcal{V}_p}{p^2} \equiv -\frac{B_q}{6}\sum_{c=a+1}^{2a-1} W_c(0,w) \pmod{p}.$$

Equation (20.3) also reduces (20.1) to $\mathcal{U}_p/p^2 \equiv -B_qB_1/12$.

Finally, for $c > a$ one has $P_c(t) = (p/3+t)U_c(t)$, so $\partial_t(P_c(t)F(c+t))|_0 \equiv W_c(0) \pmod{p}$. The reflection gives $W_{K-c}(0) \equiv W_c(0) \pmod{p}$. Consequently

$$\partial_t Z_p(0,w) \equiv B_1(w) + 2\sum_{c=a+1}^{2a-1} W_c(0,w) \pmod{p}.$$

Adding the unit and carry contributions proves the first formula. Multiplication by the exact integral factor $A_n/(1-2w)$ proves the companion-numerator formula. All numerical denominators used above are units: the hypotheses exclude $2, 3, 7$. □

The next sections determine this block derivative exactly and use its unit-pole residue to complete the third-level congruence.

21 Simple poles of the full-block derivative

Let $p = 3a-1 > 3$ be prime, so a is even. Work over $\mathbb{F}_p$. Set

$$y_c(t) = \frac{(t-1/3)_c}{(1+t)_c}.$$

$$K_c(t,w) = \prod_{j=0}^{c-1} \frac{(1-(2-6j-6t)w)(1-(3j+3t-1)w)(1-(3j+3t+1)w)}{(1-(6j+6t+6)w)(1-(-3j-3t-3)w)(1-(-3j-3t-1)w)},$$

$$F(x,w) = (6x-1)(1+(18x^2-6x-2)w), \qquad Z_p(t,w) = \sum_{c=0}^{p-1} y_c(t)K_c(t,w)F(c+t,w).$$

Every derivative in this section is taken first as a rational function in t, w and then specialized to $t=0$.

Theorem 21.1. *The rational function $\partial_t Z_p(0,w)$ has at most simple poles at every finite point. Its possible poles are $w = 1/x$ with $x \in \mathbb{F}_p^\times$. Moreover it is regular at $w = 1/2$.*

Proof. All denominators in $y_c(t)$ are nonzero at $t = 0$, since $c < p$. For $0 \le c \le a$, $y_c(0)$ is nonzero, while for $a < c < p$ it has a simple zero. Thus the latter range contributes only $y_c'(0)K_c(0,w)F(c,w)$ to the derivative.

Fix a nonzero root value $x \in \mathbb{F}_p$. In the complete index interval $0 \le j < p$, the three denominator root progressions of K hit x at unique positions D_1, D_2, D_3, and its numerator progressions hit it at positions N_1, N_2, N_3. Positions are chosen in $\{0, \dots, p-1\}$. Direct substitution gives

$$N_i \equiv a - 1 - D_i \pmod p \qquad (i = 1, 2, 3). \tag{21.1}$$

The root slopes with respect to t are opposite in each paired numerator/denominator progression. The pole order of $K_c(0,w)$ at $1/x$ is exactly

$$q_x(c) = \#\{i : D_i < c\} - \#\{i : N_i < c\}.$$

A negative value means a zero.

The range $0 \le c \le a$. Only positions in $0 \le j < a$ can contribute here. Reflection (21.1) preserves this interval, so its numerator positions are the reflections $a - 1 - D_i$ of its denominator positions. There are at most two denominator occurrences. If two occur, their positions have sum at least $a - 1$. Here are the direct checks:

coincident progressions	index equation	consequence
D_1, D_2	$2j_1 + j_2 = 3a - 4$	$j_1 + j_2 \ge 2a - 3 \ge a - 1$
D_1, D_3	$2j_1 + j_3 = 2a - 3$	$j_1 \le a - 2,\ j_1 + j_3 \ge a - 1$
D_2, D_3	$j_2 = a - 1,\ j_3 = 0$	$j_2 + j_3 = a - 1.$

The last pair has root $x = -1$, which does not occur in D_1 on this interval, excluding a triple coincidence. A two-position multiset whose sum is at least $a - 1$ never has more elements in an initial interval than its reflected multiset. Consequently $q_x(c) \le 0$ in that case. If there is exactly one denominator occurrence, $q_x(c) \le 1$.

It follows that a positive unit-range pole is simple and comes from a unique denominator factor in the entire interval $j < a$. Whenever $q_x(c) = 1$, the reflected index $c' = a - c$ also has $q_x(c') = 1$. The exact finite-field reflection gives

$$y_{a-c}(0)K_{a-c}(0,w) = y_c(0)K_c(0,w), \qquad F(a-c,w) = -F(c,w).$$

The first identity follows from $v_{a-1-j}(0,w) = v_j(0,w)^{-1}$ for $v_j = y_{j+1}K_{j+1}/(y_jK_j)$, and the resulting full product $v_0 \cdots v_{a-1} = 1$; the evenness of a ensures its constant value is one. The second uses $a = 1/3$ in the field. The two summands therefore have opposite residues at their common simple pole. Their unique denominator factor is the same factor, so under the t-deformation their pole location has the same velocity. Their derivatives' double-pole coefficients consequently cancel. If $c = a - c$, the residue is already zero because $6c - 1 = 0$. All summands with $q_x(c) \le 0$ acquire at most a simple pole upon differentiation, since their logarithmic derivative has at most simple poles. This proves the assertion for the unit range.

The range $a < c < p$. Positions $D_i < a$ cancel their reflected numerator positions in every such prefix. The remaining positions are reflected by $D \mapsto 4a - 2 - D$ inside $[a, p-1]$. Thus

$q_x(c) = q_x(p + a - c)$, and it suffices to consider $a < c \le 2a - 1$. Write $u = D_3$. The other two denominator positions are explicitly

$$D_2 = \begin{cases} u + a - 1, & u \le 2a - 1, \\ u - 2a, & u \ge 2a, \end{cases}$$

$$D_1 = \begin{cases} (5a - 4 - u)/2, & u \text{ even}, \\ (2a - 3 - u)/2, & u \text{ odd}, \ u \le 2a - 3, \\ (8a - 5 - u)/2, & u \text{ odd}, \ u \ge 2a - 1. \end{cases}$$

If u is odd, D_1 lies either below a or above $2a - 1$; and D_2, D_3 cannot both lie in $[a, 2a - 2]$. Thus $q_x(c) \le 1$. For even $u < a$, or even $u \ge 2a$, at most one denominator position lies in that interval, with the same conclusion. It remains to consider even $a \le u \le 2a - 2$. The two possible left-hand positions are u and $(5a - 4 - u)/2$. The third position is $D_2 = u + a - 1$, whose reflected numerator position is $v = 3a - 1 - u$. For both denominator positions to precede v, one must have

$$\frac{5a - 4 - u}{2} < 3a - 1 - u, \quad \text{or equivalently } u < a + 2.$$

Thus only $u = a$ can give a double pole. In this case $x = -2$, the positions are $a, 2a - 2, 2a - 1$, and $q_x(c) = 2$ only at $c = 2a - 1$ and its reflection $c = 2a$. At these two exceptional terms,

$$18c^2 - 6c - 2 = 2 \quad \text{in } \mathbb{F}_p,$$

so $F(c, w)$ has the factor $1 + 2w$, which vanishes at $w = 1/x = -1/2$. It removes one pole order. Since $y_c(0) = 0$, no derivative of K_c occurs in this range. All carry contributions therefore have at most simple poles as well.

Regularity at $w = 1/2$. For $x = 2$, the numerator positions are $0, 1, a$, and the denominator positions are $a - 2, a - 1, p - 1$. If $a \ge 4$, all unit-range prefixes have negative pole order except $c = 0, a$. The index $c = 0$ is regular. At $c = a$ differentiation can introduce only a simple pole, but $F(a, 1/2) = 0$ removes it. Every carry prefix has a numerator excess and is regular. For $a = 2$ ($p = 5$), the only additional possible unit-range pole is at $c = 1$, where $6c - 1 = 0$ in $\mathbb{F}_5$; the same cancellation proves regularity. This completes the proof. □

22 The leading infinity coefficient of the full-block derivative

Let $Z_p(t, w)$ be the full-block function defined above. We prove its leading derivative coefficient by an exact finite summation with a rational telescoping certificate.

Lemma 22.1 (A finite very-well-poised identity)**.** *For $N \ge 0$, as a rational-function identity in A, B,*

$$\sum_{k=0}^{N} (-1)^k \frac{A + 2k}{A} \frac{(A)_k (B)_k (-N)_k}{k!(1 + A - B)_k (1 + A + N)_k} = \frac{(1 + A)_N}{(1 + A - B)_N}.$$

Proof. Write the summand as $t_{N,k}$. For $N \ge 1$ put

$$r_N = \frac{A + N}{A - B + N}, \qquad R_{N,k} = -\frac{k(A - B + k)(A + N + k)}{(A + 2k)N(A - B + N)}.$$

The successive-term ratios verify the exact rational identity

$$R_{N,k+1} t_{N,k+1} - R_{N,k} t_{N,k} = t_{N,k} - r_N t_{N-1,k}.$$

Indeed the numerator simplification required is just

$$-(A+k)(B+k)(k-N)-k(A-B+k)(A+N+k)=-(A+2k)(Ak-BN+k^2).$$

At $k=0$ the boundary vanishes by the factor k, and at $k=N+1$ it vanishes by termination. Summing yields $S_N=r_NS_{N-1}$, with $S_0=1$, proving the displayed product. The argument is first made at generic parameters; the resulting rational identity extends to every specialization at which the original two sides are defined. □

Proposition 22.2. *For every prime $p>3$ with $p\equiv 2 \pmod 3$,*

$$\boxed{[w^1]_\infty\,\partial_t Z_p(0,w)=-24 \quad \text{in } \mathbb{F}_p.}$$

Here the coefficient refers to the Laurent expansion at infinity.

Proof. Put $a=(p+1)/3$, so a is even and $a\equiv 1/3 \pmod p$. For $0\le c\le a$, all leading kernel roots are units modulo p, and the leading coefficient of the product is

$$P_c(t,w)=u_c(t)+O(w^{-1}),\qquad u_c(t)=(-1)^c\left(\frac{(t-1/3)_c}{(1+t)_c}\right)^3,$$

to first order in t at zero. This follows directly from $f_x(w)=-((x-1/3)/(x+1))^2+O(w^{-1})$ in this range. For $c>a$, the scalar factor has a simple zero at $t=0$ and the kernel at zero has degree -2 at infinity. Consequently the derivative of its weighted summand has degree at most -1; these terms contribute nothing to the coefficient of w^1.

Write $x=c+t$ and $G_c(t)=x^3u_c(t)$. Directly from the ratio $u_{c+1}/u_c=-((x-1/3)/(x+1))^3$ one obtains

$$(6x-1)(18x^2-6x-2)u_c=-54(G_{c+1}-G_c)-4(6x-1)u_c.$$

After reduction modulo p, replace $1/3$ by a. The derivative of $G_0=t^3$ at zero is zero. Also u_{a+1} contains the factor t^3, so the derivative of G_{a+1} is zero. Therefore the desired coefficient is $-4L'(0)$, where

$$L(t)=\sum_{c=0}^{a}(6c+6t-3a)(-1)^c\frac{(t-a)_c^3}{(1+t)_c^3}.$$

To evaluate this derivative, use the terminating deformation

$$V(t)=\sum_{c=0}^{a}3(A+2c)(-1)^c\frac{(A)_c(B)_c(-a)_c}{c!(1+A-B)_c(1+A+a)_c},\qquad A=-a+2t,\quad B=-a+t.$$

At $t=0$, each summand equals its counterpart in $L(0)$. Its derivative also equals that counterpart's derivative: the upper-parameter velocities are $2,1,0$ and the lower velocities are $0,1,2$, yielding the same logarithmic derivative as the three identical upper and three identical lower parameters of L. The weight has derivative 6 in both sums. None of the numerical factors in these terms vanishes for $c\le a$ except possibly the explicit linear weight; equality of derivatives there follows by the product rule without division by that weight. Thus $L'(0)=V'(0)$.

Lemma 22.1 gives the exact rational identity

$$V(t)=3(-a+2t)\frac{(1-a+2t)_a}{(1+t)_a}.$$

There is exactly one zero numerator factor at $t=0$. As a is even,

$$V'(0)=3(-a)\frac{2(-1)^{a-1}(a-1)!}{a!}=6.$$

All denominators used at zero are units modulo p. It follows that the leading coefficient is $-4\cdot 6=-24$, as asserted. □

23 An integral residue functional for the unit poles

The following lemma justifies passage from a coefficientwise congruence to the total residue at the unit poles, even when their reductions collide.

Lemma 23.1 (Integral unit-pole residue functional)**.** *Let $B, C \in \mathbb{Z}_p[w]$ have constant term one. Suppose the leading coefficient of B is a unit and all its roots are p-adic units, while $C(w) = \prod_i (1-u_i w)$ with $u_i \in p\mathbb{Z}_p$. For $P \in \mathbb{Z}_p[w]$, the sum of the residues of $P/(BC)$ at the unit poles belongs to $\mathbb{Z}_p$. This functional preserves coefficientwise congruences modulo p^a after retaining the same denominator factors.*

For $a = 3$, replace every factor involving a root divisible by p, including inverse factors, by its expansion modulo p^3. The resulting expression has only unit poles. The sum of its finite residues, equivalently its coefficient of w^{-1} in the Laurent expansion at infinity, agrees modulo p^3 with the original unit-pole residue sum.

Proof. Since $C \equiv 1 \pmod{p}$ and B has unit leading coefficient, their resultant is a unit. Bezout decomposition separates the parts with denominators B and C over $\mathbb{Z}_p[w]$. The part with denominator B has an integral Laurent expansion at infinity, proving integrality of its total finite residue. The other part has no unit pole.

The coefficient of w^j in C^{-1} is divisible by p^j. Thus its truncation through degree two differs from C^{-1} by $p^3 H/C$ for an integral polynomial H. Numerator factors with roots in $p\mathbb{Z}_p$ have the same divisibility property. Apply the first assertion to the resulting errors. Repeated poles and cancelled factors cause no change to this argument. □

24 Closure of the third odd level

For a prime $p > 3$ with $p \equiv 2 \pmod{3}$, put

$$q = (p-2)/3, \qquad a = q+1, \qquad b_q = (1/3)_q/q!,$$

and retain the exact primary and companion interpolation families. The result of this section is unconditional; its rational-function ingredients are the full-block derivative theorem, the simple-pole theorem, and the first Laurent-coefficient theorem proved above. No finite unit projector or large-prime restriction is used here.

Theorem 24.1. *For every prime $p > 3$, $p \equiv 2 \pmod{3}$,*

$$\boxed{\sum_{k=0}^{(p^3-2)/3} (-1)^k (6k+1) \left(\frac{(1/3)_k}{k!}\right)^3 \equiv p^2 \sum_{k=0}^{(p-2)/3} (-1)^k (6k+1) \left(\frac{(1/3)_k}{k!}\right)^3 \pmod{p^8}.}$$

24.1 The exact block derivative

Let $J_p(w) = \partial_t Z_p(0, w)$ be the rational block derivative. The simple-pole theorem says that its only finite poles are simple poles at $w = 1/x$, $x \in \mathbb{F}_p^\times$, and that it is regular at $w = 1/2$. The first Laurent-coefficient theorem gives $\deg_\infty J_p \le 1$ and $[w^1]_\infty J_p = -24$. We record explicitly why its required zero at the origin holds also for $p = 5$ and $p = 11$.

Set $M = p^2$, $n = (p^2-1)/3$, and $c_n = (2/3)_n/n!$. We first record its normalizing unit explicitly. Since $n = pq + 2q + 1$, its numerator has $q+1$ factors divisible by p, with quotients $1, 4, \ldots, 3q+1$,

whereas its denominator has q, with quotients $1,\dots,q$; none is divisible by p^2. Thus $v_p(c_n)=1$. The complete unit blocks cancel modulo p. The remaining partial unit quotient is

$$\frac{3^q q!\prod_{j=0}^{q-1}(3j+2)}{(2q+1)!}.$$

Combining it with the stripped factors gives

$$\frac{c_n}{p}\equiv 3^{-n}\frac{(p-1)!}{q!(2q+1)!}\equiv -1 \pmod p.$$

Indeed $n=(p-1)(p+1)/3$ is a multiple of $p-1$; Wilson's theorem and odd q give $q!(2q+1)!\equiv 1 \pmod p$. The exact moment and numerator normalizations give

$$\mathcal{N}_n(w)=-Mc_n\sum_{j\ge 0}R_j(M)w^j.$$

For $j<p-1$, the degree of R_j in M^2 is at most $\lfloor (2j+1)/3\rfloor \le 2q$. The exact triangular moment operator has diagonal entries $3k+1$; all entries with $0\le k\le 2q$ are p-adic units. Its integral recurrence and integral initial moments therefore give $R_j\in\mathbb{Z}_p[M^2]$ in this range. Since $v_p(c_n)=1$, $\mathcal{N}_n/p^2$ has zero coefficients modulo p in degrees less than $p-1$.

Write

$$A_n(w)=\frac{\prod_{D\in\mathcal{D}_n}(1-Dw)}{\prod_{H\in\mathcal{H}_n}(1-Hw)}.$$

This notation denotes the common reversed-root quotient, not the adjacent-cutoff polynomial also sometimes denoted by A_n. The full-block derivative theorem gives

$$\frac{\mathcal{N}_n(w)}{p^2}\equiv -\frac{b_q A_n(w)}{12(1-2w)}J_p(w) \pmod p.$$

All factors outside J_p are invertible at zero, so J_p has a zero of order at least $p-1$. Consequently $(w^{p-1}-1)J_p(w)$ is a polynomial of degree at most p, divisible by w^{p-1}. The coefficient at infinity and regularity at $1/2$ determine its two remaining coefficients:

$$\boxed{J_p(w)=12(1-2w)\frac{w^{p-1}}{w^{p-1}-1}\quad \text{in } \mathbb{F}_p(w).} \tag{24.1}$$

In particular,

$$\overline{\mathcal{N}_n/p^2}=-b_q\overline{A}_n(w)\frac{w^{p-1}}{w^{p-1}-1}. \tag{24.2}$$

24.2 The unit residue and its normalization

Let $\widetilde{\mu}_M(H)$ denote the companion masses and $\nu=(-1)^n Mc_n^2$. The exact partial fractions give

$$\sum_{p\nmid H}\widetilde{\mu}_M(H)=-\frac{\nu}{2}\sum_{\rho\ \text{unit pole}}\operatorname{Res}_{w=\rho}\mathcal{N}_n(w).$$

The full-block theorem proves coefficientwise divisibility of $\mathcal{N}_n$ by p^2. This is also an integral rational divisibility statement with its original denominator: multiply the series by that denominator, which is integral with constant term one. Every coefficient of the resulting polynomial is divisible by p^2. The integral unit-pole residue lemma may therefore be applied directly to $\mathcal{N}_n/p^2$ modulo p. Its reduction

has only unit poles, and its total residue is its coefficient of w^{-1} at infinity. Since n is even and $c_n/p \equiv -1 \pmod p$, $\nu/p^4 \equiv 1 \pmod p$. Formula (24.2) yields

$$p^{-6}\sum_{p\nmid H}\widetilde{\mu}_{p^2}(H) \equiv \frac{b_q}{2}[w^{-1}]_\infty\left(\overline{A}_n(w)\frac{w^{p-1}}{w^{p-1}-1}\right) \pmod p. \tag{24.3}$$

For completeness the leading normalization of $\overline{A}_n$ can be found without any harmonic evaluation. A complete block of p indices has the same three residue progressions in the numerator and denominator of A_n; their products cancel modulo p. As $n = pq + (2p-1)/3$, the remaining quotient is the companion quotient at cutoff $M_* = 2p$, with $n_* = (2p-1)/3 = 2q+1$. Its only nonunit nodes are $p, 2p, -2p$, and all its denominator roots are units. Put $c_* = (2/3)_{n_*}/n_*!$. Direct multiplication of the root products gives

$$\frac{\prod_{H\in\mathcal{H}_{n_*}} H}{\prod_{D\in\mathcal{D}_{n_*}} D} = (-1)^{n_*}(2p)c_*^2 = -2pc_*^2.$$

Removing the three nonunit nodes, whose product is $-4p^3$, shows that the leading coefficient of its degree-two unit quotient is

$$\lambda = \frac{2}{(c_*/p)^2} \pmod p.$$

The elementary Wilson normalization

$$(c_*/p)b_q \equiv 1 \pmod p$$

then gives $\lambda = 2b_q^2$. Indeed, after deleting the unique factor p in c_*, the remaining numerator residues from c_*/p and b_q together are exactly $1, \dots, p-2$. Their denominator is $3^{p-1}(2q+1)!q!$, and both $(p-2)!$ and $(2q+1)!q!$ equal 1 modulo p, by Wilson's theorem and the oddness of q.

Define the reciprocal unit-root quotient by

$$Q(X) = \frac{\prod_{D\in\mathcal{D}_n,\ p\nmid D}(1-X/D)}{\prod_{H\in\mathcal{H}_n,\ p\nmid H}(1-X/H)} = 1 + \sum_{j\ge 1} e_j X^j \in \mathbb{Z}_p[[X]].$$

After reduction, the preceding normalization says

$$\overline{A}_n(w) = 2b_q^2 w^2 \overline{Q}(w^{-1}).$$

Since $p-1 \ge 4$, the factor $w^{p-1}/(w^{p-1}-1) = 1 + O(w^{-(p-1)})$ cannot affect the coefficient of w^{-1} in a function of degree two. Equation (24.3) therefore becomes

$$\boxed{\sum_{p\nmid H}\widetilde{\mu}_{p^2}(H) \equiv p^6 b_q^3 e_3 \pmod{p^7}.} \tag{24.4}$$

This computation applies uniformly to $p = 5$, $p = 11$, and all larger primes in the specified residue class.

24.3 Cancellation with the exact transport correction

The companion-to-primary weight identity has kernel $Q(pX)$:

$$\widetilde{\mu}_{p^2}(pH) = p\mu_p(H)Q(pH).$$

Summing over the nodes and expanding gives the exact convergent identity

$$\mathcal{A}(p^2) - pE(p) = \sum_{p\nmid H} \widetilde{\mu}_{p^2}(H) + p\sum_{j\geq 1} p^j e_j M_j(p).$$

The first two primary moments vanish. The proved uniform estimate $M_j(p) \in p^2\mathbb{Z}_p$ for every $j \geq 1$ puts all terms with $j \geq 4$ in $p^7\mathbb{Z}_p$, and the valuations tend to infinity. The exact cubic moment is

$$M_3(p) = p^2(p-1)^3 b_q^3 \equiv -p^2 b_q^3 \pmod{p^3}.$$

Thus the cubic transport contribution is $-p^6 e_3 b_q^3 \pmod{p^7}$, cancelling (24.4). Hence

$$\mathcal{A}(p^2) \equiv pE(p) \pmod{p^7}.$$

Finally the independently proved first-step congruence $E(p^3) \equiv p\mathcal{A}(p^2) \pmod{p^8}$, valid for every prime in the present domain, proves Theorem 24.1.

25 The completed odd branch and verification

The higher-level arguments and their exhaustive finite certificates prove the target for every odd $r \geq 5$. Theorem 24.1 proves it for $r = 3$ for every permitted prime. Together they establish Theorem 1.1 without changing the sum, cutoff, recurrence multiplier, or target modulus.

The working mechanism in the final step is specific: reflection cancels the potential double poles of a parameter derivative; a first-resonance argument supplies a high-order zero; rational uniqueness then determines the whole function. This proves the needed unit residue, which equals the negative of the cubic transport term. It is not an assumption that either term vanishes separately. The elementary rational uniqueness argument is classical; no claim that its ingredients or this application have historical priority is made. Priority and comparison with every relevant preceding proof require a separate literature assessment.

The new root classification was also checked by exact arithmetic for all 48 primes $p \leq 500$ in the stated residue class. The original, unchanged third-level sums were independently evaluated modulo p^{10}, keeping valuations separate from unit inversions. The following are supplementary checks, not the uniform proof:

p	5	11	17	23	29	41	47	53	59
$v_p(E(p^3) - p^2E(p))$	8	8	9	8	8	8	8	8	8

The higher-level proof retains its 158 and 56 exact moment certificates. The accompanying archive contains the derivation fragments, exact verification programs, their recorded outputs, and a checksum manifest. This single source embeds the principal finite-certificate evaluators and the new direct sum evaluator.

A Finite certificates on the two positive companion grids

A.1 A finite certificate on the grid $H = 3h + 1$

Retain the rational deformation $V_n(z)$ defined in the companion construction. For every $h \geq 0$ and $n \geq h$,

$$V_n(3h+1) = -\frac{(-1)^h(3h+1)}{2}. \tag{A.1}$$

All lower parameters at $z = 3h + 1$ are regular: in particular $(5 - 3h)/6$ is never an integer. The factor $(-h)_k$ makes the sum terminate at $k = h$, so it suffices to prove the identity for $n = h$.

Set

$$P(h,k) = 18k^2 - 6k + 3h - 1,$$

$$u(h,k) = \frac{(-1)^{k-h}(6k-1)P(h,k)}{(1-3h)(3h+1)} \frac{(-1/3)_k((3h-1)/6)_k(-h-2/3)_k(-h)_k}{k!((5-3h)/6)_k(h+4/3)_k(h+2/3)_k}.$$

This is exactly the summand of $V_h(3h+1)$ divided by the right side of (A.1). We prove $\sum_{k=0}^{h} u(h,k) = 1$ by a recurrence with step two in h.

Define the polynomial

$$\begin{aligned} A(h,k) = {} & 486h^5 + (1458k^2 - 1944k + 3564)h^4 \\ & + (7857k^2 - 10476k + 10233)h^3 \\ & + (-972k^4 + 2592k^3 + 13149k^2 - 19836k + 14229)h^2 \\ & + (-2511k^4 + 6696k^3 + 7137k^2 - 15468k + 9486)h \\ & + 243k^6 - 972k^5 - 189k^4 + 3384k^3 + 414k^2 - 4072k + 2392. \end{aligned}$$

The rational certificate is

$$R(h,k) = \frac{-k(3k-2)(3h-6k+1)A(h,k)}{D_+(h,k)},$$

where

$$\begin{aligned} D_+(h,k) = {} & (6k-1)(h-k+1)(h-k+2)(3h-3k+5)(3h-3k+8) \\ & \cdot (3h+3k+2)(3h+3k+4)P(h,k). \end{aligned}$$

Cancellation of consecutive rising factors and expansion of the remaining polynomials verifies the rational identity

$$u(h+2,k) - u(h,k) = R(h,k+1)u(h,k+1) - R(h,k)u(h,k). \tag{A.2}$$

For an independently checkable form, if $q(h,k) = u(h,k+1)/u(h,k)$ and $r(h,k) = u(h+2,k)/u(h,k)$, then the certificate is the polynomial equality obtained by clearing all denominators in $q(h,k)R(h,k+1) - R(h,k) = r(h,k) - 1$. The ratio q is explicitly

$$q(h,k) = -\frac{(6k+5)P(h,k+1)}{(6k-1)P(h,k)} \frac{(k-1/3)(k+(3h-1)/6)(k-h-2/3)(k-h)}{(k+1)(k+(5-3h)/6)(k+h+4/3)(k+h+2/3)}.$$

Here are the upper-boundary details. Put $g(h,k) = R(h,k)u(h,k)$. The apparent poles at $k = h+1, h+2$ cancel because

$$\frac{(-h)_k}{(h-k+1)(h-k+2)} = \frac{(-h-2)_k}{(h+1)(h+2)}.$$

Thus g has a finite value at both boundary indices. Also $g(h,0) = 0$. Direct cancellation gives

$$\left.\frac{R(h,k)}{r(h,k)}\right|_{k=h+1} = -\frac{4(3h+1)(9h^3+39h^2+46h+15)}{(h+2)(6h+5)(18h^2+33h+17)},$$

$$q(h+2,h+1) = -\frac{5(3h+2)(9h+11)}{(h+2)(6h+5)(18h^2+33h+17)}.$$

The first displayed quotient plus $1+q(h+2,h+1)$ is identically zero. Multiplication by $u(h+2,h+1)$ therefore gives

$$g(h,h+1) = -u(h+2,h+1) - u(h+2,h+2).$$

Summing (A.2) for $0 \le k \le h$ and adding these two final terms proves

$$\sum_{k=0}^{h+2} u(h+2,k) = \sum_{k=0}^{h} u(h,k).$$

At $h=0$ the sum is 1; at $h=1$ its terms are $-1/4$ and $5/4$, again summing to one. Induction separately on the two parity classes proves (A.1) for every h.

This proves one exact node family of the companion interpolation. It does not, by itself, prove any prime-power lifting congruence.

A.2 The second positive grid: a complete finite certificate

For $h \ge 1$, let $t_h(k)$ be the summand of $V_{h+1}(3h+2)$ and normalize it by $u_h(k) = t_h(k)/(3h+2)$. Explicitly,

$$u_h(k) = \frac{(-1)^k(6k-1)(h+6k^2-2k)}{2h(3h+2)} \frac{(-1/3,h/2,-h-1,-h-1/3)_k}{k!(2/3-h/2,h+5/3,h+1)_k}.$$

The sum terminates at $h+1$. The desired exact assertion is

$$\sum_{k=0}^{h+1} u_h(k) = \begin{cases} 1, & h \text{ odd}, \\ 0, & h \text{ even}. \end{cases}$$

All denominator Pochhammers are nonzero on this range.

Put

$$\begin{aligned} J(h,k) = {} & 54h^5 + 162h^4k^2 - 216h^4k + 486h^4 + 1089h^3k^2 - 1452h^3k + 1725h^3 \\ & - 108h^2k^4 + 288h^2k^3 + 2442h^2k^2 - 3512h^2k + 3002h^2 \\ & - 351hk^4 + 936hk^3 + 2082hk^2 - 3608hk + 2549h \\ & + 27k^6 - 108k^5 - 126k^4 + 656k^3 + 507k^2 - 1316k + 840, \end{aligned}$$

$$R(h,k) = -\frac{k(3k-2)(3h-6k+2)J(h,k)}{D_2(h,k)},$$

where

$$\begin{aligned} D_2(h,k) = {} & (6k-1)(h-k+2)(h-k+3)(h+k+1)(h+6k^2-2k) \\ & \cdot (3h-3k+4)(3h-3k+7)(3h+3k+5). \end{aligned}$$

Direct rational cancellation proves the finite WZ certificate

$$u_{h+2}(k) - u_h(k) = R(h,k+1)u_h(k+1) - R(h,k)u_h(k). \tag{A.3}$$

For a compact independently reproducible verification, the consecutive ratios are

$$q(h,k) = \frac{u_h(k+1)}{u_h(k)} = \frac{(h+2k)(3k-1)(6k+5)(h-k+1)(3h-3k+1)(h+6k^2+10k+4)}{(k+1)(6k-1)(h+k+1)(h+6k^2-2k)(3h-6k-4)(3h+3k+5)},$$

$$\begin{aligned} r(h,k) &= \frac{u_{h+2}(k)}{u_h(k)} \\ &= \frac{(h+1)(h+2)(h+3)(h+2k)(3h+4)(3h+5)(3h+7)}{(h-k+2)(h-k+3)(h+k+1)(h+k+2)(h+6k^2-2k)} \\ &\quad \times \frac{(3h-6k+2)(h+6k^2-2k+2)}{(3h-3k+4)(3h-3k+7)(3h+3k+5)(3h+3k+8)}. \end{aligned}$$

Substitution in $q(h,k)R(h,k+1) - R(h,k) = r(h,k) - 1$ leaves identically zero after clearing denominators.

At the lower boundary $R(h,0)u_h(0) = 0$. At the upper boundary, cancel the apparent zeros and poles before evaluating. For instance,

$$\frac{(-h-1)_k}{(h-k+2)(h-k+3)} = \frac{(-h-3)_k}{(h+2)(h+3)}.$$

The needed endpoint can also be checked entirely from rational ratios:

$$\frac{R(h,k)}{r(h,k)}\Big|_{k=h+2} = -\frac{4(9h^3+60h^2+127h+87)}{(h+3)(6h+11)^2},$$

$$\frac{u_{h+2}(h+3)}{u_{h+2}(h+2)} = -\frac{3(3h+5)}{(h+3)(6h+11)^2}.$$

The first displayed ratio plus one plus the second ratio is exactly zero. Hence the upper boundary is $-u_{h+2}(h+2) - u_{h+2}(h+3)$. Summing (A.3) over $0 \le k \le h+1$ therefore proves that the normalized sum is unchanged when h increases by two. The initial normalized terms are

$$h = 1: \quad -\frac{1}{10},\ \frac{5}{4},\ -\frac{3}{20}, \qquad h = 2: \quad -\frac{1}{16},\ -\frac{105}{176},\ \frac{11}{16},\ -\frac{5}{176}.$$

Their sums are respectively one and zero, proving both parity cases. At $h = 0$ the removable value $V_n(2) = 0$ for $n \ge 1$ supplies the remaining case. For an ambient truncation $n \ge h+1$, all subsequent terms vanish and their denominators remain nonzero; the same node value therefore holds for every admissible ambient cutoff.

B A finite certificate for the negative companion grid

The following establishes the negative grid of the proposed companion interpolation. It is an exact finite identity and makes no claim about the remaining prime-power error estimate.

For $h \ge 0$, set

$$t(h,k) = -\frac{(-1)^k(6k-1)(3k^2-k-h-1)}{2(h+1)} \frac{(-1/3)_k(-h-1)_k(2h+1)_k(2h+5/3)_k}{k!(h+5/3)_k(-2h-1/3)_k(-2h-1)_k}.$$

All denominator factors are nonzero for $0 \le k \le h+1$. We prove

$$\sum_{k=0}^{h+1} t(h,k) = 3h+2. \tag{B.1}$$

Put $u(h,k) = t(h,k)/(3h+2)$ and define the polynomial

$$\begin{aligned} A(h,k) = {} & 1728h^5 - 2592h^4k^2 + 3456h^4k + 9504h^4 \\ & - 12024h^3k^2 + 16032h^3k + 20712h^3 \\ & + 432h^2k^4 - 1152h^2k^3 - 19704h^2k^2 + 27296h^2k + 22408h^2 \\ & + 1026hk^4 - 2736hk^3 - 13308hk^2 + 20176hk + 12074h \\ & - 27k^6 + 108k^5 + 450k^4 - 1520k^3 - 3027k^2 + 5444k + 2604. \end{aligned}$$

The certificate is

$$R(h,k) = -\frac{k(3k-2)(2h-k+2)(6h-3k+4)A(h,k)}{D_-(h,k)},$$

where

$$D_-(h,k) = 16(h+1)(2h+1)(2h+3)(3h+4)(6h+5)(6h+7) \cdot (6k-1)(h-k+2)(h-3k^2+k+1).$$

Direct rational cancellation gives

$$u(h+1,k) - u(h,k) = R(h,k+1)u(h,k+1) - R(h,k)u(h,k). \tag{B.2}$$

For reproducibility, this is the polynomial identity obtained after substituting the displayed R and the consecutive-factor ratios of u; the exact residual is zero. The quadratic factor $h-3k^2+k+1$ cancels from the product Ru, so roots of that factor cause no singularity.

We specify the upper boundary without interpreting an undefined terminating summand. Write $r(h,k) = u(h+1,k)/u(h,k)$ as its rational consecutive-parameter ratio. Cancellation gives

$$\left.\frac{R(h,k)}{r(h,k)}\right|_{k=h+2} = -1.$$

The canceled quotient has no pole at this substitution for any integer $h \geq 0$. Thus the continued product at the upper boundary is

$$[R(h,k)u(h,k)]_{k=h+2} = -u(h+1,h+2).$$

This can equivalently be obtained by cancelling $(-h-1)_k/(h-k+2) = (-h-2)_k/(h+2)$ before specializing; the quotient R/r also covers the coincident zero factors at $h=0$. At the lower boundary $R(h,0)u(h,0) = 0$. Sum (B.2) over $0 \leq k \leq h+1$ and add the new endpoint $u(h+1,h+2)$. The sums of u at successive h are equal. For $h=0$, the unnormalized terms are $-1/2$ and $5/2$, whose sum is two. Hence the normalized sum equals one, proving (B.1).

B.1 Embedding in the cleared interpolation

At the negative node $H = -2(3h+2)$, the candidate $V_n(z)$ has exactly the summand $t(h,k)$ above. Its numerator parameter $-h-1$ terminates the finite sum at $k = h+1$. For this node to belong to the companion grid one has $n \geq h+1$. The companion denominator is

$$D_n(z) = \prod_{j=1}^{n}(1 - z/(6j))(1 + z/(3j)) \prod_{j=0}^{n-1}(1 + z/(3j+1)).$$

The only possible zero of $D_n(H)$ is the last product's factor at $j = 2h+1$. It occurs precisely when $n \geq 2h+2$. If $h+1 \leq n \leq 2h+1$, all original summand denominators are nonzero, the sum terminates at $h+1$, and (B.1) gives $V_n(H) = 3h+2$.

If $n \geq 2h+2$, treat $D_n(z)V_n(z)$ term by term after clearing its finite denominator factors. In a summand $k \leq h+1$, the factor indexed by $j = 2h+1$ remains in the denominator-clearing tail and vanishes at H. In a summand $k \geq h+2$, the numerator contains the factor at $j = h+1$ in $1 + z/[2(3j-1)]$, and that factor vanishes at H. The additional factor $2-z$ in V_n is nonzero at H. Thus every cleared summand is zero, giving

$$[D_n(z)V_n(z)]_{z=H} = 0 = D_n(H)(3h+2).$$

Combining the two ranges proves the full cleared negative-grid identity

$$\boxed{[D_n(z)V_n(z)]_{z=-2(3h+2)} = D_n(-2(3h+2))(3h+2) \qquad (0 \leq h < n).} \tag{B.3}$$

No raw $0/0$ hypergeometric summand is used in this argument.

C Reproducible integer certificate evaluator

The following complete program is the evaluator used for the 158 higher-level moment certificates. For the 56 fifth-level certificates, the exhaustive case list is instead $[(5,2),(11,2),(17,2),(23,2)]$; all other calculations are identical. These lists supplement the uniform proofs and do not extrapolate from a finite sample to an infinite family.

```
"""Exact finite certificates for exceptional polynomial-projector parameters.

The proved degree and logarithmic denominator bounds reduce each required
prime-power moment to Newton interpolation from small integer cutoffs.
All calculations use Python integers modulo explicit prime powers; no
floating point, symbolic heuristic, or original enormous truncated sum.
"""
from pathlib import Path
from math import comb
import json


def strip(x,p):
    assert x
    v=0
    while x%p==0:x//=p;v+=1
    return x,v


def ilog(x,p):
    k=0
    while x>=p:x//=p;k+=1
    return k


def vmod(x,p,cap):
    return cap if x==0 else strip(x,p)[1]


def poch_step(unit,val,num,den,p,mod):
    nu,nv=strip(num,p);du,dv=strip(den,p)
    return unit*nu*pow(du,-1,mod)%mod,val+nv-dv


def solve_series(a,b,previous,source,J,mod):
    out=[0]*(J+1)
    for j in range(J+1):
        rhs=sum(b[i]*previous[j-i] for i in range(min(3,j)+1))
        if j<len(source):rhs+=source[j]
        rhs-=sum(a[i]*out[j-i] for i in range(1,min(3,j)+1))
        out[j]=rhs%mod
    return out


def newton_value(values,X,mod):
    delta=values[:];ans=0;binom=1
    for k in range(len(values)):
        if k:binom=binom*(X-k+1)//k
        ans=(ans+delta[0]*(binom%mod))%mod
        delta=[(delta[i+1]-delta[i])%mod for i in range(len(delta)-1)]
    return ans


def certify(p,s,family):
    indices=[i*(p-1)-(1 if family=='companion' else 3) for i in range(1,3*s+2)]
    degrees={j:2*((2*j+1)//3) if family=='companion' else 2*((2*j)//3) for j in indices}
    J=max(indices);nmax=max(degrees.values())
    log=ilog(3*nmax+(1 if family=='companion' else 2),p)
    A=(3*log+1)//2 if family=='companion' else (3*log)//2
    cap=A+2;mod=p**cap
    sample={j:[] for j in indices}
```

```
    previous=[0]*(J+1)
    du=cu=bu=1;dv=cv=bv=0
    max_norm_v=0
    for n in range(nmax+1):
        if n:
            du,dv=poch_step(du,dv,3*n-4,3*n,p,mod)
            cu,cv=poch_step(cu,cv,3*n-1,3*n,p,mod)
            bu,bv=poch_step(bu,bv,3*n-2,3*n,p,mod)
        assert min(dv,cv,bv)>=0
        if family=='companion':
            M=3*n+1
            if n==0:
                current=[mod-1]*(J+1)
            else:
                a=[1,-2,-(M-2)*(3*M-4),2*M*(M-2)**2]
                b=[1,-2,-(M-1)*(3*M-5),-2*(M-3)*(M-1)**2]
                t=(6*n-1)*du*pow(p,dv,mod)%mod
                alpha=18*n*n-6*n-2
                current=solve_series(a,b,previous,[t,t*(alpha+1),t*alpha],J,mod)
            mu,mv=strip(M,p);norm_v=mv+cv;norm_unit=mu*cu%mod
            sign=-1
        else:
            L=3*n+2
            a=[1,-1,-(L-1)*(3*L-2),2*L*(L-1)**2]
            b=[1,-1,-(L-2)*(3*L-7),-2*(L-2)**2*(L-3)]
            t=(6*n+1)*bu*pow(p,bv,mod)%mod
            current=solve_series(a,b,previous,[t],J,mod)
            lu,lv=strip(L,p);vu,vv=strip(L-1,p)
            norm_v=lv+vv+bv;norm_unit=lu*vu*bu%mod;sign=2
        assert norm_v<=A,(family,p,s,n,norm_v,A)
        max_norm_v=max(max_norm_v,norm_v)
        scale=sign*pow(p,A-norm_v,mod)*pow(norm_unit,-1,mod)%mod
        for j in indices:
            if n<=degrees[j]:sample[j].append(current[j]*scale%mod)
        previous=current
    X=(p**(2*s)-1)//3 if family=='companion' else (p**(2*s+1)-2)//3
    rows=[]
    for j in indices:
        result=newton_value(sample[j],X,mod)
        valuation=vmod(result,p,cap)-A
        target=-(s-1) if family=='companion' else -s
        assert valuation>=target,(p,s,family,j,valuation,target)
        rows.append({'normalized_index':j,'polynomial_degree_bound':degrees[j],
            'scaled_value_residue':result,'valuation_lower_bound':valuation,
            'required_lower_bound':target,'passed':True})
    return {'p':p,'s':s,'r':2*s+1,'family':family,'precision_exponent':cap,
        'common_scale_exponent':A,'max_sample_cutoff':nmax,
        'max_normalizer_valuation':max_norm_v,'number_of_moments':len(rows),
        'checks':rows}


if __name__=='__main__':
    cases=[(5,3),(11,3),(17,3),(23,3),(29,3),(5,4),(5,5)]
    records=[]
    for p,s in cases:
        for family in ['companion','primary']:
            row=certify(p,s,family);records.append(row)
            print(json.dumps({k:v for k,v in row.items() if k!='checks'}),flush=True)
    result={'status':'All exact finite projector-moment certificates passed.',
        'pair_count':len(cases),'family_pair_count':len(records),
        'moment_count':sum(r['number_of_moments'] for r in records),
        'method':'Proved small-cutoff Laurent recurrence, common p-power scaling, exact Newton interpolation at
      the prime-power cutoff.',
        'records':records}
    Path(__file__).with_suffix('.json').write_text(json.dumps(result,indent=2)+'\n')
```

D Reproducible third-level unit-mass evaluator

This evaluator supplies the five finite exceptions in the proof of the third-level first-step congruence. The final companion step at that level is not asserted by this computation.

```
"""Exact modular interpolation masses, using interval products in residue classes.
No floating arithmetic; products remove powers of p before inversion.
"""
from pathlib import Path
from collections import Counter
import json,time

class Products:
    def __init__(self,p,L,N):
        self.p=p;self.mod=p**L;self.tables={}
        B=4*N
        for step in (3,6):
            for residue in range(step):
                low=(-B-residue+step-1)//step;high=(B-residue)//step
                vs=[0];us=[1]; factors=[]
                for j in range(low,high+1):
                    x=residue+step*j;v=0
                    if x==0:x=1
                    else:
                        while x%p==0:x//=p;v+=1
                    factors.append(x%self.mod)
                    vs.append(vs[-1]+v);us.append(us[-1]*x%self.mod)
                inv=[0]*len(us);inv[-1]=pow(us[-1],-1,self.mod)
                for j in range(len(factors)-1,-1,-1):
                    inv[j]=inv[j+1]*factors[j]%self.mod
                self.tables[(step,residue)]=(low,vs,us,inv)
    def product(self,lo,hi,step):
        assert (hi-lo)%step==0 and hi>=lo
        residue=lo%step;base,vs,us,inv=self.tables[(step,residue)]
        a=(lo-residue)//step-base;b=(hi-residue)//step-base+1
        return vs[b]-vs[a],us[b]*inv[a]%self.mod

def analyze(p,r,L=10):
    N=p**r;m=(N-2)//3;mod=p**L;pr=Products(p,L,N)
    def prods(H,numer):
        ranges=([(6-H,6*m-H,6),(-3*m-H,-3-H,3),(-3*m+1-H,-2-H,3)] if numer
                else [(1-H,3*m+1-H,3),(2-H,3*m+2-H,3),(-6*m-2-H,-2-H,6)])
        v=0;u=1
        for lo,hi,step in ranges:
            a,b=pr.product(lo,hi,step);v+=a;u=u*b%mod
        return v,u
    vq,uq=prods(0,False);vd,ud=prods(0,True)
    cv=vq-vd;cu=uq*pow(ud,-1,mod)%mod
    denominator_roots=set(z for j in range(1,m+1) for z in (6*j,-3*j,-3*j+1))
    mass=[0]*(r+1);counts=[0]*(r+1);minimum=[None]*(r+1);dist=[Counter() for _ in range(r+1)]
    def vp(x):
        n=0
        if not x:return L
        while x%p==0:n+=1;x//=p
        return n
    half=pow(2,-1,mod)
    for h in range(m+1):
        for H,c in [(3*h+1,1 if h%2==0 else 0),(3*h+2,(-1)**h*half),(-6*h-2,-half)]:
            if not c or H in denominator_roots:continue
            a,b=prods(H,True);d,e=prods(H,False)
            v=cv+a-d;assert v>=0
            u=cu*c*b*pow(e,-1,mod)%mod
            term=u*p**v%mod;t=vp(H)
            mass[t]=(mass[t]+term)%mod;counts[t]+=1;dist[t][v]+=1
            minimum[t]=v if minimum[t] is None else min(minimum[t],v)
    return {'p':p,'r':r,'modulus_exponent':L,'mass_mod':mass,
            'mass_valuations':[vp(z) for z in mass],'counts':counts,
            'minimum_individual_valuations':minimum,
            'individual_valuation_counts':[dict(z) for z in dist]}
```

```
if __name__==’__main__’:
    rows=[]
    for p in (5,11,17,23,29):
        start=time.time();row=analyze(p,3);rows.append(row)
        print(p,row[’mass_valuations’],row[’minimum_individual_valuations’],’seconds’,round(time.time()-start,2)
     ,flush=True)
    Path(__file__).with_suffix(’.json’).write_text(json.dumps(rows,indent=2))
```

E Direct checks of the original third-level sum

The following standalone standard-library program verifies the original cutoff and target modulus. It does not use the block identity in its computation.

```
"""Supplementary exact modular checks of the unchanged Swisher E.3 sum.

This program uses no block identity or pole formula from the proof.
It keeps p-valuations separately, so it never inverts a multiple of p.
"""
import json
from pathlib import Path


def strip(x, p):
    v = 0
    while x % p == 0:
        x //= p
        v += 1
    return x, v


def e_sum(p, r, precision):
    modulus = p ** precision
    unit, valuation, total = 1, 0, 0
    cutoff = (p ** r - 2) // 3
    for k in range(cutoff + 1):
        assert valuation >= 0
        if 3 * valuation < precision:
            term = (6 * k + 1) * pow(unit, 3, modulus)
            term *= p ** (3 * valuation)
            total = (total + (-term if k % 2 else term)) % modulus
        if k < cutoff:
            numerator, nv = strip(3 * k + 1, p)
            denominator, dv = strip(3 * (k + 1), p)
            unit = unit * numerator * pow(denominator, -1, modulus) % modulus
            valuation += nv - dv
    return total


def run():
    rows = []
    for p in (5, 11, 17, 23, 29, 41, 47, 53, 59):
        precision = 10
        defect = (e_sum(p, 3, precision) - p * p * e_sum(p, 1, precision)) % p ** precision
        valuation = precision if defect == 0 else strip(defect, p)[1]
        assert valuation >= 8, (p, defect, valuation)
        rows.append({"p": p, "r": 3, "cutoff": (p ** 3 - 2) // 3,
                     "target_exponent": 8, "precision": precision,
                     "defect_residue": defect, "defect_valuation_at_least": valuation,
                     "passed": True})
    return {"method": "direct original sum; exact modular arithmetic", "cases": rows}


if __name__ == "__main__":
    result = run()
    Path(__file__).with_suffix(".json").write_text(json.dumps(result, indent=2) + "\n")
    print(json.dumps(result, indent=2))
```

References

[1] H. Swisher, *On the supercongruence conjectures of van Hamme*, Research in the Mathematical Sciences **2** (2015), Article 18, Section 7, Table 2. https://doi.org/10.1186/s40687-015-0037-6.

[2] W. N. Bailey, *Generalized Hypergeometric Series*, Cambridge University Press, 1935. Classical Whipple and Dixon transformations.

[3] V. J. W. Guo and M. J. Schlosser, *New q-Analogues of Van Hamme's (E.2) Supercongruence and of a Supercongruence by Swisher*, Results in Mathematics (2023). https://doi.org/10.1007/s00025-023-01885-8.

[4] QEEDDD, *Verifiable Mathematical Reasoning Corpus*, research context supplied for this investigation. https://github.com/QEEDDD/Verifiable-Mathematical-Reasoning-Corpus.

SUPPLEMENT S

A short-cutoff creative-microscope proof of the odd branch of F.3

Working proof for independent audit 10 November 2026

Let $(x;q)_k = \prod_{i=0}^{k-1}(1-xq^i)$ and $[m]_q = (1-q^m)/(1-q)$. Products with several first arguments mean the product of the respective shifted factorials. Define

$$T_L(a,q) = \sum_{k=0}^{L} c_k(a,q), \qquad c_k(a,q) = (-1)^k[8k+1]_q q^k \frac{(aq, q/a, q, -q^2; q^4)_k}{(aq^4, q^4/a, q^4, -q^3; q^4)_k}.$$

The parameter a is an indeterminate throughout every root-of-unity calculation. In particular, parameter denominators do not vanish there.

Theorem 1. *Let $p \equiv 3 \pmod 4$ be prime and $r = 2s+1 \geq 3$. Put $N = (p^r-3)/4$ and $M = (p^{r-2}-3)/4$. Then*

$$\sum_{k=0}^{N}(-1)^k(8k+1)\frac{(1/4)_k^3}{k!^3} \equiv p^2 \sum_{k=0}^{M}(-1)^k(8k+1)\frac{(1/4)_k^3}{k!^3} \pmod{p^{3s}}.$$

Since $3s \geq 2s+1 = r$, this implies the odd branch of Swisher F.3.

The terminating evaluation. We use the following terminating very-well-poised basic hypergeometric evaluation, with $v = 4t+1$:

$$T_t(q^v, q) = T_t(q^{-v}, q) = (-1)^t q^{-2t}[v]_q. \tag{1}$$

Here is an exact general identity from which it follows:

$$\sum_{k=0}^{t}[2mk+1]_q \frac{(q^{1-v}, q^{1+v}, q/c, q; q^m)_k}{(q^{m-v}, q^{m+v}, cq^m, q^m; q^m)_k} c^k q^{(m-2)k} = (c/q)^t \frac{(q^2/c; q^m)_t}{(cq^m; q^m)_t}[v]_q, \quad v = mt+1. \tag{2}$$

To specify the input identity completely, the terminating very-well-poised ${}_6\phi_5$ summation is

$$\sum_{k=0}^{t}\frac{1-AQ^{2k}}{1-A}\frac{(A,B,C,Q^{-t};Q)_k}{(Q, AQ/B, AQ/C, AQ^{t+1};Q)_k}\left(\frac{AQ^{t+1}}{BC}\right)^k = \frac{(AQ, AQ/(BC);Q)_t}{(AQ/B, AQ/C;Q)_t}.$$

Take $Q = q^m$, $A = q$, $B = q^{1+v}$, $C = q/c$, and $v = mt+1$. The argument becomes cq^{m-2}. On the right, reversal of the two finite products with negative powers gives

$$\frac{(q^{m+1};q^m)_t}{(q^{m-v};q^m)_t} = (-1)^t q^{mt(t-1)/2+t}[v]_q,$$

$$(cq^{m-v-1};q^m)_t = (-c)^t q^{-mt(t-1)/2-2t}(q^2/c;q^m)_t,$$

which proves (2) from that standard summation. This is also the specialization used in Guo–Zudilin, *Dwork-type supercongruences through a creative q-microscope*, arXiv:2001.02311, formula (3.23) and the proof of Theorem 3.6. Its substitution $m = 4, c = -q^{-1}$ gives (1). Equation (2) is an exact identity at the two parameter specializations, rather than a congruence.

Lemma 1 (Root blocks)**.** *Suppose $h > 1$, $h \equiv 1 \pmod 4$, and ζ is a primitive hth root of unity. Write $t = (h-1)/4$. For $b \geq 0$ and $0 \leq j < h$,*

$$\lim_{q\to\zeta} c_{bh+j}(a,q) = \begin{cases} (-1)^b \dfrac{(1/4)_b}{b!} c_j(a,\zeta), & 0 \leq j \leq t, \\ 0, & t < j < h. \end{cases} \tag{3}$$

Furthermore $c_{t-j}(a,\zeta) = -c_j(a,\zeta)$ for $0 \leq j \leq t$. Consequently, if $L \bmod h \geq t$, then $T_L(a,q)$ vanishes modulo $\Phi_h(q)$, in the local ring with coefficient field $\mathbb{Q}(a)$.

Proof. Set

$$U_k = (aq, q/a; q^4)_k/(aq^4, q^4/a; q^4)_k,$$
$$Q_k = (q;q^4)_k/(q^4;q^4)_k, \qquad V_k = (-q^2;q^4)_k/(-q^3;q^4)_k.$$

In a complete block of length h, the numerator and denominator of U each run over the same root residues, so their ratio is one. The analogous full-block ratio for V is one because both products equal 2. In Q, the numerator zeros occur at indices $t+uh$ and the denominator zeros at indices $(u+1)h-1$. Their limiting ratio in the uth block is $(4u+1)/(4u+4)$; the products of nonzero factors cancel by permutation of the nontrivial roots. Thus b full blocks contribute $(1/4)_b/b!$. The external $(-1)^k q^k$ contributes $(-1)^b$. A final partial block with $j > t$ has one extra numerator zero. This proves (3).

For completeness, reversing the products in the first short block gives, with all the following expressions evaluated at $q = \zeta$,

$$\frac{U_{t-j}}{U_j} = q^{6j-4t(t+1)}, \qquad \frac{Q_{t-j}}{Q_j} = (-1)^t q^{3j-2t(t+1)}, \qquad \frac{V_{t-j}}{V_j} = q^{j-2t^2-t}.$$

These follow by replacing the reversed factors q^{h-u} by q^{-u}; all denominators here are nonzero for generic a. Also $[8(t-j)+1]_q = -q^{-8j-1}[8j+1]_q$. Combining these identities and the factor $(-1)^{t-2j}q^{t-2j}$ from the external sign and power yields

$$c_{t-j}(a,q) = -q^{-(2t+1)h} c_j(a,q) = -c_j(a,q).$$

The short sum is zero, and (3) proves the assertion for every cutoff whose last block includes this short block. □

Parameter interpolation and the common denominator. Set $n = p^2$, so $n \equiv 1 \pmod 8$, and write

$$N = nM + \frac{3(n-1)}{4}, \qquad \Delta(a,q) = T_N(a,q) - q^{(1-n)/2}[n]_q T_M(a,q^n).$$

For $0 \leq j \leq M$, let $A_j = (4j+1)n$. At $a = q^{\pm A_j}$, the left series terminates at $nj + (n-1)/4 \leq N$, while the right series terminates at j. By (1), the left value is $(-1)^{nj+(n-1)/4} q^{(1-A_j)/2}[A_j]_q$. The right value is $q^{(1-n)/2}[n]_q(-1)^j q^{-2nj}[4j+1]_{q^n}$. They agree since $(n-1)/4$ is even and $[n]_q[4j+1]_{q^n} = [A_j]_q$. Therefore

$$P(a,q) = \prod_{j=0}^{M} (1 - aq^{A_j})(a - q^{A_j})$$

divides the parameter numerator of Δ. Explicitly a common parameter denominator is

$$D(a,q) = (aq^4, q^4/a; q^4)_N.$$

It clears the right denominator too, because each exponent $4n, 8n, \ldots, 4nM$ occurs among $4, 8, \ldots, 4N$. Hence $a^N D\Delta$ is a polynomial in a with coefficients in $\mathbb{Q}(q)$ and is divisible by P in $\mathbb{Q}(q)[a]$. The leading coefficient of P is a signed power of q.

An extra cyclotomic factor survives interpolation. Fix an even $\ell \in \{2, 4, \dots, 2s\}$, and put $h = p^\ell$. Let $R_h = \mathbb{Q}[q, q^{-1}]_{(\Phi_h)}$. For any odd integer d, the ratio $Q_k = (q; q^4)_k/(q^4; q^4)_k$ has nonnegative Φ_d valuation: in each period d its numerator zero occurs at some index $0 \le j_0 < d$, whereas its denominator zero occurs at index $d-1$. The factors $1+q^{4i+3}$ have no zero at an odd-order root. The same facts apply to q^n at a primitive p^ℓ root (at $\ell = 2$ the limit is $q^n = 1$ and Q_k has the finite value $(1/4)_k/k!$). Thus $a^N D\Delta \in R_h[a]$.

At a primitive hth root, the left sum vanishes by Lemma 1, since

$$N \bmod h = \frac{3(h-1)}{4} \ge \frac{h-1}{4}.$$

For $\ell = 2$, the right side vanishes due to $[n]_q$ and the remaining sum is finite for generic a. For $\ell > 2$, its base q^n has order h/n, and $M \bmod (h/n) = 3(h/n-1)/4$, so Lemma 1 again applies. Consequently $a^N D\Delta \in \Phi_h R_h[a]$.

Since P has unit leading coefficient in R_h, polynomial division shows that $H = a^N D\Delta/P$ belongs to $R_h[a]$. Reducing the equality $PH = a^N D\Delta$ modulo Φ_h, the polynomial P is nonzero (indeed its leading coefficient remains a unit). The residue ring is a polynomial ring over a field and has no zero divisors; hence $H \in \Phi_h R_h[a]$. Specializing $a = 1$ gives

$$\operatorname{ord}_{\Phi_h} \Delta(1, q) \ge 1 + \operatorname{ord}_{\Phi_h} P(1, q) - \operatorname{ord}_{\Phi_h} D(1, q). \tag{4}$$

Exact root counts. Put $u = h/n = p^{\ell-2}$. Then $u \equiv 1 \pmod 4$ (including $u = 1$), and

$$\#\{0 \le j \le M : u \mid 4j+1\} = \lfloor M/u \rfloor + 1 = \lfloor N/h \rfloor + 1.$$

Indeed the first solution in a period is $(u-1)/4$, while $M \bmod u = 3(u-1)/4$; for $u = 1$ both residues are zero. The last floor equality follows from $N = nM + 3(n-1)/4$ and $n \cdot 3(u-1)/4 + 3(n-1)/4 = 3(h-1)/4 < h$. Therefore

$$\operatorname{ord}_{\Phi_h} P(1, q) = 2(\lfloor N/h \rfloor + 1), \qquad \operatorname{ord}_{\Phi_h} D(1, q) = 2\lfloor N/h \rfloor.$$

Equation (4) gives at least three factors for every even ℓ, i.e.

$$\operatorname{ord}_{\Phi_{p^{2j}}} \Delta(1, q) \ge 3 \quad (1 \le j \le s). \tag{5}$$

The specialization $q = 1$ has a p-unit denominator. This point is needed in addition to the local multiplicities. At $a = 1$ the Pochhammer portion of each summand is $Q_k^3 V_k$. The preceding zero-count argument shows that after cancellation Q_k has no odd-order cyclotomic denominator. Its remaining denominator factors come from $1 - q^{4i}$, and so are even-order cyclotomic polynomials; their values at 1 are 1 or 2. Likewise every factor of $1 + q^{4i+3}$ has even order and value at 1 equal to 1 or 2. Thus there is a Laurent polynomial denominator $B(q) \in \mathbb{Z}[q, q^{-1}]$ such that $B(1)$ is a power of 2, $A(q) = B(q)\Delta(1, q) \in \mathbb{Z}[q, q^{-1}]$, and B has no odd-order cyclotomic factor. This remains true for the right side after substituting $q \mapsto q^n$ because n is odd: an odd power cannot turn an even-order root into an odd-order root, and the value at $q = 1$ of the substituted denominator is unchanged. By (5), A is divisible over $\mathbb{Q}[q, q^{-1}]$ by $\prod_{j=1}^{s} \Phi_{p^{2j}}(q)^3$. Since these factors are monic, Gauss's lemma gives the same divisibility over $\mathbb{Z}[q, q^{-1}]$. Their values at $q = 1$ are all p. Finally,

$$T_L(1, 1) = \sum_{k=0}^{L} (-1)^k (8k+1) \frac{(1/4)_k^3}{k!^3}, \qquad q^{(1-n)/2}[n]_q\big|_{q=1} = p^2.$$

Since $B(1)$ is a p-adic unit, (5) proves Theorem 1.